CHE301 – MIDTERM MERGED PPT (LECTURE 01 TO 64) Analytical Chemistry

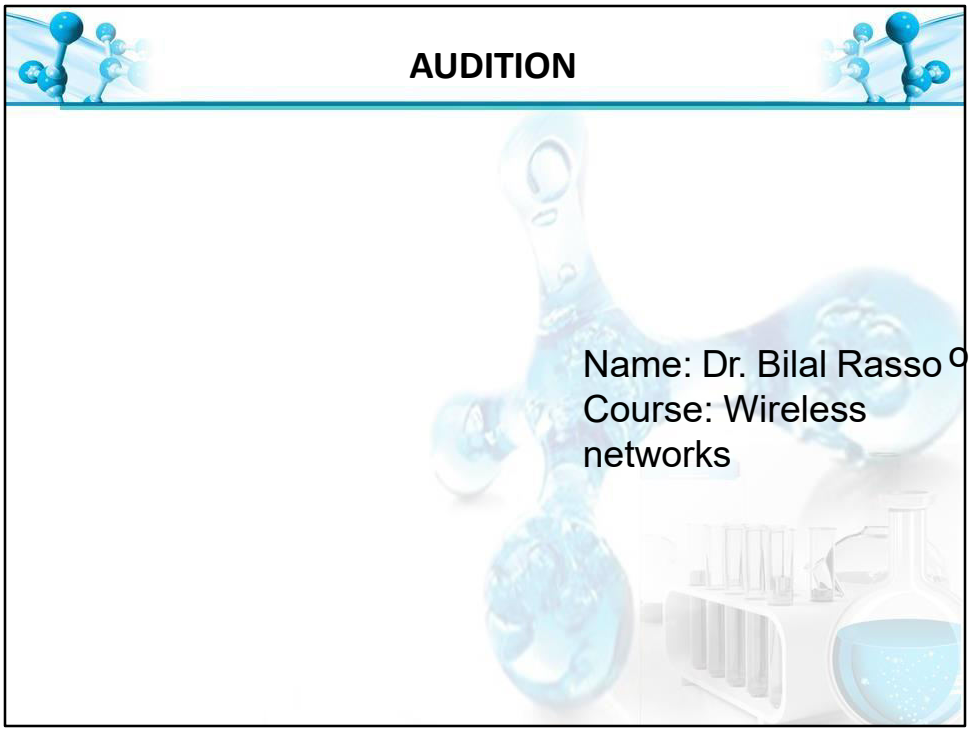
Prepared By:

All About Biotech

AARISH ALI & TEAM (+96569002641)

Note:

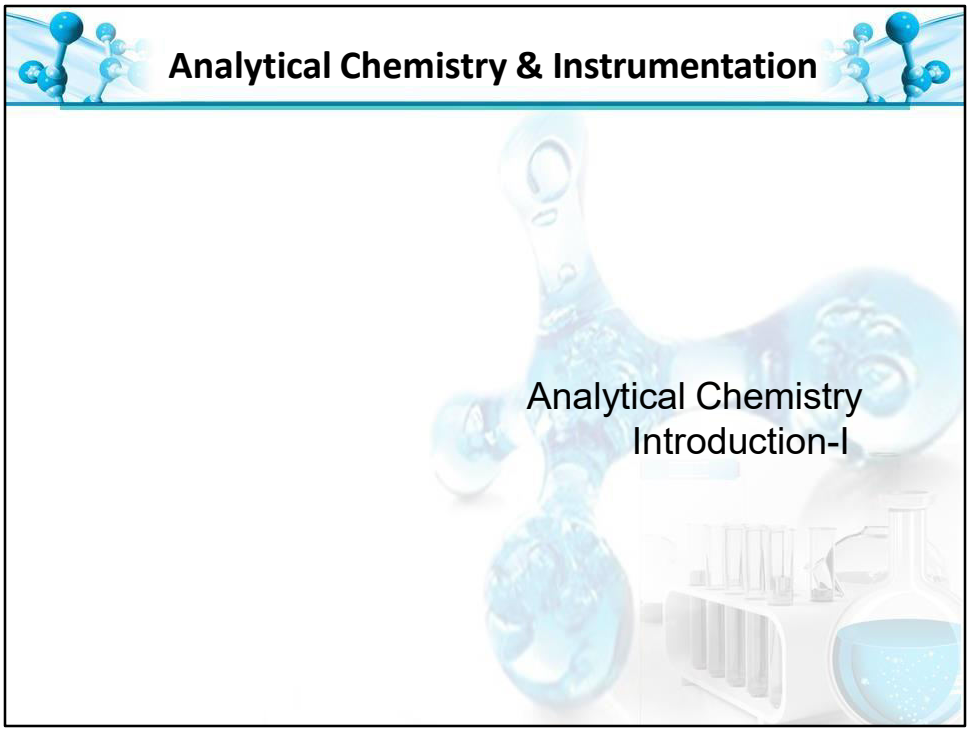
Students must follow the course announcement for midterm syllabus & any up-gradation in the course.



In the name of Allah, the most beneficent, the most merciful
We are starting this new course on Analytical Chemistry and Instrumentation.

In chemistry whatever you are working on, synthesizing, manufacturing, you must have to come across the evaluation of your product or results.
That's where you have to use your analytical skills.
Most of the time even at the beginning we need a reverse engineering, means first you have to analyze a product before starting your own work.
If you fail to determine the attribute of the product, it is impossible to create a new thing.
If you fail to evaluate your product, it is impossible to maintain the quality or specs.
If you fail to maintain the quality, that simply spoil your product or technique.

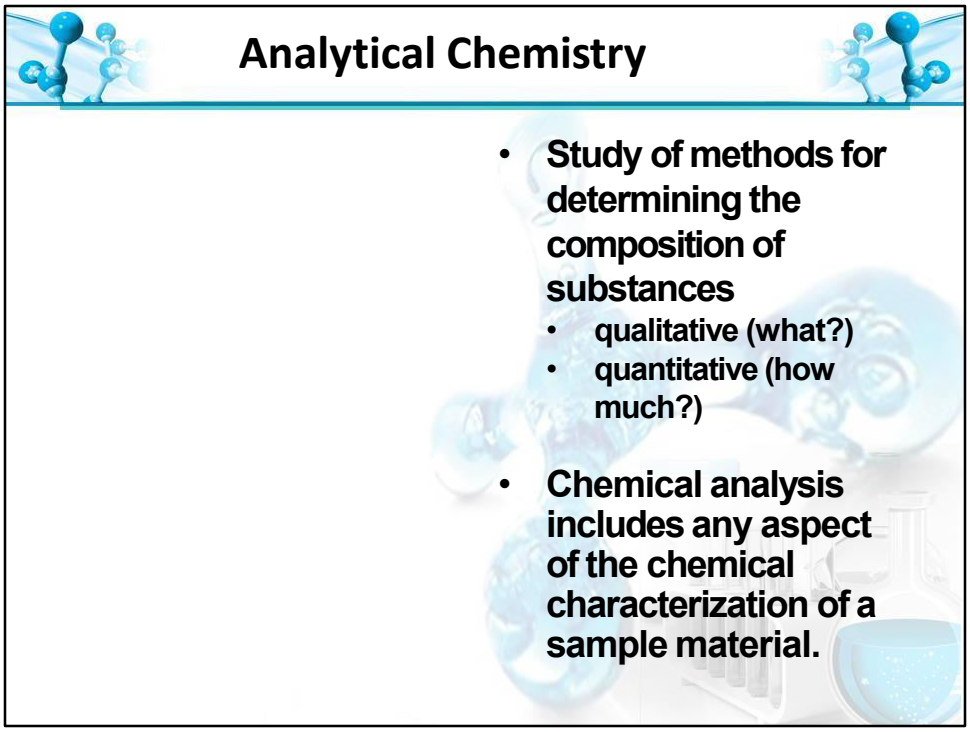
In any analytical work we emphasize on three things, i,.e. Analyst, Equipment and Methods
And to get appropriate results we must have Trained Analyst, Calibrated Equipment and Validated test methods



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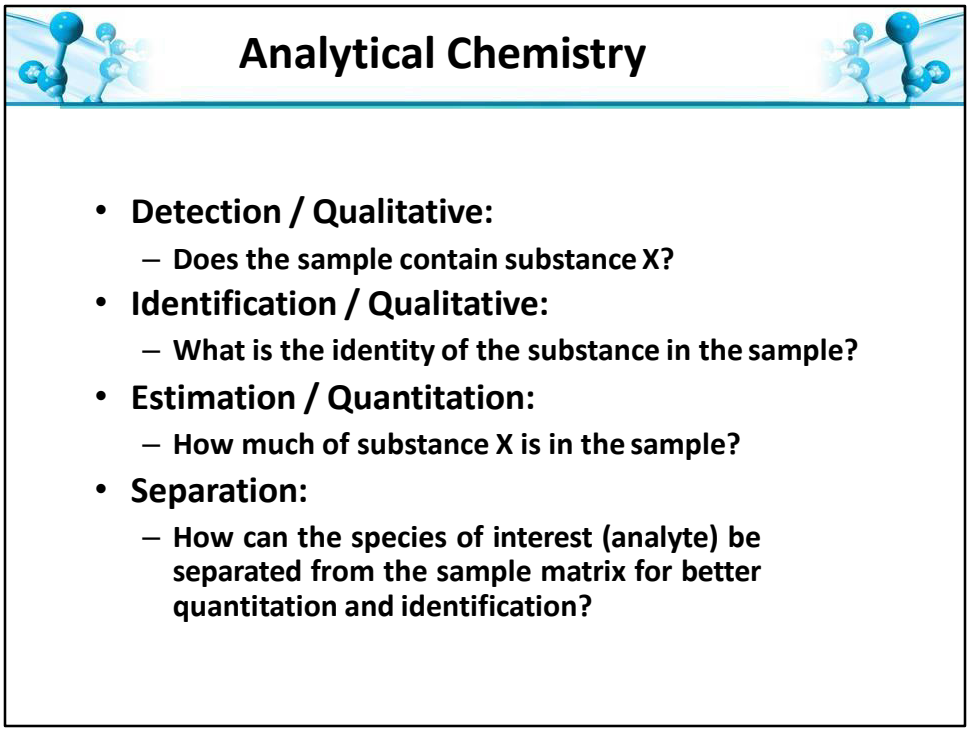
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Analytical Chemistry is basically study of testing methods for determining composition of substances. This determination may be limited to qualitative results or quantitative estimation. Qualitative means what substances are present in composition.
Whereas the percentage of components in a mixture is called quantitative estimation.

Chemical analysis includes any aspect of chemical characterization, starting from collection of sample, extraction, sample preparation, instrumental analysis to statistical evaluation of collected data.



Qualitative analysis is detection and may be identification.
Does your sample contain pesticide residues?
Does your sample contain elements (may be essential of non-essential, major or toxic elements)
Does your sample bearing load of microbes?

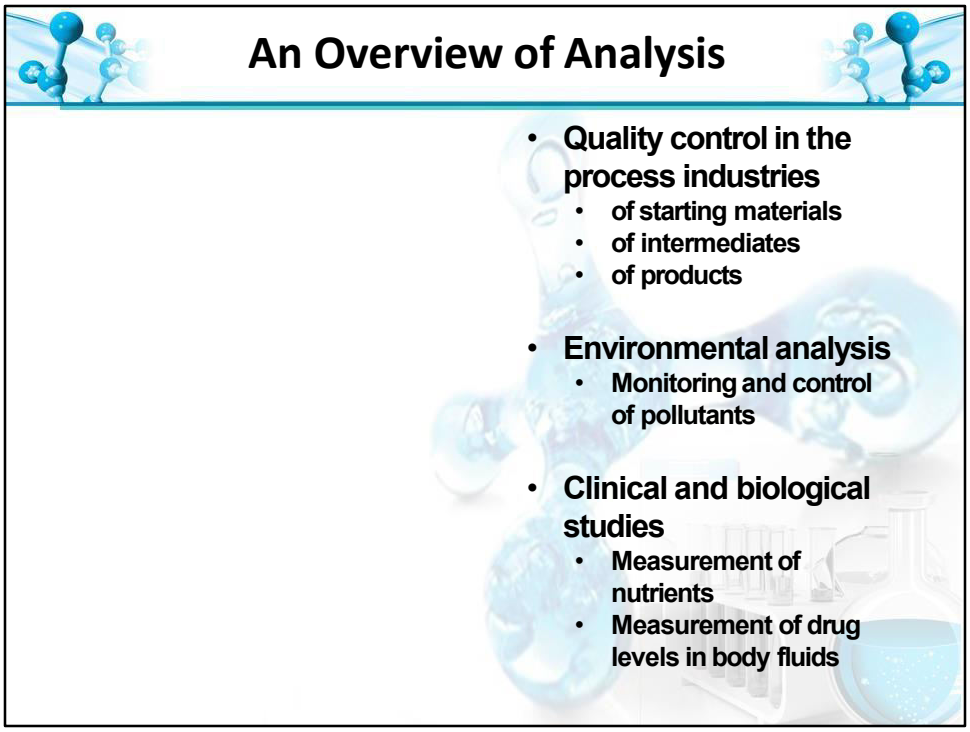
Identification of a perticular substance in presence of other related substances may also called qualitative analysis. For example
Does your sample contain a particular pesticide or not?
Does your sample contain calcium or magnesium?
Does your sample contain pathogenic bacteria?

Estimation or quantification or quantitation means to find out amount. For example in previous examples
If we quantify pesticide residues, metal concentration or number colony forming units in a contaminated sample is included.

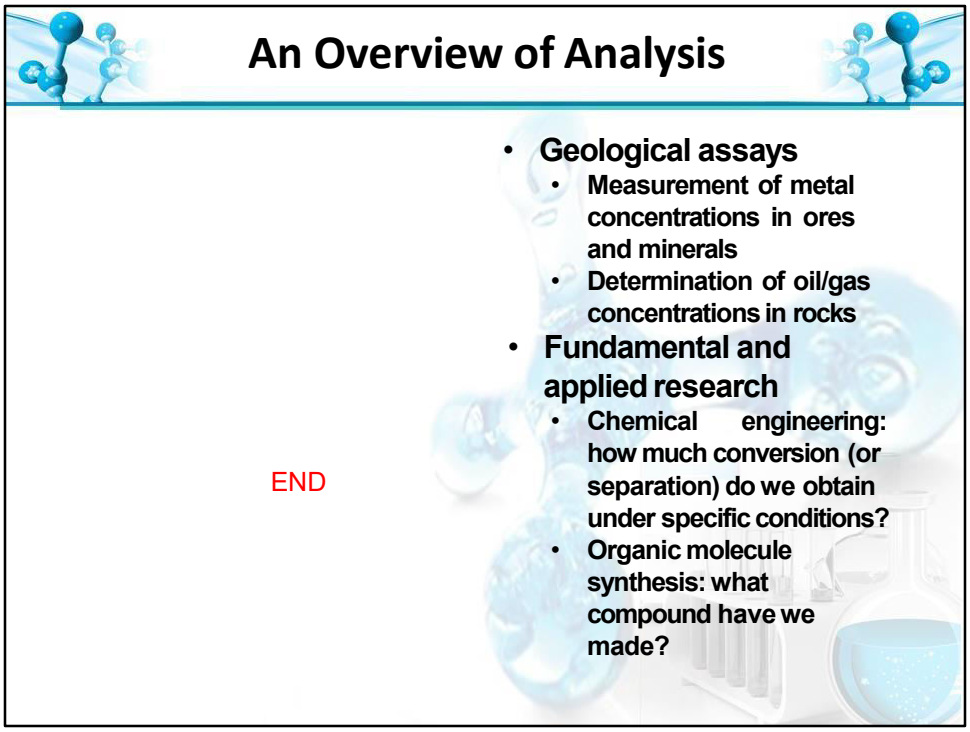
In analytical chemistry another important feature is to isolate or separate species of interest that are called analytes.
The sample from which we separate these analytes is called matrix.
For example we can estimate amount of pesticides in water, soil or vegetables.
Water, soil and vegetable are our matrix while pesticide is an analyte.

Similarly from one matrix we could estimate concentration of different analytes.

Actually the texture, state of matrix vary and extraction of particular analytes also changed.
Water can be used as such for analysis of lead or mercury concentration as such, but we have to digest sample of soil in water and have to use wet digestion using HNO_3 to extract out available metals in water before analysing our sample.
We can use organic solvents and techniques like filteration, distillation, solid phase and solvent extraction before start analysis.

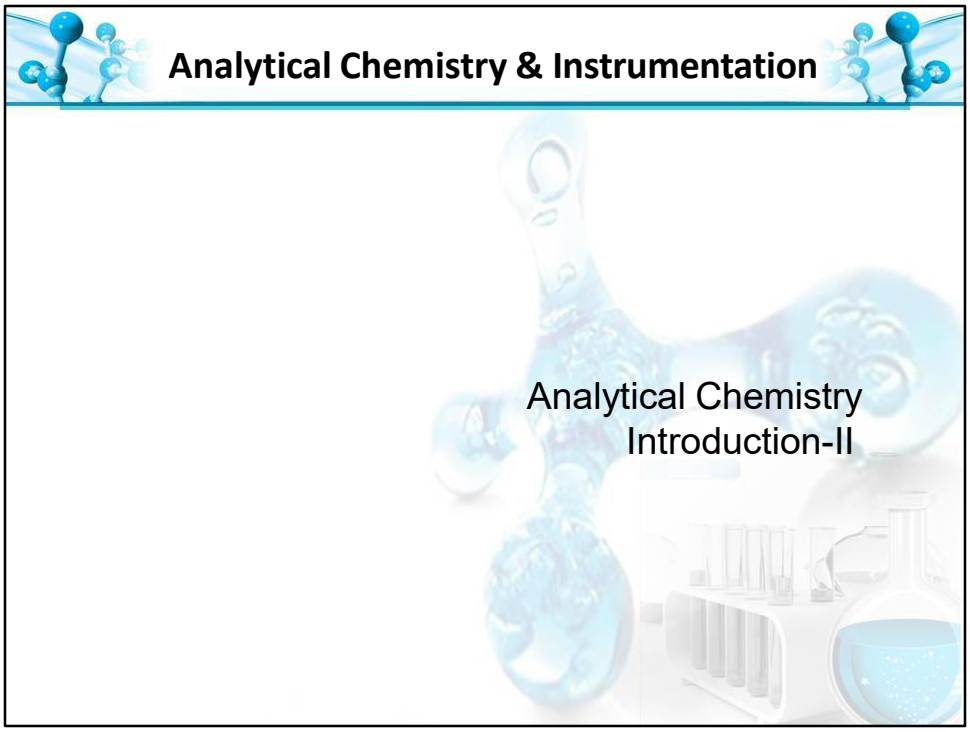


Here is an overview of testing and analysis requirement in different sectors.
In process industries who are involved in manufacturing of food, pharmaceutical, polymer and other materials. We have to analyze raw material, intermediate quality control checks and finished goods.
In environmental analysis the polutents like effluent or waste water, effluent gases or stack emmissions, soil and water resources may be monitored to apply controls.
In clinical side body fluids including blood, urins, mucus and body tissues may be analysed for measurement of microbial load, drug or other attributes and certain enzymatic levels may be assessed.



In geology we check the components of soil and earth. Therefore measurement of metal in ores oil/gas concentration in rocks may be checked.
We use these analytical techniques in academic research. In Organic synthesis to learn the chemistry of synthesized product. In inorganic chemistry to estimate metal concentration in a sample. Separation techniques used to obtain certain substances etc.

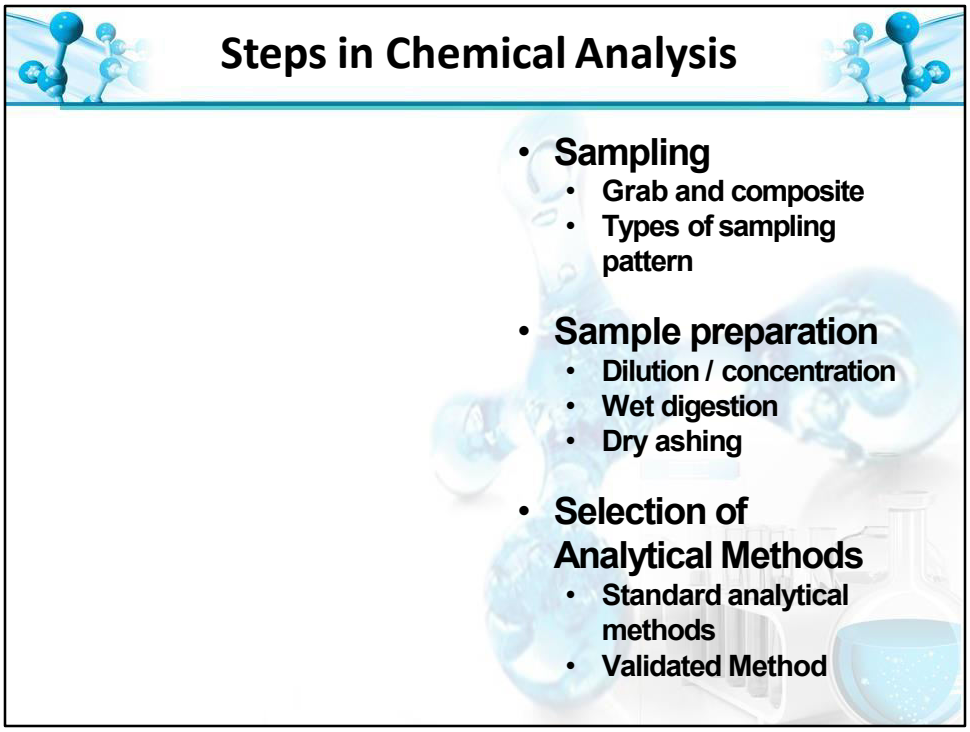
In next lecture we will learn about steps for analyzing a sample.



In this module we will discuss about steps in analysis.
We must understand here the two terms; matrix and analyte

An **analyte**, **component** (in clinical chemistry) or **chemical species**, is a substance or chemical constituent that is of interest in an analytical procedure.

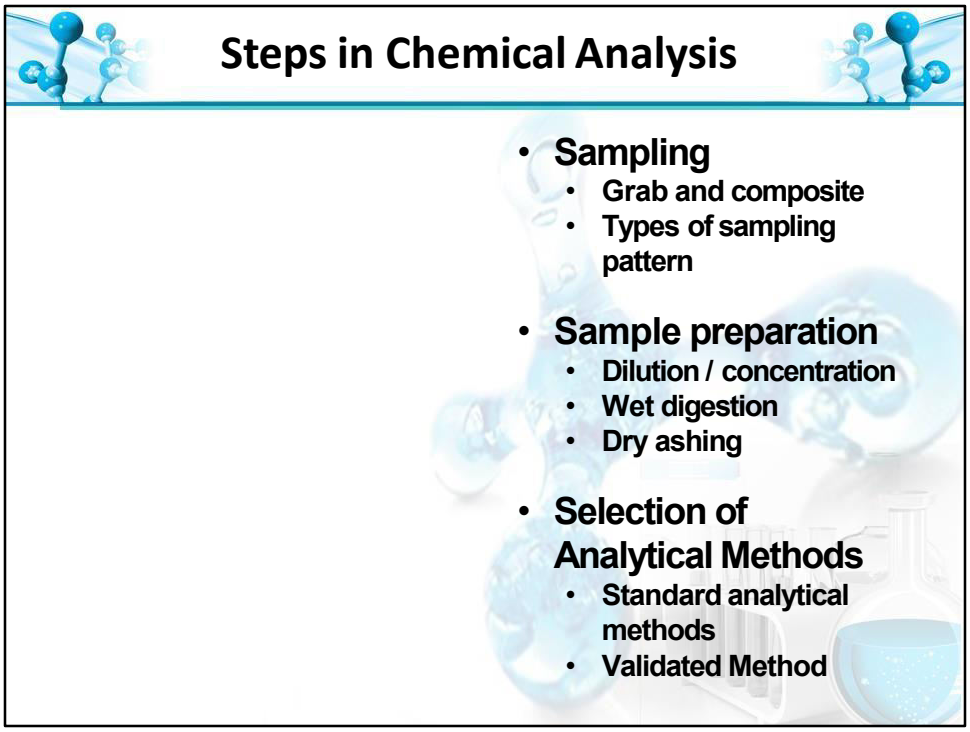
While in chemical analysis, **matrix** refers to the components of a sample other than the analyte of interest. The matrix can have a considerable effect on the way the analysis is conducted and the quality of the results obtained; such effects are called matrix effects.



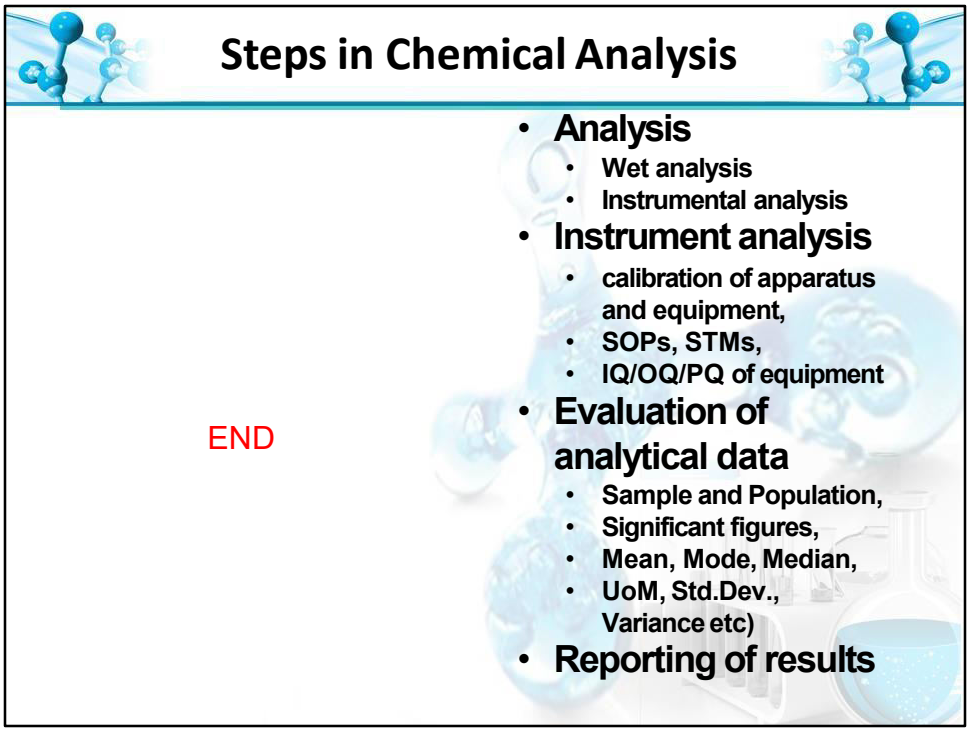
First step in analysis is Sampling. The results of your analysis simply depend upon the sampling. Sampling technique, sampling type and sampling pattern depend upon the matrix, location, variability of constituents, bulk of bags or product etc.
Sample may collect at some particular time and location are called grab sample or many grab samples may combine to form composites.

From 100 of Kg, 1000 of litters or acres of land we collect the sample in grams. And to analyse we have to finalize just 2-3 gram of sample, or maximum 100 mL of liquid. Some time sample need dilution while some time we have to concentrate a sample. To analyse we have to digest or extract our analyte from the sample. This is called sample preparation.

Then we have to select an analytical method, that may be a standard method or a non-standard laboratory developed validated method. Selection of method depend upon variability of constituents in matrix, concentration of analyte in the sample, and availability of proper instrumentation.



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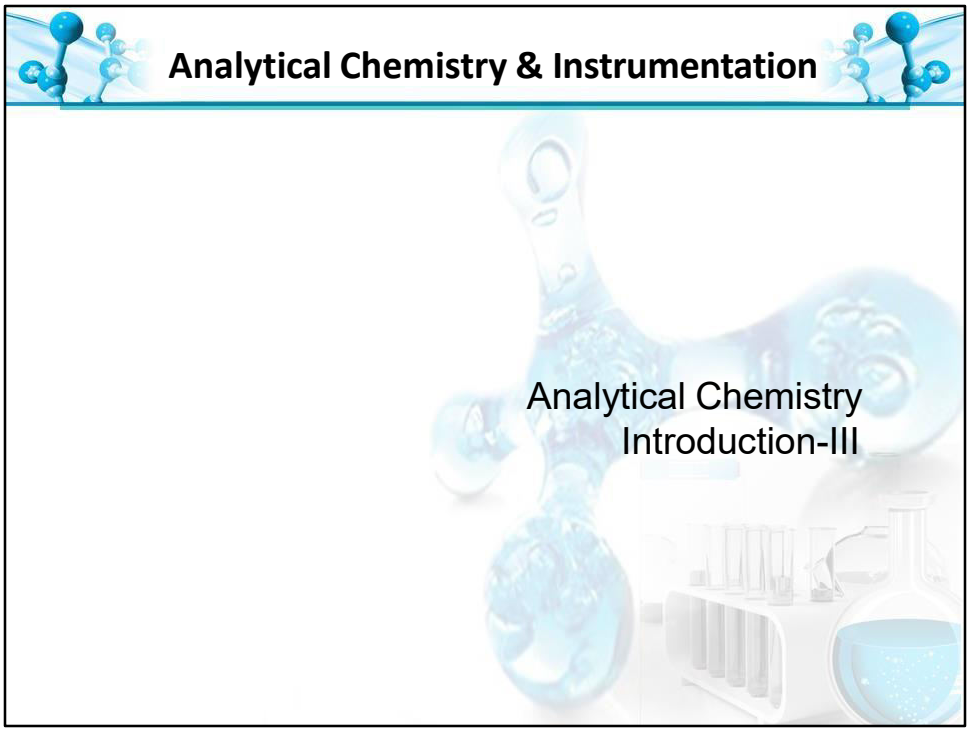


We may use an instrumental method of analysis or wet analysis. Certain wet analysis methods are acid base titration, redox titration, complexometric titration or gravemetric analysis methods.

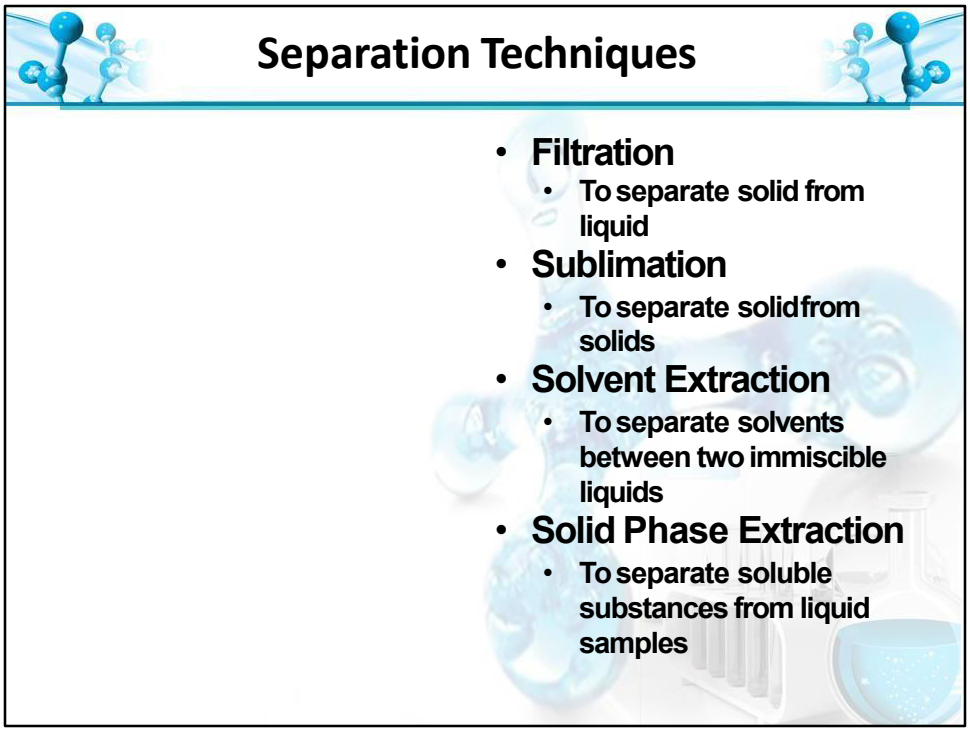
If we choose instrumental method, the instrument should be calibrated. Should follow proper standard operating procedure and standard testingmethods. Instrument should be installed, operated and performed according to standard specifications.

And the last but not least is statistical evaluation of analytical data. We will learn in future modules about sample and population of analytical data. Significance of significant figures. Calculation of mean, median and mode of available data and measurement of uncertainty, standard deviation and variance etc.

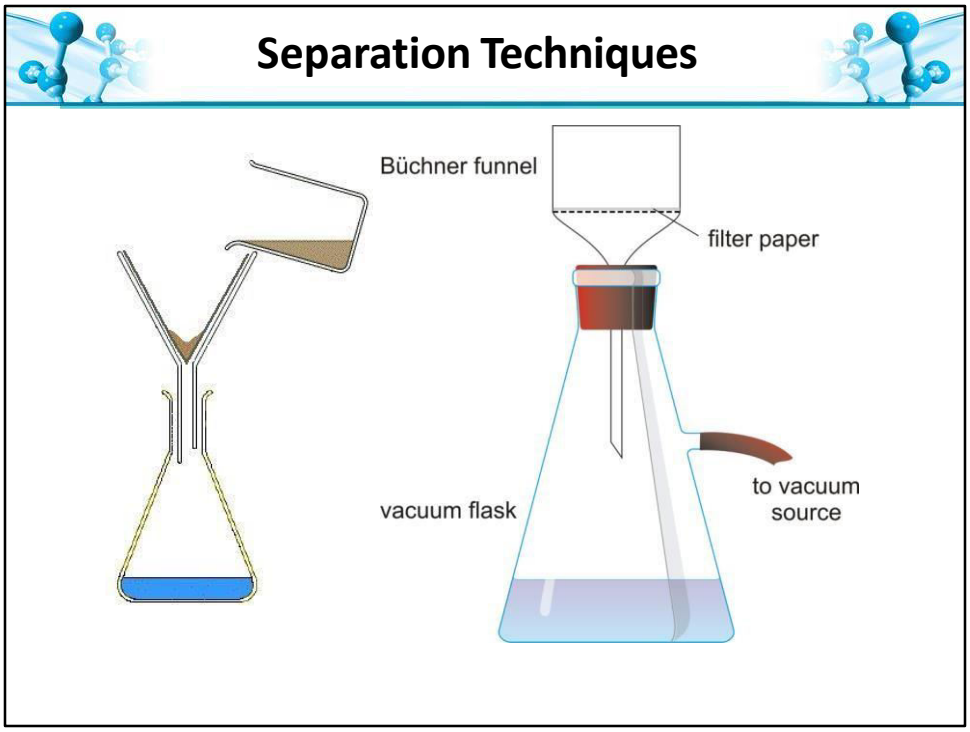
Finally we report our analytical results along with standard specifications, using proper units of measurements and all related informations or logos on report.



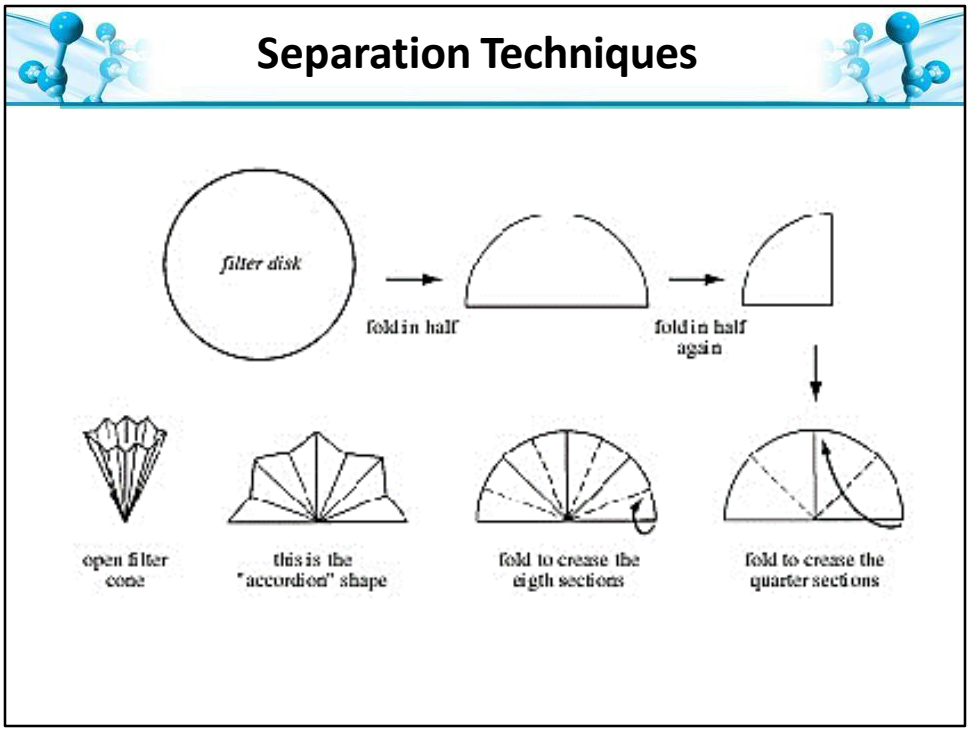
In current module we will discuss about separation techniques; a brief introduction. As a part of sample preparation we have to separate our analyte in first phase. Our matrix may be gas, solid, liquid or semi solid. Our analyte may be soluble or suspended in matrix. First we have to start with separation or semi purification.



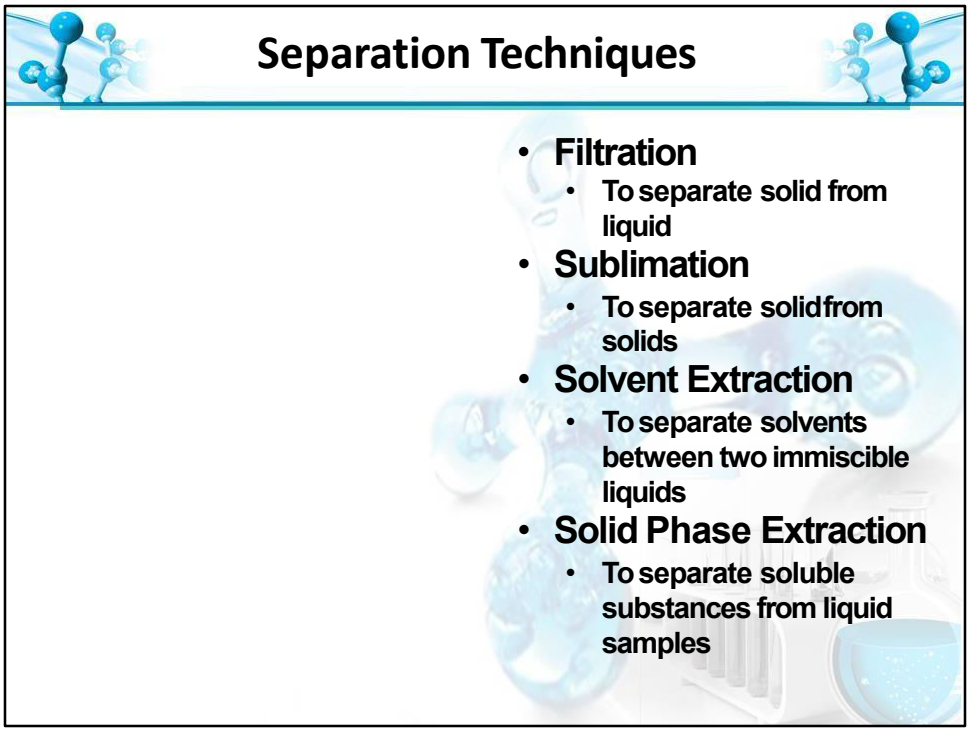
Filtration is very basic analytical technique that is used to separate solid insoluble particles from liquid.
Sometime matrix is in suspension form and we have to coagulate these suspended material to separate.



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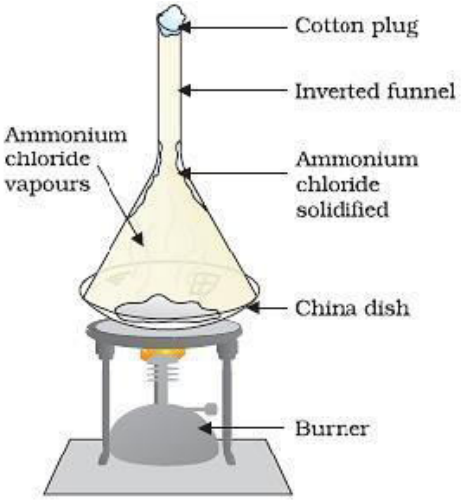


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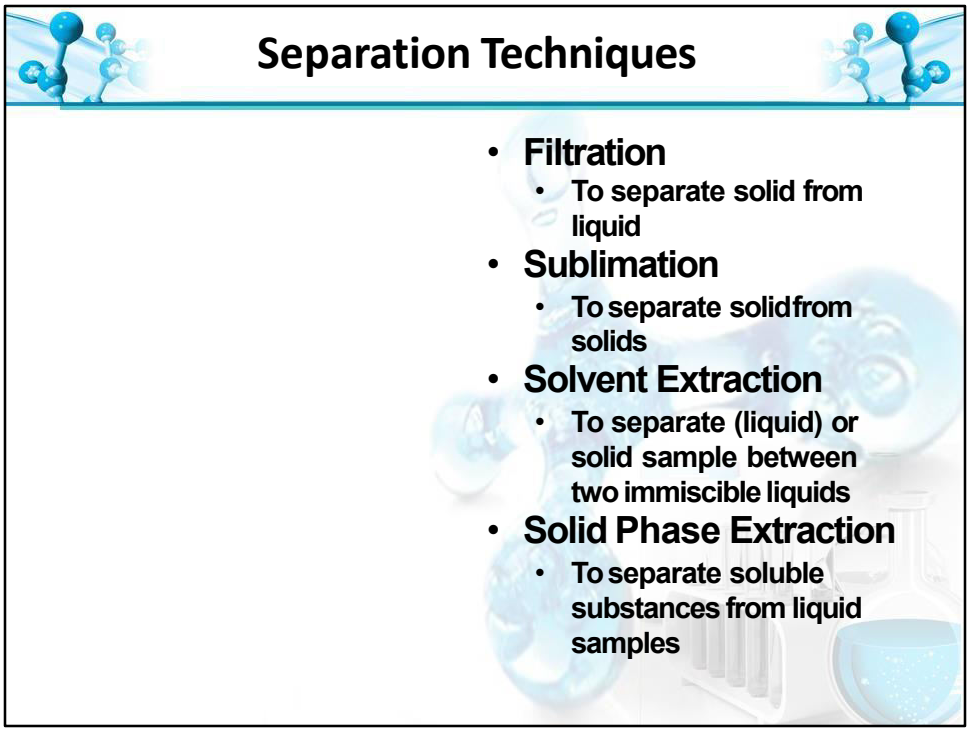


To separate solid from solid sublimation can be used as an analytical technique.

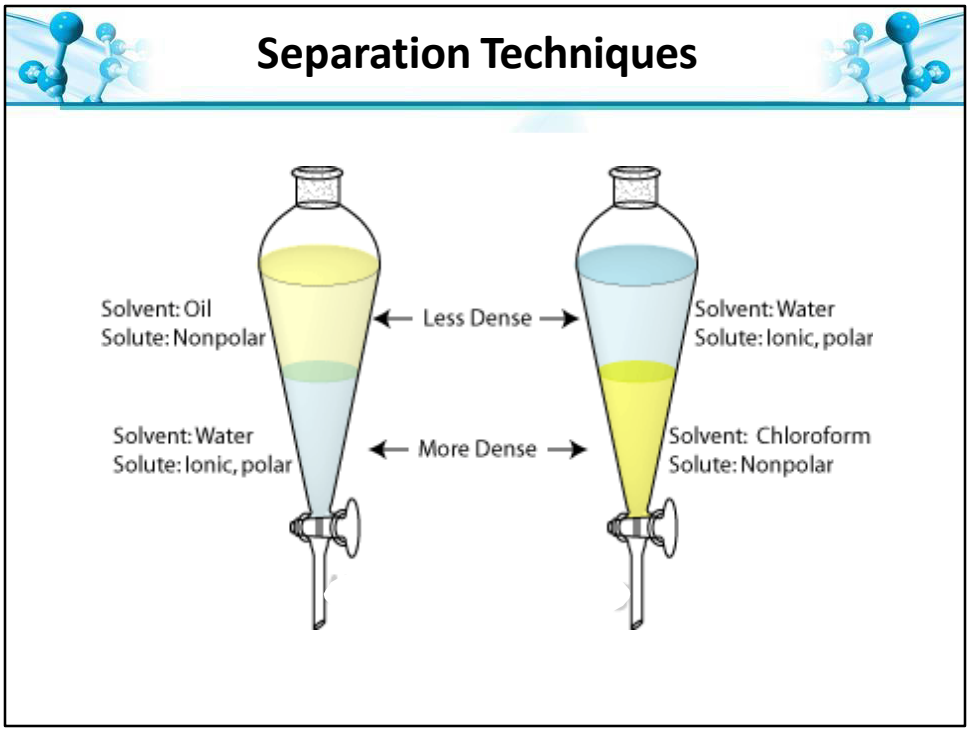
Separation Techniques



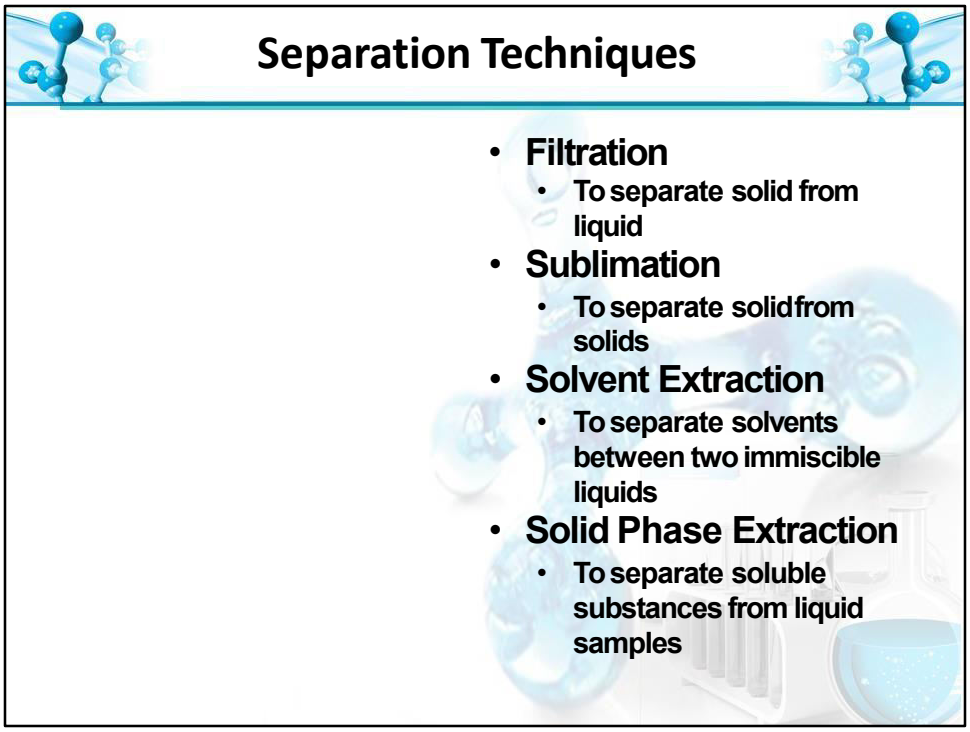
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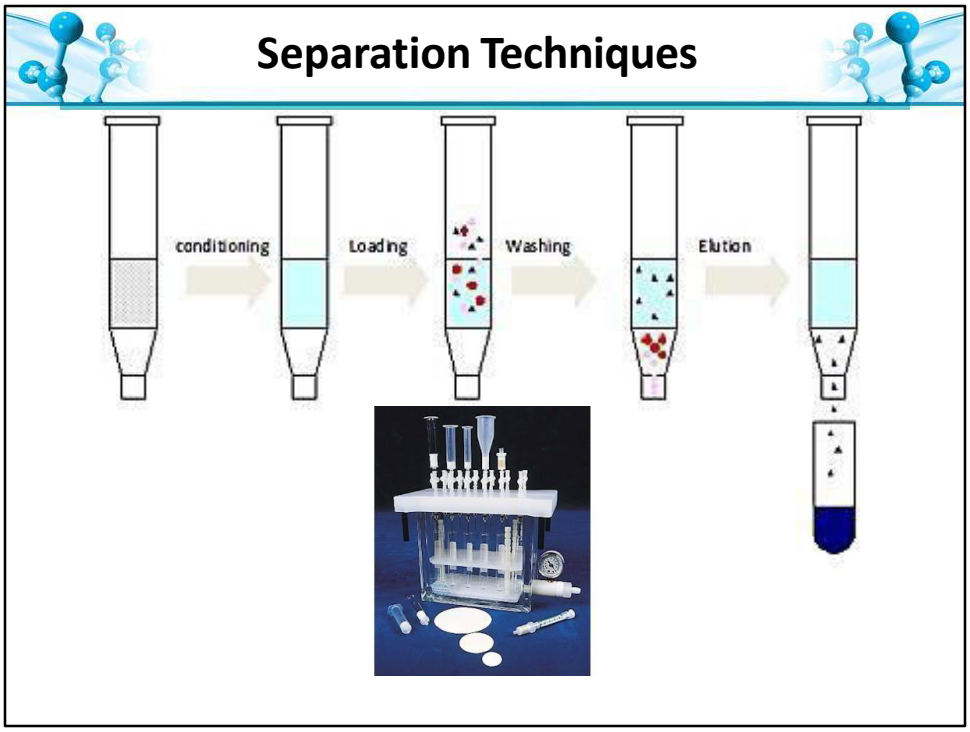
To separate liquid or solid sample between two immiscible liquids is called solvent extraction.



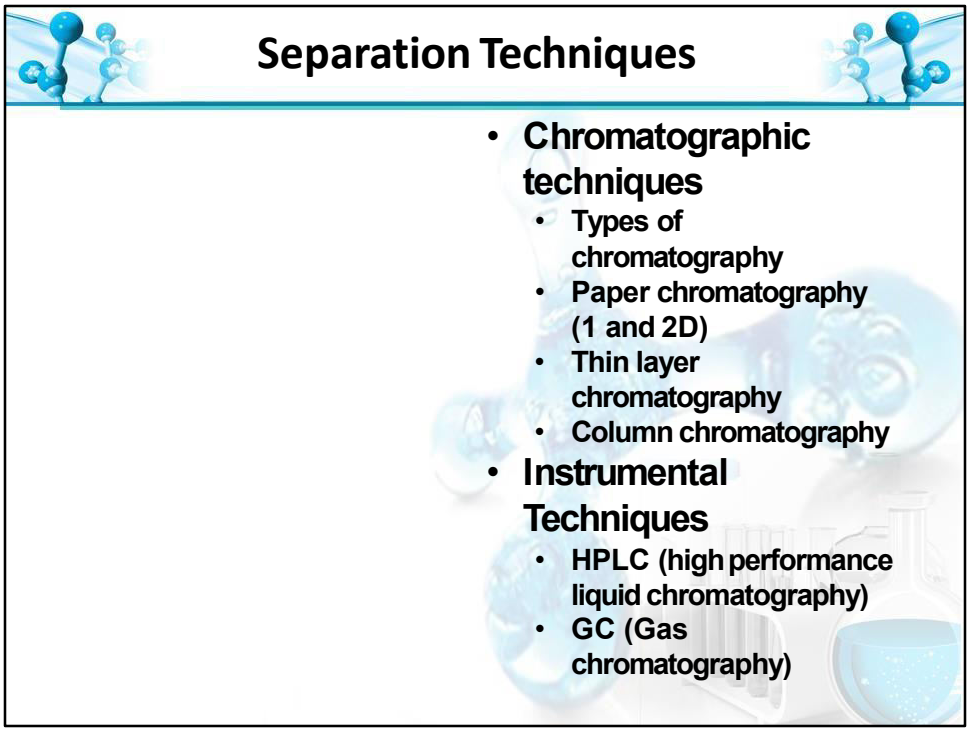
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To separate soluble substances (which are in very low concentration) in presence of other interfering compounds we can use solid phase extraction.

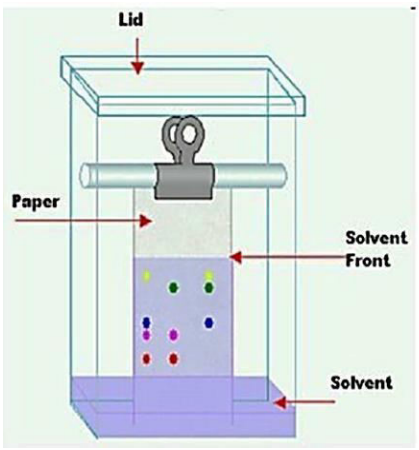


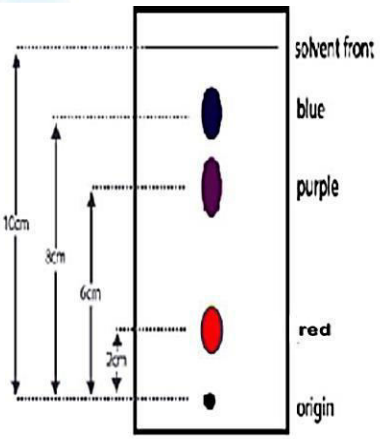
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Different chromatographic techniques may use to separate the analyte.
On basis of stationary and mobile phase we divide them into many types.

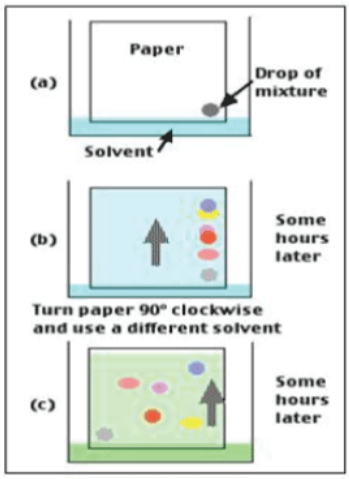
Separation Techniques





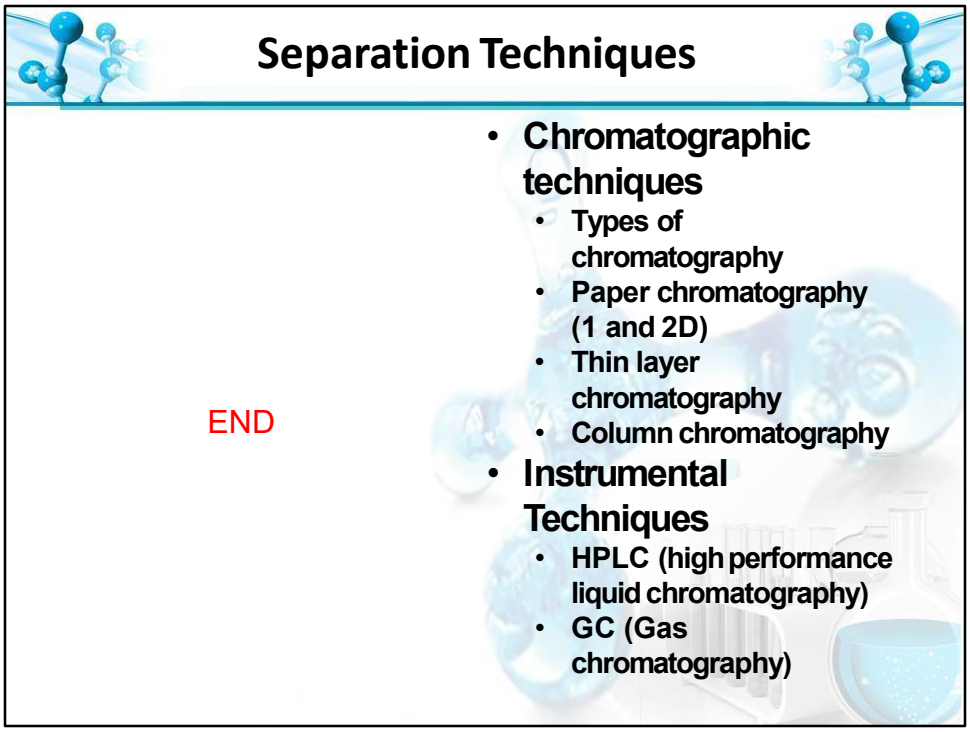
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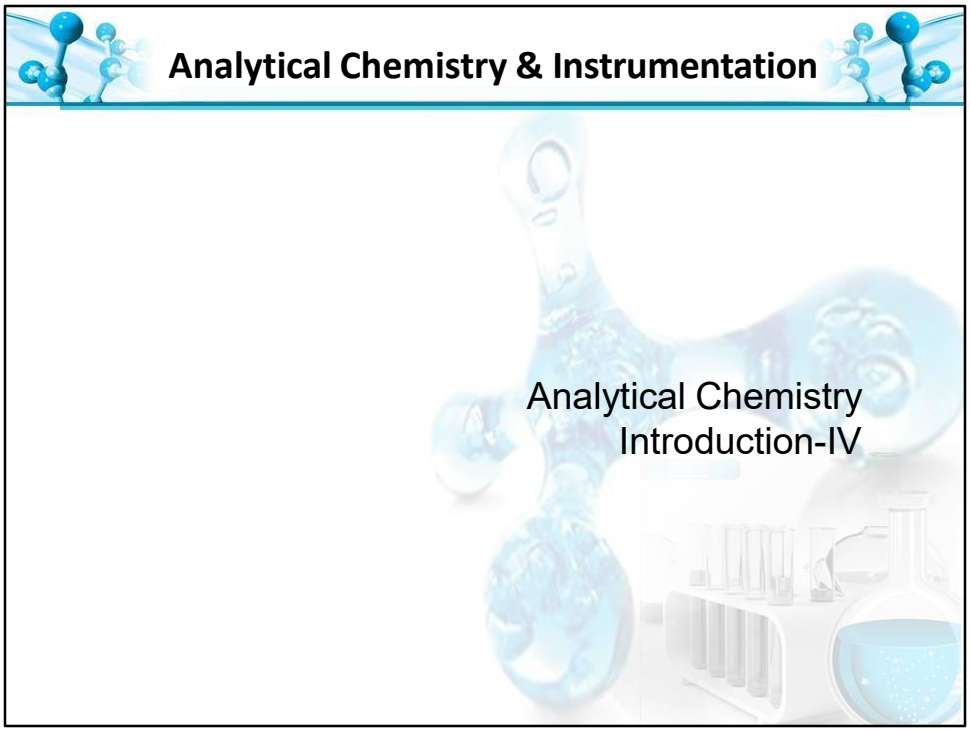




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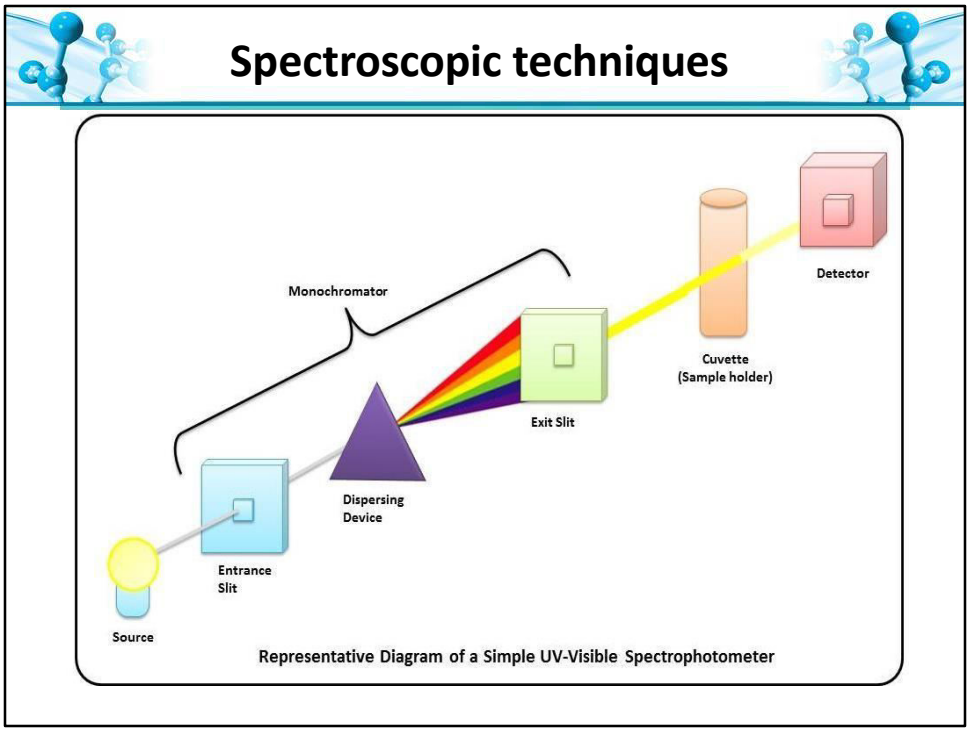


Now we will take a tour to some spectroscopic techniques.

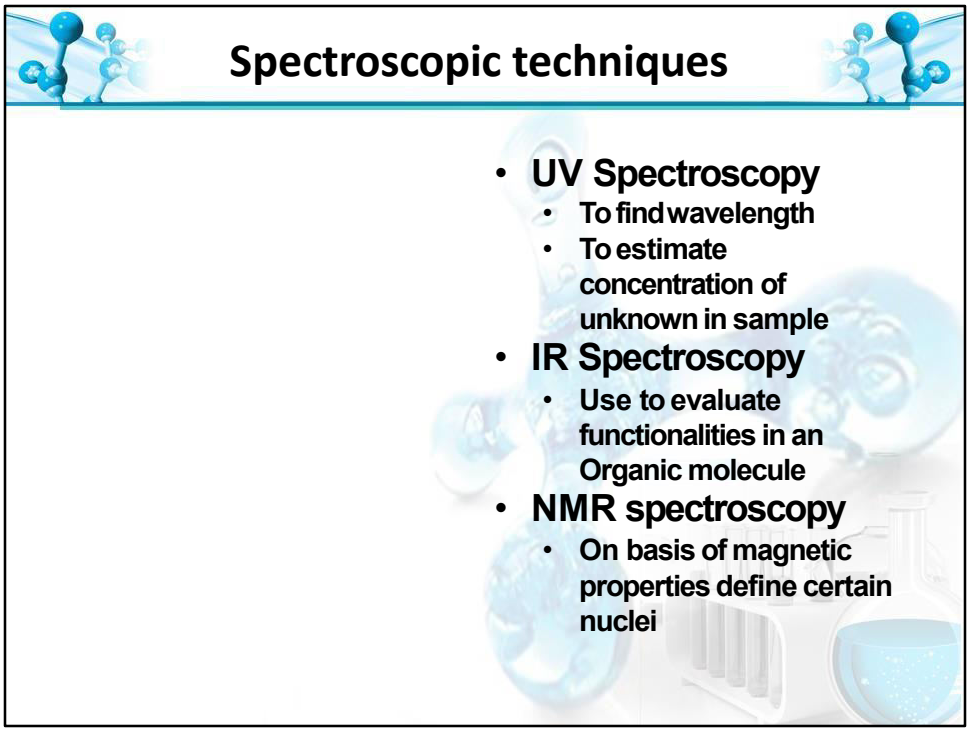
Spectroscopic techniques

$$A = log_{10} \frac{I_o}{I} = \varepsilon lc$$

All the spectroscopic techniques use certain kind of electromagnetic radiations and are based on Beer Lambert law. These are basically two laws; one connected to absorbance with Transmittance; the lambert law and second discuss relation of absorbance with concentration of sample; that is Beer's law.

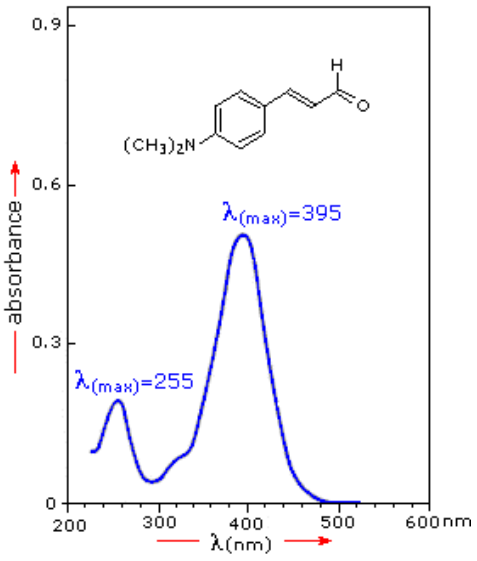


A light beam of particular wavelength when passed through a sample decreased in intensity because it either absorb or distorted and don't reach to the detectors. The particular part of the radiation that absorbed reflect certain characteristics of our sample and molecule.



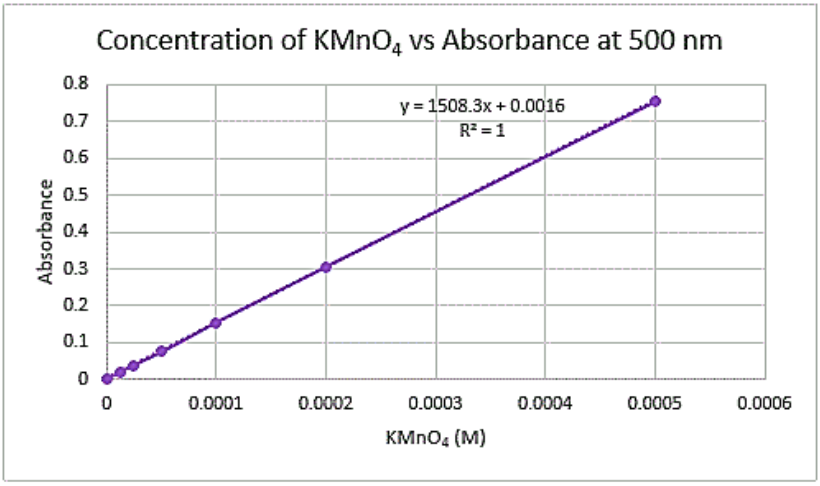
There are different spectroscopic techniques available in market to solve our different analytical problems. The most widely used technique is UV Spectroscopy. That is basic instrument found in food and drug analysis laboratories.

Spectroscopic techniques

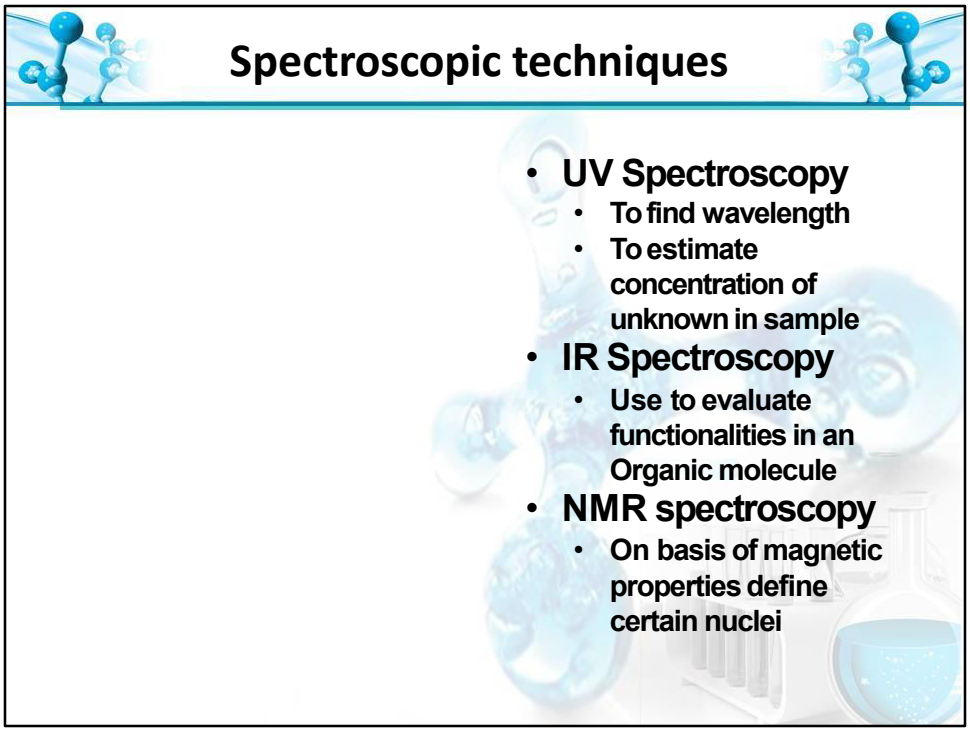


We use this technique for either finding out wavelength maximum of any molecule or use it for evaluation of concentration. N,N-dimethyl cinnamaldehyde

Spectroscopic techniques

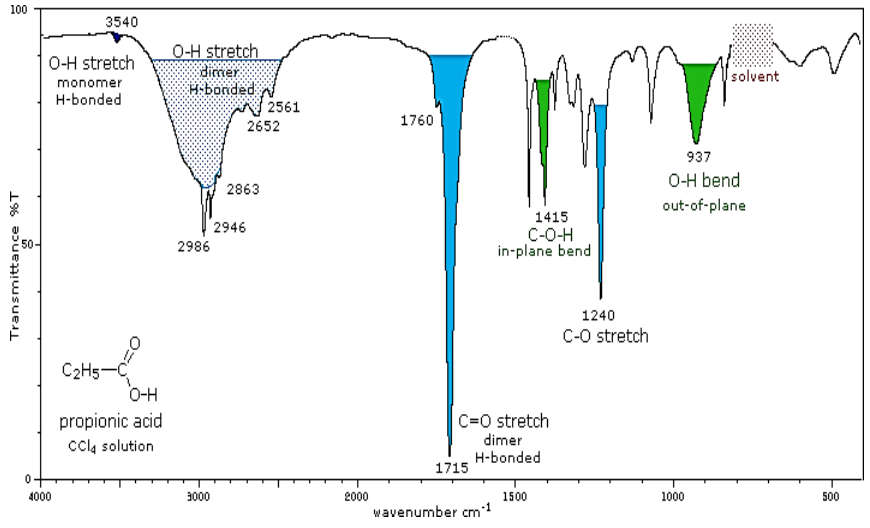


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Another spectroscopic technique is IR Spec. We

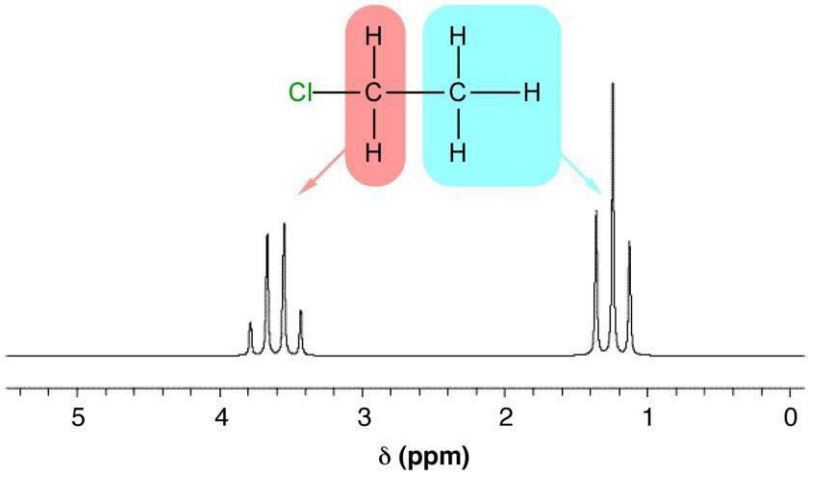
Spectroscopic techniques



Spectroscopic techniques

- **UV Spectroscopy**
  - To find wavelength
  - To estimate concentration of unknown in sample
- **IR Spectroscopy**
  - Use to evaluate functionalities in an Organic molecule
- **NMR spectroscopy**
  - On basis of magnetic properties define certain nuclei

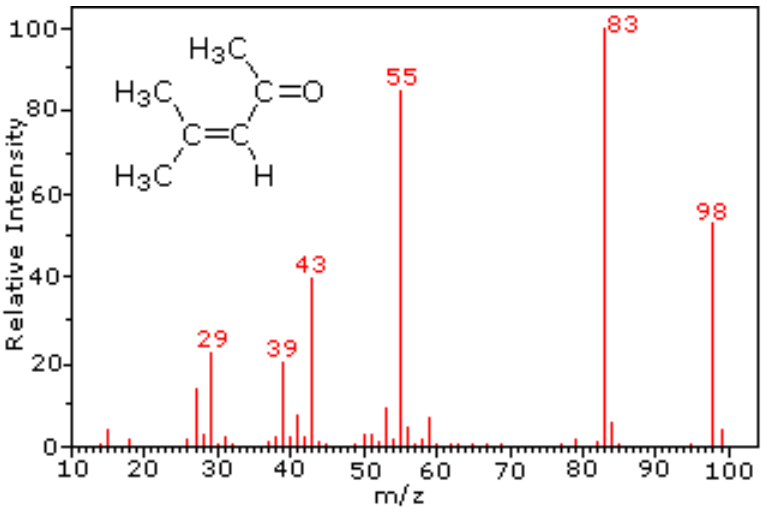
Spectroscopic techniques



Spectroscopic techniques

- Mass Spectrometry
  - Used to evaluate mass of separated molecules
  - Used as a detector
- Atomic Absorption spectroscopy
  - FAAS, GFAAS
- Atomic flouresence spectroscopy
  - Evaluate certain organic molecules
- Atomic Emission spectroscopic
  - ICP OES, ICP MS

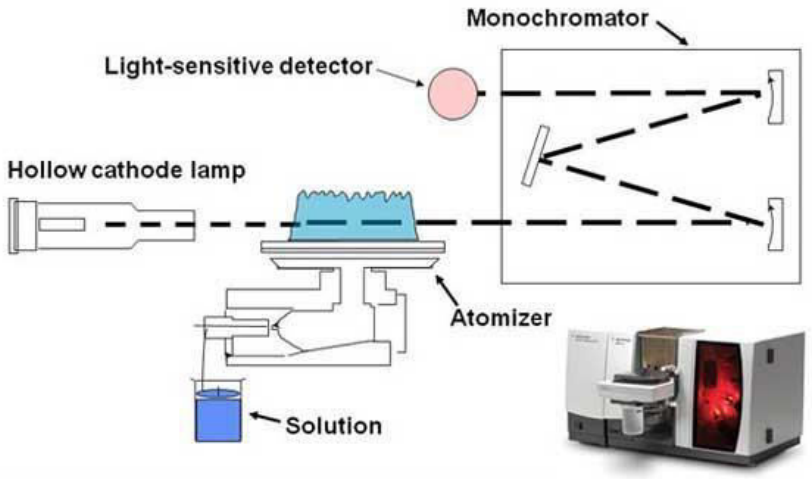
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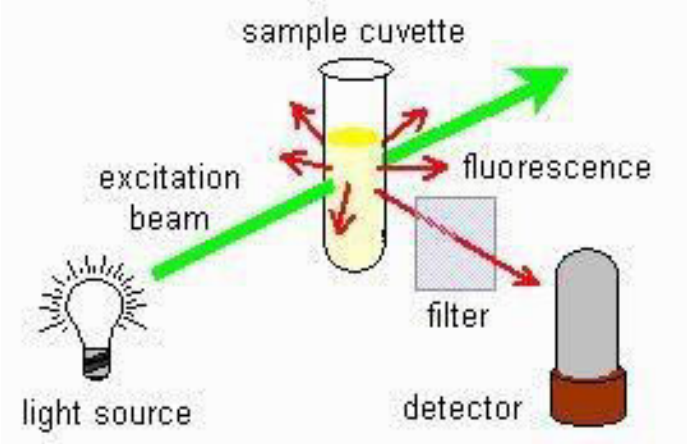
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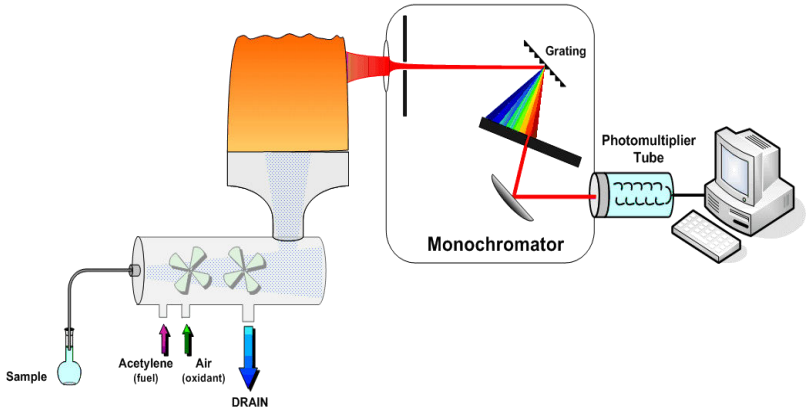


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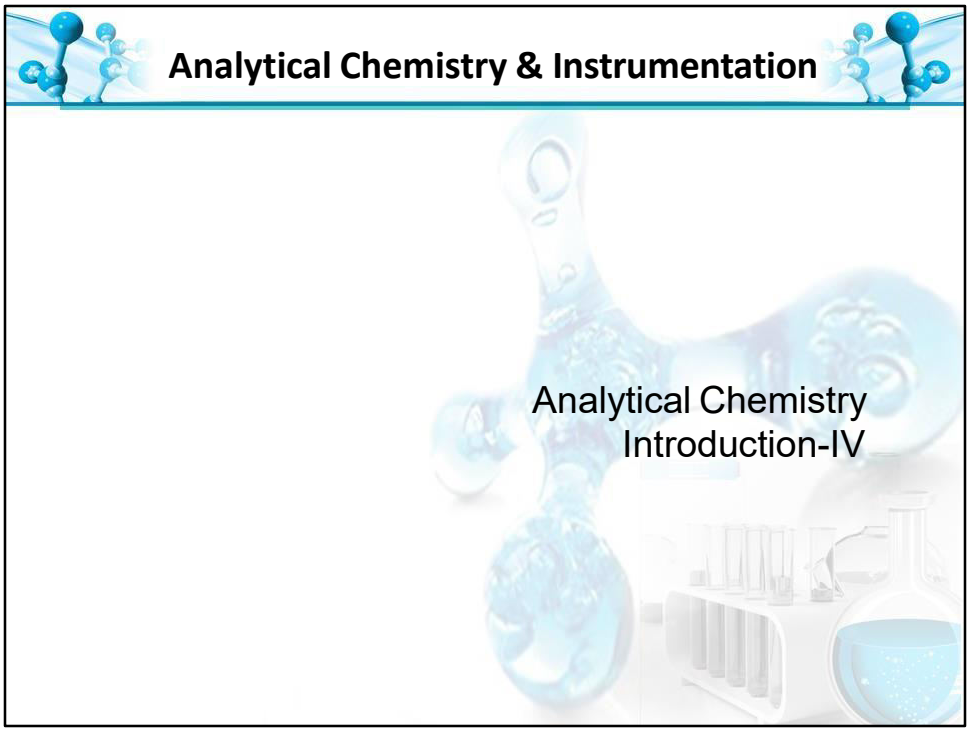
Atomic Emission Spectroscopy



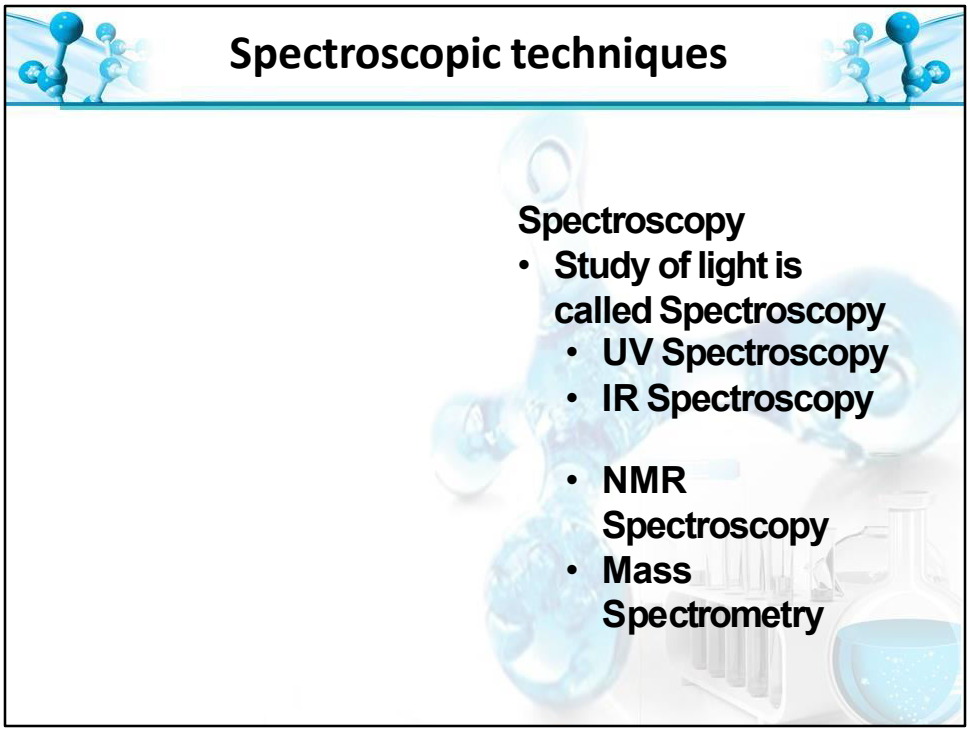
Spectroscopic techniques

END

- Mass Spectrometry
  - Used to evaluate mass of separated molecules
  - Used as a detector
- Atomic Absorption spectroscopy
  - FAAS, GFAAS
- Atomic fluorescence spectroscopy
  - Evaluate certain organic molecules
- Atomic Emission spectroscopic
  - ICP OES, ICP MS



Now we will take a tour to some spectroscopic techniques.



There are different spectroscopic techniques available in market to solve our different analytical problems. The most widely used technique is UV Spectroscopy. That is basic instrument found in food and drug analysis laboratories.

Spectroscopic techniques

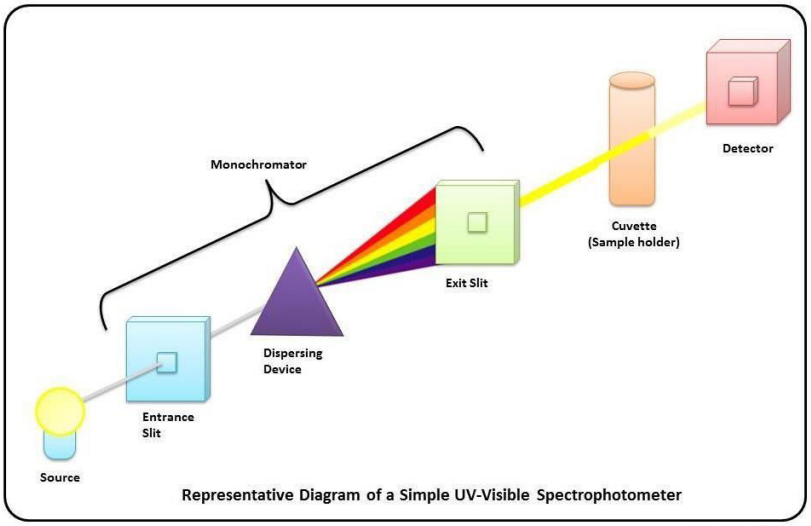
Beer Lambert Law

$$A = log_{10} \frac{I_o}{I} = \varepsilon lc$$

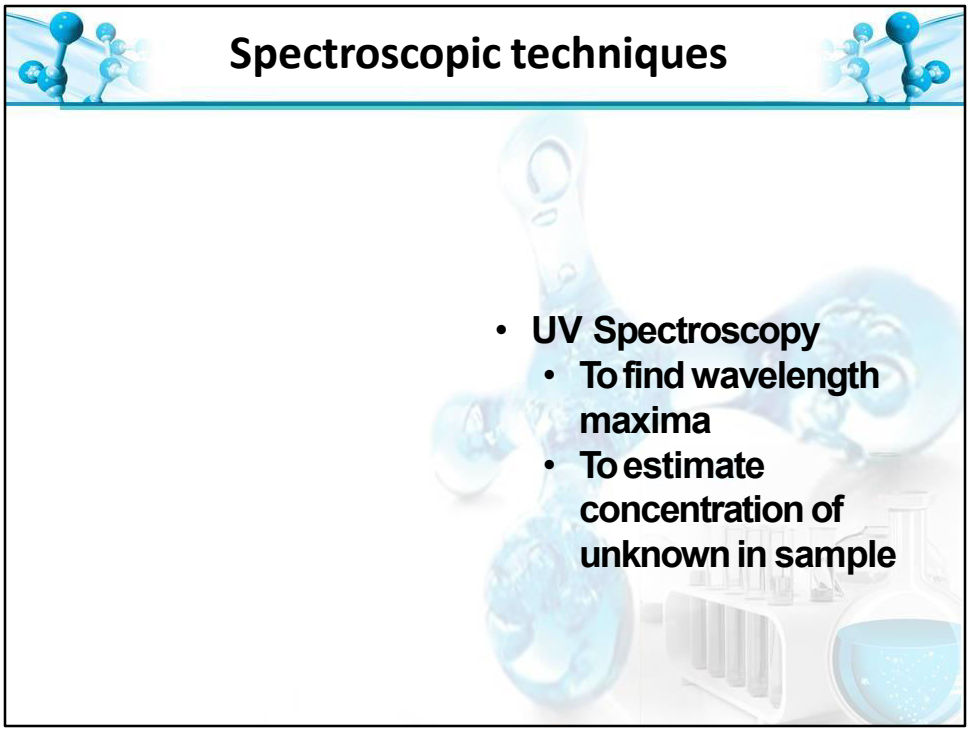
I_0 = Incident light
I = Transmitted light
l = path length
C = Concentration
ε = Molar Absorptivity Constant

All the spectroscopic techniques use certain kind of electromagnetic radiations and are based on Beer Lambert law. These are basically two laws; one connected to absorbance with Transmittance; the lambert law and second discuss relation of absorbance with concentration of sample; that is Beer's law.

Spectroscopic techniques

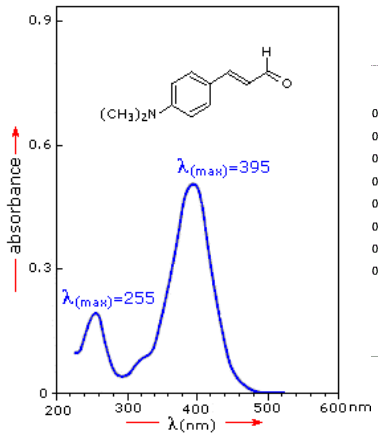


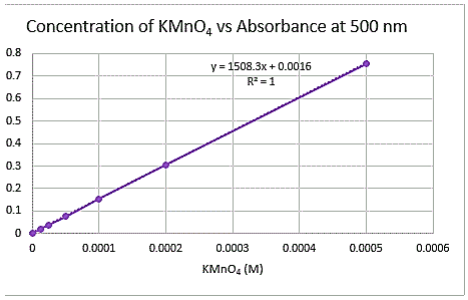
A light beam of particular wavelength when passed through a sample decreased in intensity because it either absorb or distorted and don't reach to the detectors. The particular part of the radiation that absorbed reflect certain characteristics of our sample and molecule.



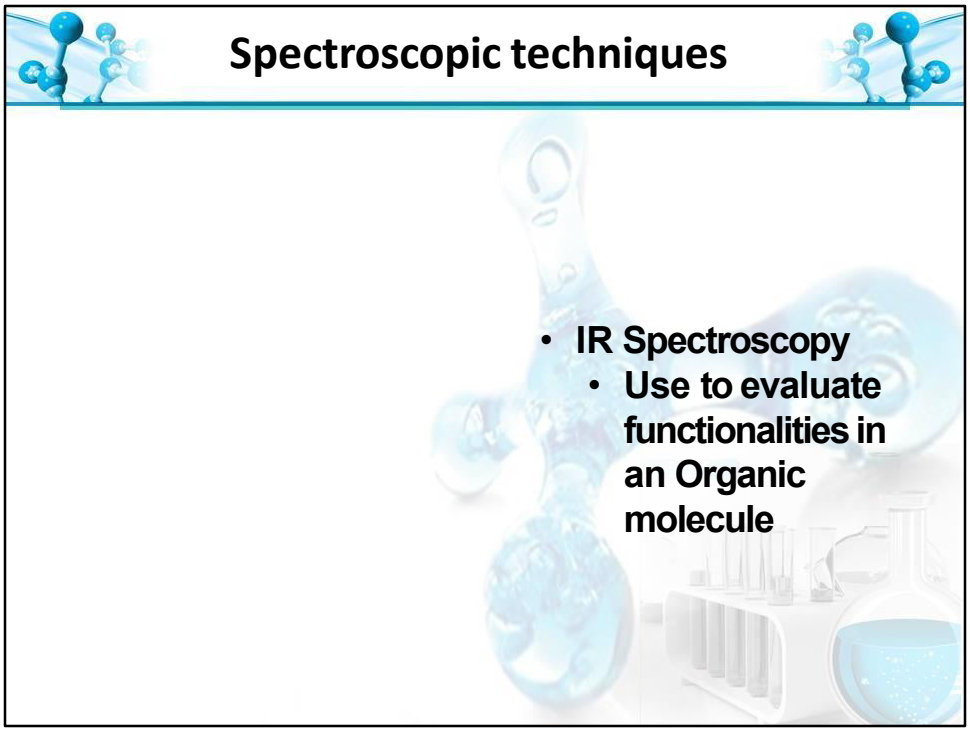
There are different spectroscopic techniques available in market to solve our different analytical problems. The most widely used technique is UV Spectroscopy. That is basic instrument found in food and drug analysis laboratories.

Spectroscopic techniques



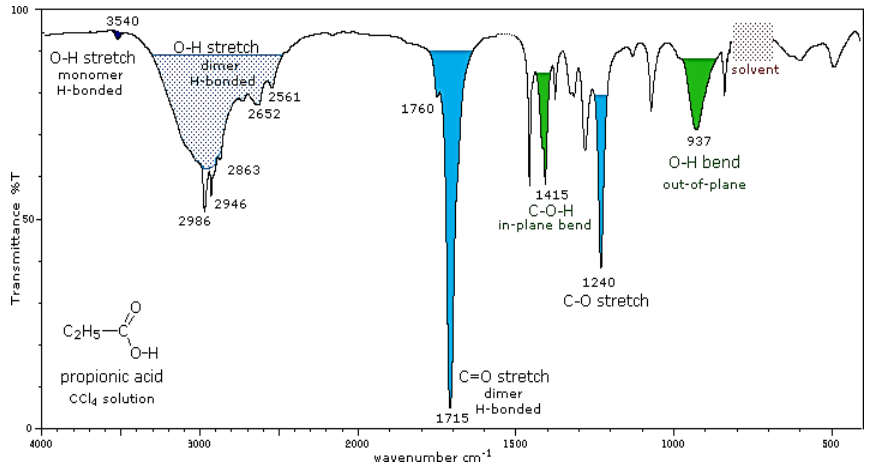


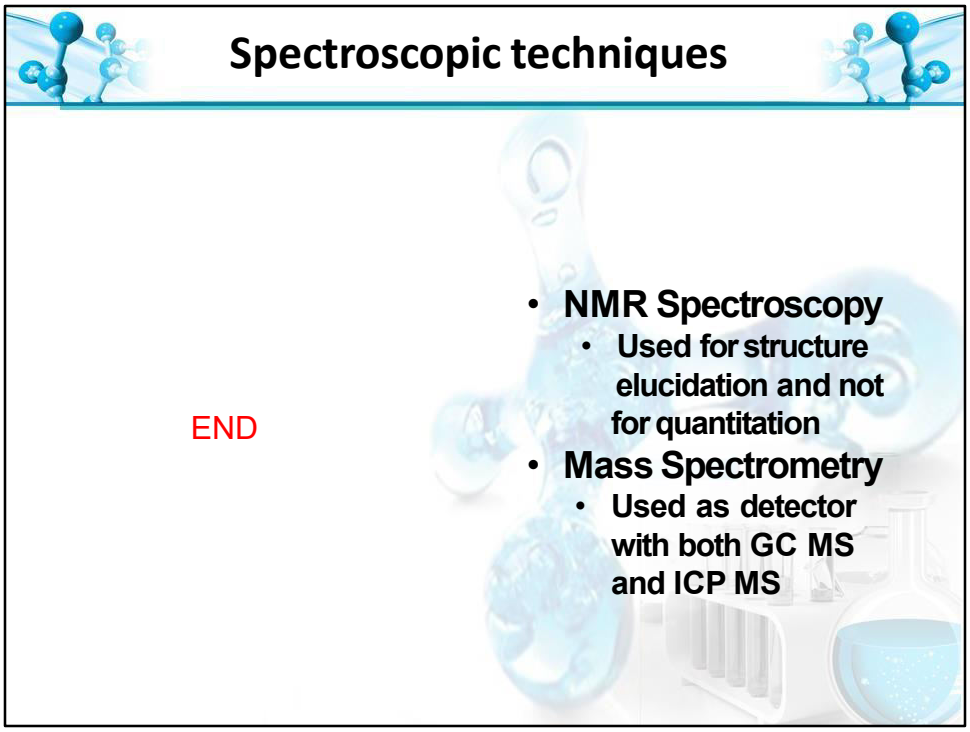
We use this technique for either finding out wavelength maximum of any molecule or use it for evaluation of concentration. N,N-dimethyl cinnamaldehyde



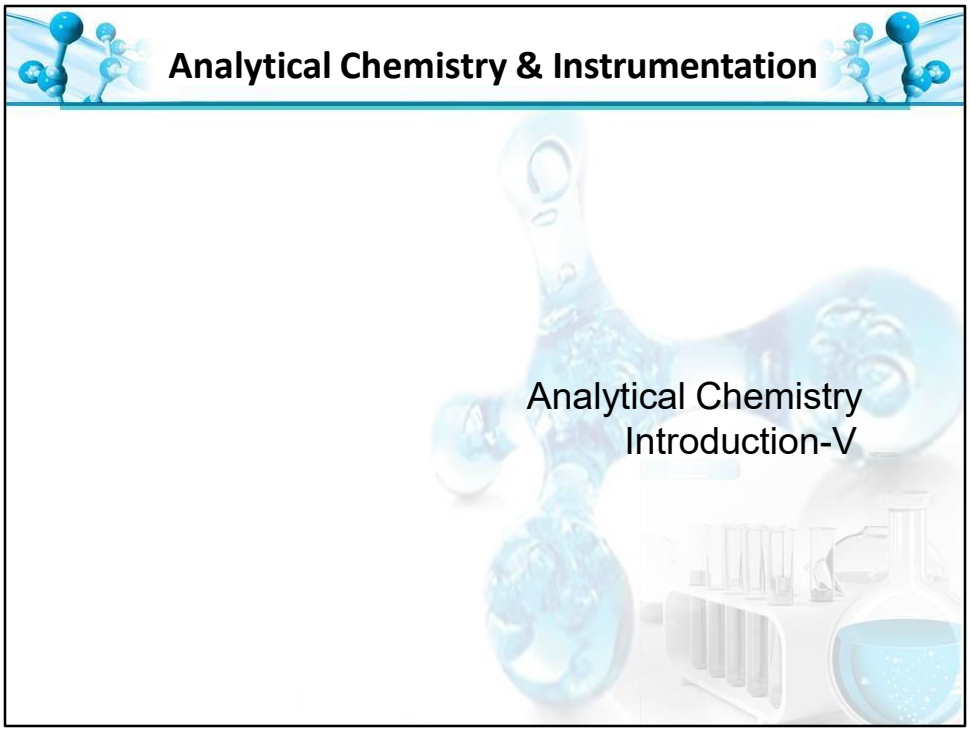
Another spectroscopic technique is IR Spec. We

Spectroscopic techniques





Another spectroscopic technique is IR Spec. We



Now we will take a tour to some spectroscopic techniques.

Spectroscopic techniques

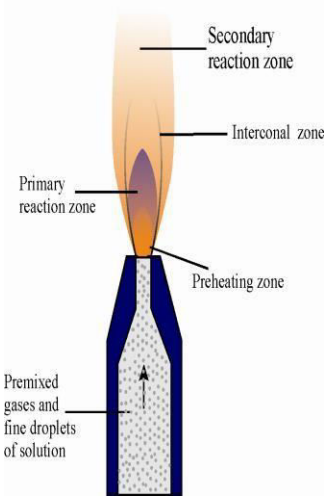
Atomic Spectroscopy

- **Atoms of different type absorb light of a certain wavelength**
- **They emit the same wavelength as they absorb**

Spectroscopic techniques

Flame photometry

Elements	Emission Wavel-engthh (nm)	Flame Colour
Sodium (Na)	589	Yellow
Potassium (K)	766	Violet
Barium (Ba)	554	Lime green
Calcium (Ca)	662	Orange
Lithium (Li)	670	Red

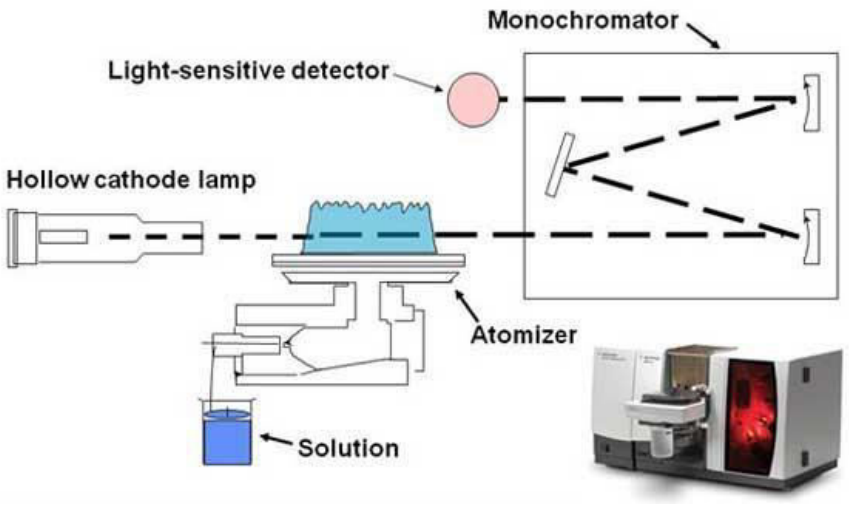


Spectroscopic techniques

Atomic Absorption spectroscopy

- **Flame Atomic Absorption Spectroscopy (FAAS)**
- **Graphite Furnace Atomic Absorption Spectroscopy (GFAAS)**

Spectroscopic techniques



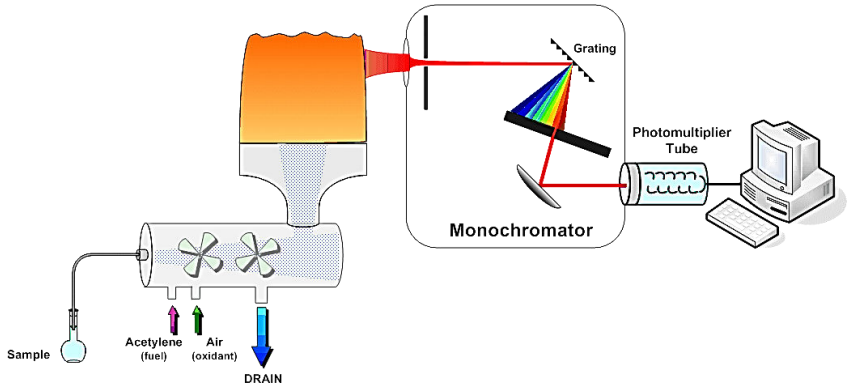
Spectroscopic techniques

Atomic Emission spectroscopy

- **Inductive Coupled Plasma – Optical Emission Spectroscopy (ICP / OES)**
- **Inductive Coupled Plasma - Mass Spectrometry (ICP MS)**

Spectroscopic techniques

Atomic Emission Spectroscopy

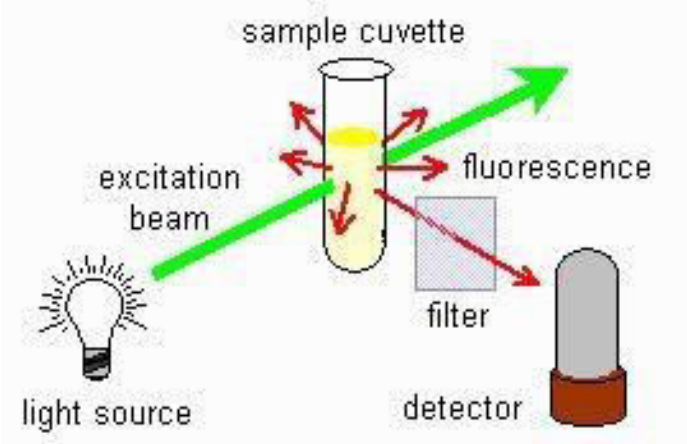


Spectroscopic techniques

Atomic fluorescence spectroscopy

- **Evaluate certain organic molecules and metals in different oxidation states**

Spectroscopic techniques

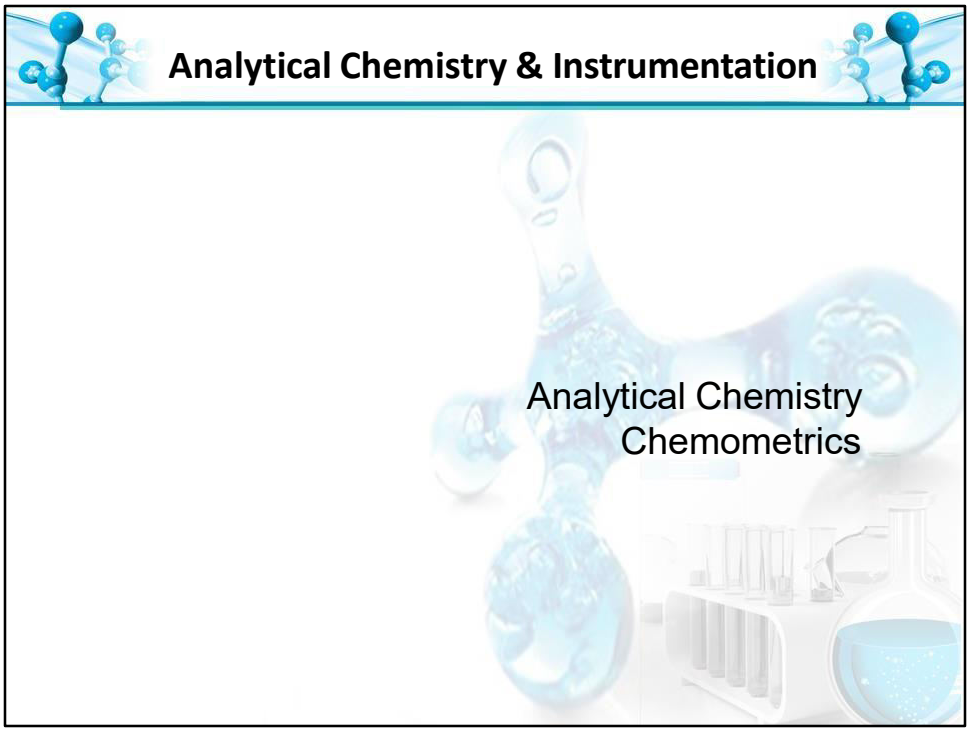


Spectroscopic techniques

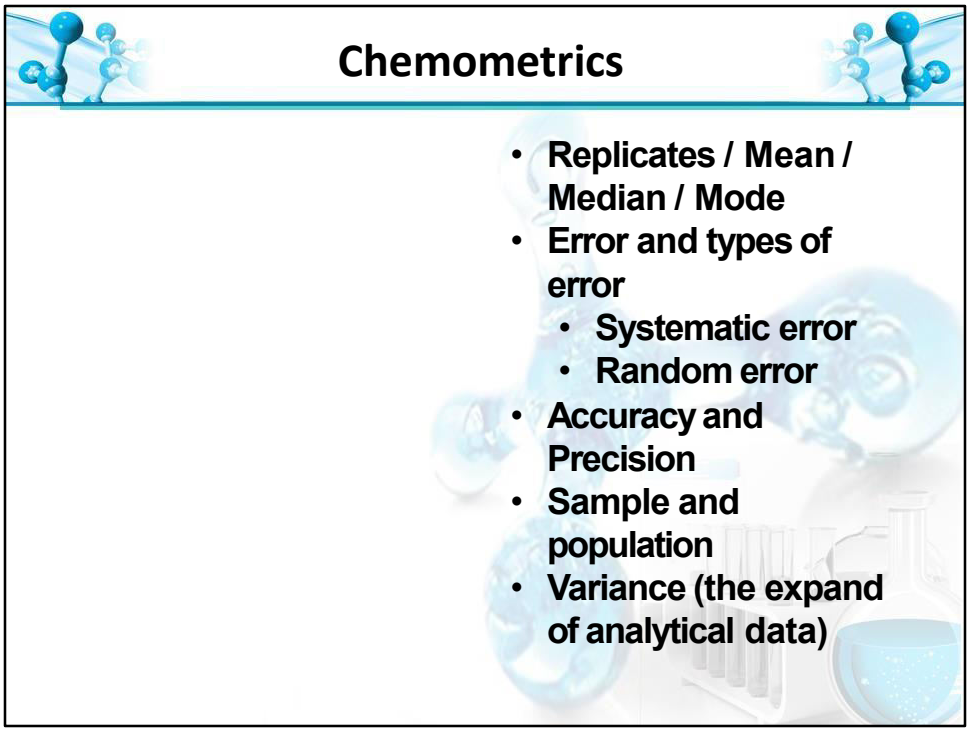
END

Atomic Spectroscopy

- Thus we can determine qualitatively the metal atom
- We can estimate concentration of particular metal atoms



In this module we will discuss about Chemometrics. **Chemometrics** is the use of mathematical and statistical methods to improve the understanding of chemical information and to correlate quality parameters or physical properties to analytical instrument data.



The statistical techniques what we discuss here in this module will be concept of replicates. Arithmetic mean, median and mode of data.
Then we will discuss about errors, types of error, systematic and random error in the data.
Error free data is called accurate. Accurate data may or may not be precise.

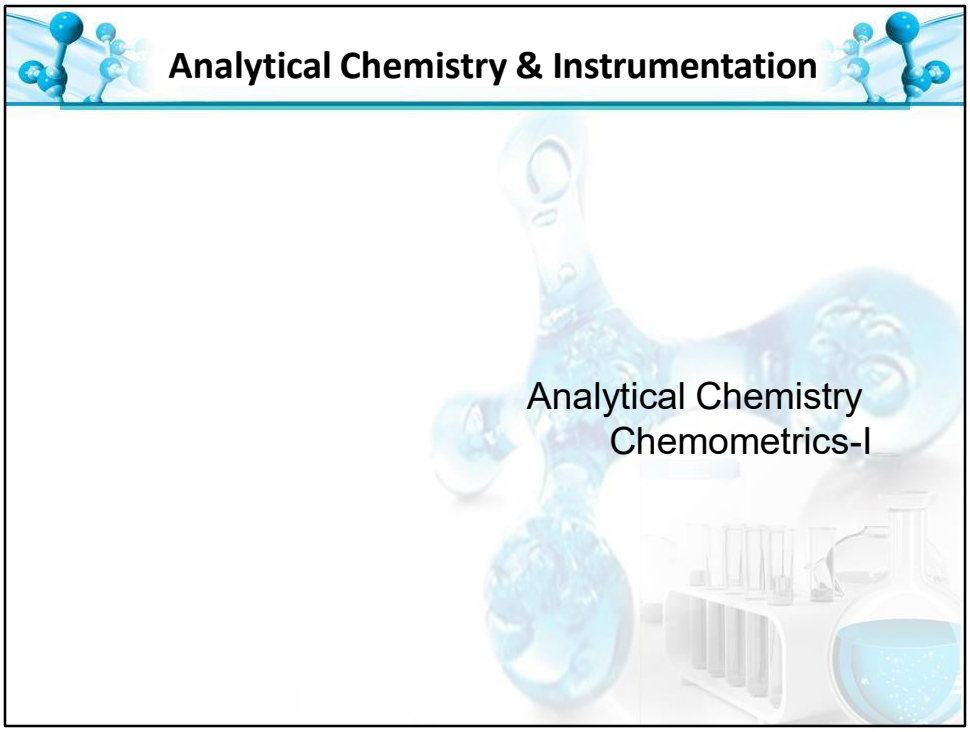
Chemometrics

- Standard deviation (the measure of precision)
- Reporting of data
  - Significant figures
  - Treating significant figures in calculations
  - Rounding off data
- Confidence Interval
- CV and SEM

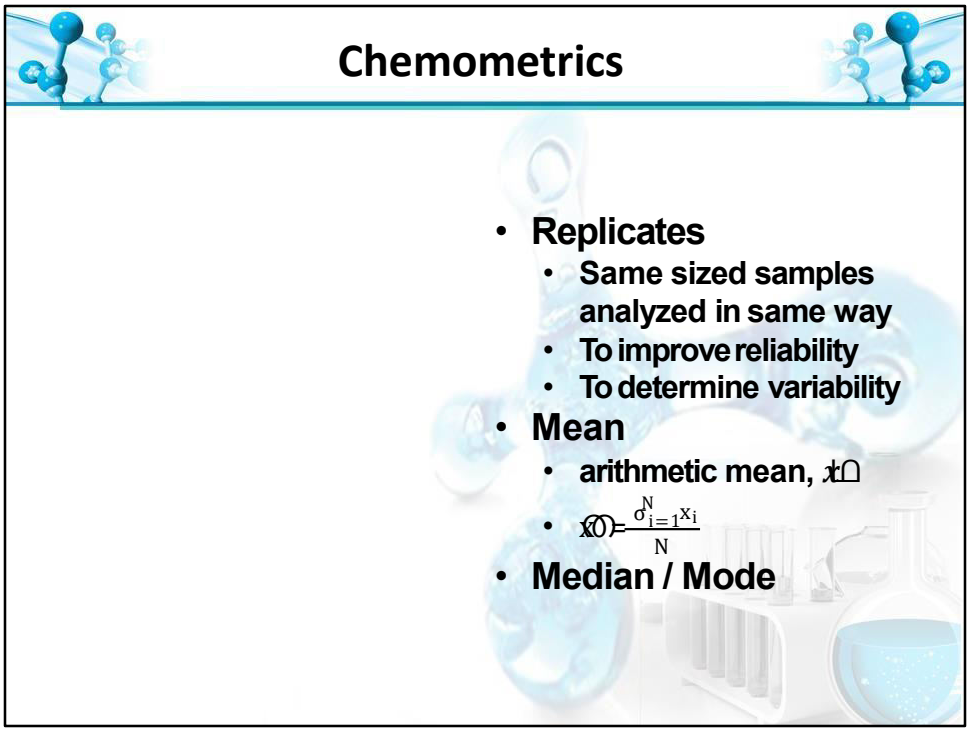
Chemometrics

END

- Analysis of Variance
  - Z-test
  - T-test
  - F-Test
- Standard preparations
- Sampling
  - Sampling types
  - Sampling techniques
  - Sample preparations

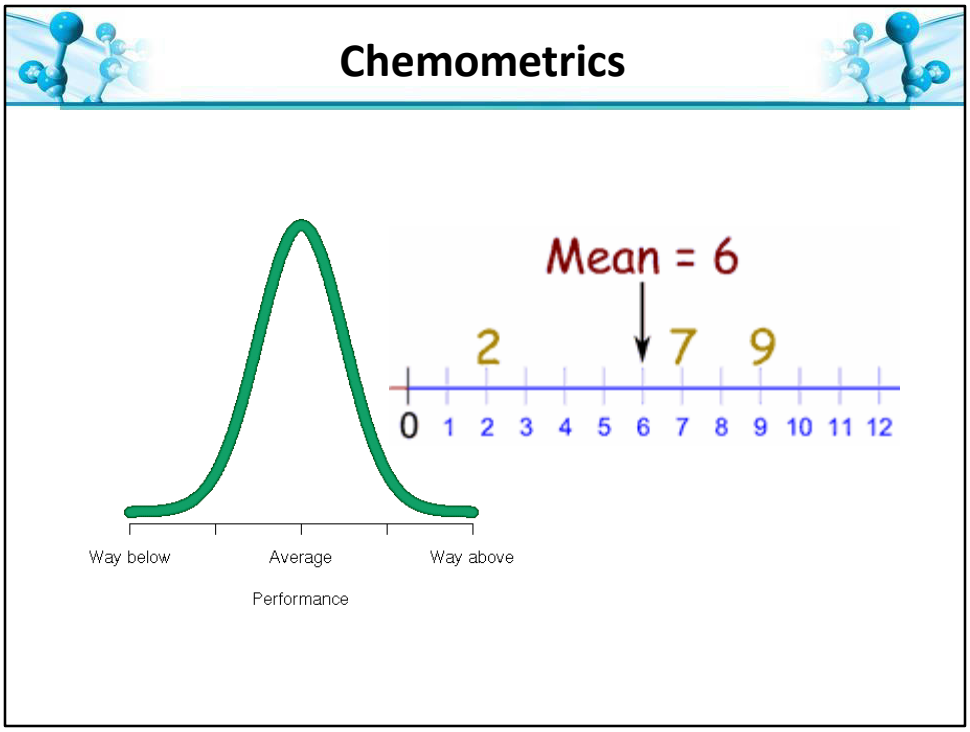


Now we will discuss about replicate measurements.

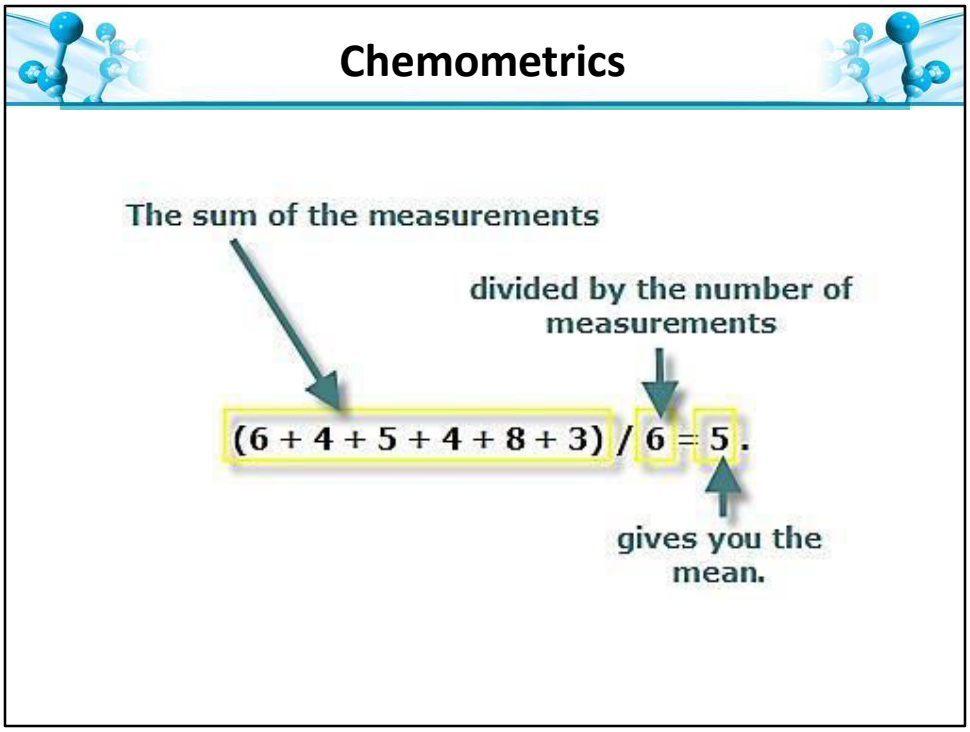


The replicates mean analysis of same size of sample under similar analytical conditions. Without changing reagents, analyst, environment etc. Only variable is the number of samples. This is done to improve reliability. Like we determine best tennis player in best of five games.
The second objective may be to determine variability in the samples. Like variability in land quality during soil analysis.

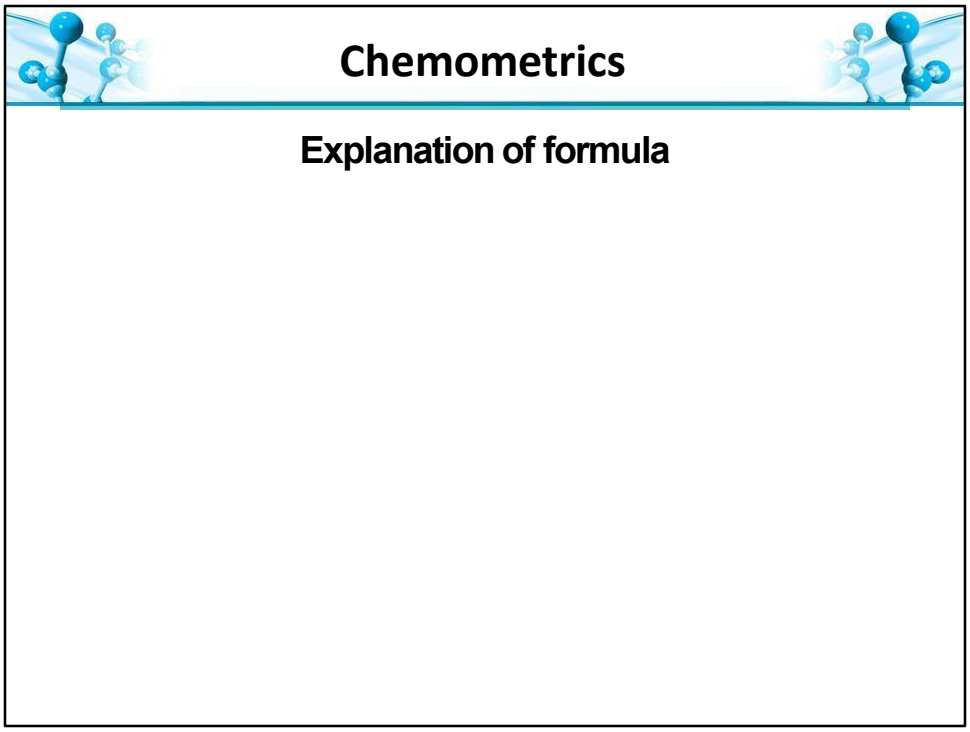
We analyse till concordant readings. Means with minimum variability in between of analytical results.
We may take average of all the readings. This average may also called arithmetic mean in statistics.



Add the three readings and divide by three.
Average represents the peak of this gaussian curve.



Add the number of replicates and divide by number of readings gives us mean ofthe data.

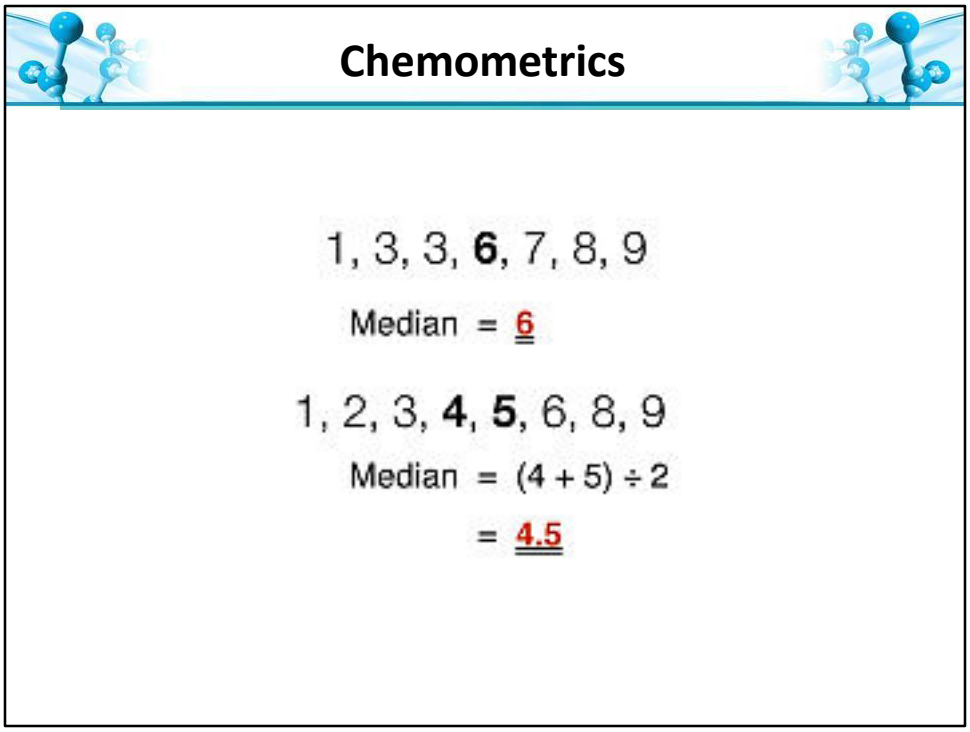


Explain the formula

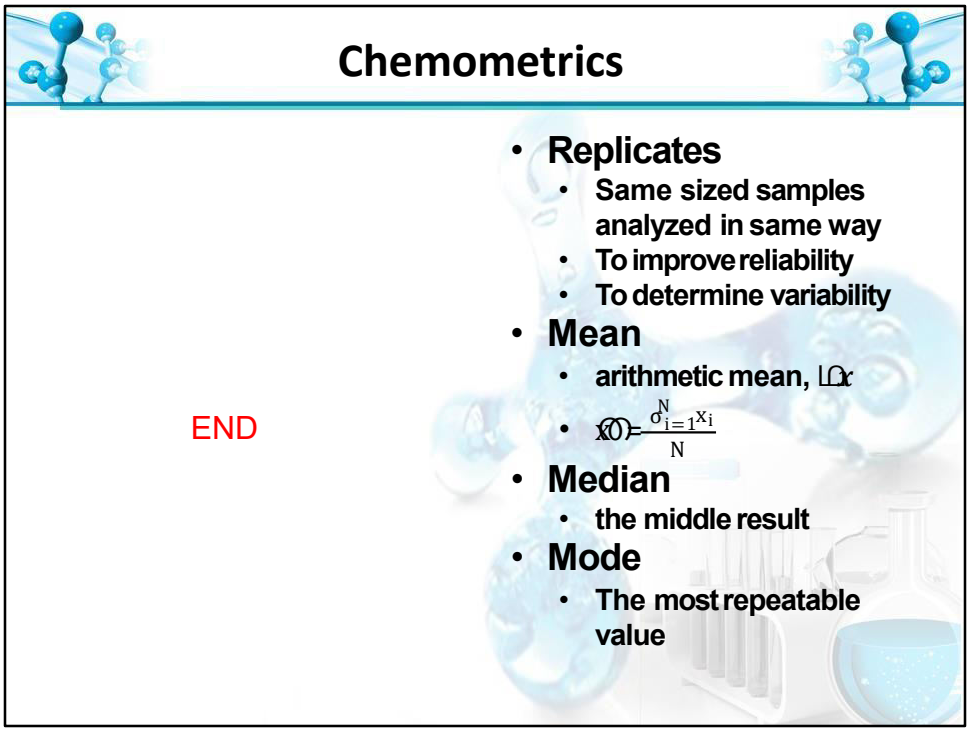
Chemometrics

- **Replicates**
  - Same sized samples analyzed in same way
  - To improve reliability
  - To determine variability
- **Mean**
  - arithmetic mean, $\bar{x}$
  - $\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$
- **Median**
  - the middle result
- **Mode**
  - The most repeatable value

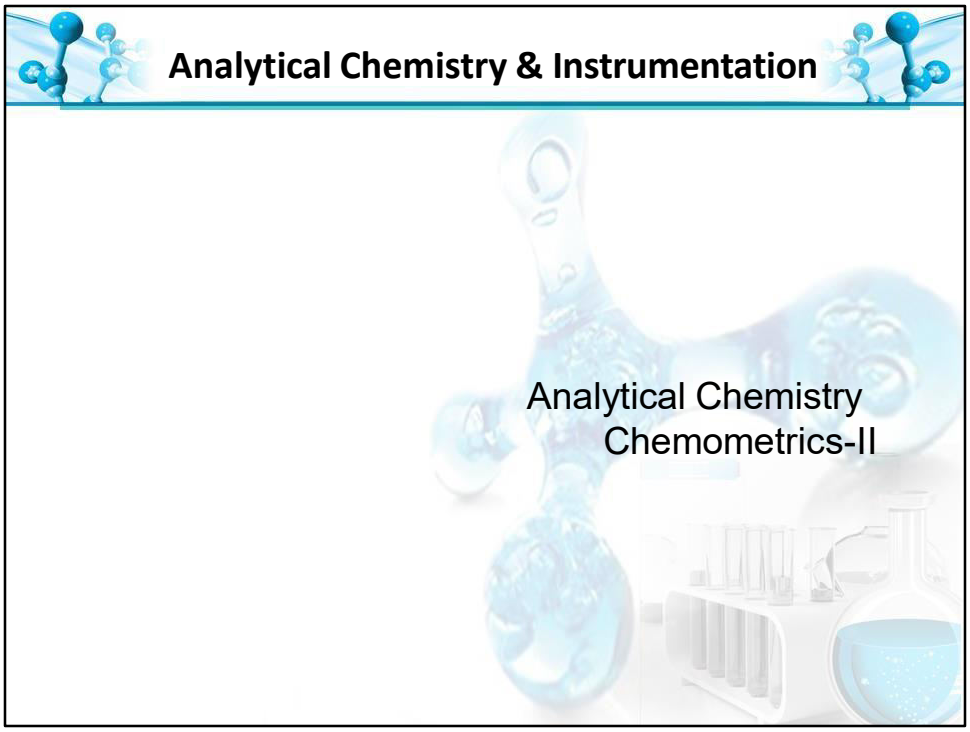
To calculate Median arrange the replicates or your data in ascending or descending order.



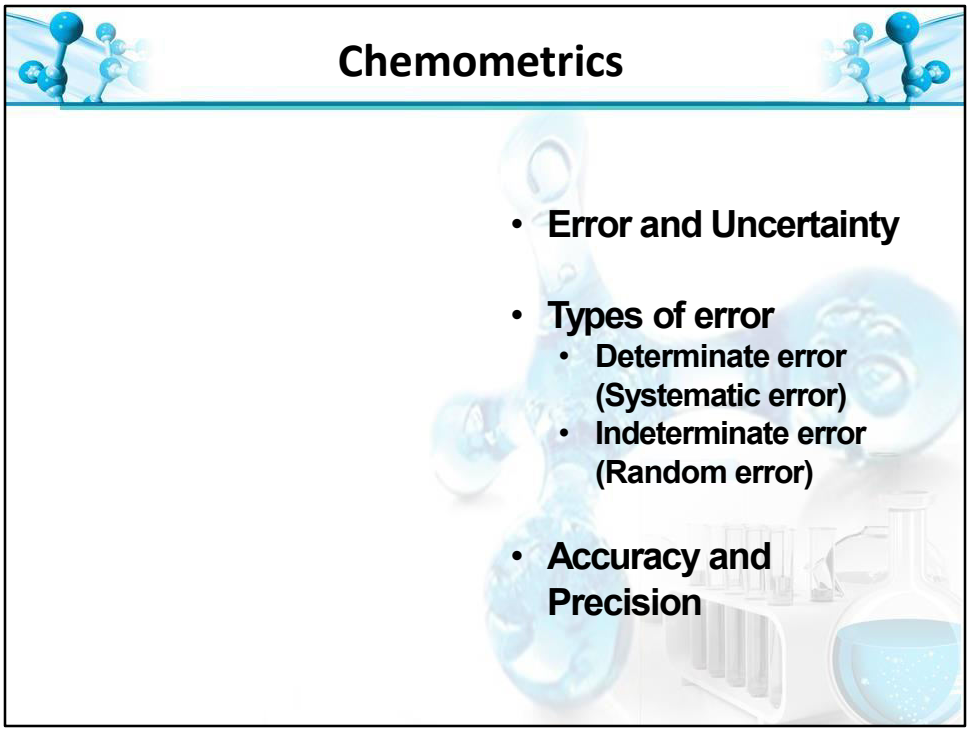
The middle of the reading in odd number of data is called median. If you have even number of data points then add the two mid readings and divide by two. Means take mean of the two middle reading.



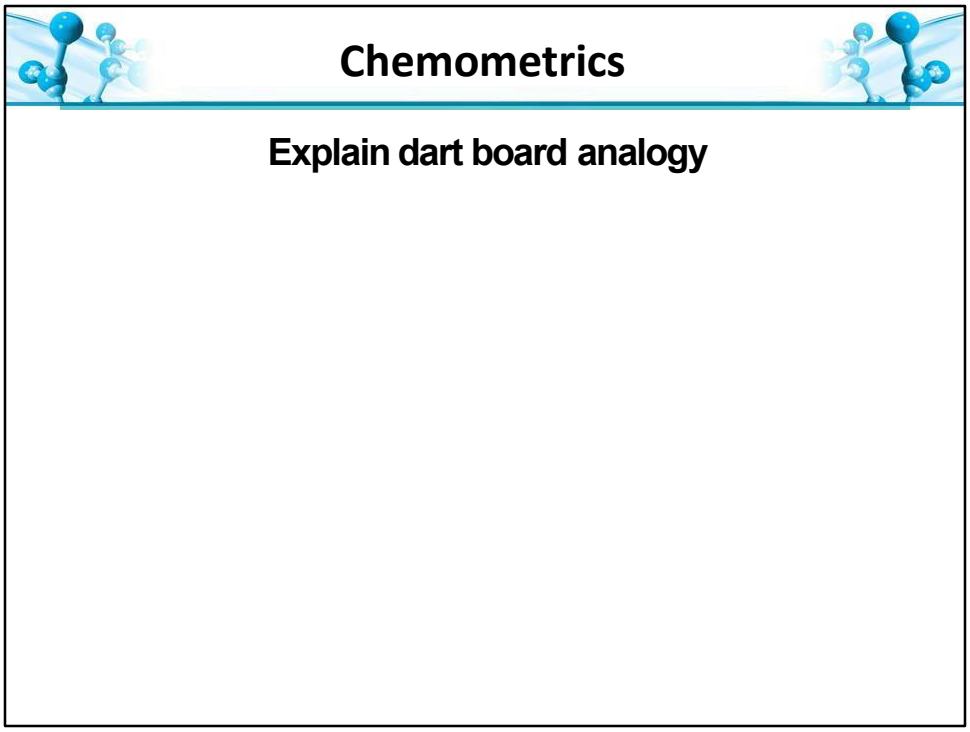
Mode means the most repeatable data point



Now we will discuss about some terms regarding errors.
Error simply means not correct. But mathematically it may have some other meanings as well.



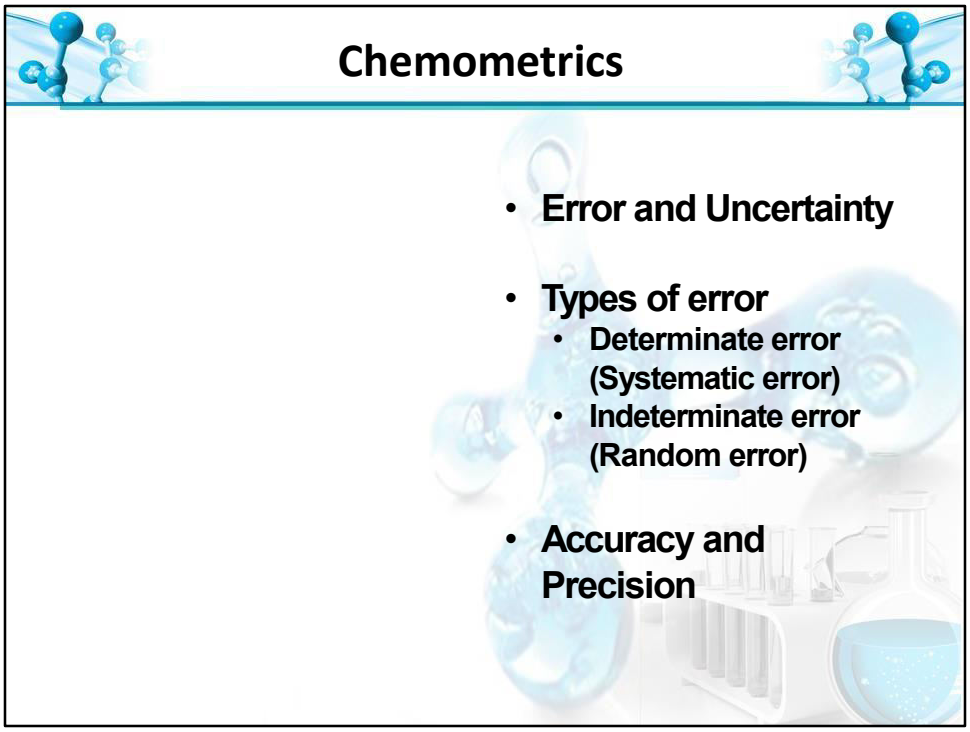
Normally during an analysis we can not have a single repeating reading. For example during an acid base titration we have 10.0 mL as titrant reading every time. It shows the careless behaviour of analyst or manufactured data. Otherwise it may be 10.5 mL or 9.5 mL etc.



In dart board one can score maximum points if target the mid circle with plus minus on above and below and right to left

Example heart patient lifeline

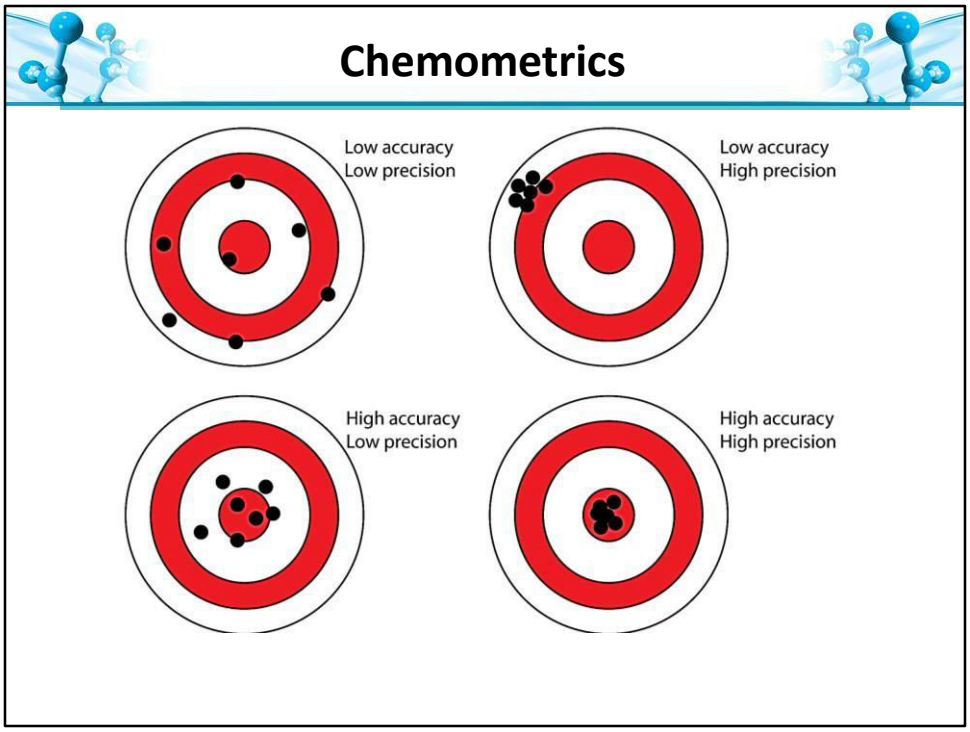
The life line of a heart patient as appeared in ECG is random movement. The linearity means death.
We will explain quantification of uncertainty later on, here we just want to have some understanding.



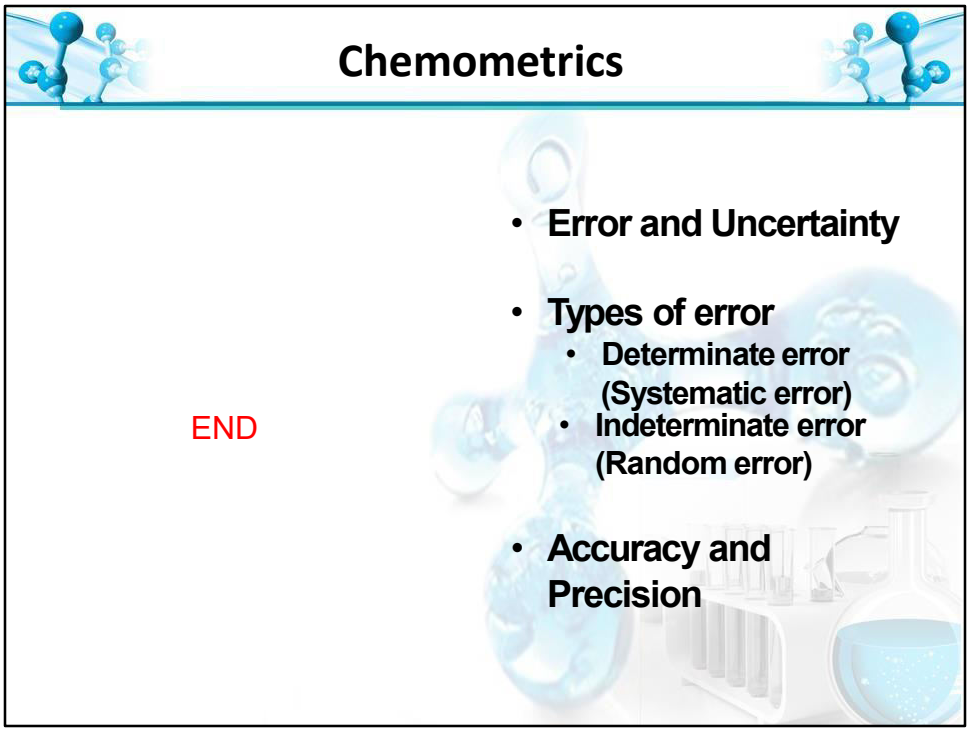
Above type of uncertainty is called random error, that can be minimized but not eliminated. Therefore it is called indetrminate error, i.e. that can not be determined exactly
The second type of error is determinate error.

For example the fault in a weighing balance, whenever you place 1 g it read 1.1 g. Determinate error may be positive or negative while the indeterminate error always appeared to be above and below the average, therefore show + and – results.

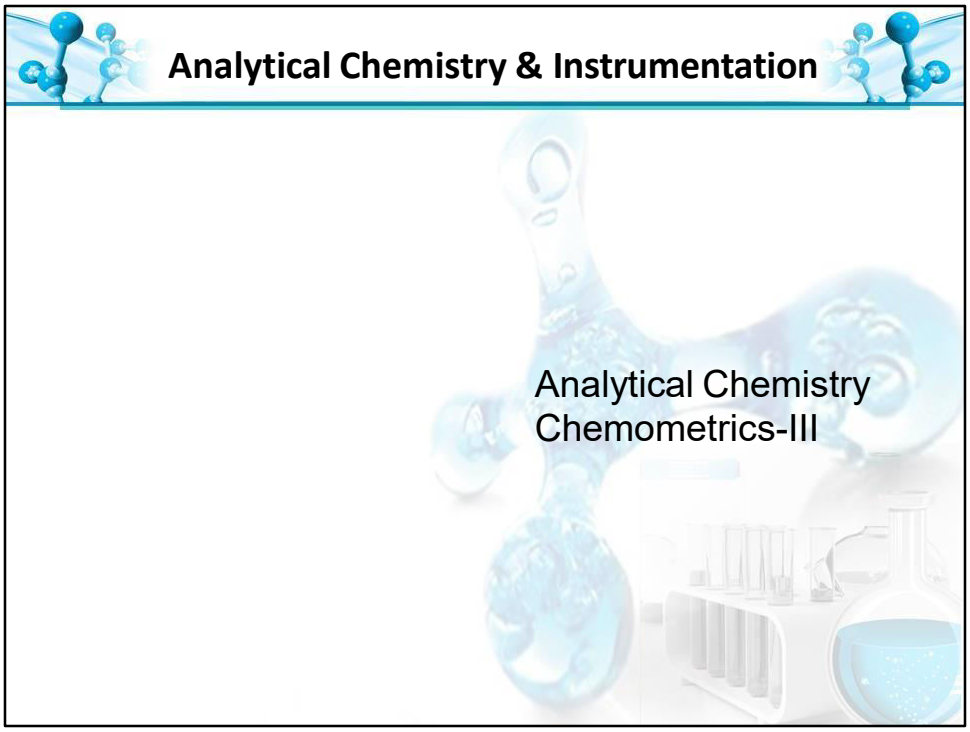
The third concept is accuracy and precision. Accuracy is closeness of data to the actual results. While precision is understood as closeness of data within itself. We can learn it in a better way with dart board analogy.



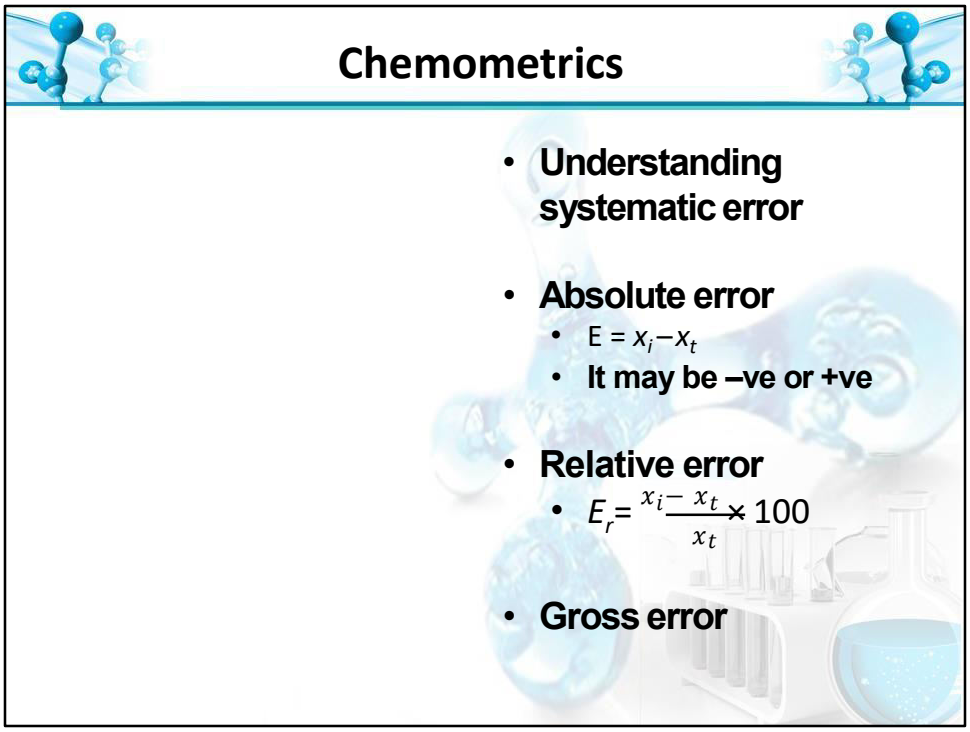
This dart board analogy representing the concept of accuracy and precision. The best suited posture is lower right bottom representing high accuracy and high precision. And the least one is above left corner with lowest accuracy and low precision.



For good analytical result the best suited position is high precision, means least variation among the results. And close to the actual value



In these slides we will build our understanding about systematic error.



Keep in my mind that systematic errors are those which could be estimated, therefore also called determinate error.
These error will be either in positive or negative side.

Chemometrics

How to calculate these errors?

- **Consider we want to check the true value of our pipette.**
  - **First we have to generate data, fill the pipette and make it empty in a beaker placed on a balance and already tarred**
  - **Collect replicate data**
  - **Calculate mean of that**
  - **Difference in actual volume and observed volume may consider as an error**

How to calculate these errors?

Chemometrics

- **Difference between experimental value and true value**
- **Shows whether the value is high or low**
  - **-ve error; measurement result is low**
  - **+ve error; measurement result is high**
- **no relation with sample size**

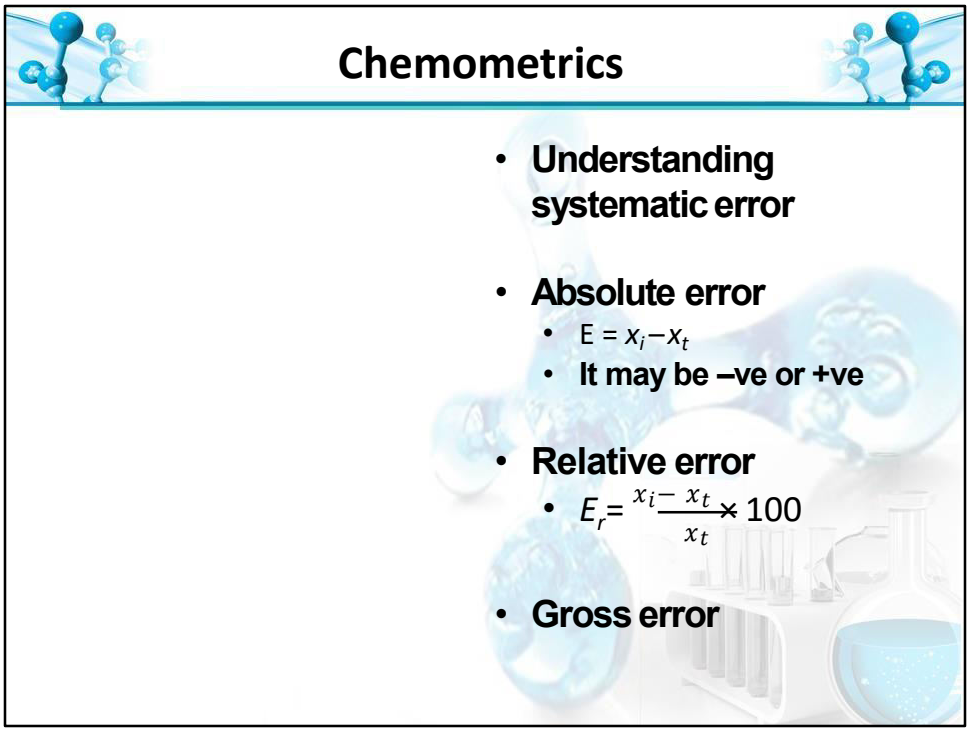
For individual data:

$$\text{AbsoluteError } (E_a) = X_i - X_t$$

For average of data:

$$\text{AbsoluteError } (E_a) = X_i - X_t$$

Absolute errors
It may appear as constant error or proportional errors
It may be calculated in individual values or average of data
Or malfunctioning of balance as positive error (may be constant error)



Relative error give another aspect of errors.
Calculation of error relative to actual value is called relative error.

error may rise with sample volume, but relative error will remain same

Chemometrics

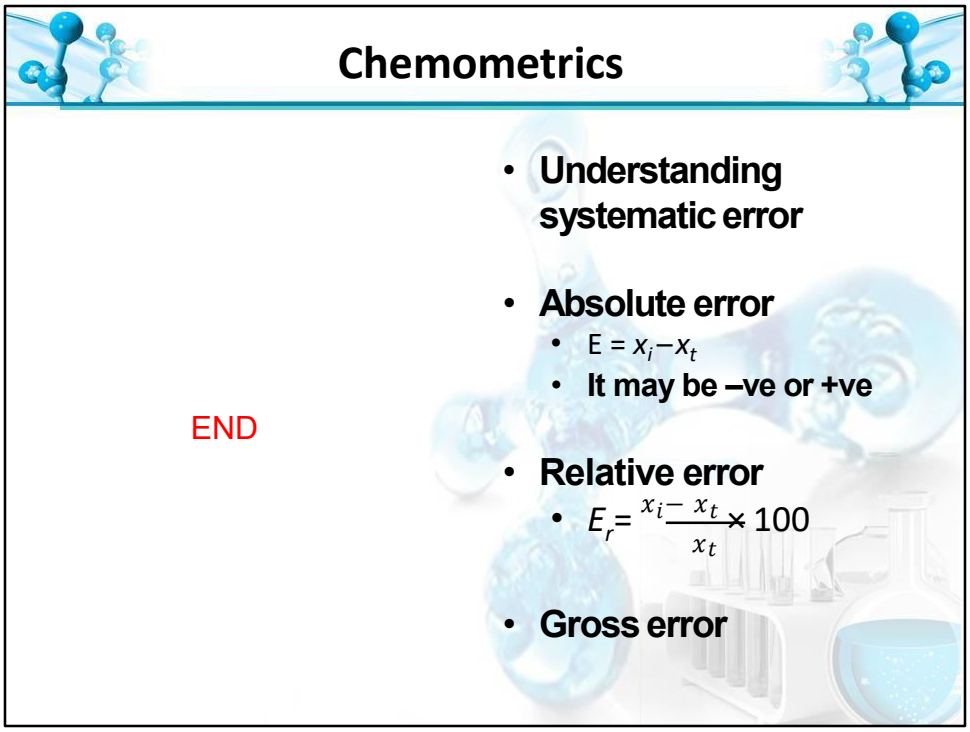
$$\text{Relative Error } (E_r) = \frac{(X_i - X_t)}{X_t} \times 100; \quad \text{In } \%$$

$$= \frac{(X_i - X_t)}{X_t} \times 1000; \text{ In PPT}$$

Problem

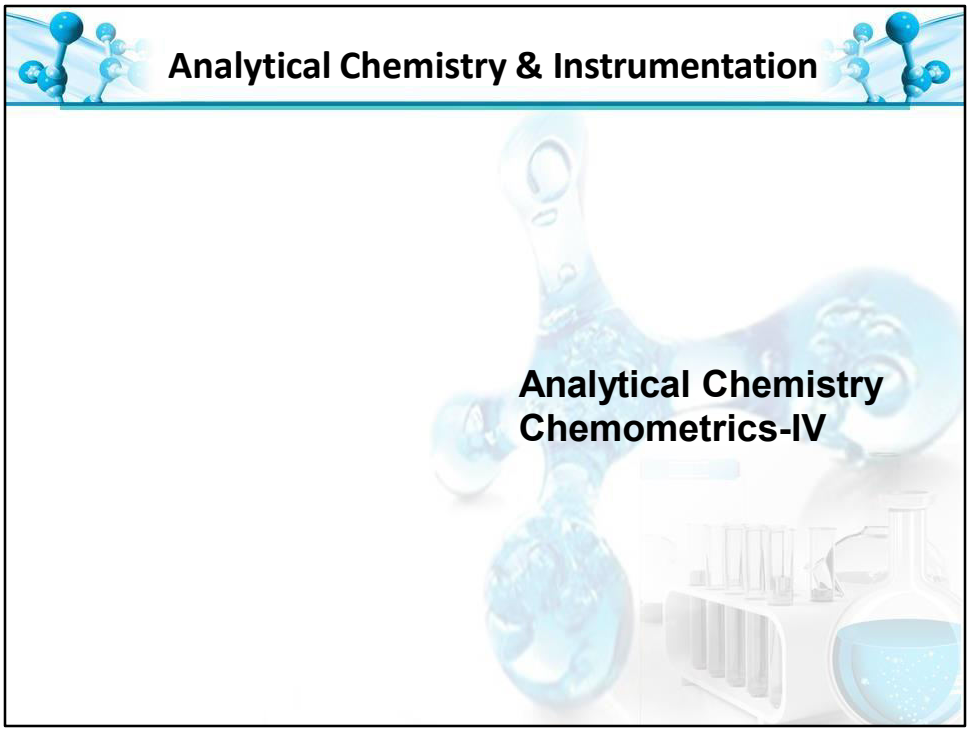
If 0.5mg of precipitate is lost as a result of washing and the precipitate actually weighs 500mg , what will be the relative error due to solubility.

It is calculated in percentage. Or part per thousands PPT.
0.5/5000*100= 0.1 % = 1 PPT

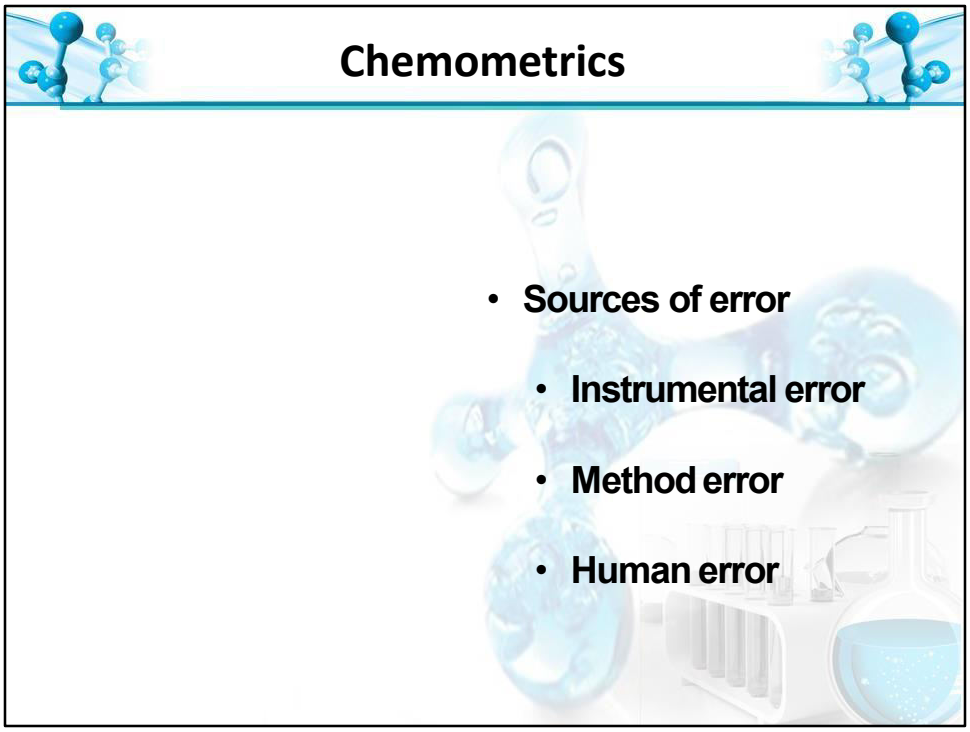


Similarly in pipette if instead of 10 mL it shows 9.5 mL average or an error of 0.5 mL.
So relative error may calculate 0.5/10*100 =5%
We can compare one pipette from other in accuracy.
Or we can compare and declare that a burette is more reliable than a pipette

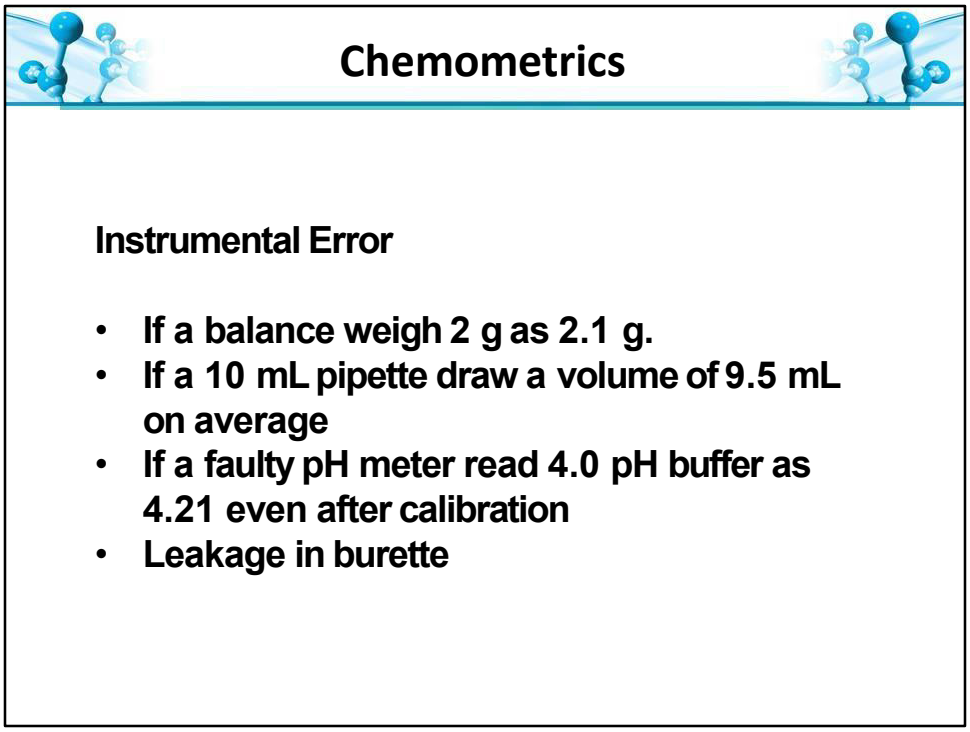
Another term is used as Gross error
They are often the product of human errors. It may be single outliying value in data set



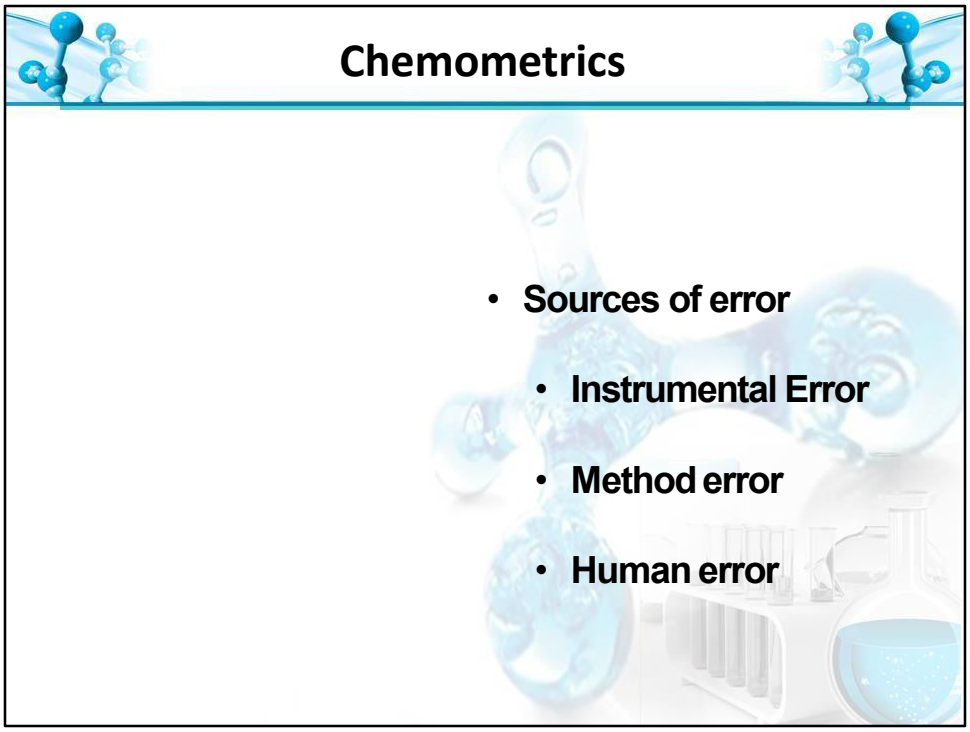
How an error occur? What are possible sources or errors is our topic for this session.



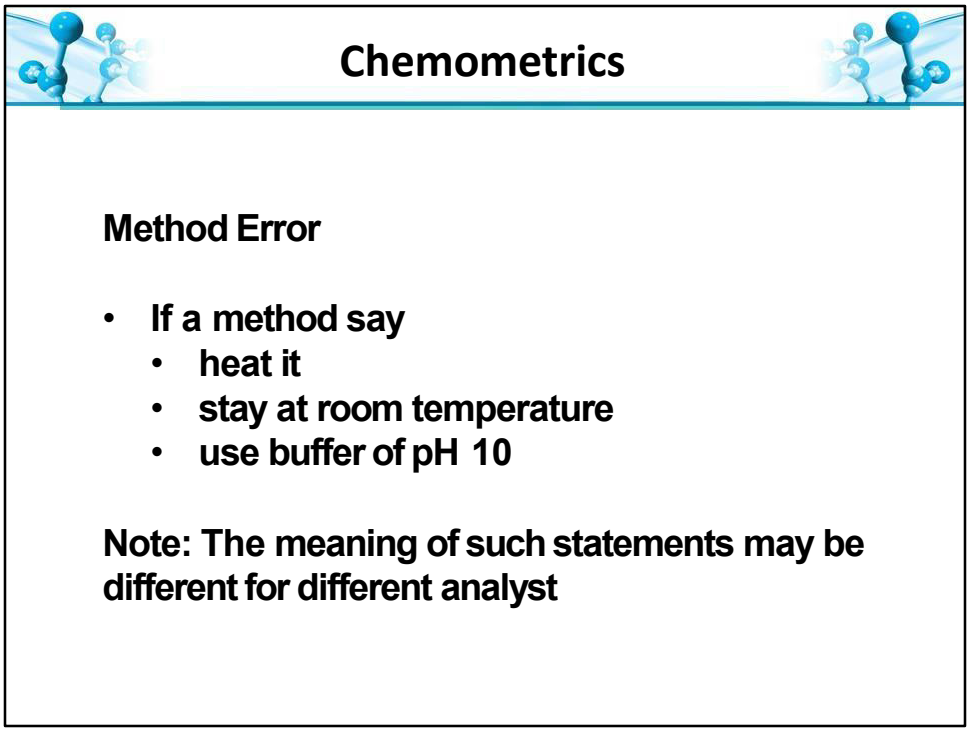
There are just three main factors that can be named as sources of error.



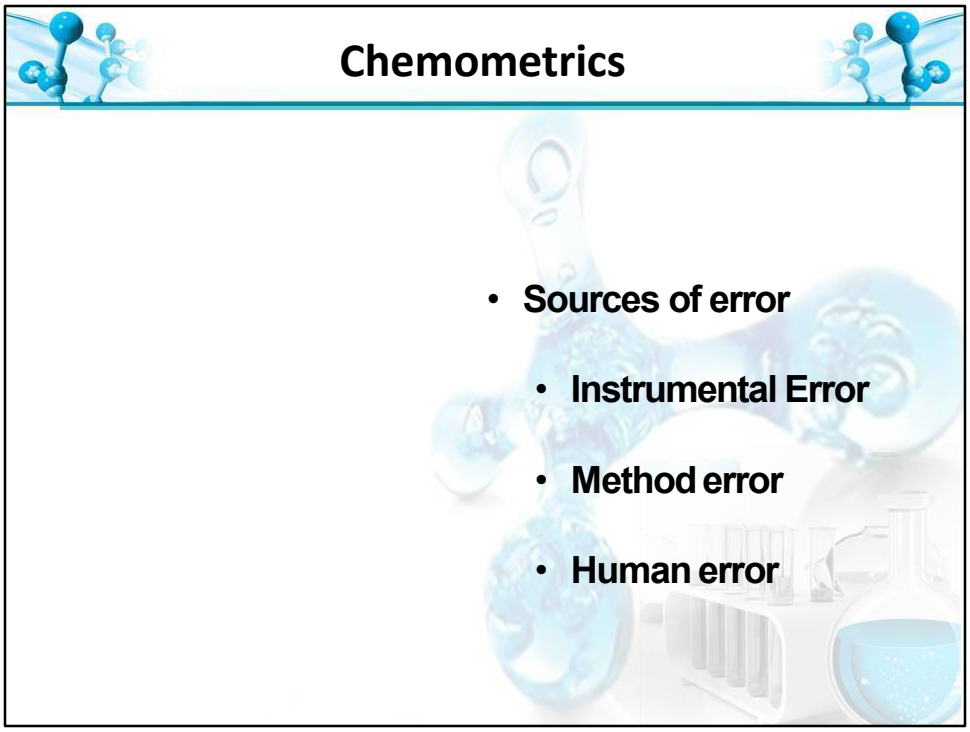
Examples or instrumental error



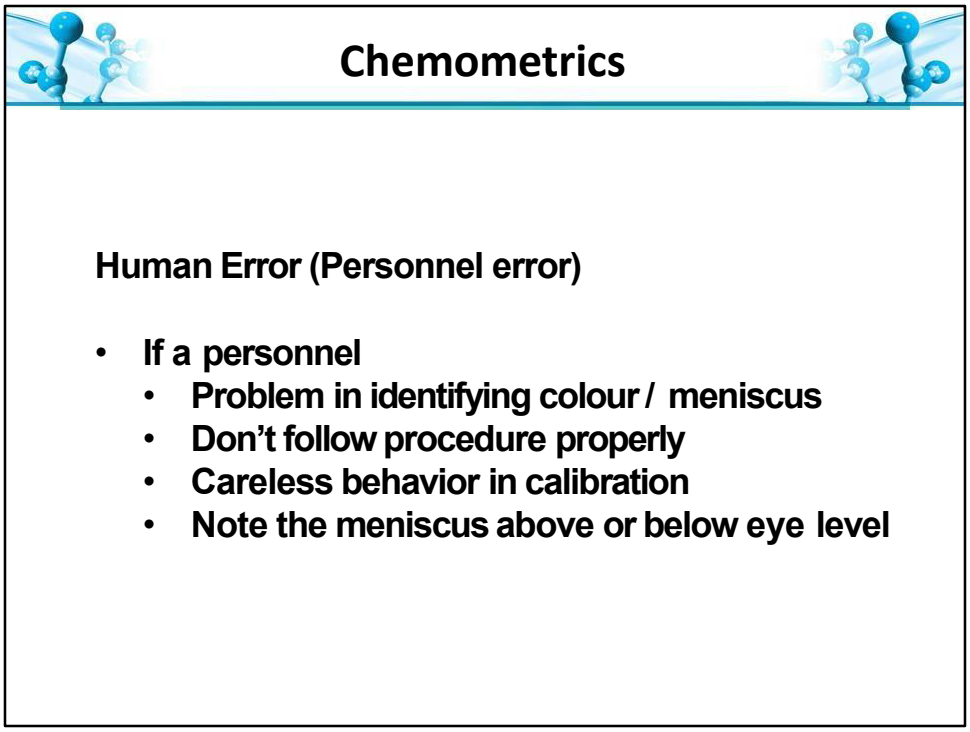
The second error type is method error. If your analytical method have some limitations then it may generate error.
We have to segregate method error from human error



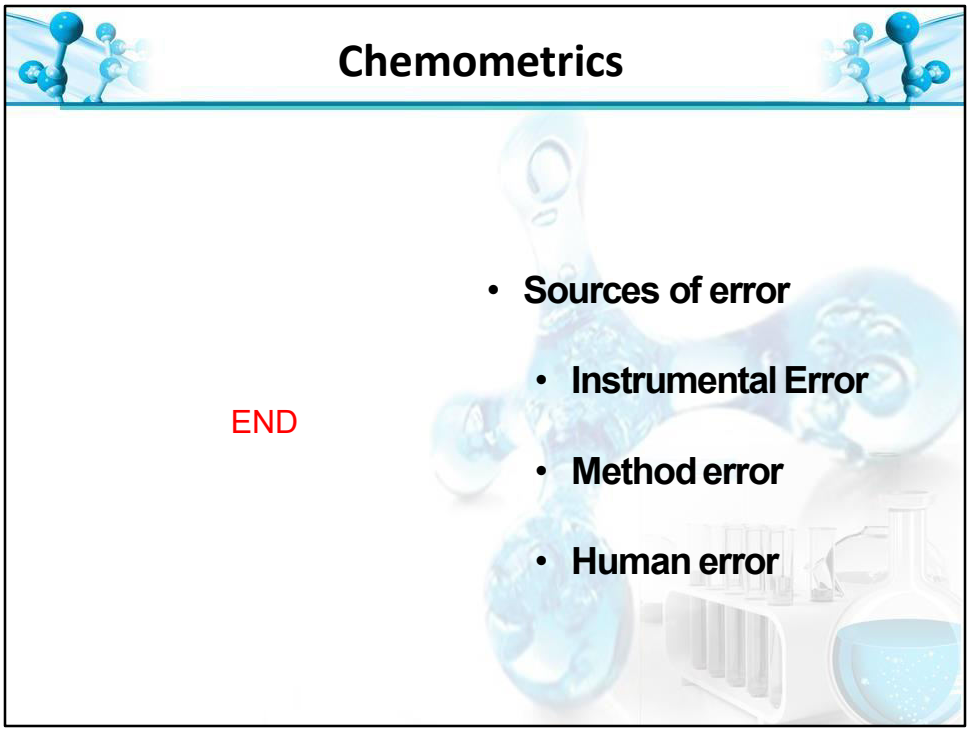
It depends on sensitivity of your method.



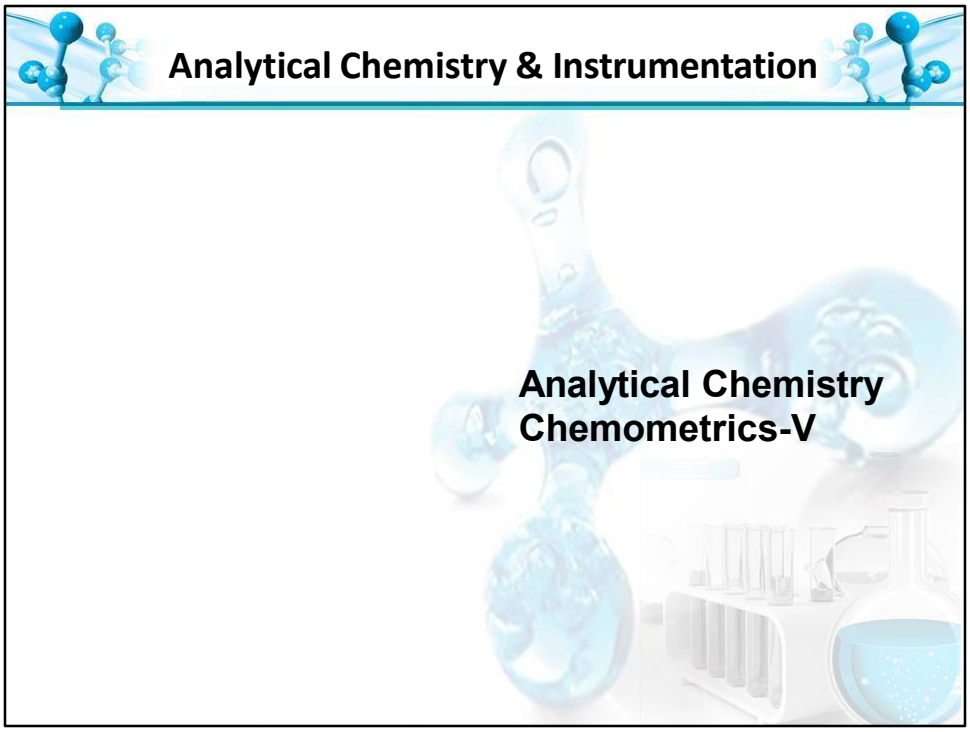
Third type of errors are personnel error. Analyst error



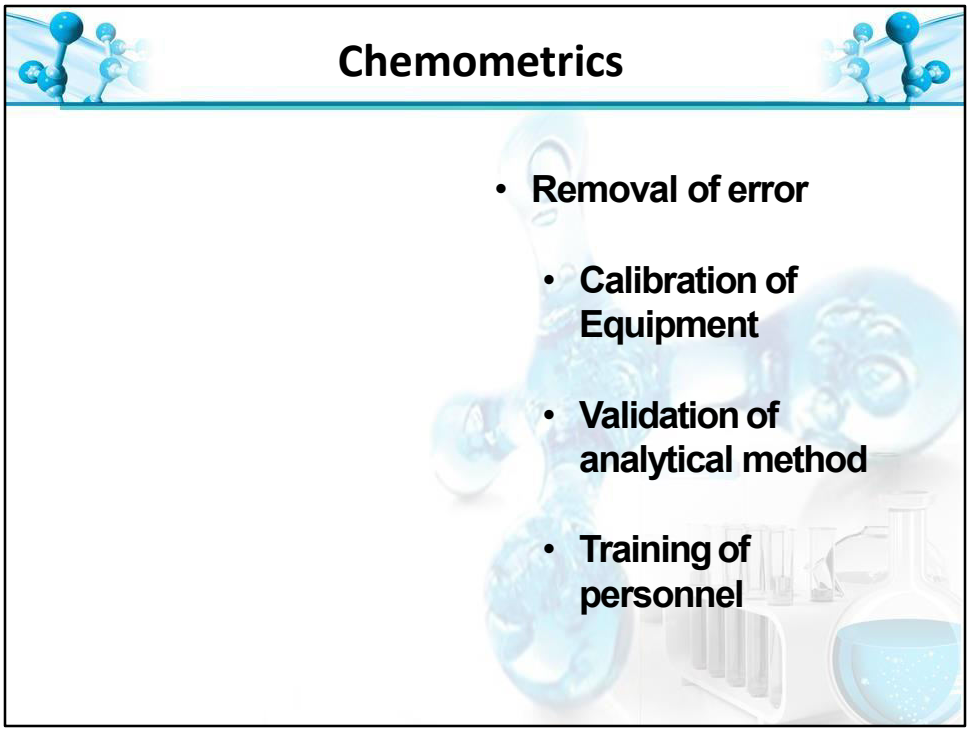
Personnel error



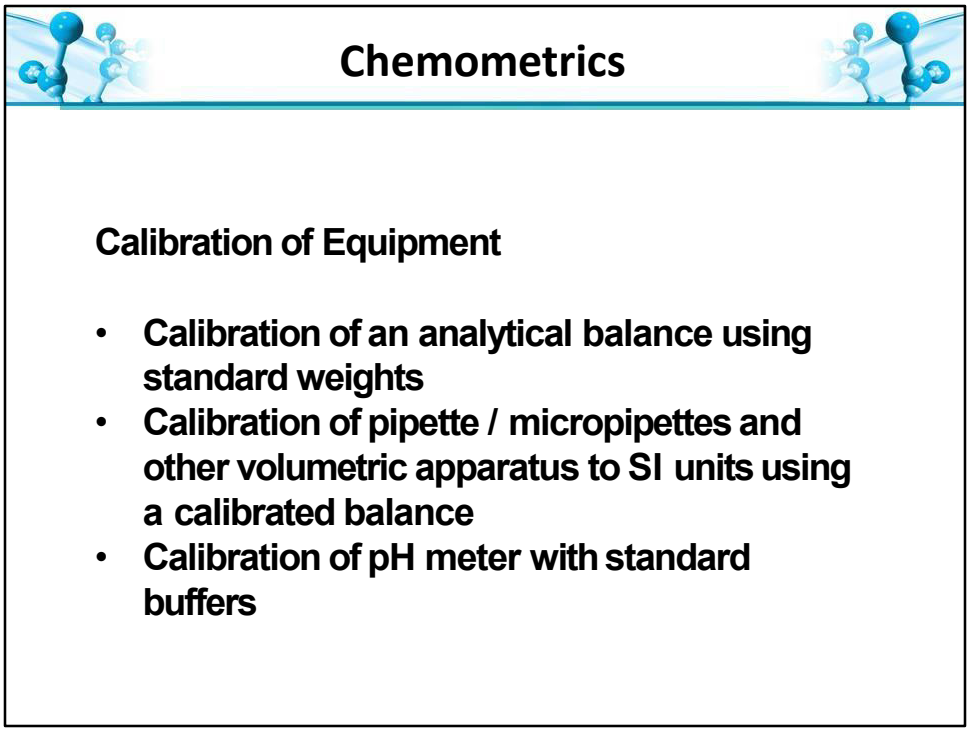
Like if he watch the meniscus above or below eye level.
Or from filling of pipette to pouring of sample it drop the sample



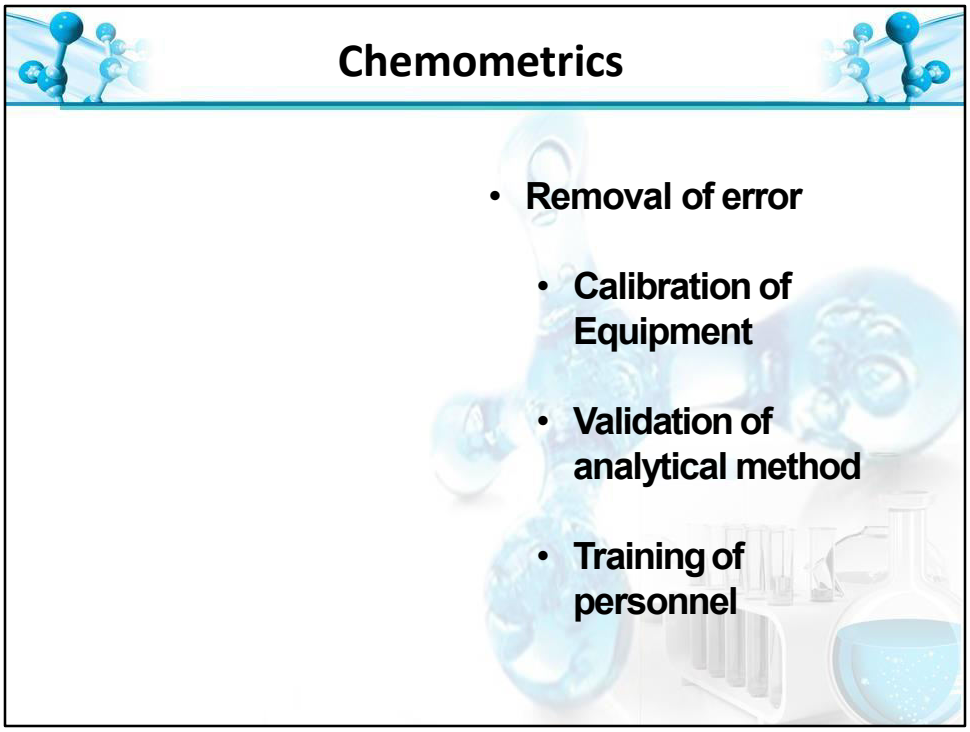
After knowing the sources of error, we must know how to remove error.



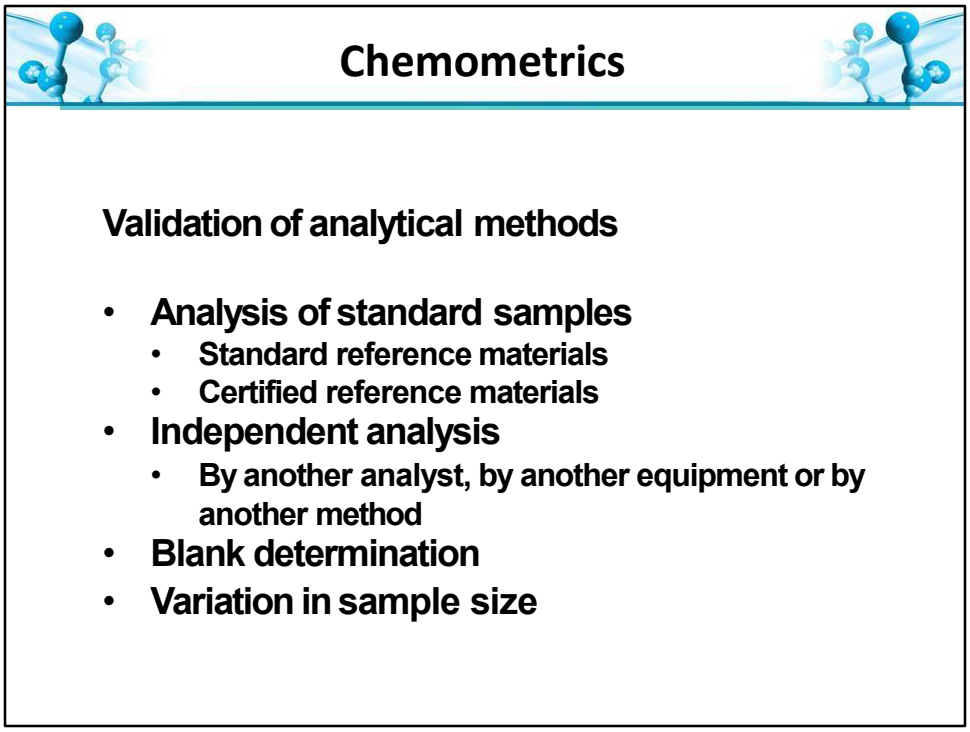
We have discussed three possible causes of error. Equipment, method and personnel. So three possible solutions. Equipment can be used properly after calibration.



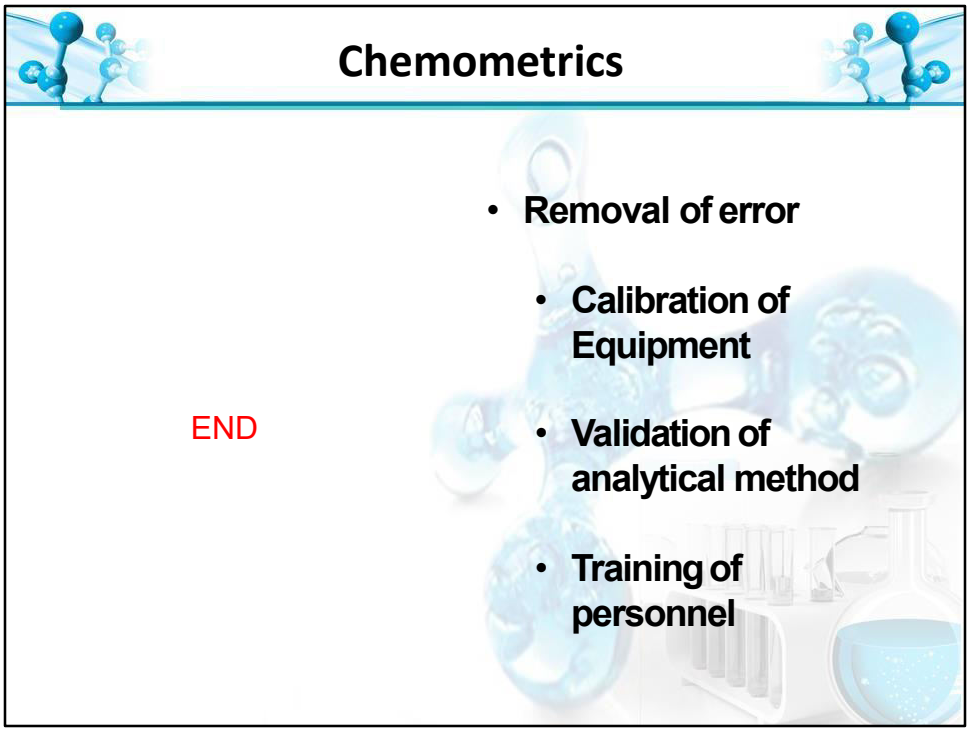
For example, we will learn calibration of volumetric equipment later on in quality control module.



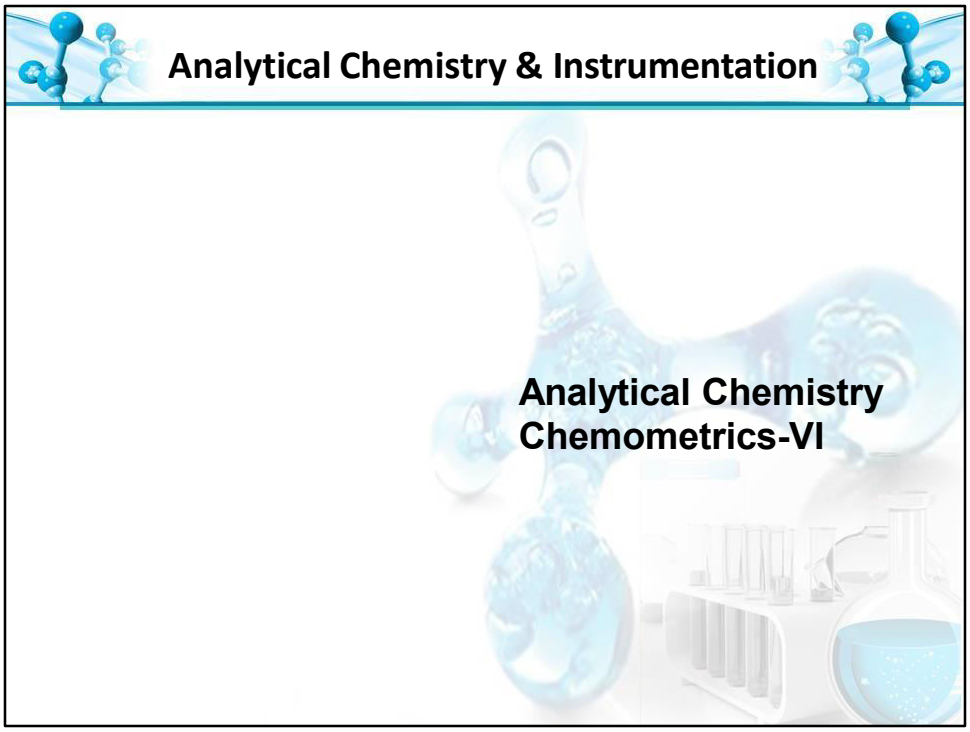
The analytical methods can be used properly after validation.
The detail of this term will also be discussed in detail in QC module, however it may be understood as



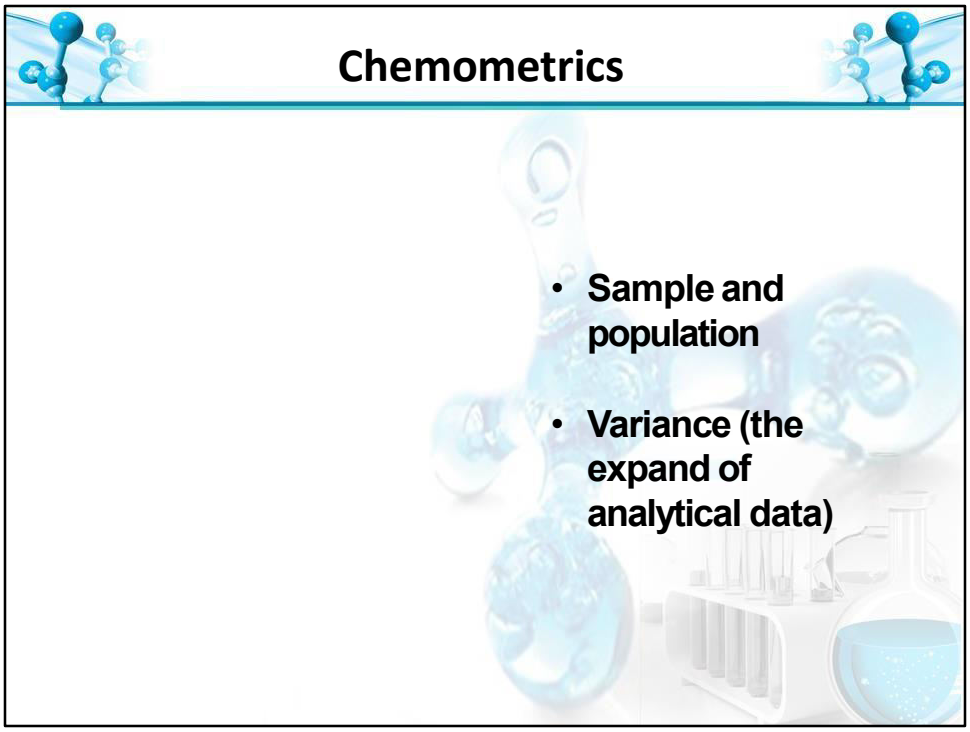
For ways to check or ammend your method



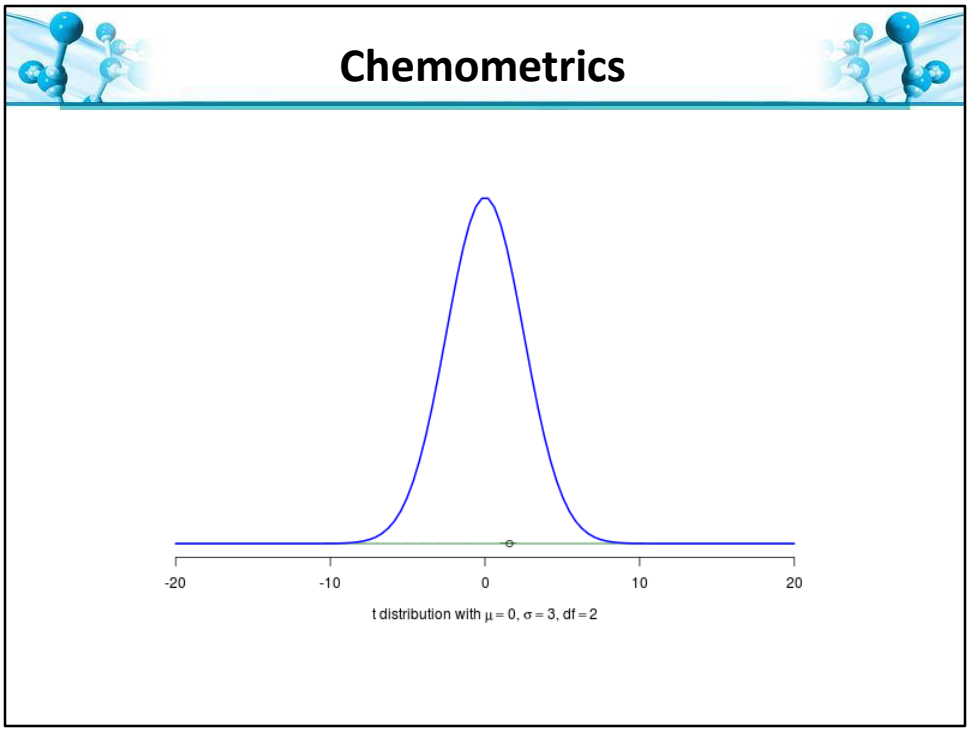
Personnel can be trained properly. If they trained improperly than practice can make a mistake perfect.
Normally we condemned the personnel and make him/her responsible. But in actual sense no one want to do imperfect.
We have to remove causes……
You can understand that if regular POLICE force is trained properly as motorway POLICE, ammended the method (means salaries and other benefits), give them proper equipment of monitoring then no one wish to enjoy bribe.
In absence of these just be writing in bold you can't change the attitude.



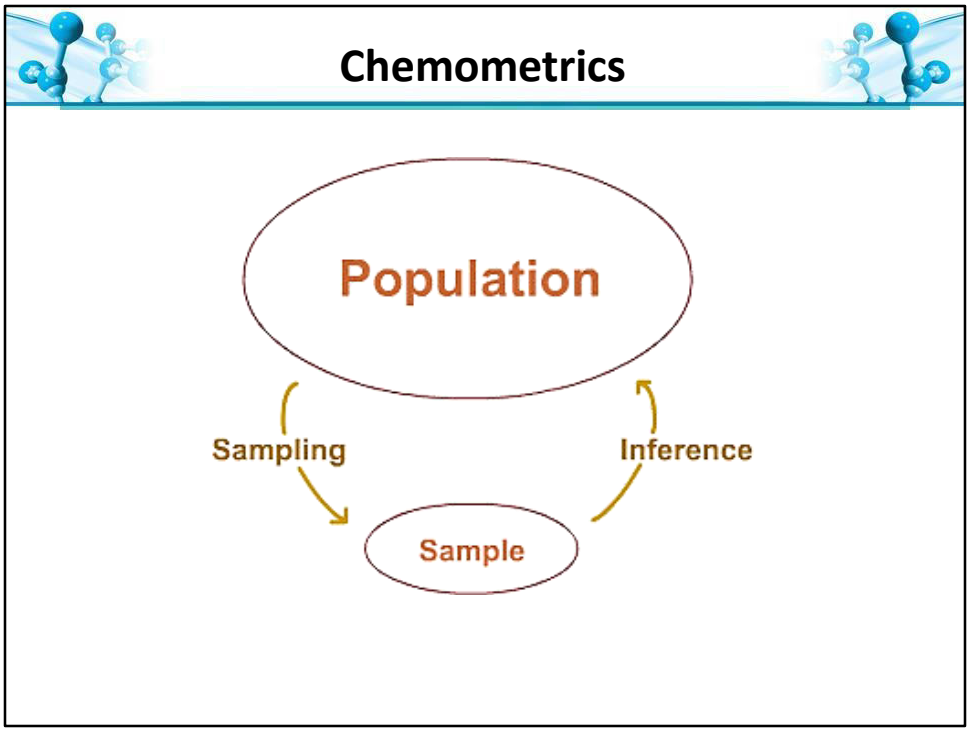
In coming slides we will discuss calculation of random error.



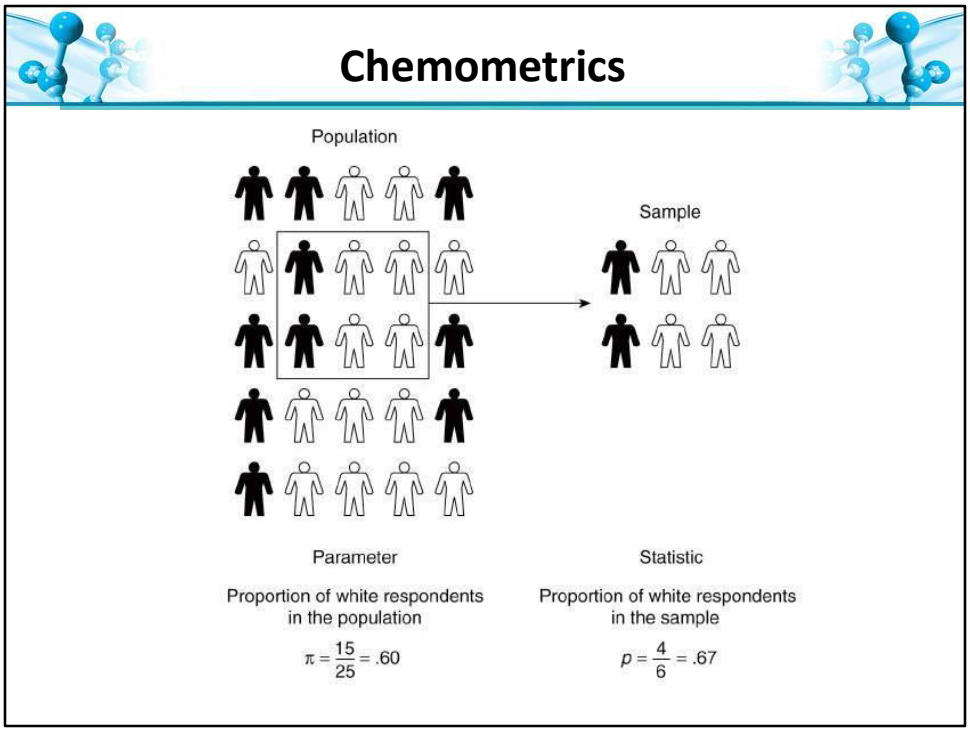
But before starting this we must have an understanding of sample and population. Sample is a part of whole population. So gather analytical data with the sample may be close to actual one or may be different a little bit.
It may vary a lot if sample is not correct.



Here we can see distribution of data and formation of normal distribution curve.



We draw a sample from population and infer our results to the population.



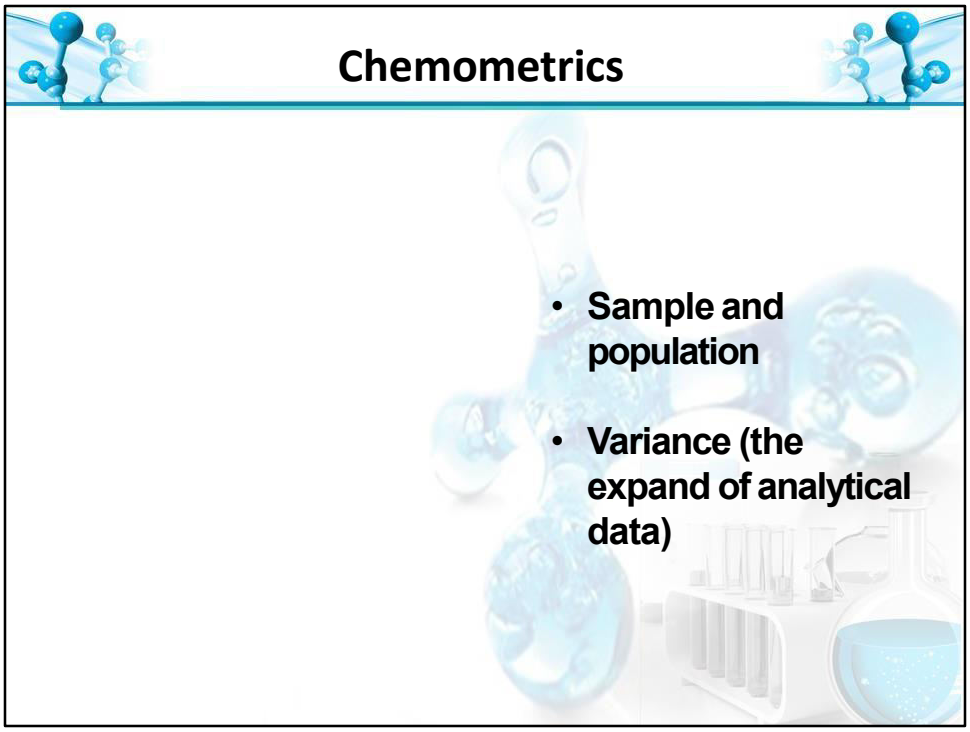
We can see that ratios in our sample can be close enough to population. However a sample from some other portion may miss guide, or biased. In a smaller sample size the chances of biased results increased.

Chemometrics

Sample and Population mean:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Mean representing population and sample.



Variance shows the expand of analytical data. The average of the **squared** differences from the Mean.

Chemometrics

Formula of sample variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{N - 1}$$

Explain the formula

Chemometrics

sample and population variance:

$$s^2 = \frac{\Sigma (x - \bar{x})^2}{n - 1}$$ Sample Variance

$$\sigma^2 = \frac{\Sigma (x - \mu)^2}{N}$$ Population Variance

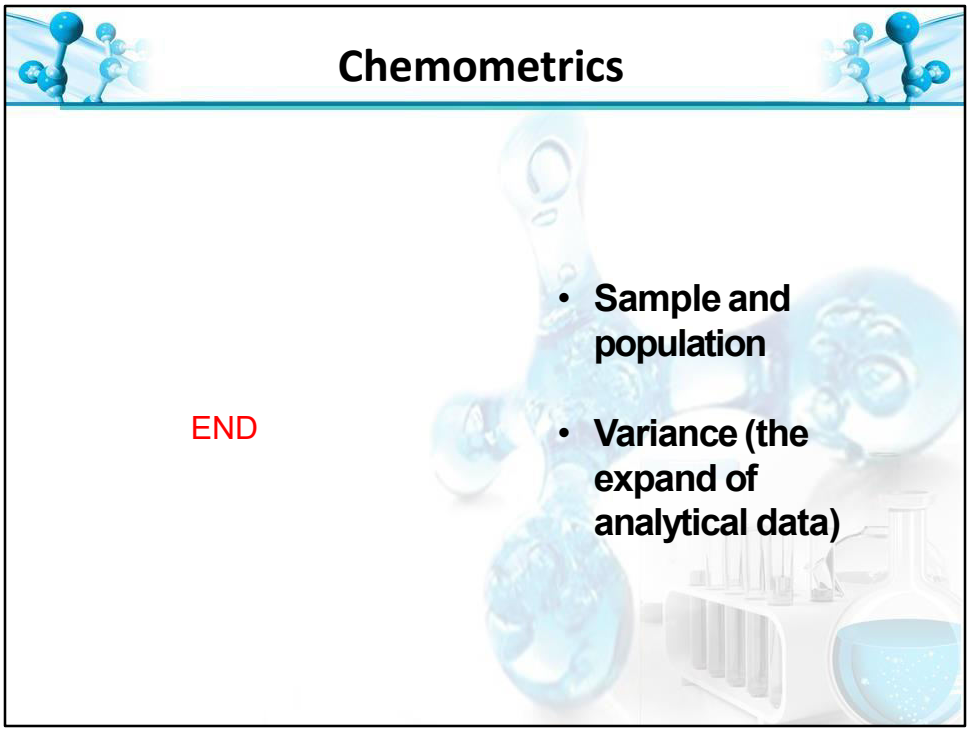
54

To remove the sample bias we divide with n-1 instead of n.

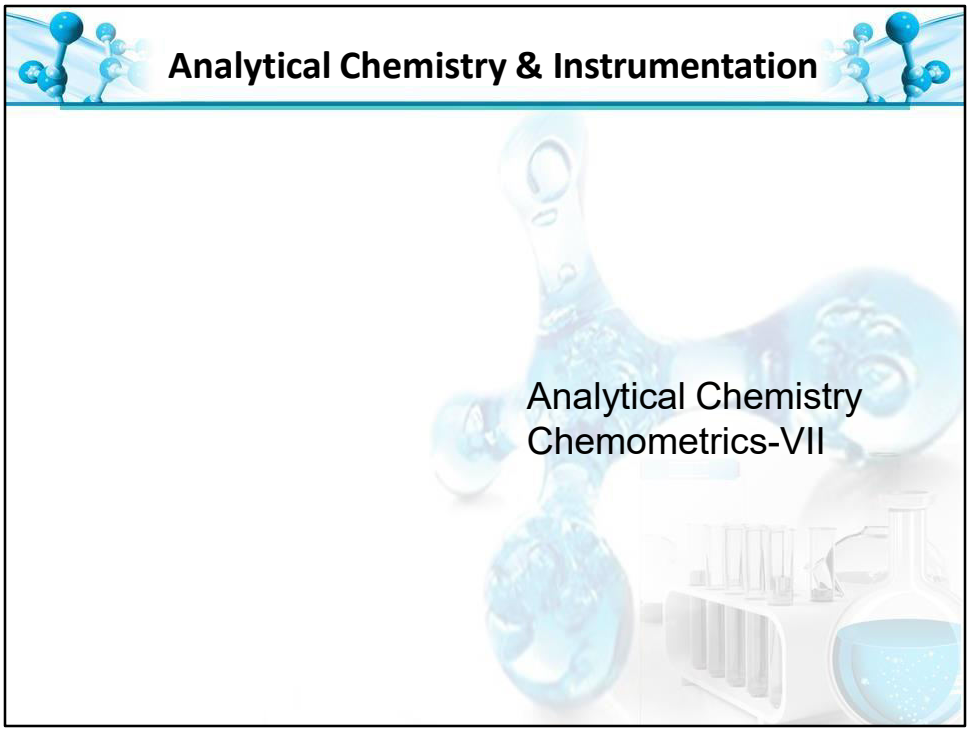
Chemometrics

Understanding unbiased variance:

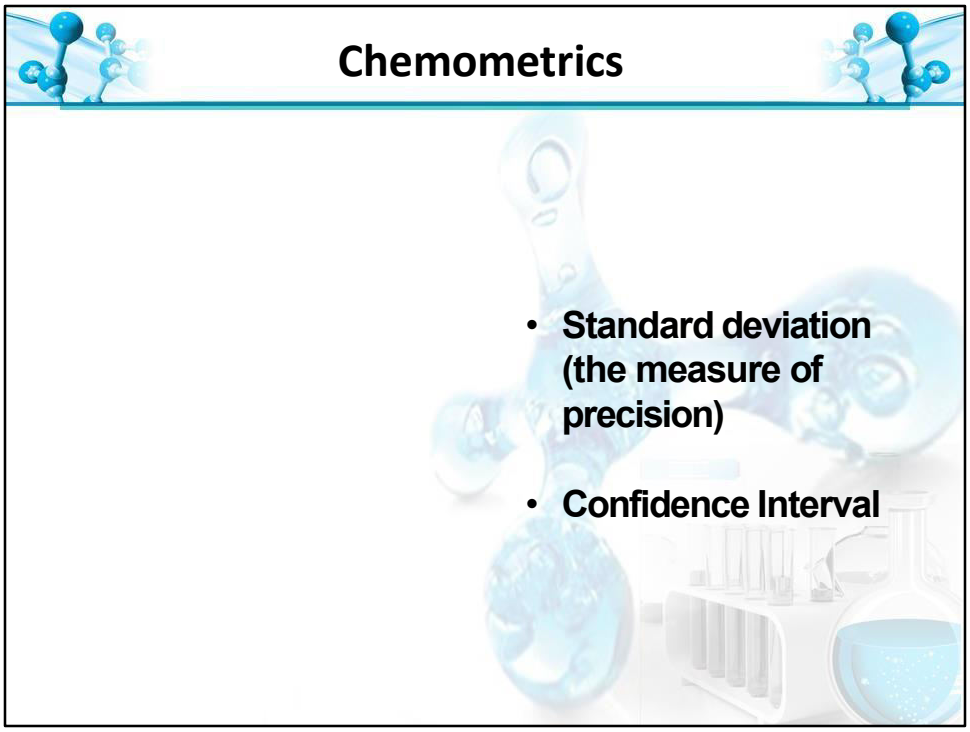
May understand from same gaussian curve. The part of the curve may cover the mean of actual or may not be.
Also show less and more distributed data.



So we learned that Variance is actually distribution of data around mean position.



The standard deviation is square root of variance. We do some exercise to understand it in a better way



Std Dev is measure of precision. It shows how close is your data in a data set

Chemometrics

Standard Deviation formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

It denote with sigma. Explain formula a bit

Chemometrics

How to calculate Std. Dev.?

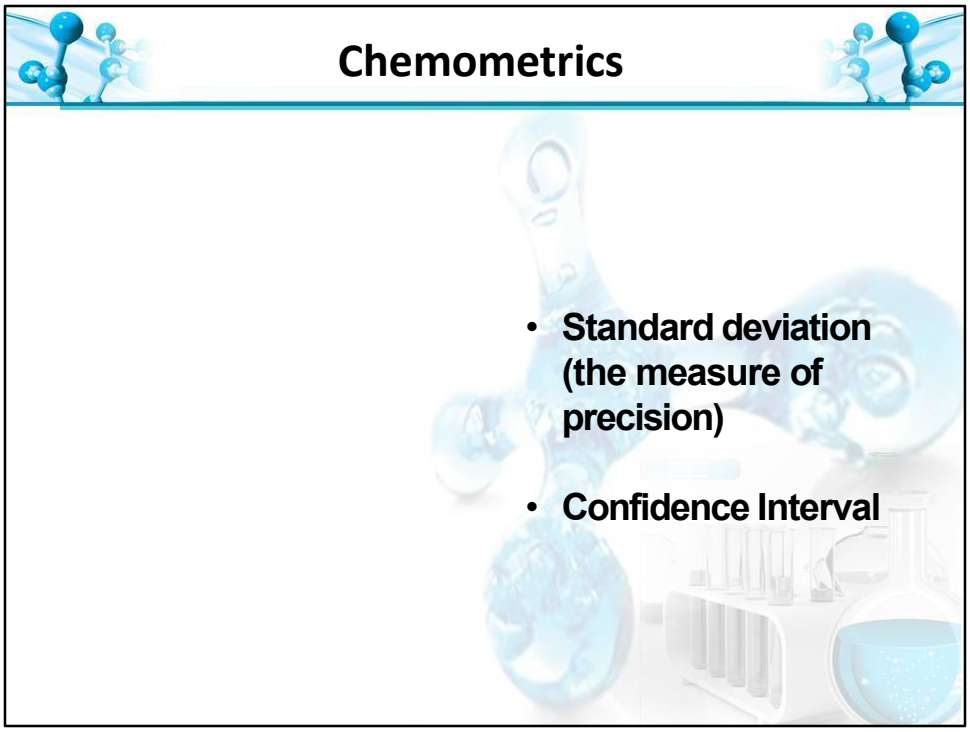
X_i	*Mean*	$X_i - X$	$(X_i - X)^2$
81	81	0	0
83	81	2	4
80	81	-1	1
83	81	2	4
79	81	-2	4
81	81	0	0
78	81	-3	9

Variance = 22/6
= 3.6
~ 4
Std. Dev. = 2

Reporting of Results = mean $\pm$ std dev

Mean = (81+83+80+83+79+81+78)/7

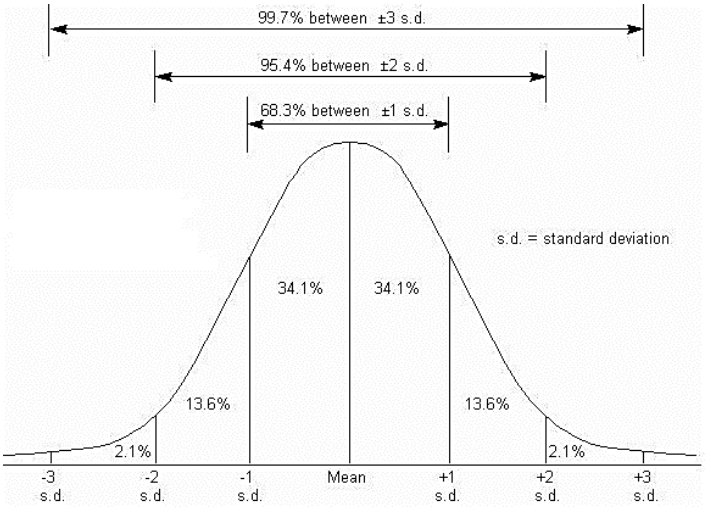
Here we show, how to calculate std dev?



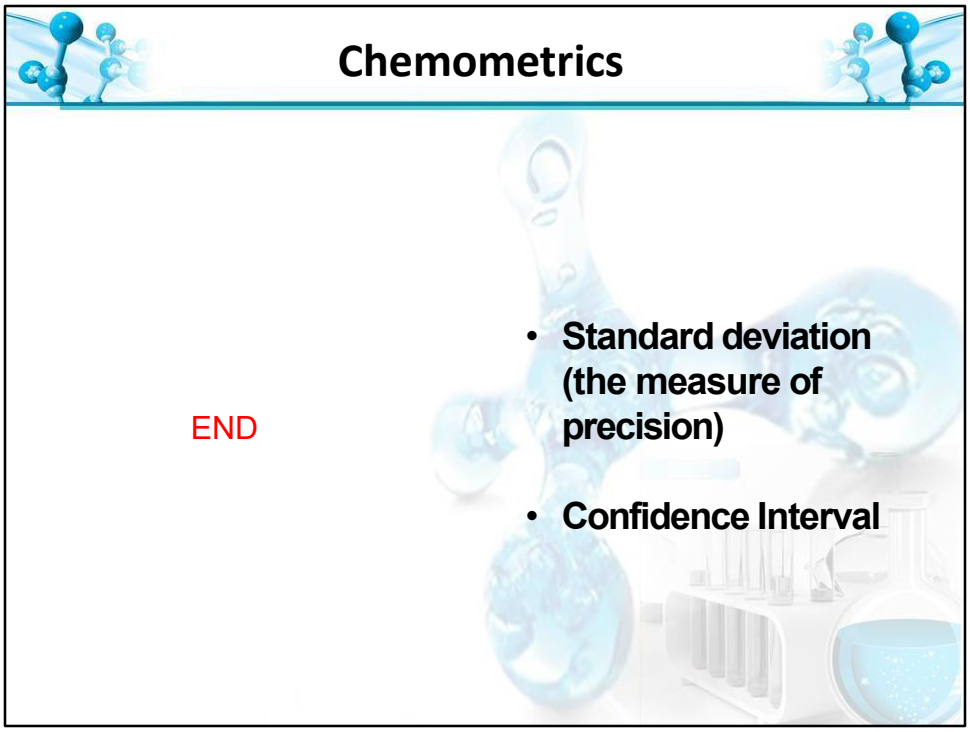
And the next concept is confidence interval. That may be understood with following diagram

Chemometrics

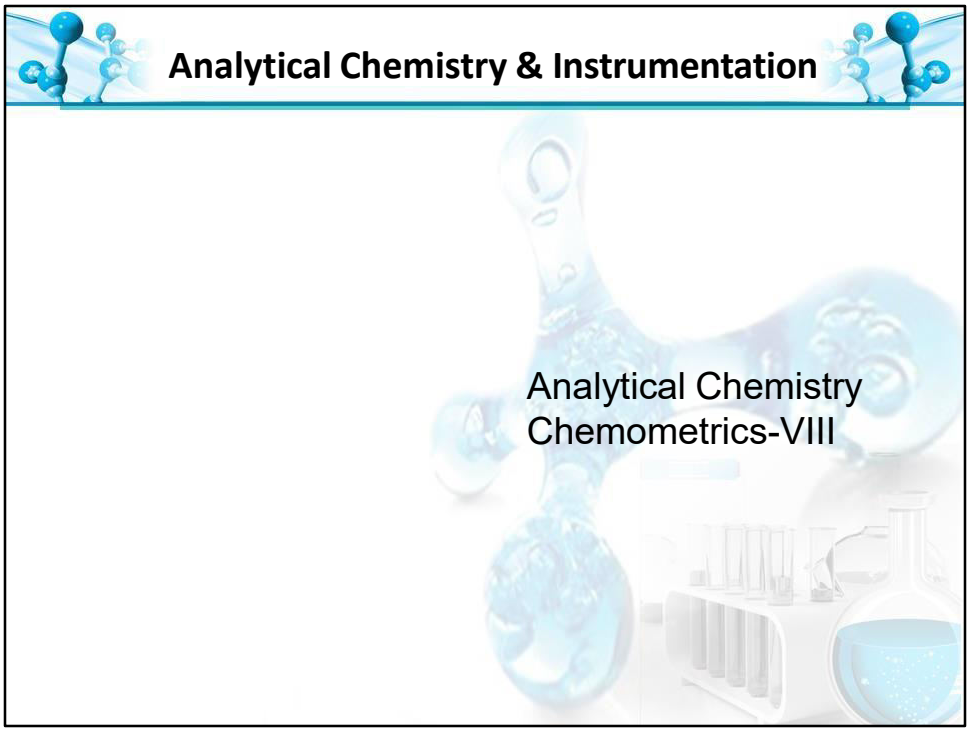
Confidence Interval:



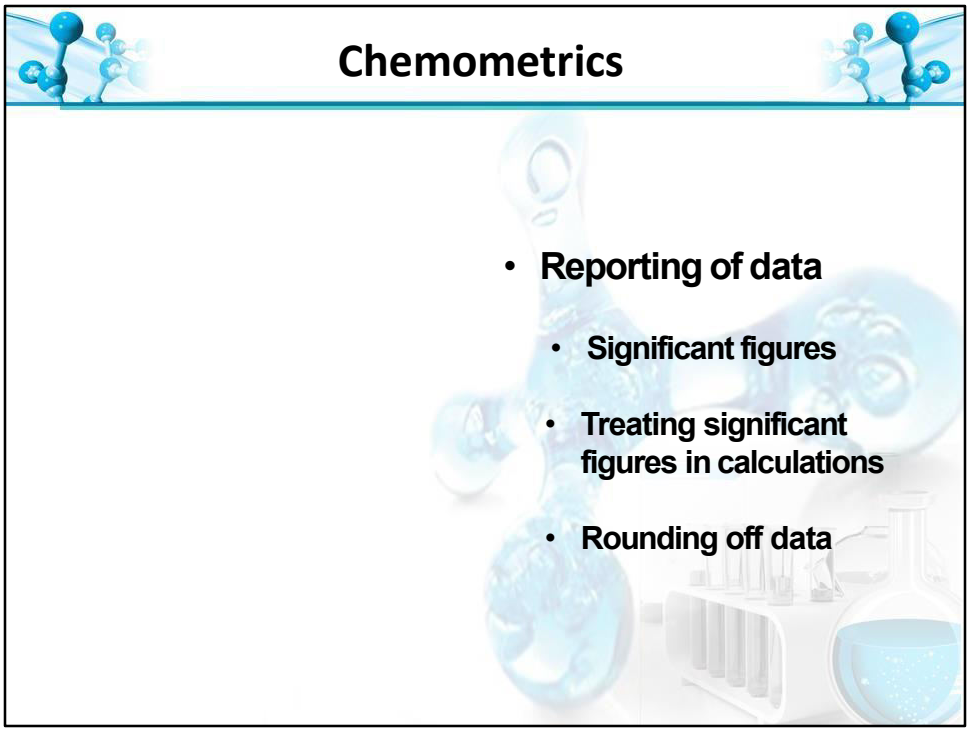
The data covers



So by increasing number of values we can improve distribution and remove bias.

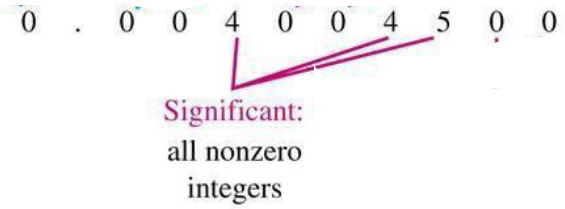


In previous slides we learn to calculate standard deviation. And to report them
But in reporting of results we must understand the significance of significant figures.

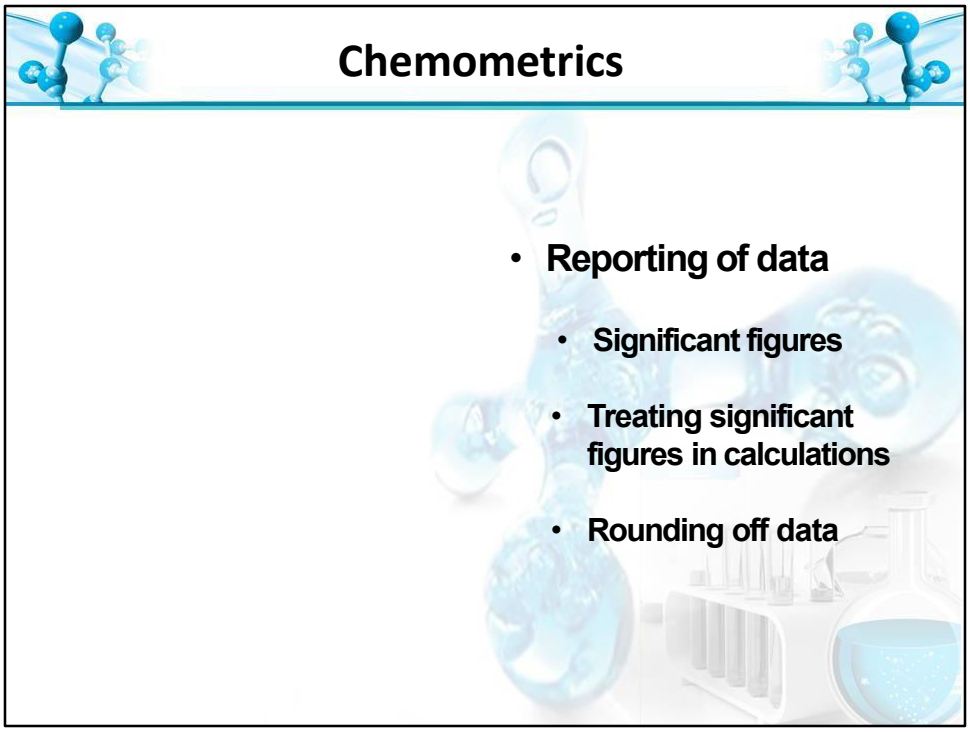


Significant figures should be placed with care. Just write a lengthy figure never make your analytical results precise.
For example if you take titrant readings from a burette, and take an average of seven readings, you might end up with six digits but you must understand that the minimum readable point on burette was 0.1 mL so your result must show same number of digits.

Significant Figures



As general rules

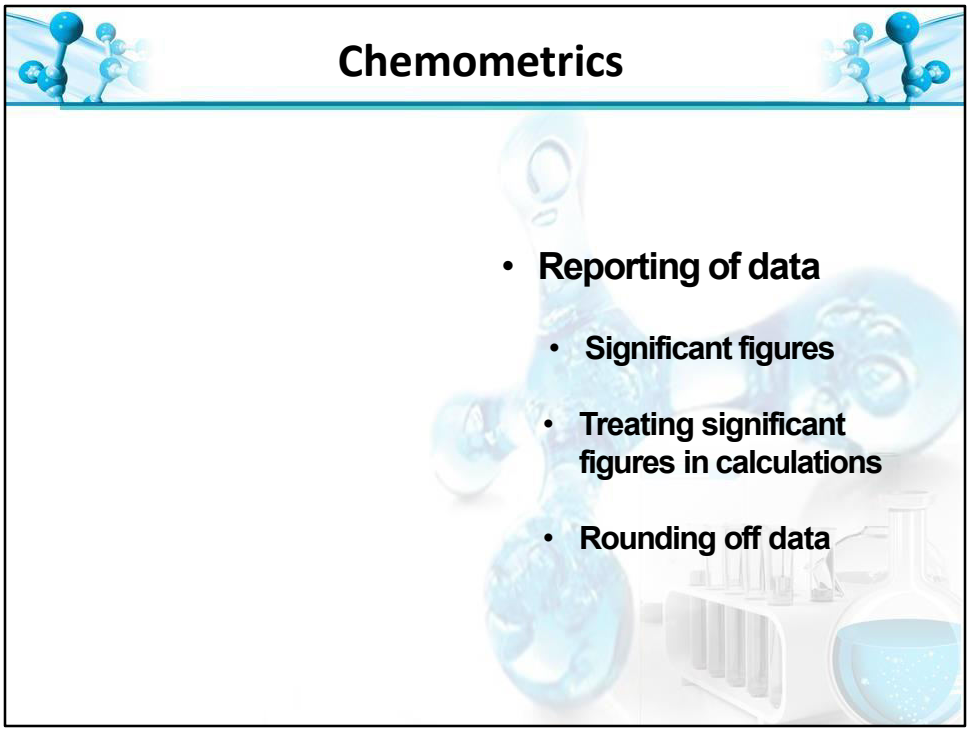


So when ever add, subtract, multiply or divide the data we must follow certain rules

Mathematical Treatment of Significant figures

- In addition or subtraction, the answer should have the least digits after decimal,
  e.g., 3.4 + 0.020 + 7.31 = 10.7.

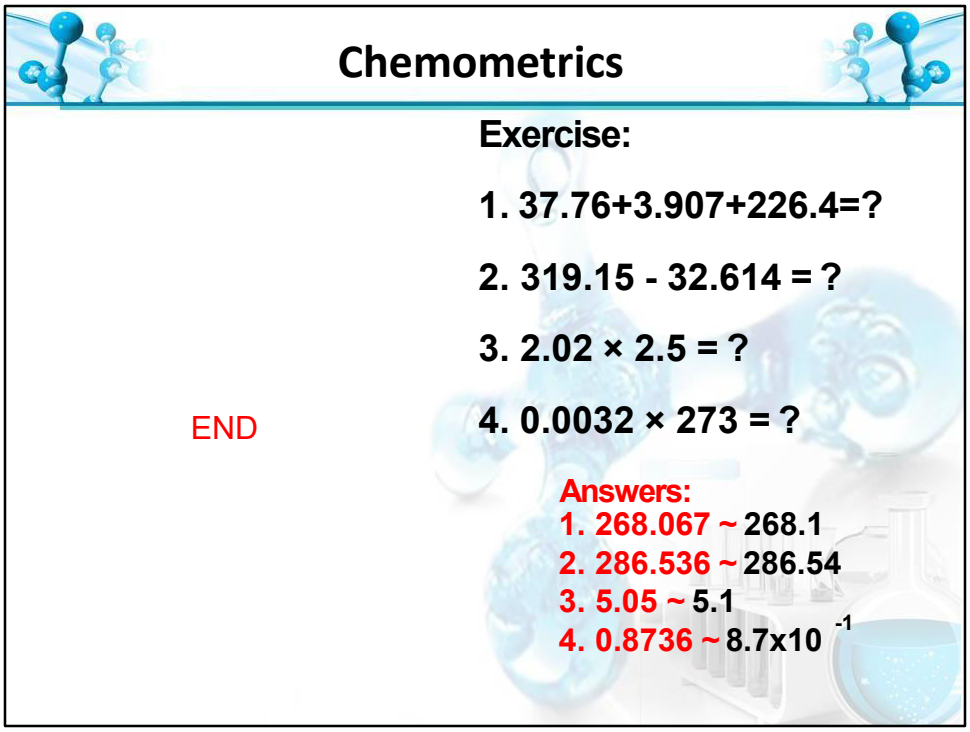
- For multiplication and division answer should be rounded so that it contains the same number of significant digits as the original number with the smallest number of significant digits.



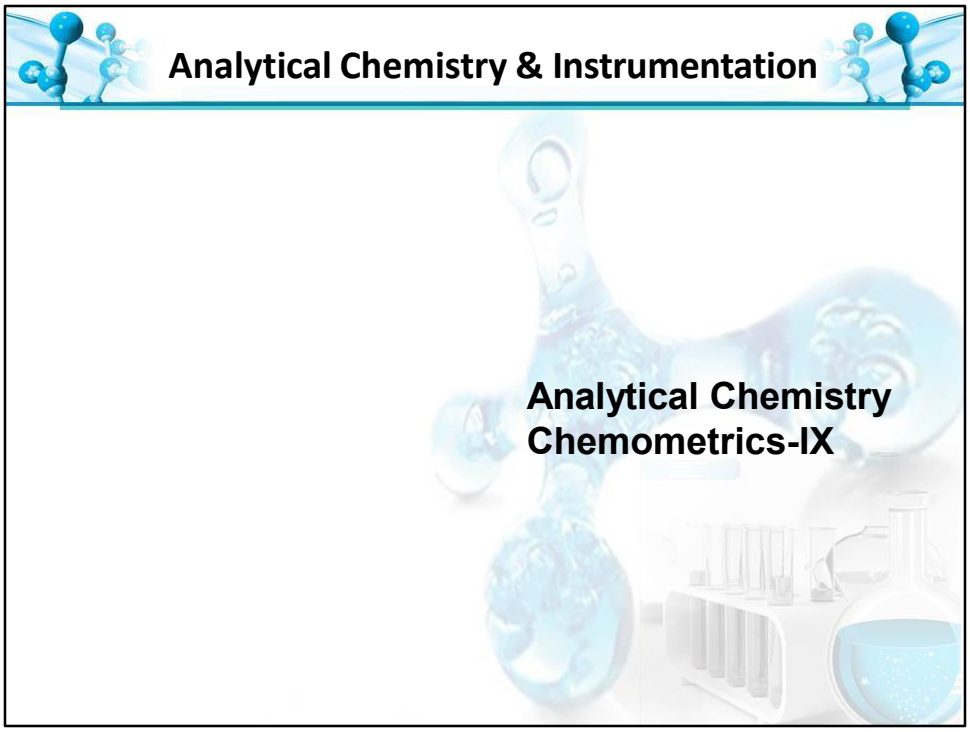
And the additional non significant digits must be rounded off as per followingrules

- In rounding a number greater than 5, last digit is increased by one.
- In rounding a digit less than 5, last digit is retained.
- In rounding a number ending in 5, always round so that the result ends with an even number.

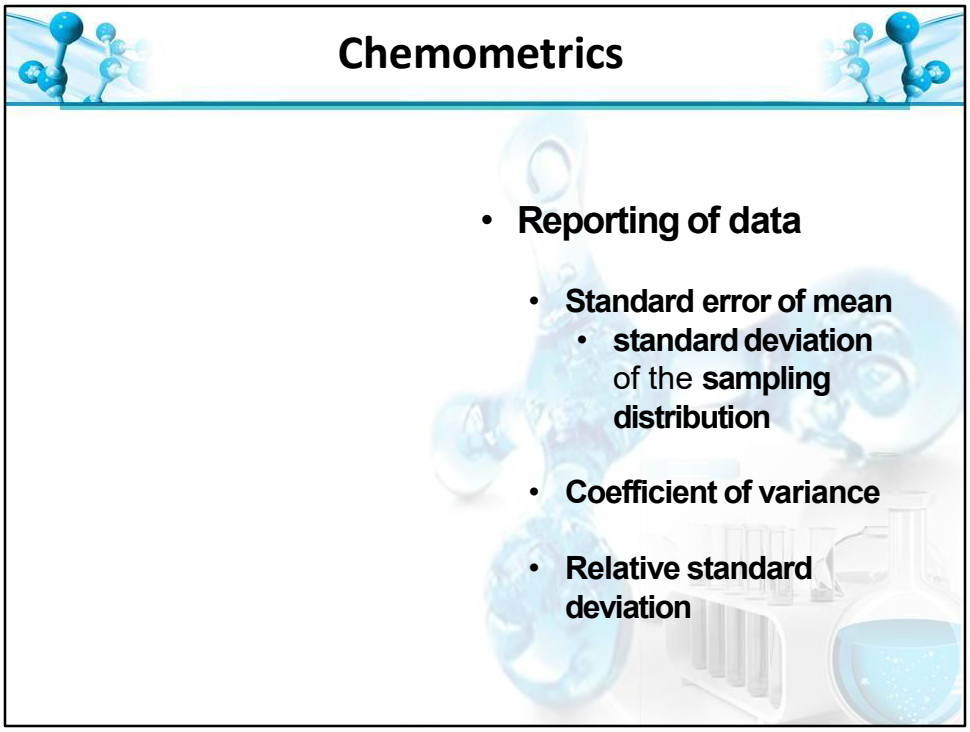
Thus, 0.635 rounds to 0.64 and 0.625 rounds to 0.62.



Here are some questions for practice……..



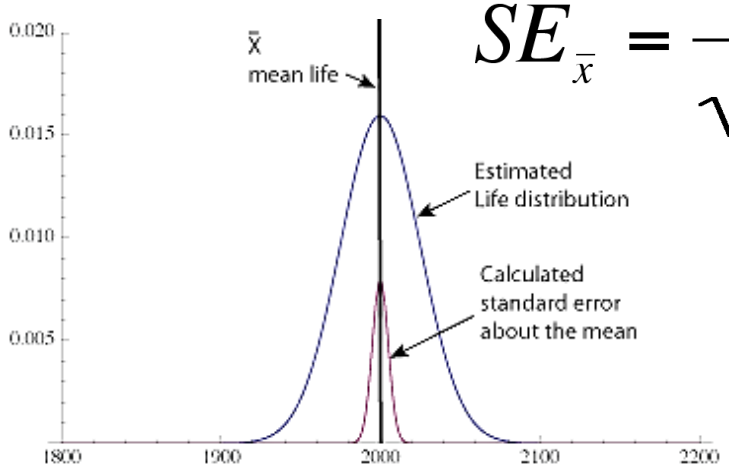
Now we will discuss some other aspects of reporting results



A **standard error** is the **standard** deviation of the **sampling distribution** of a statistic. **Standard error** is a statistical term that measures the accuracy with which a sample represents a population. In statistics, a sample **mean** deviates from the actual **mean** of a population; this deviation is the **standard error**.

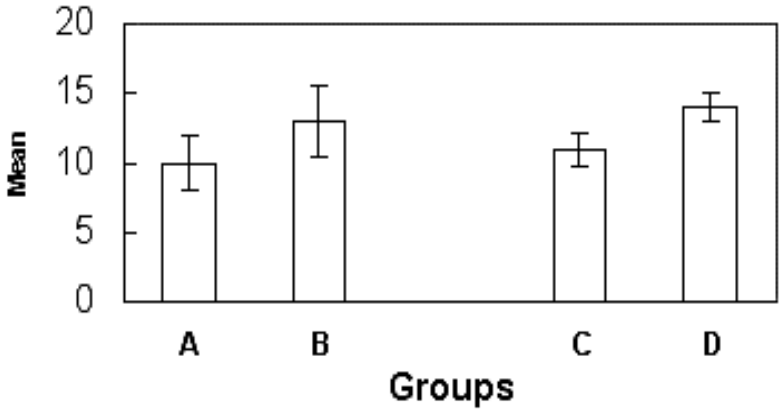
Formula for SEM:

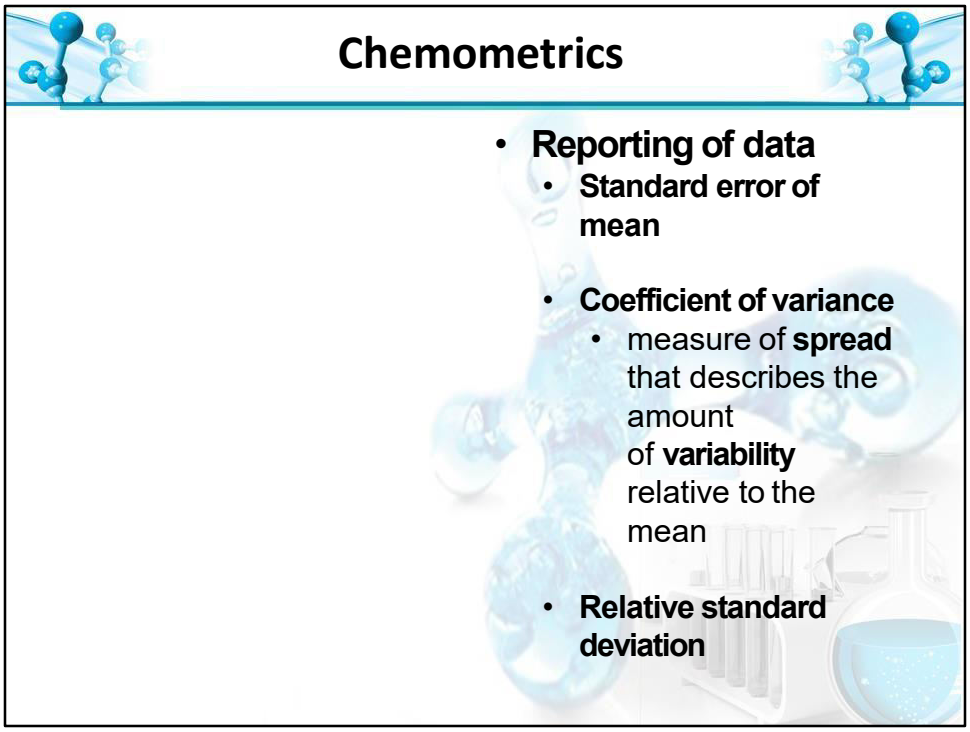
$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$



Chemometrics

Group mean and Standard errors:





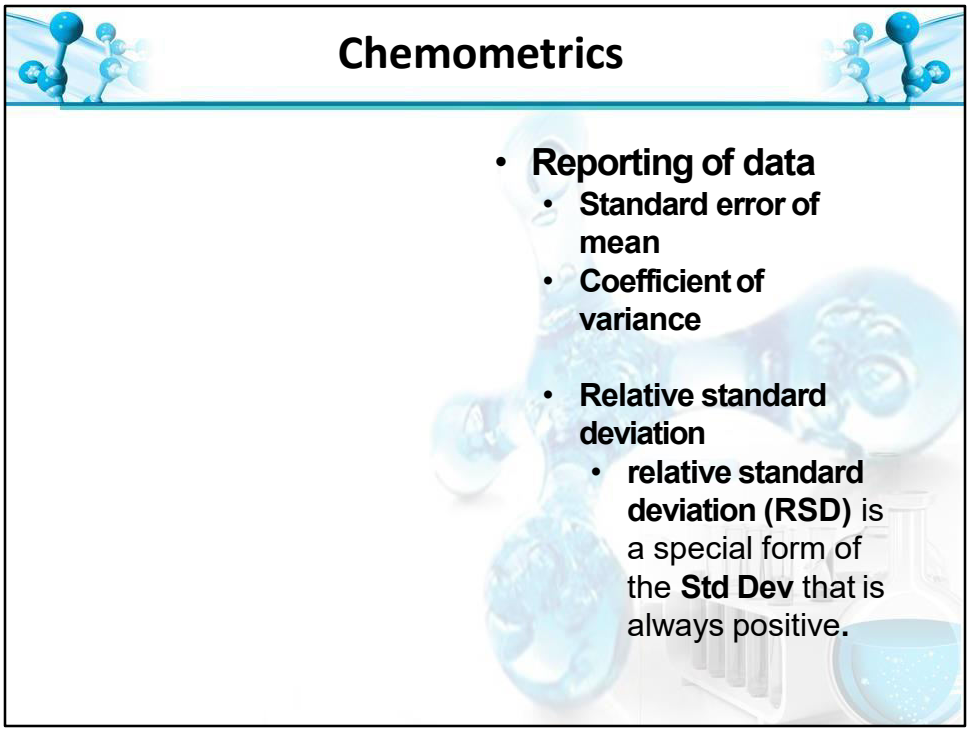
The **coefficient of variation** is a measure of spread that describes the amount of **variability** relative to the mean. Because the **coefficient of variation** is unitless, you can use it instead of the standard deviation to compare the spread of data sets that have different units or different means.

Chemometrics

Formula % CV:

$$CV = \frac{s}{\bar{x}} \times 100$$

Example:



The **relative standard deviation** (RSD) is a special form of the **standard deviation** (std dev). It's generally reported to two decimal places (i.e. an RSD of 2.9587878 becomes 2.96). As the denominator is the absolute value of the mean, the RSD will always be positive

Chemometrics

Formula RSD:

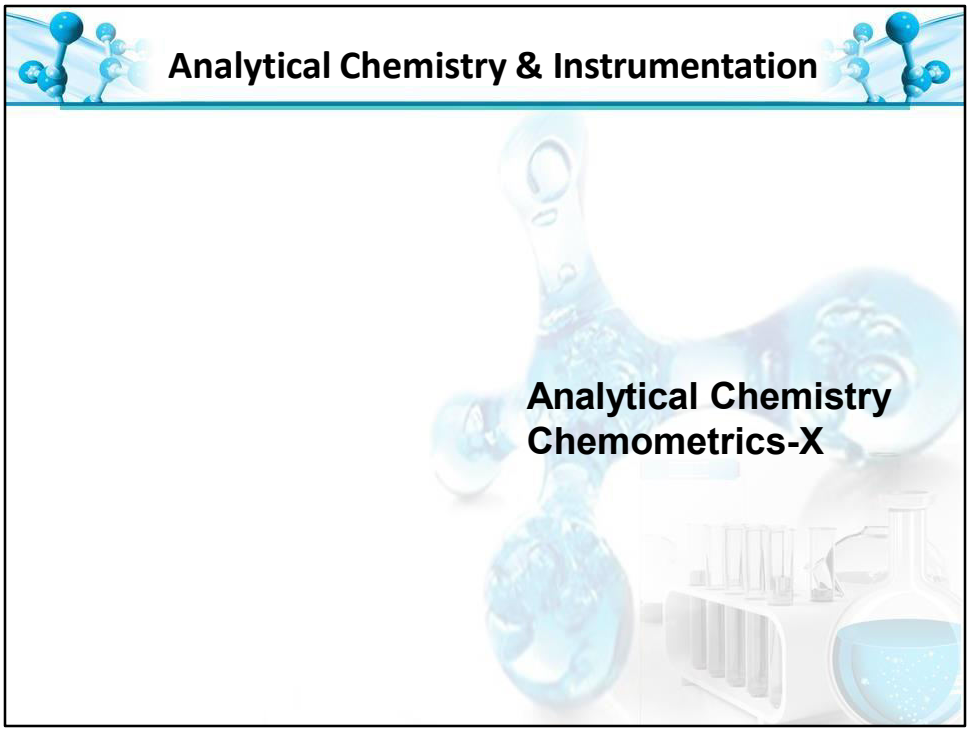
$$RSD = \frac{s}{x} \cdot 100\%$$

However, the RSD cannot be negative while the **Coefficient of Variation** can be positive or negative. This is because the two formulas differ in a minor way: the **Coefficient of Variation** divides by the mean while the RSD divides by the absolute value of the mean

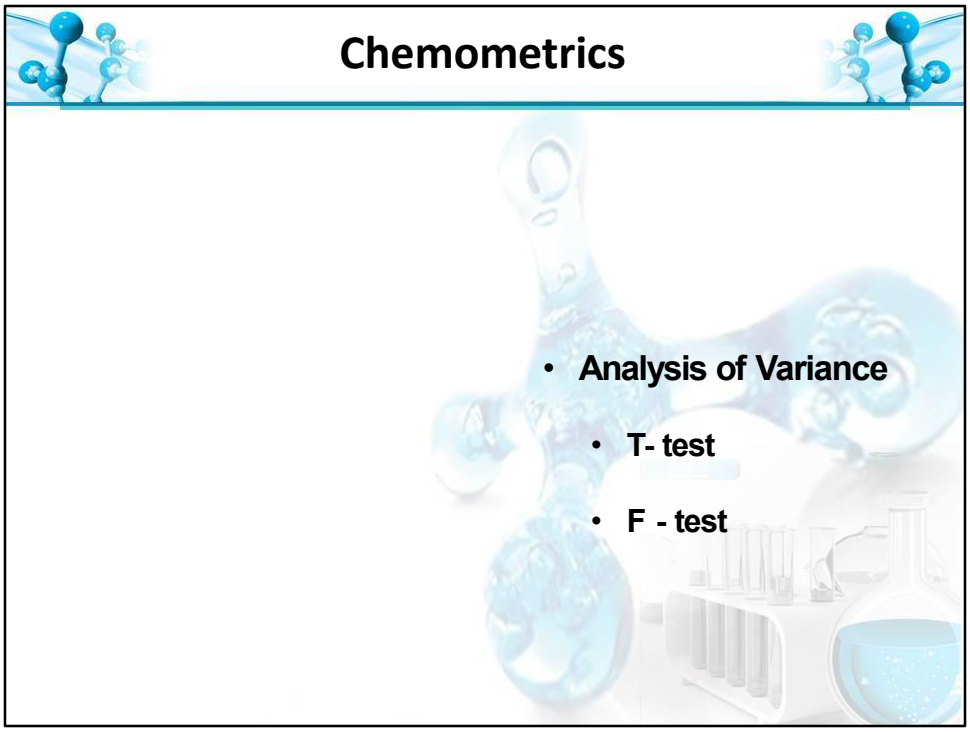
Chemometrics

END

- Reporting of data
  - Standard error of mean
  - Coefficient of variance
  - Relative standard deviation



In previous slides we learn to calculate standard deviation. And to report them
But in reporting of results we must understand the significance of significant figures.

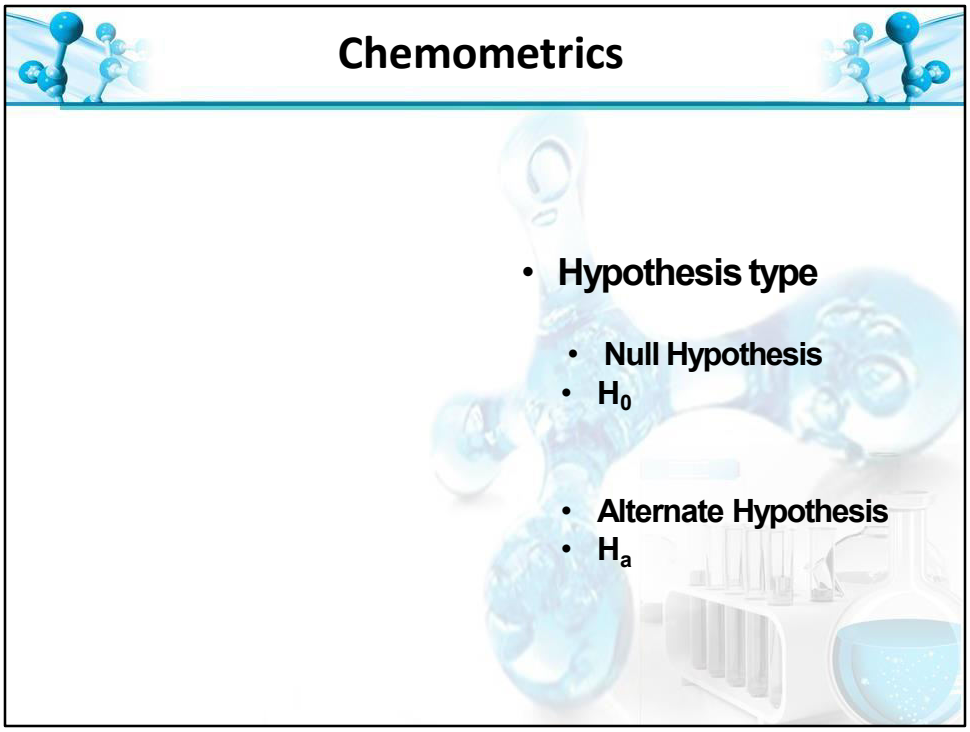


Discuss the example and explain the null and alternate hypothesis

Chemometrics

Discuss and example:

Juice	Tea	Coffee
29	28	25
30	29	28
31	27	29
31	30	27
29	29	29

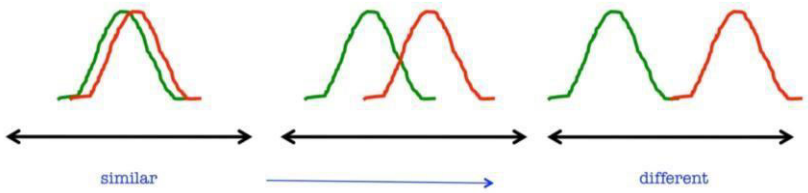


Discuss the example and explain the null and alternate hypothesis

Chemometrics

T test:

Inferential > 1 Variable > 2 Samples > T-Test

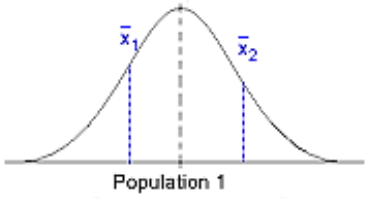


Chemometrics

T test:

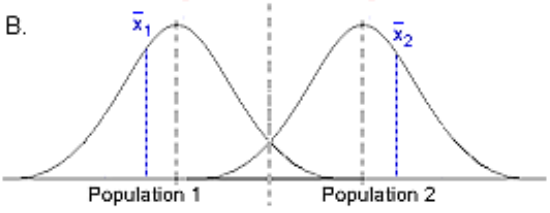
Inferential > 1 Variable > 2 Samples > T-Test

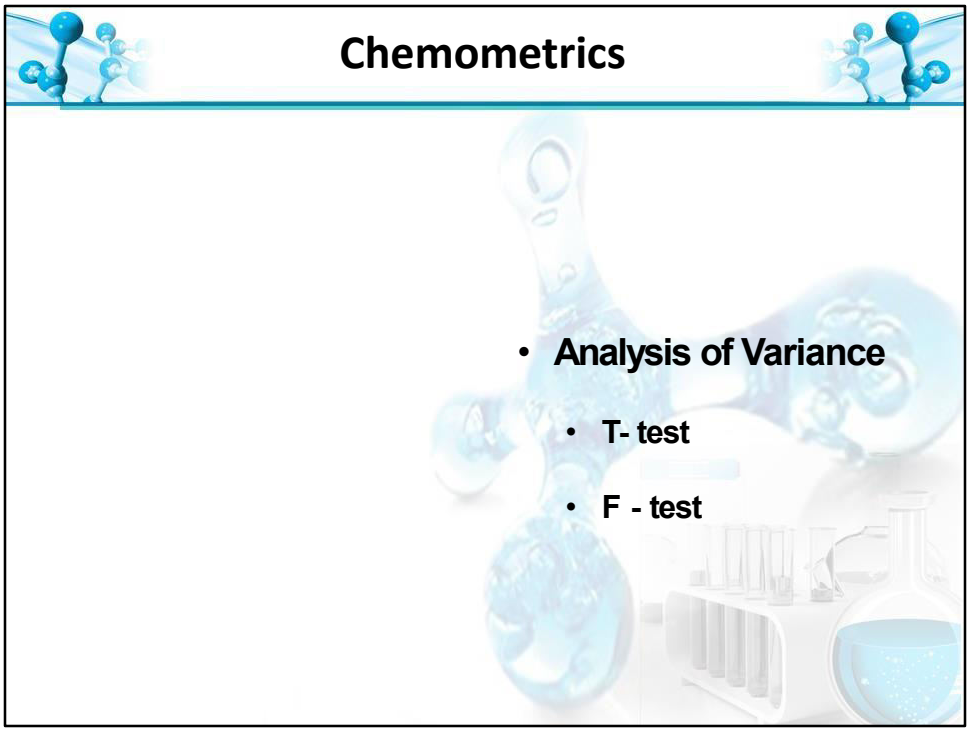
A.



$$t = \frac{\overline{X} - \mu}{\frac{s}{\sqrt{N}}}$$

B.





Discuss the example and explain the null and alternate hypothesis

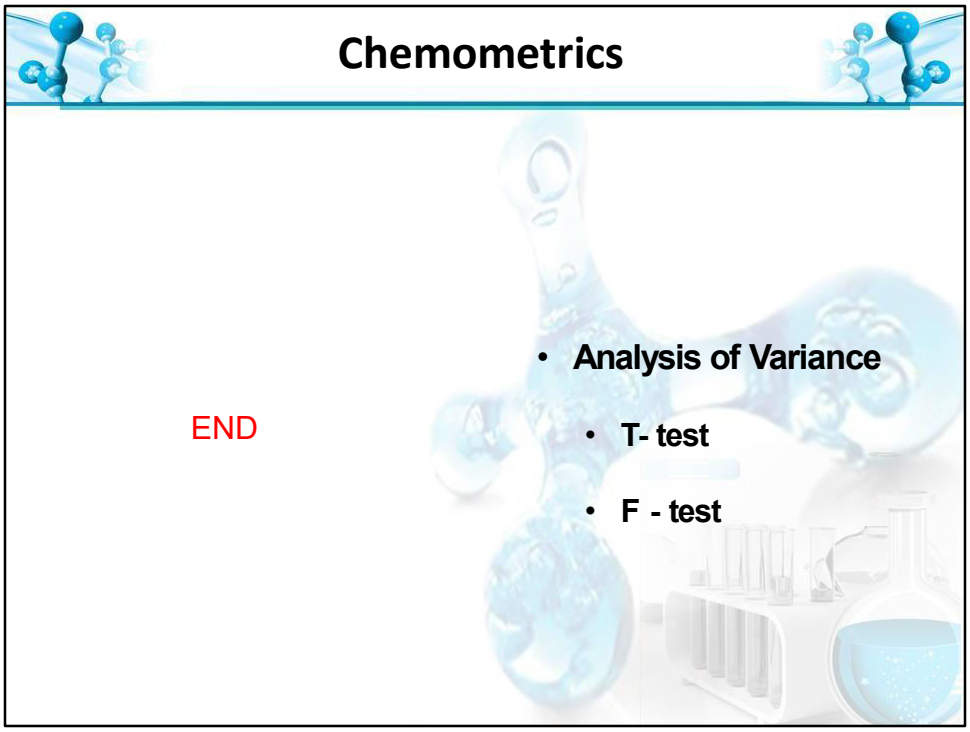
Chemometrics

Discuss and example:

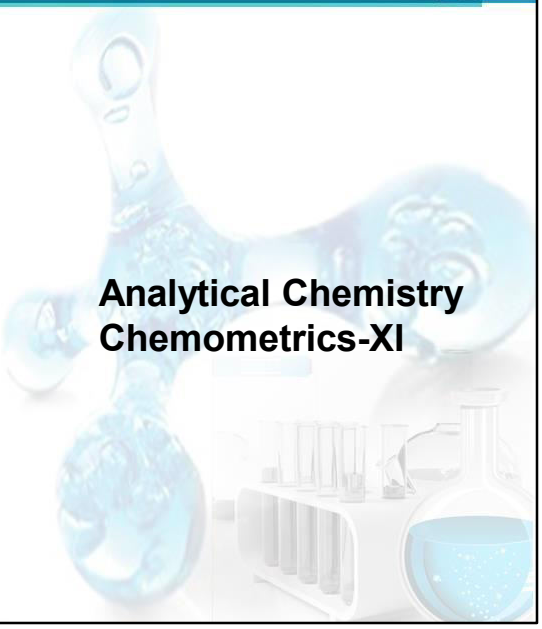
Juice	Tea	Coffee	Juice	Tea	Coffee
29	28	25	10	11	12
30	29	28	12	14	13
31	27	29	18	19	17
31	30	27	24	23	25
29	29	29	36	38	37

F test:

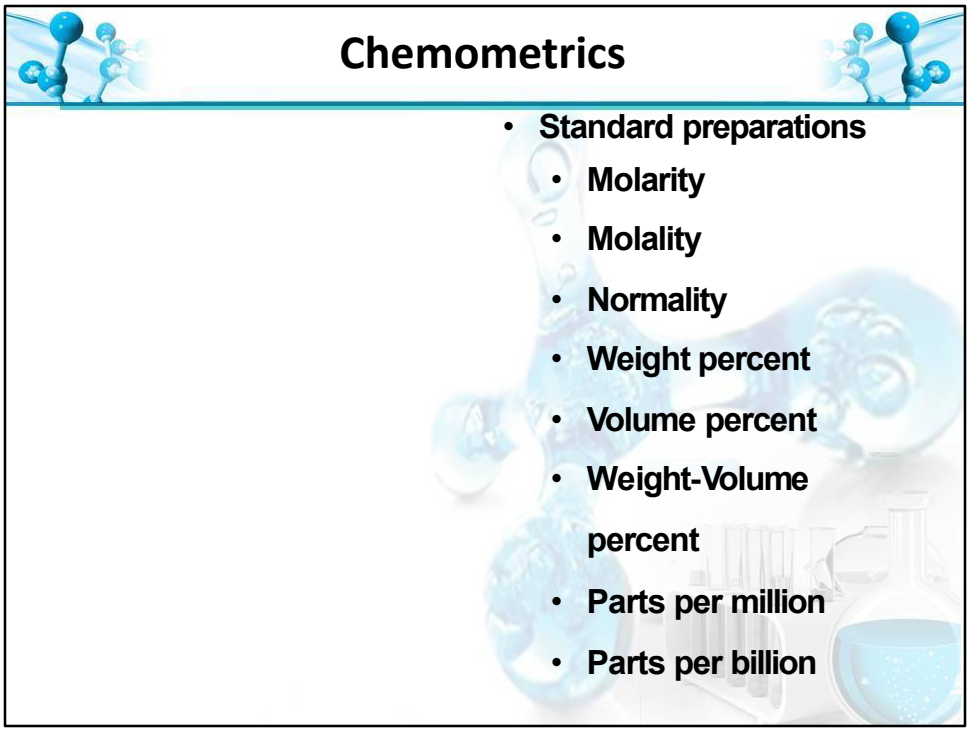
$$\text{F test} = \frac{\text{Variance between}}{\text{Variance within}}$$



Discuss the example and explain the null and alternate hypothesis



Analytical Chemistry
Chemometrics-XI



Molarity is the concentration of a particular chemical species. Formality, on the other hand, is a substance's total concentration without regard to its specific chemical form.

Chemometrics

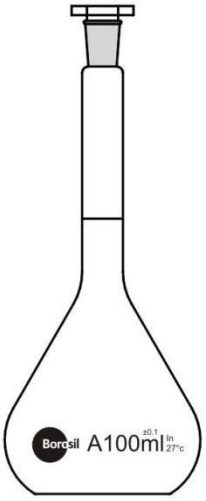
Concept of mole:



To take gram weight we use balances

Chemometrics





Not all the Volumetric apparatus used to take volumes.

Chemometrics

Molarity of solution:

$$M = \frac{\text{moles solute}}{\text{Liters of solution}}$$

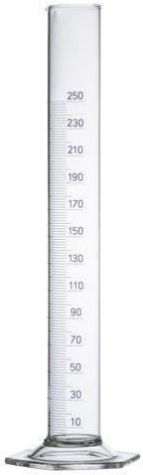
$$\text{Moles of unknown} = \frac{\text{mass of unknown}}{\text{molar mass}}$$

Molality is used in thermodynamic calculations where a temperature independent unit of concentration is needed. Molarity is based on the volume of solution containing the solute. Since density is a temperature dependent property a solution's volume, and thus its molar concentration, changes with temperature. By using the solvent's mass in place of the solution's volume, the resulting concentration becomes independent of temperature.

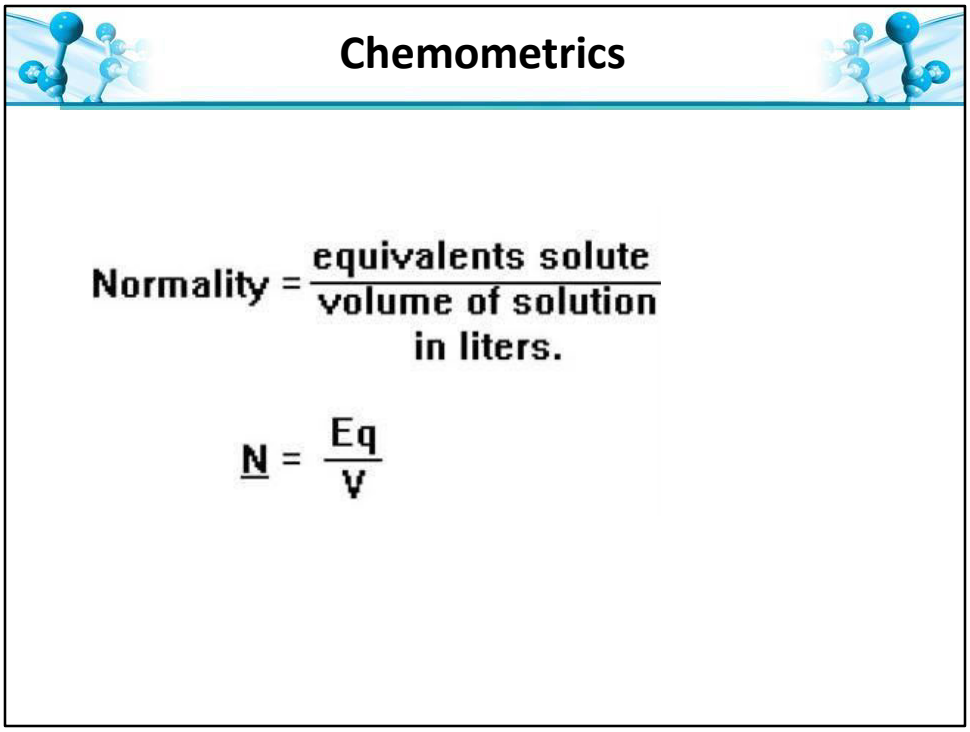
Chemometrics

$$Molality = m = \frac{moles\ of\ solute}{kg\ of\ solvent}$$

$$m = \frac{mol}{kg}$$



How to take Kg of solution….. May use balance or measuring cylinder



Normality is a concentration unit that is no longer in common use. Because you may encounter normality in older handbooks of analytical methods, it can be helpful to understand its meaning. Normality defines concentration in terms of an equivalent, which is the amount of one chemical species reacting stoichiometrically with another chemical species.

Chemometrics

$$\text{Equivalent Weight} = \frac{\text{Molar Mass}}{H^+ \text{ ion per mole}}$$

$$\text{Eq. wt. of } \mathbf{O.A.} = \frac{\text{Molecular weight}}{\text{No. of electrons gained by one molecule}} = \frac{\text{Molecular weight}}{\text{Change in O.N. per mole}}$$

$$\text{Eq. wt. of } \mathbf{R.A.} = \frac{\text{Molecular weight}}{\text{No. of electrons lost by one molecule}} = \frac{\text{Molecular weight}}{\text{Change in O.N. per mole}}$$

Note that this definition makes an equivalent, and thus normality, a function of the chemical reaction in which the species participates. Although a solution of H2SO4 has a fixed molarity, its normality depends on how it reacts.

Chemometrics

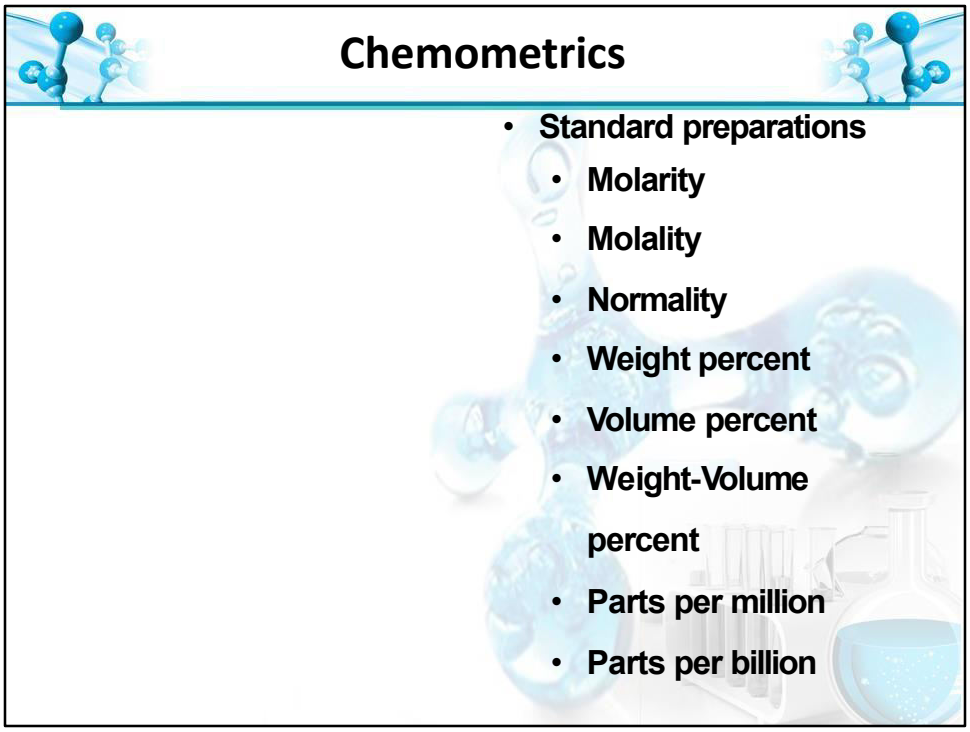
weight percent	$\frac{\text{grams solute}}{\text{100 grams solution}}$	% w/w
volume percent	$\frac{\text{mL solute}}{\text{100 mL solution}}$	% v/v
weight-to-volume percent	$\frac{\text{grams solute}}{\text{100 mL solution}}$	% w/v

Weight percent (% w/w), volume percent (% v/v) and weight-to-volume percent (% w/v) express concentration as the units of solute present in 100 units of solution. A solution of 1.5% w/v NH_4NO_3, for example, contains 1.5 gram of NH_4NO_3 in 100 mL of solution.

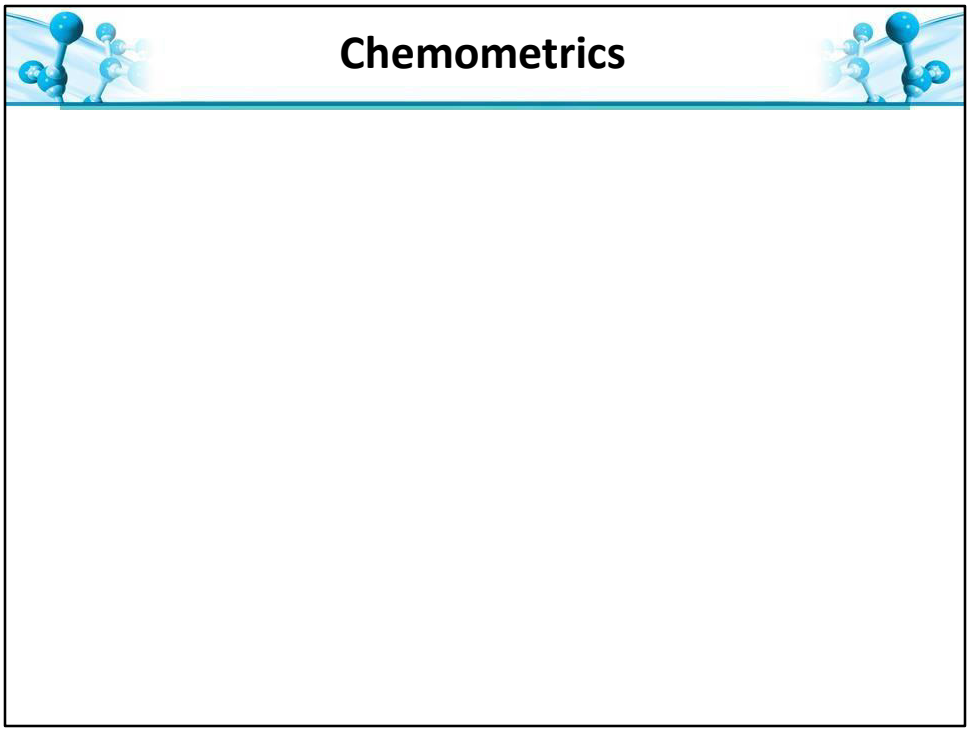
Chemometrics

parts per million	$\frac{\text{grams solute}}{10^6 \text{ grams solution}}$	ppm
parts per billion	$\frac{\text{grams solute}}{10^9 \text{ grams solution}}$	ppb

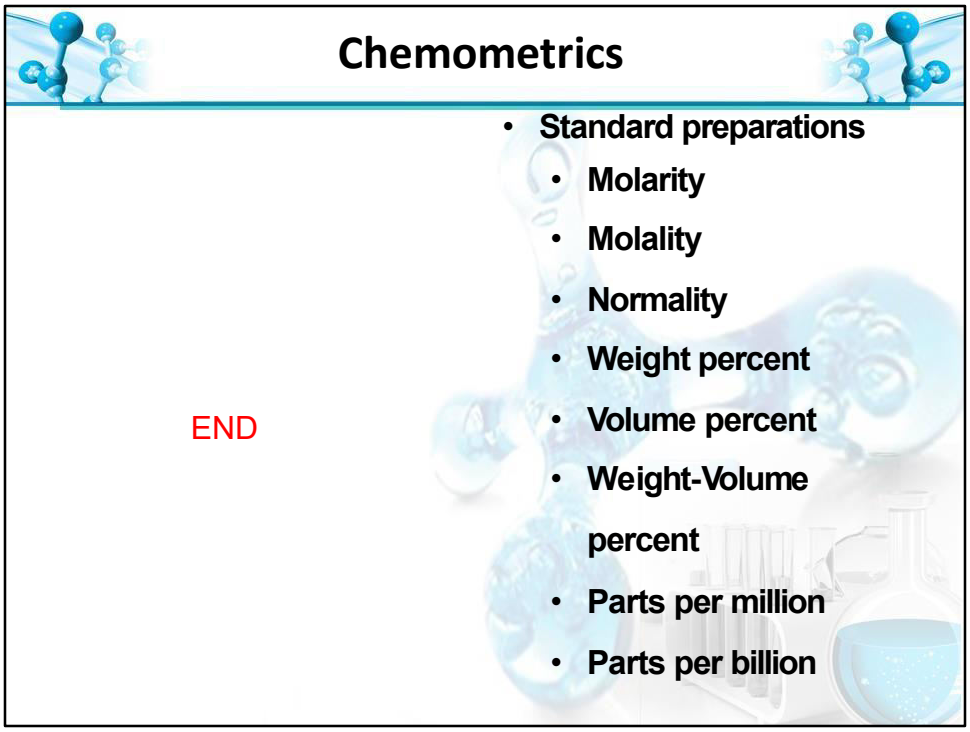
Parts per million (ppm) and parts per billion (ppb) are ratios giving the grams of solute to, respectively, one million or one billion grams of sample. For example, a steel that is 450 ppm in Mn contains 450 µg of Mn for every gram of steel. If we approximate the density of an aqueous solution as 1.00 g/mL, then solution concentrations can be express in ppm
or ppb using the following relationships.



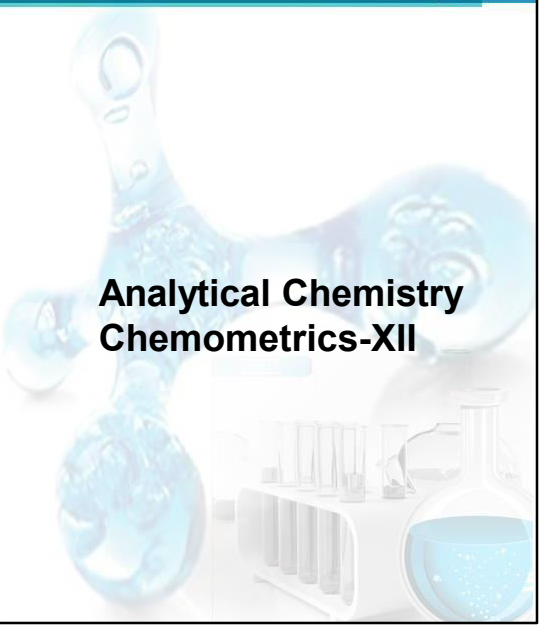
Molarity is the concentration of a particular chemical species. Formality, on the other hand, is a substance's total concentration without regard to its specific chemical form.



Discuss example of converting units and making dilutions from stock solutions……



Molarity is the concentration of a particular chemical species. Formality, on the other hand, is a substance's total concentration without regard to its specific chemical form.



Analytical Chemistry
Chemometrics-XII

Chemometrics

- Goals of sampling
- Sample Size
- Sampling types
- Sampling techniques
- Sample preparations

Chemometrics

- **Goals of Sampling**
  - **To obtain a mean analyte concentration**
  - **Unbiased estimate of the population mean.**
  - **Every possible sample is equally and likely to be drawn.**

Chemometrics

- **Gross Sample Size is determined by**
  - **the uncertainty that can be tolerated between the composition of the gross sample**
  - **the degree of heterogeneity**
  - **the level of particle size at which heterogeneity begins.**

Chemometrics

- Sampling types
  - Grab sample
  - Composite Samples
  - Gross samples

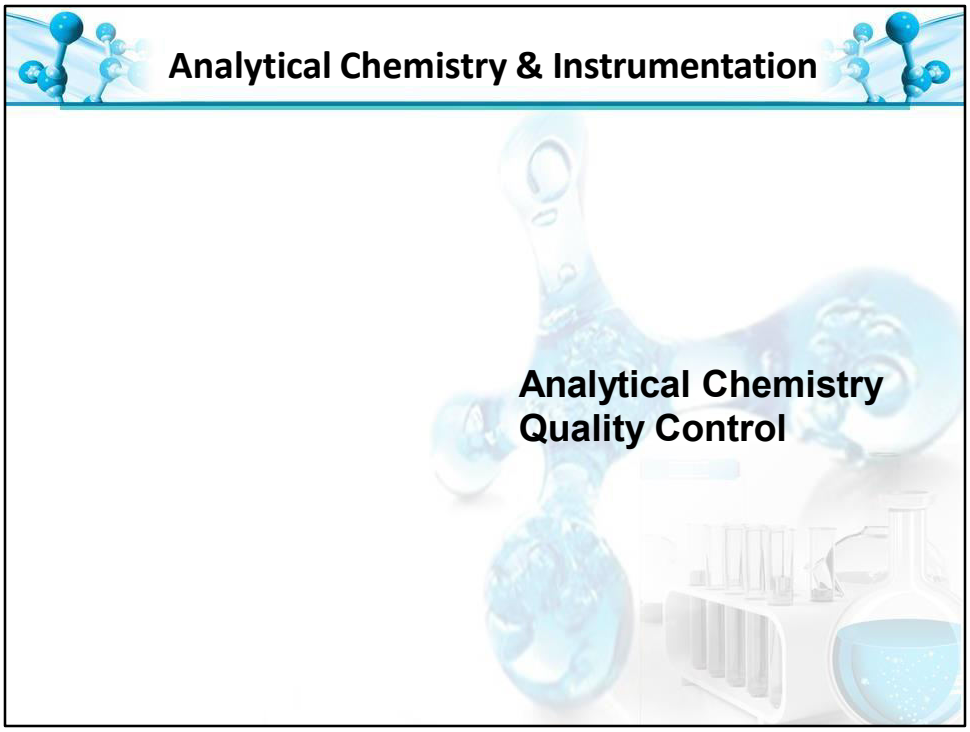
Chemometrics

- **Sampling techniques**
  - **Sampling Homogeneous Solutions (Liq. & Gases)**
  - **Sampling solids (sawing, milling, grinding)**

Chemometrics

END

- **Sample preparations**
  - **Gross samples may be huge**
  - **Crushing, grinding, sieving, mixing, and dividing the sample (often into halves) to reduce its mass**
  - **Digestion and Extraction techniques**



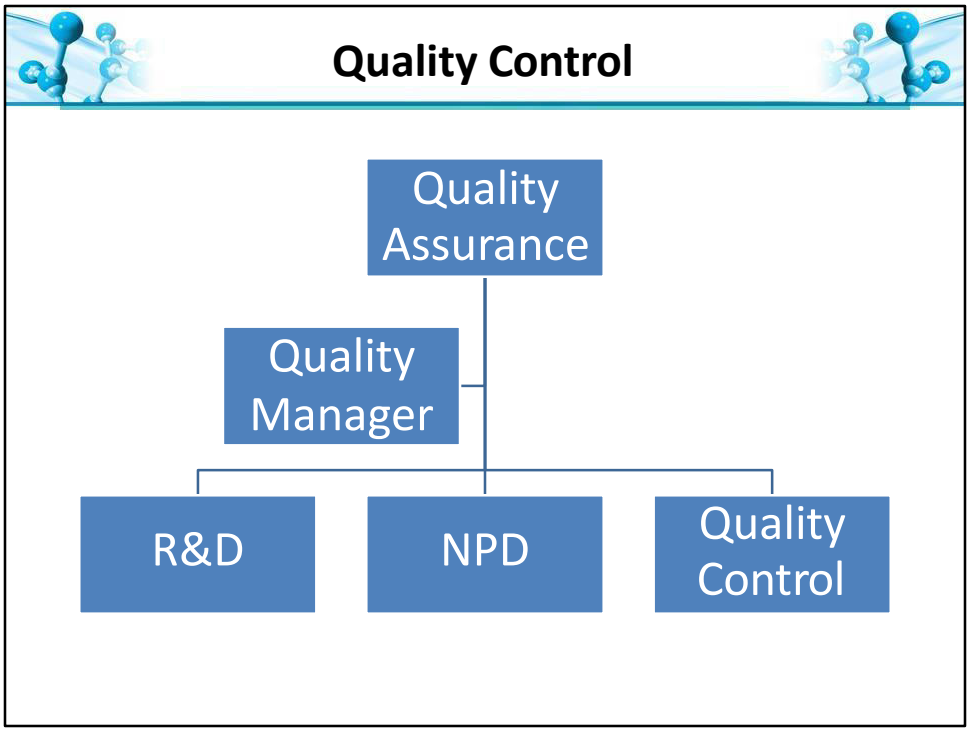
In this module we will discuss about Quality Control and Quality Assurance



Here we have some definitions of Quality



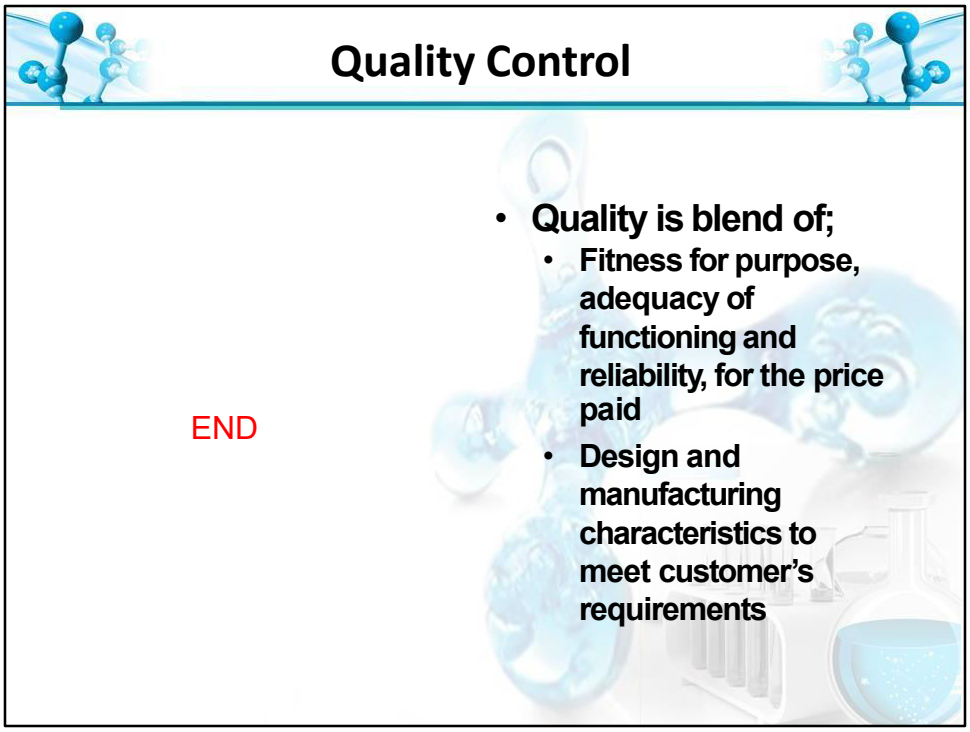
Quality is not residing in any particular field of an organization but it is at par and a part of every



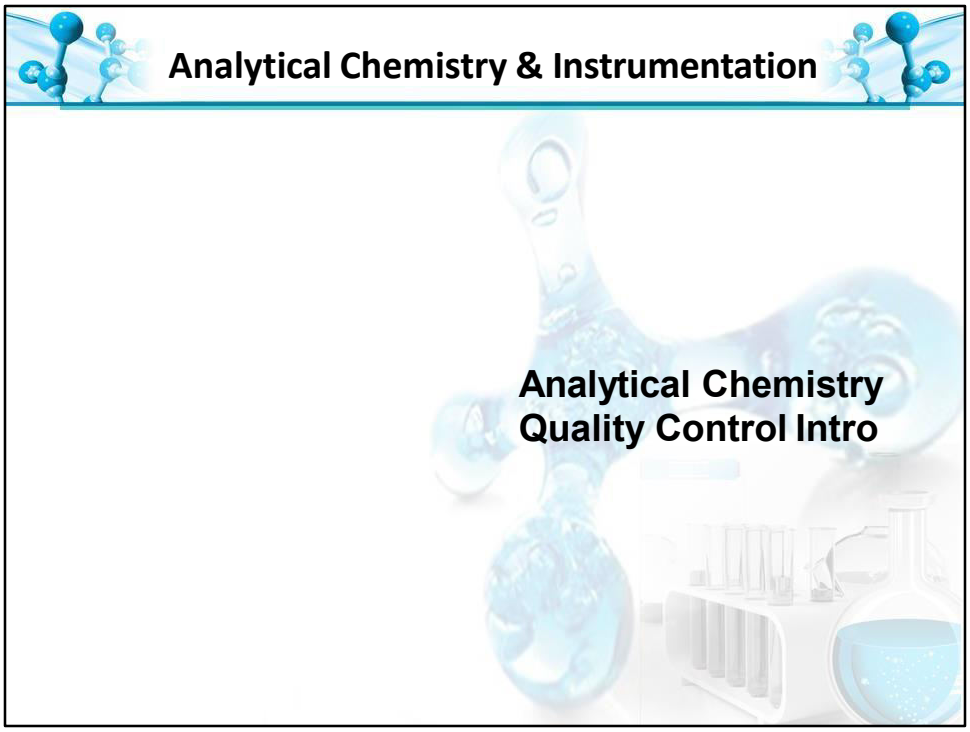
Quality Control is a part of Quality Assurance. Quality Control is part of testing while QA is overall system



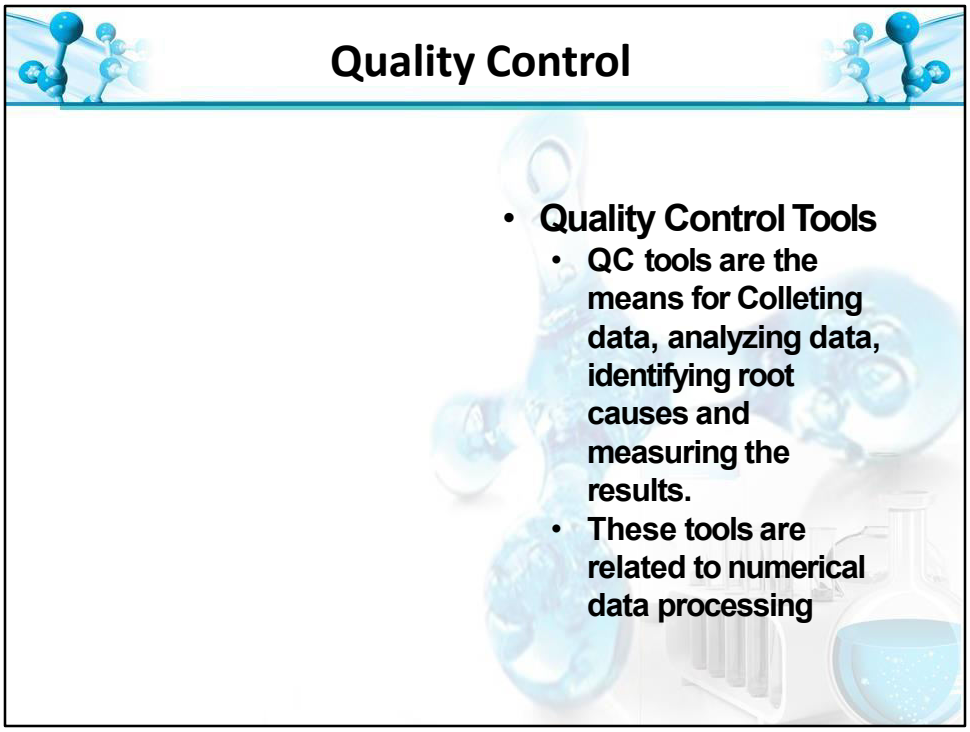
Here we can understand the difference in QC and QA



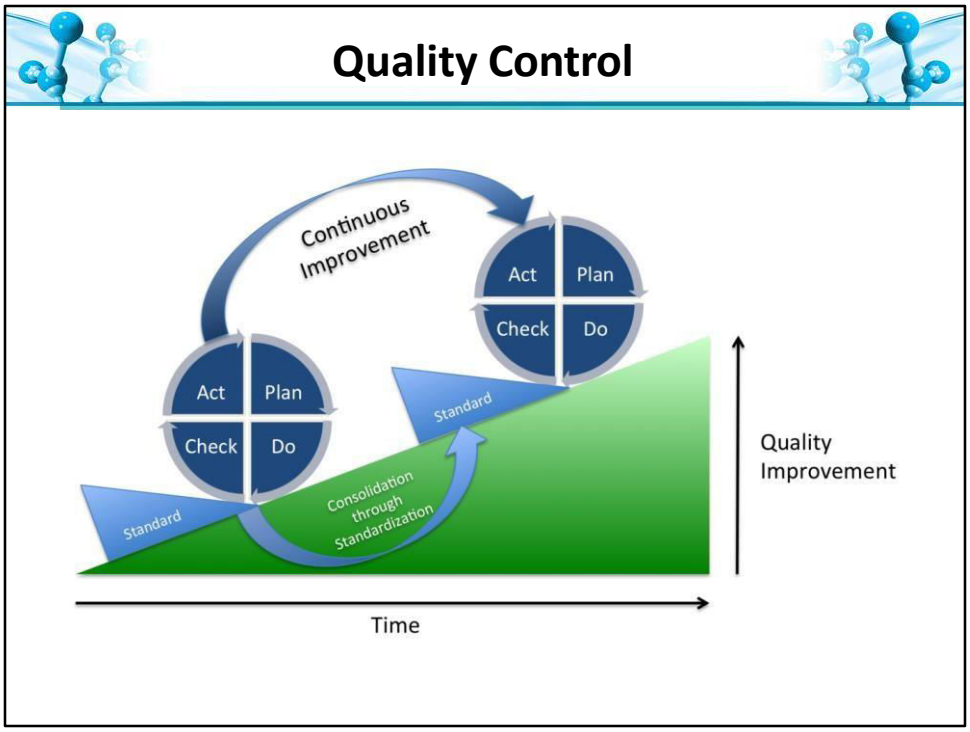
Quality is blend of……



Quality Control Tools



QC tools are means to ……..



Continuous improvement cycle Plan-do-check and act



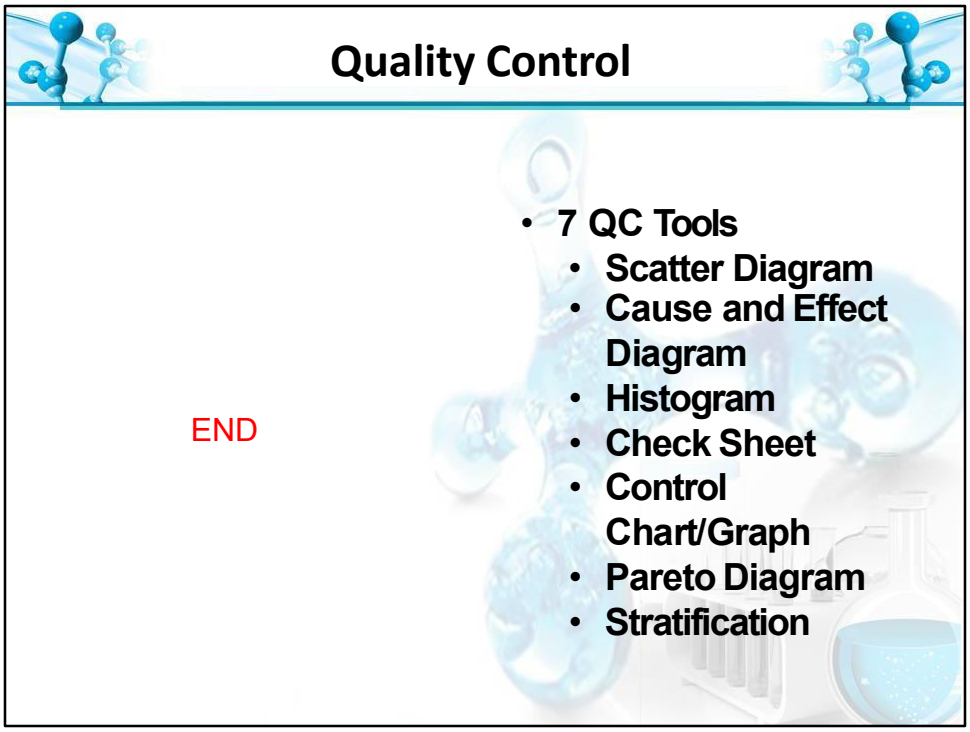
How to check the improvements? To monitor we have to use QC tools…..

Quality Control

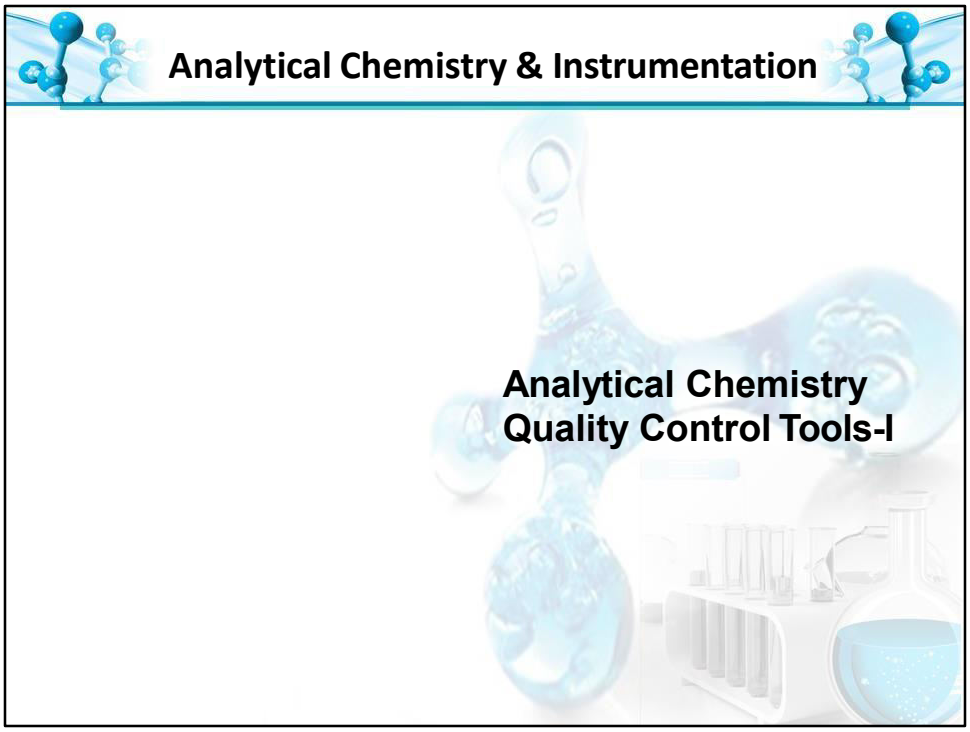
Application of QC tools in Problem Solving

	Graphs	Check sheet	Stratification	Pareto Diagram	Cause & Effect Diagram	Histogram	Scatter Diagram	Control Chart
Identification of problem	◎			◎				
Defining the problem	○	◎	◎	◎		○		
Record of facts		◎	◎			◎		◎
Detecting causes of problem			○		◎	◎	◎	
Develop Improvement method (Solution)								
Implementation								
Evaluation of result	○		○		◎	◎	◎	
Process control (Standardization)								◎

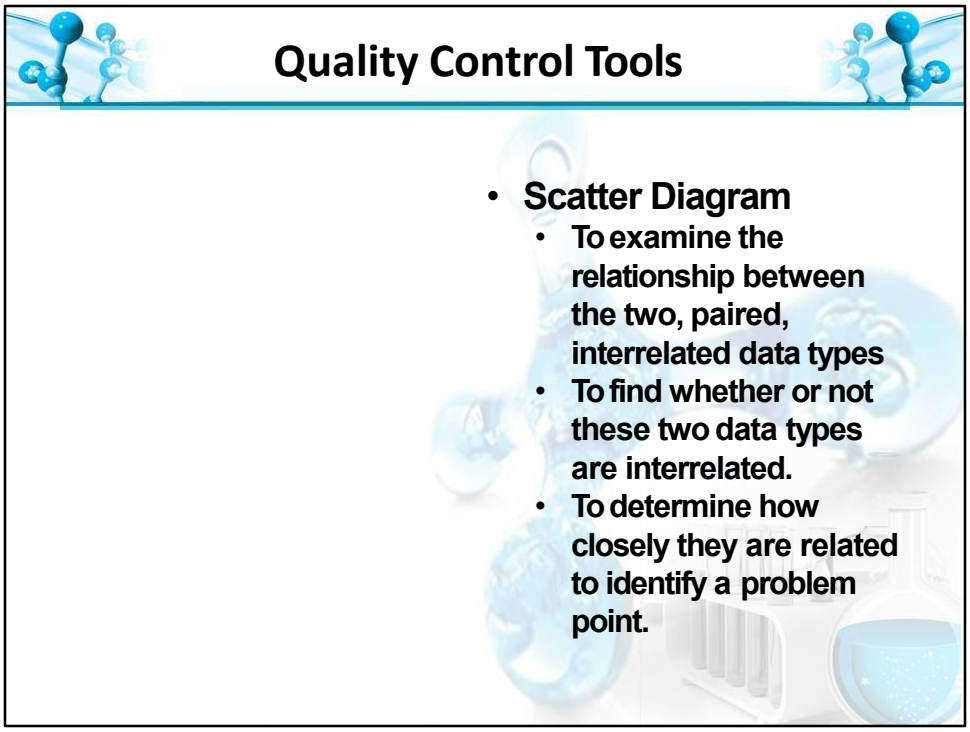
Application of QC tools in problem solving…..



In next presentation we will learn in detail the QC tools one by one.



In QC tools we will discuss scatter diagram.



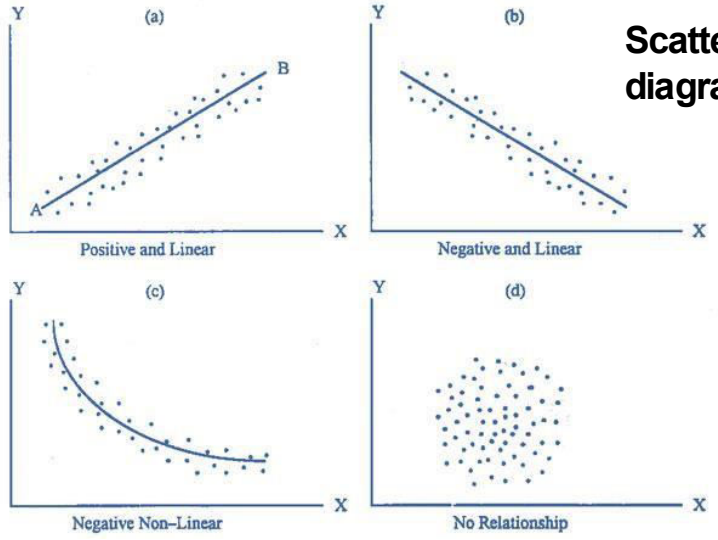
Scatter diagram is to……

Quality Control Tools

- **Examples of Scatter diagram**
  - **Relationship between thermal treatment temperature of a steel material and its tensile strengths**
  - **Relationship between visit made by a salesman and volume of sales**
  - **Relationship between the number of persons visiting a department store and volume of sales**
  - **Relationship between absorbance and concentration of a coloured sample (UV spectroscopy); calibration curve**
  - **Relationship of volume to pressure or temperature**

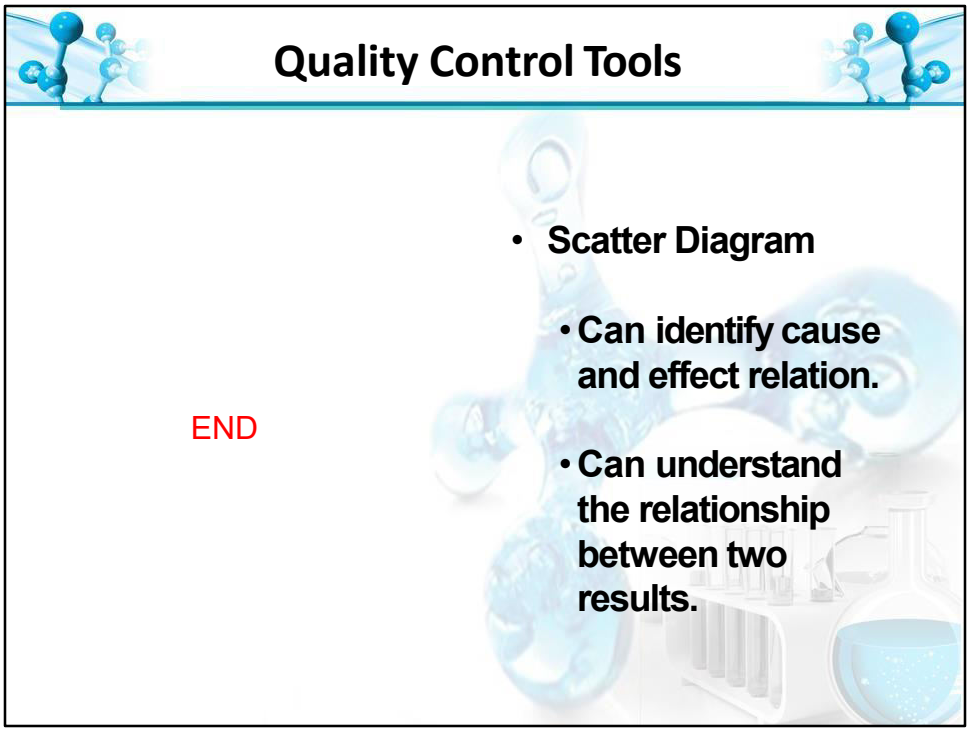
Examples of scatter diagram……

Quality Control Tools

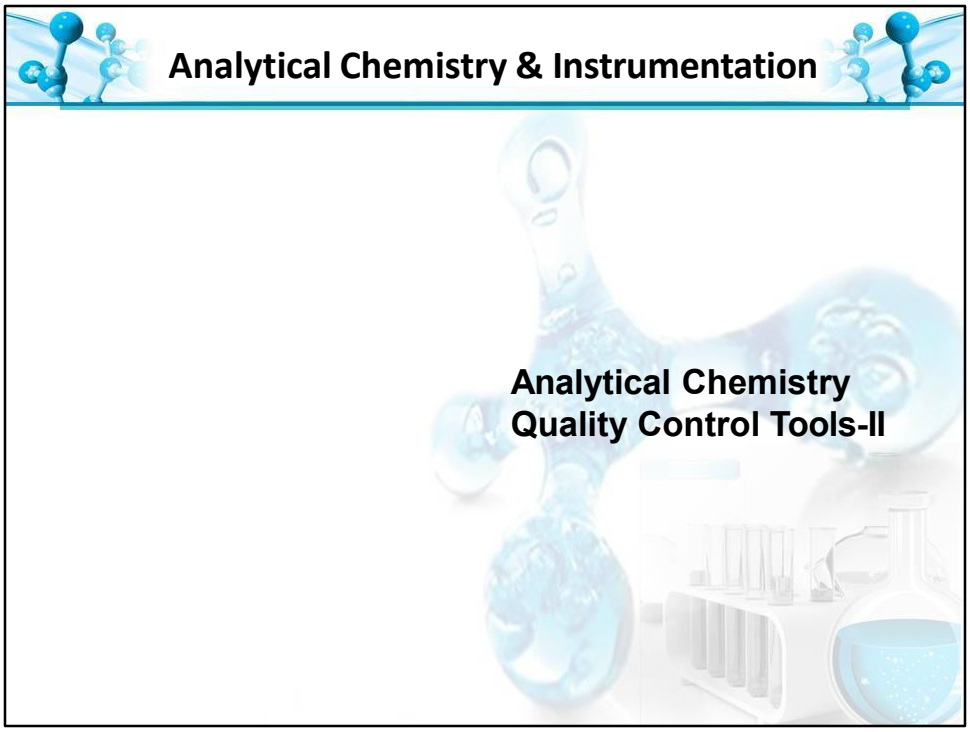


Scatter diagrams

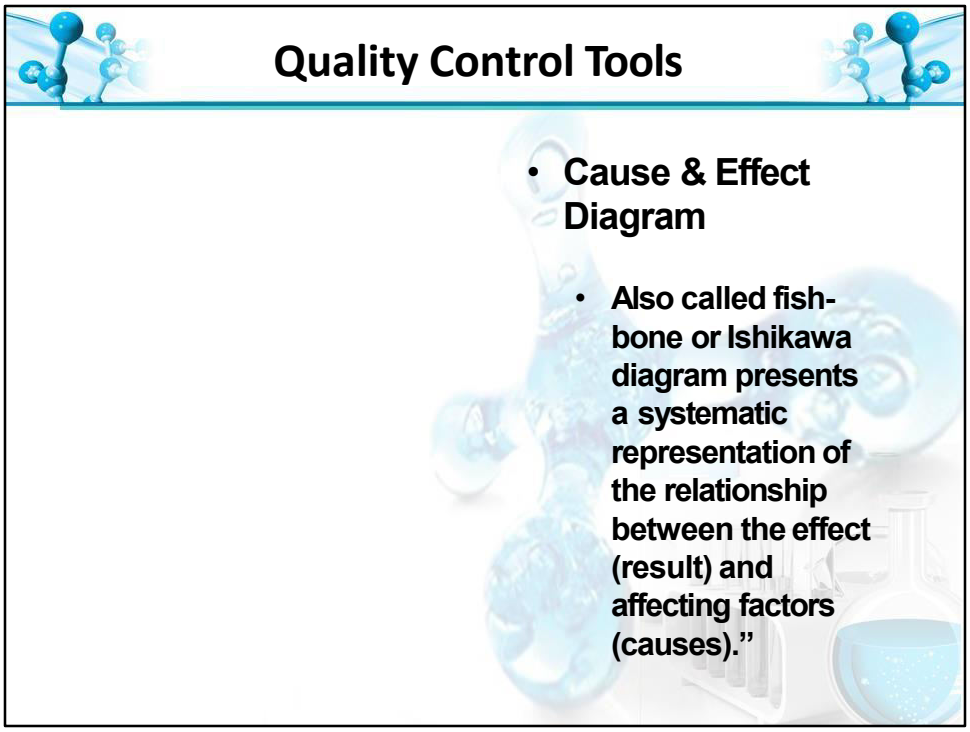
Understanding different types of scatter diagram……



Use of scatter diagram……



The second QC tool is cause and effect diagram…

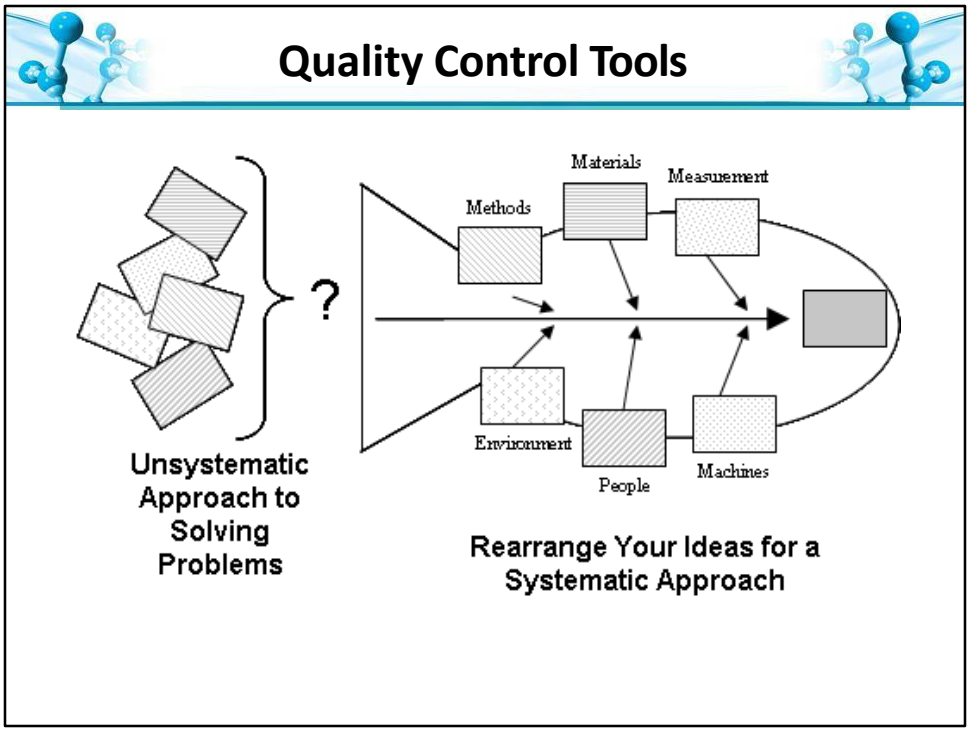


It is also called fish bone diagram, or Ishikawa diagram is a tool to identify the root cause that is responsible for an effect.

Quality Control Tools

- **Cause & Effect Diagram**
  - **Solving a problem requires clarification of a cause and effect relationship, where the effect (e.g., the result of work) varies according to factors (e.g., facilities, machines used, method of work, workers, materials and parts used etc.).**
  - **To obtain a good work result, we must identify the effects of various factors and develop measures to improve the result accordingly.**

The purpose of this diagram is……



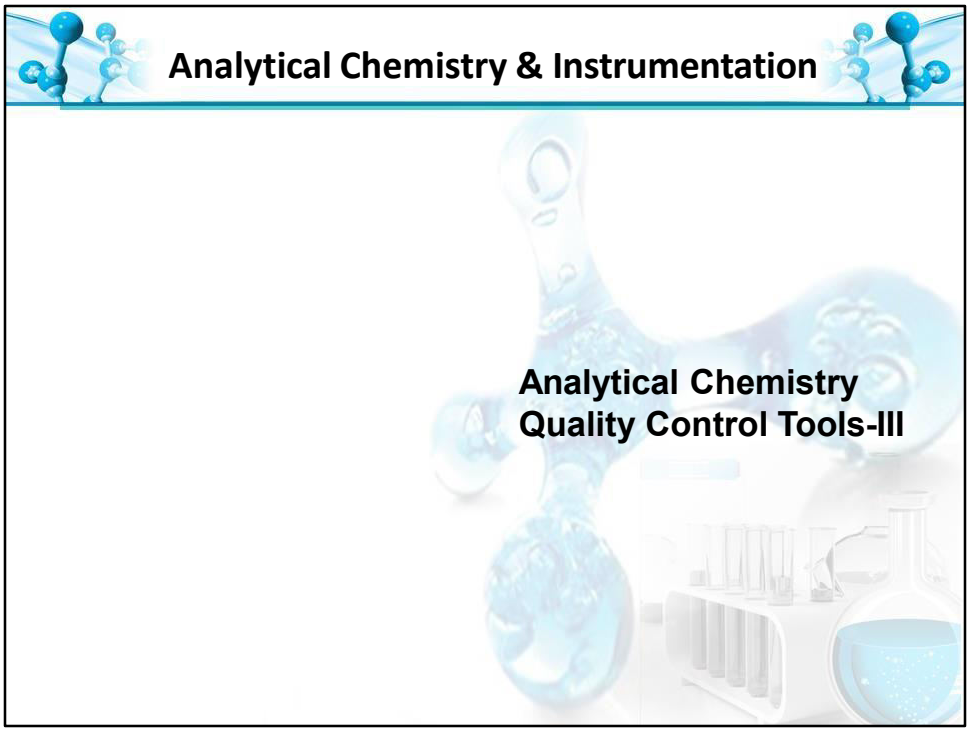
Causes and effects……

Quality Control Tools

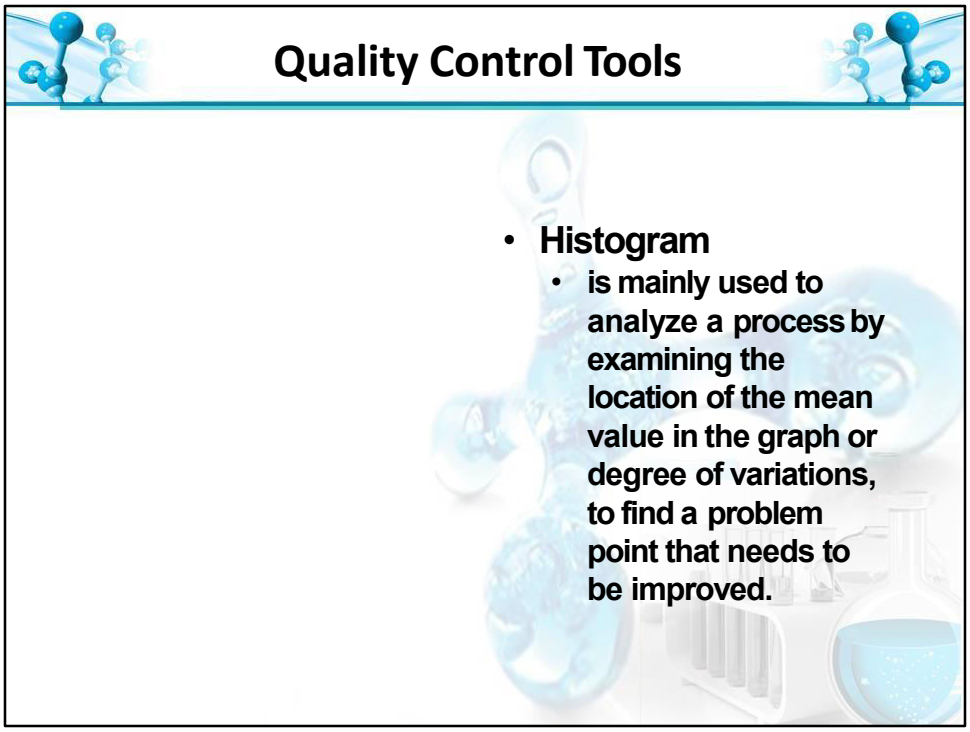
END

- **Cause & Effect Diagram**
  - **Can obtain a clear overall picture of causal relation. (A change in the cause triggers a variation in the result.)**
  - **Can list up all causes to identify important causes.**
  - **Can determine the direction of action (counter measure).**

Can be used ……



After knowing the sources of error, we must know how to remove error.



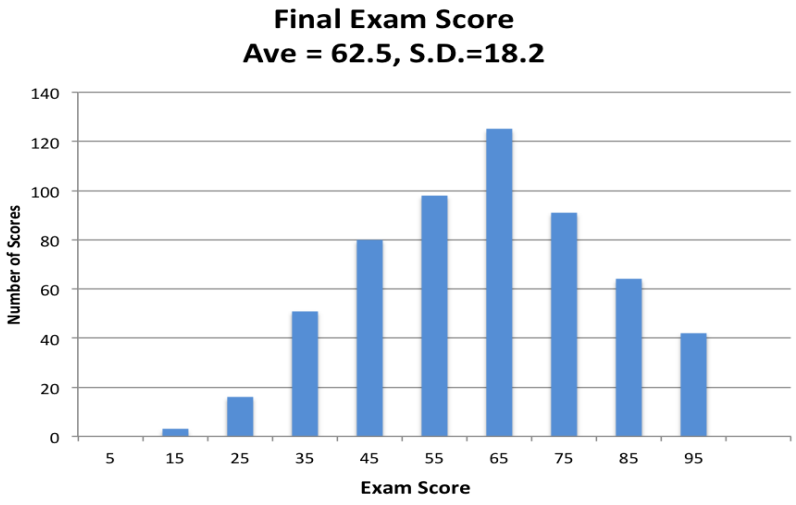
Use of histogram

Quality Control Tools

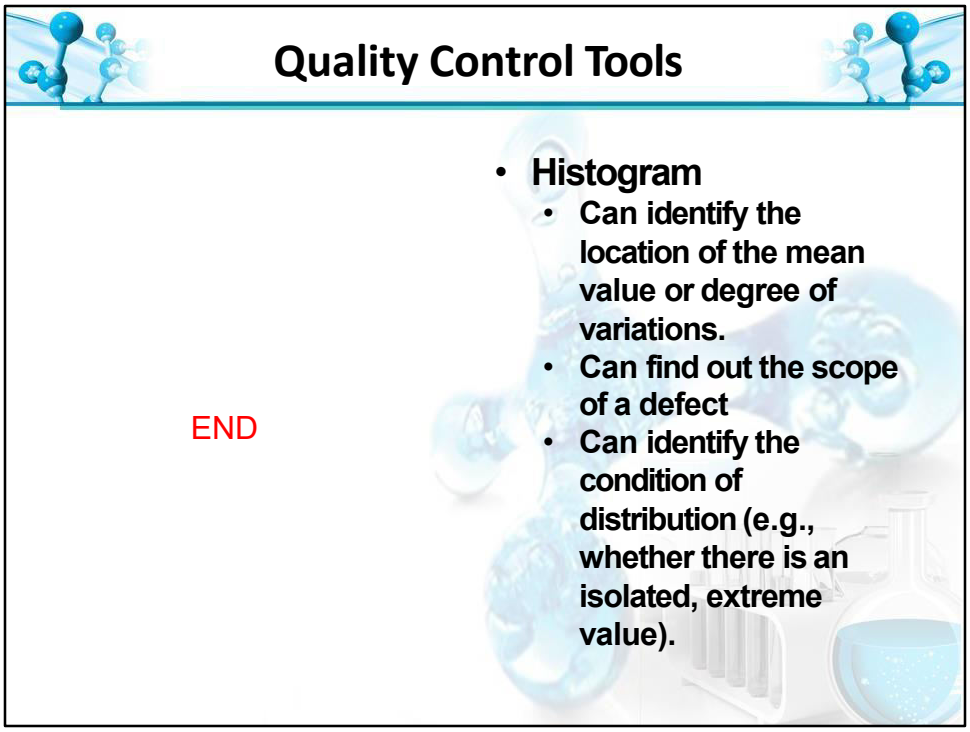
- **Histogram**
  - **When creating a histogram, "a range of data is divided into smaller sections having a uniform span and the number of data contained in each section (the number of occurrences) is counted to develop a frequency distribution table."**
  - **Then, "a graph is formed from this table by using vertical bars, each having the height proportional to the number of occurrences in each section."**
  - **Used to analyze a process to identify a problem point that needs to be improved by finding the location of the mean value or degree of variations in the graph.**

How to generate histogram

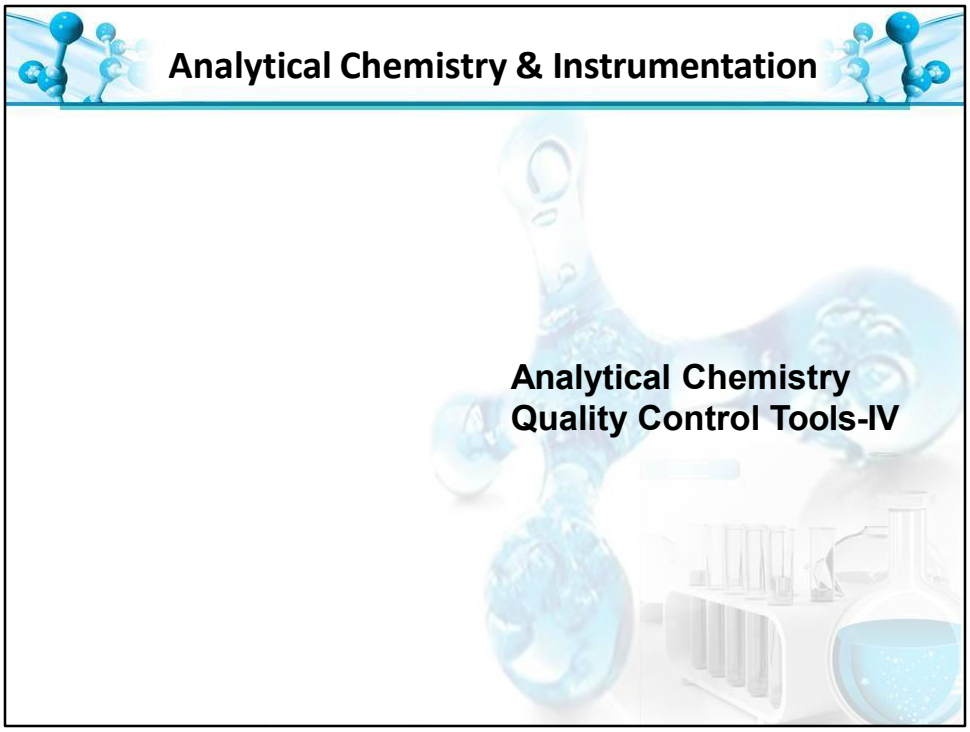
Quality Control Tools



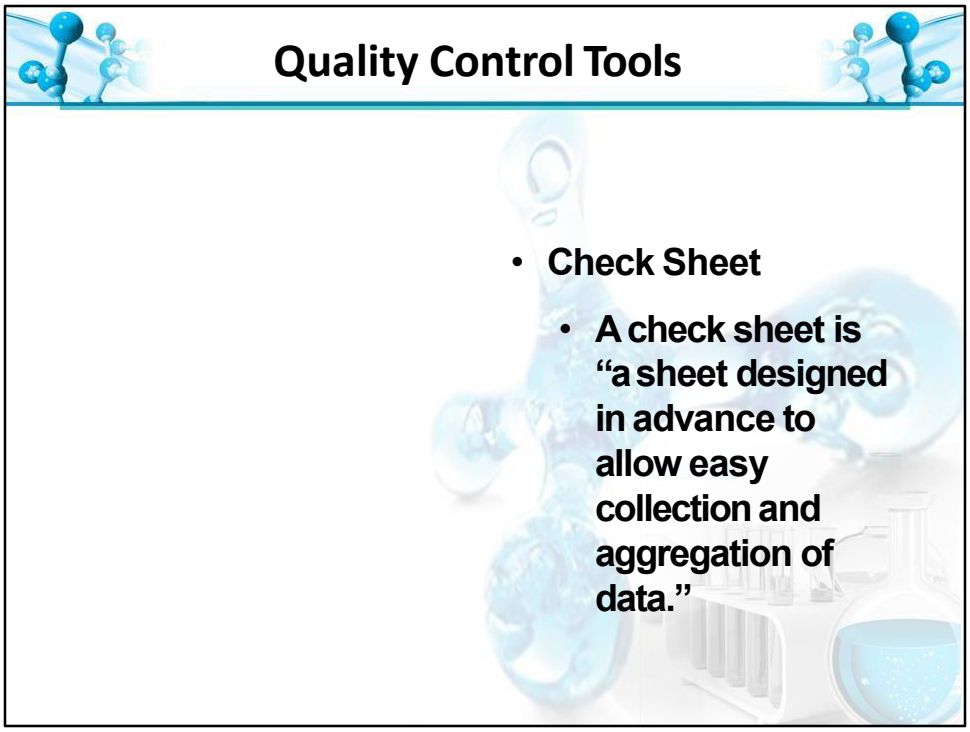
Explain histogram



What can be assessed from histogram



The fourth tool is Check sheets



A check sheet is sheet to collect the data….

Quality Control Tools

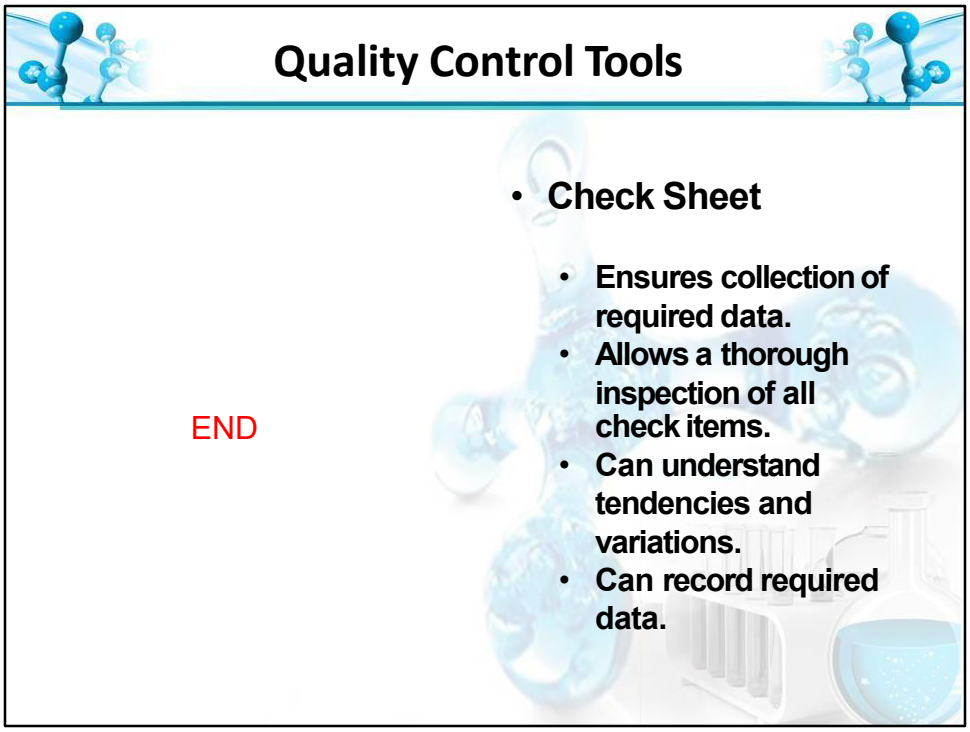
- **Check Sheet**
  - **To check that procedure is followed effectively .**
  - **To eliminate possibility of skipping any of the required inspection items or step.**
  - **It is a self check.**
  - **Development of check sheet is effective QA tool, filling the data is Quality Control check**
  - **A check sheet is also effective in performing stratification (categorization).**

Purpose of use of check sheets……

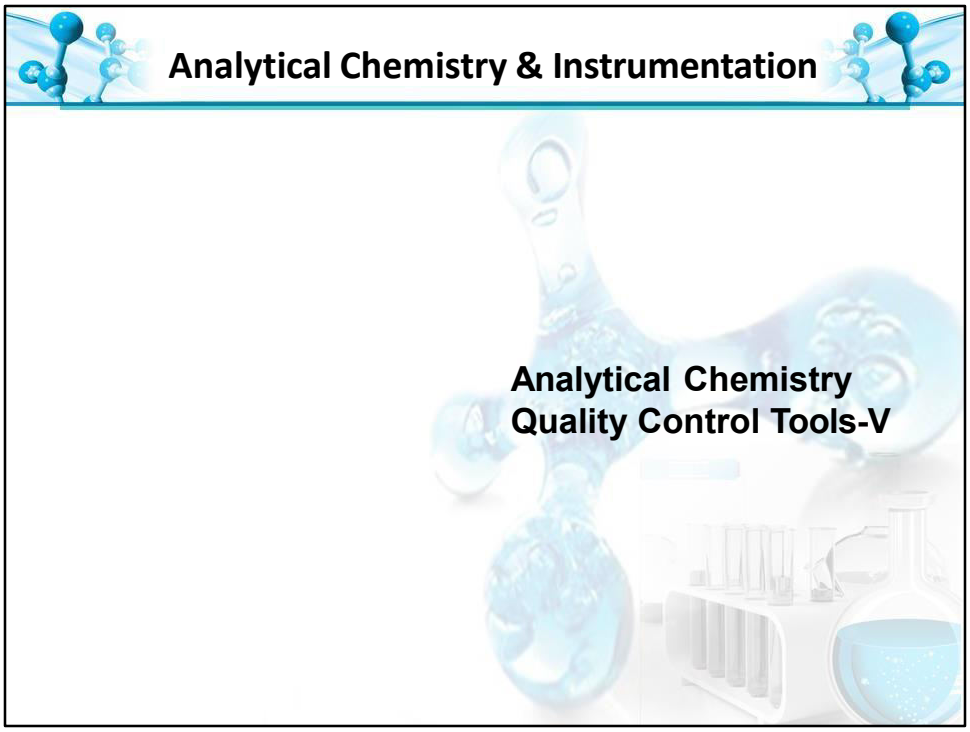
Quality Control Tools

What a Clean Bathroom Looks Like	
Trash can is emptied.	
Toilet is scrubbed inside.	
Toilet is wiped down on the outside, including the base.	
The mirrors are shiny, no spots	
The sink is scrubbed clean.	
The counters are wiped down.	
The floor is swept and/or mopped.	
The soap container is at least half full.	
There are fresh towels.	
The shower or tub has been scrubbed.	
There is soap and shampoo in the shower or tub.	
The dirty laundry is where it belongs.	

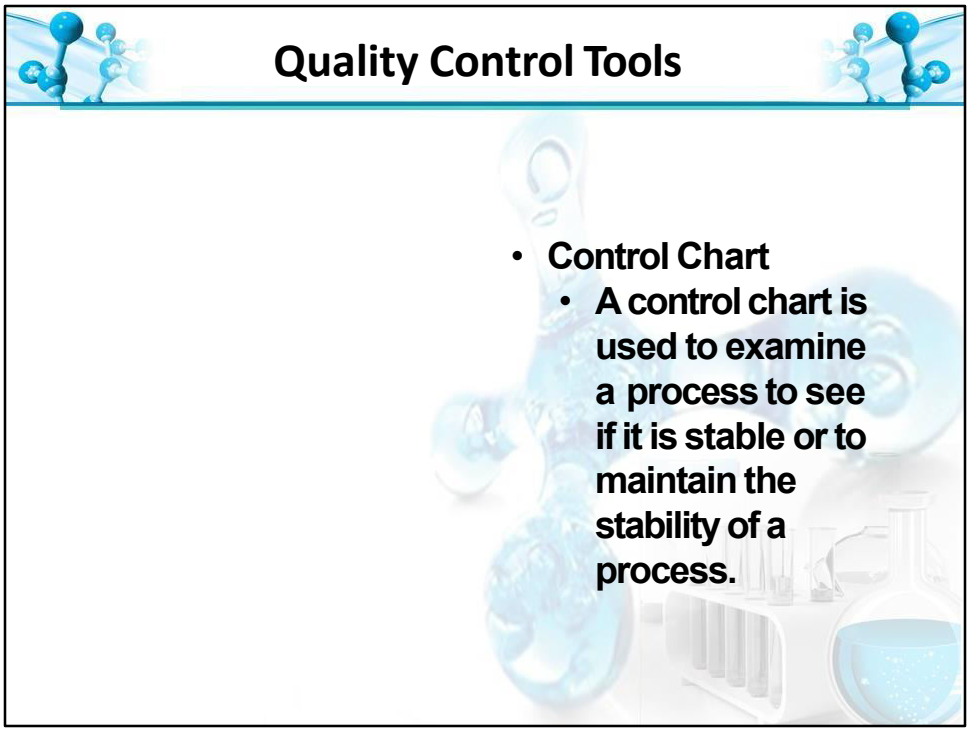
A simplest form of check sheet to monitor cleaning of bathroom. Other possible examples……



Benefits of using this tool….



The fifth tool is control chart or other type of graphs. First we will discuss control charts….



A control chart is used …….

Quality Control Tools

- **Control Chart**
  - **There are two types of control charts: one used for managerial purposes and the other for analytical purposes.**
  - **A control chart is used to identify dots that are outside the control limit, which indicate some anomalies in a process.**
  - **Seven consecutive dots below or above the mean (central) value, or an increment or a decline represented by seven consecutive dots also indicate "a problem in a process."**
  - **It needs to examine what has caused such a tendency or an increment/decline.**

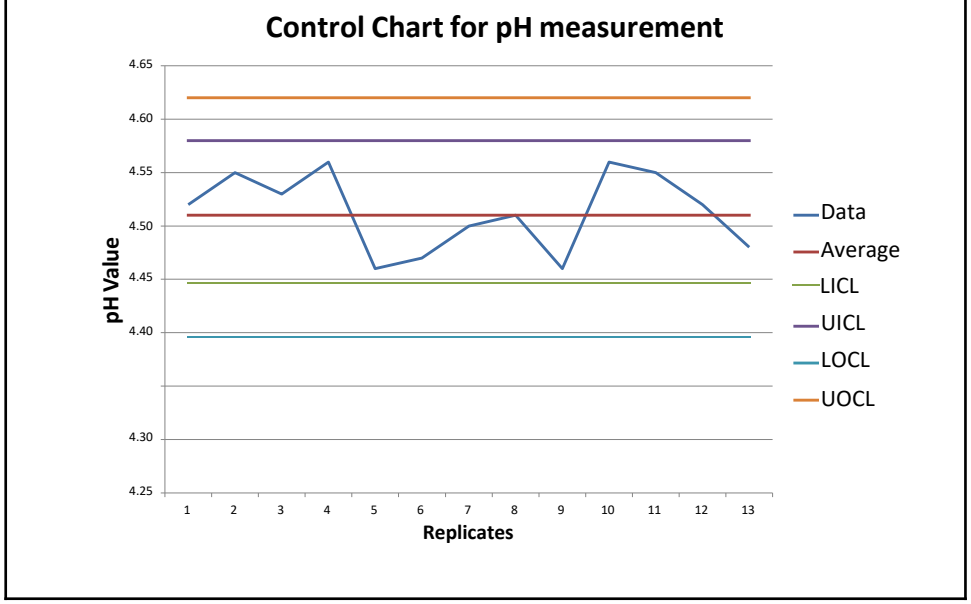
What is a control chart…..

Quality Control Tools

	Data	Average	LICL	UICL	LOCL	UOCL
	4.52	4.51	4.44	4.58	4.40	4.62
	4.55	4.51	4.44	4.58	4.40	4.62
	4.53	4.51	4.44	4.58	4.40	4.62
	4.56	4.51	4.44	4.58	4.40	4.62
	4.46	4.51	4.44	4.58	4.40	4.62
	4.47	4.51	4.44	4.58	4.40	4.62
	4.50	4.51	4.44	4.58	4.40	4.62
	4.51	4.51	4.44	4.58	4.40	4.62
	4.46	4.51	4.44	4.58	4.40	4.62
	4.56	4.51	4.44	4.58	4.40	4.62
	4.55	4.51	4.44	4.58	4.40	4.62
	4.52	4.51	4.44	4.58	4.40	4.62
	4.48	4.51	4.44	4.58	4.40	4.62
Mean	4.51		4.44	4.58	4.40	4.62
Std Dev	0.04					

Explain the data…..

Quality Control Tools



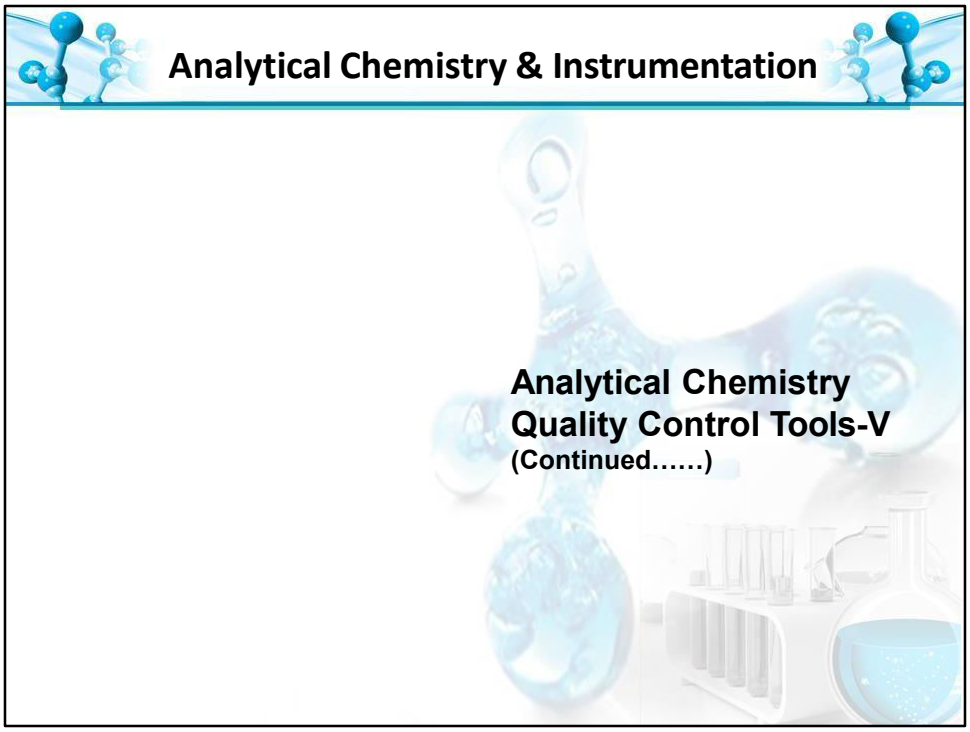
Explain the chart

Quality Control Tools

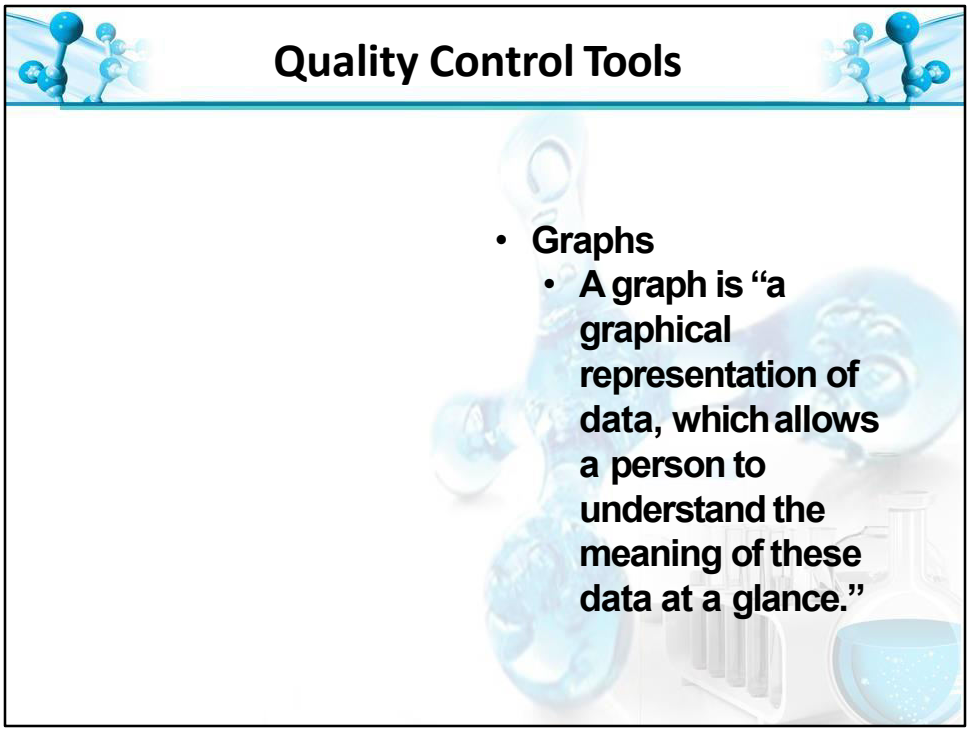
END

- **Control Chart**
  - **Can identify a change caused by elapse of time**
  - **Can judge the process if it is in its normal state or there are some anomalies by examining the dots plotted on the chart.**

What is it used for?



Other type of control is to monitor through graphs

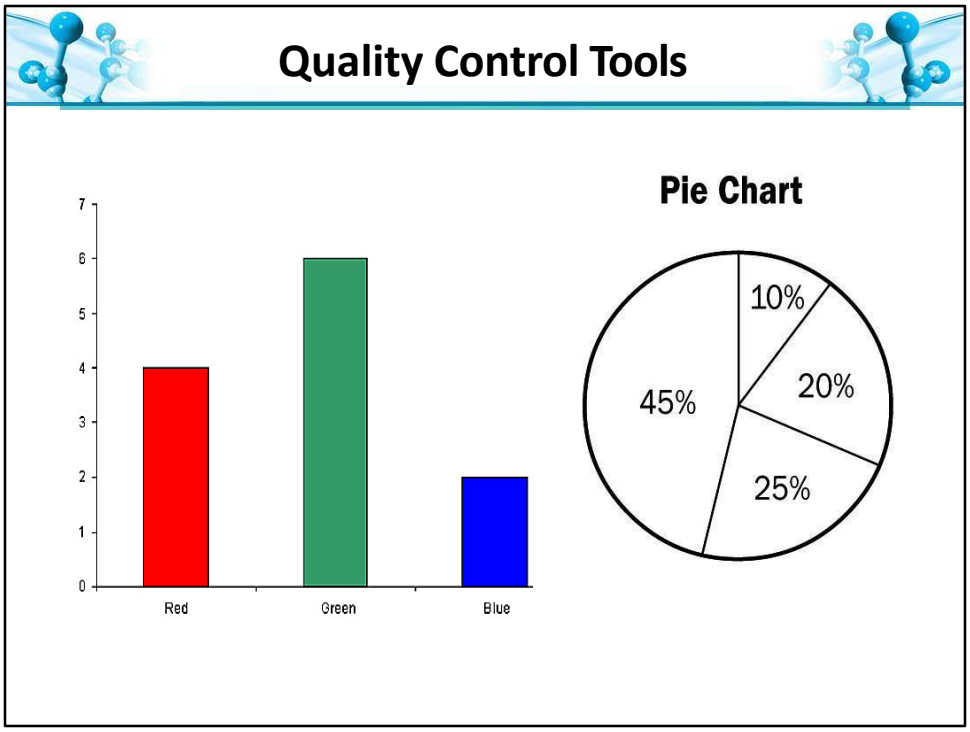

Quality Control Tools
• Graphs
• A graph is "a graphical representation of data, which allows a person to understand the meaning of these data at a glance."

A graph is…..

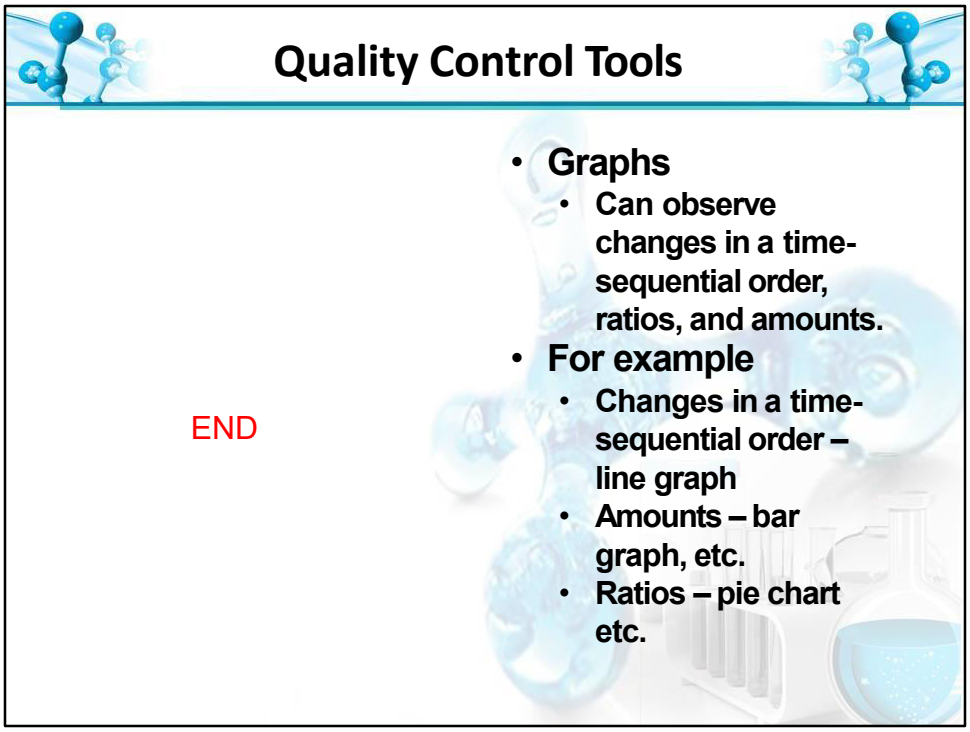
Quality Control Tools

- **Graphs**
  - **Unprocessed data simply represent a list of numbers, and finding certain tendencies or magnitude of situation from these numbers is difficult, sometimes resulting in an interpretational error.**
  - **A graph is a effective means to monitor or judge the situation, allowing quick and precise understanding of the current or actual situation.**
  - **A graph is a visual and summarized representation of data that need to be quickly and precisely conveyed to others.**

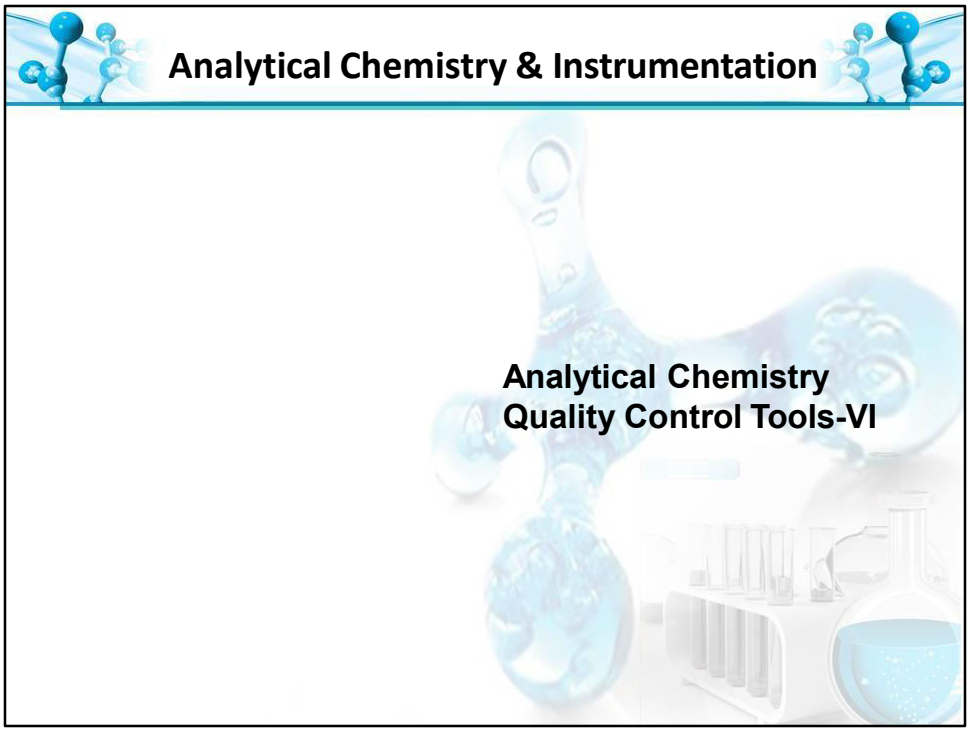
Use of graphical presentation….


Quality Control Tools
7
6
5
4
3
2
1
0
Red
Green
Blue
Pie Chart
10%
20%
45%
25%

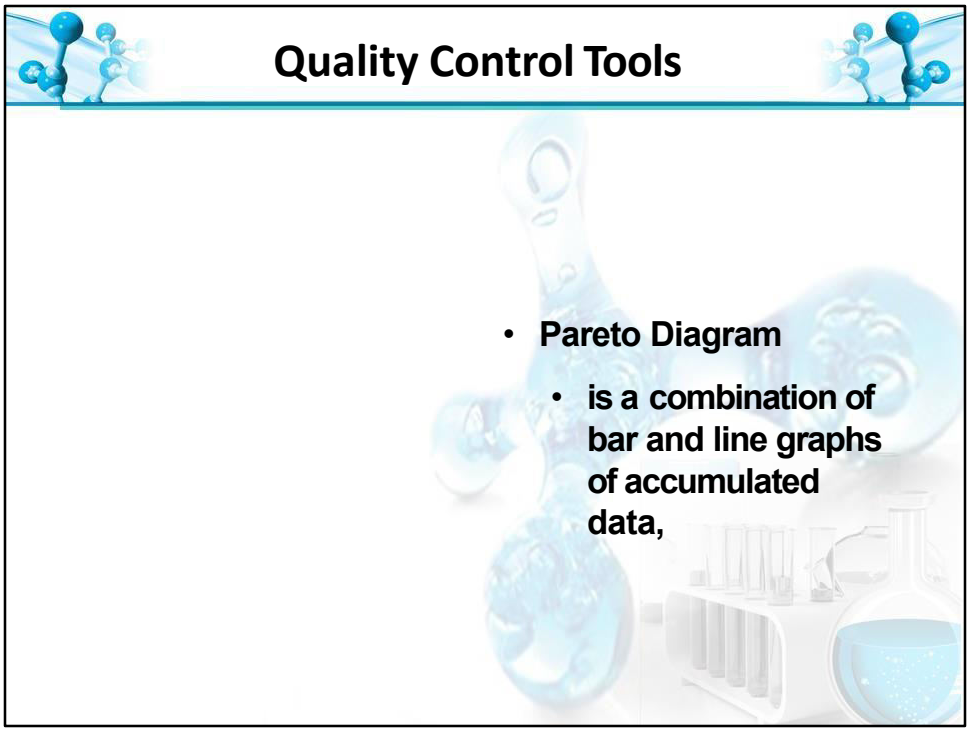
Examples…..



Use of graphs



The sixth QC tool is….



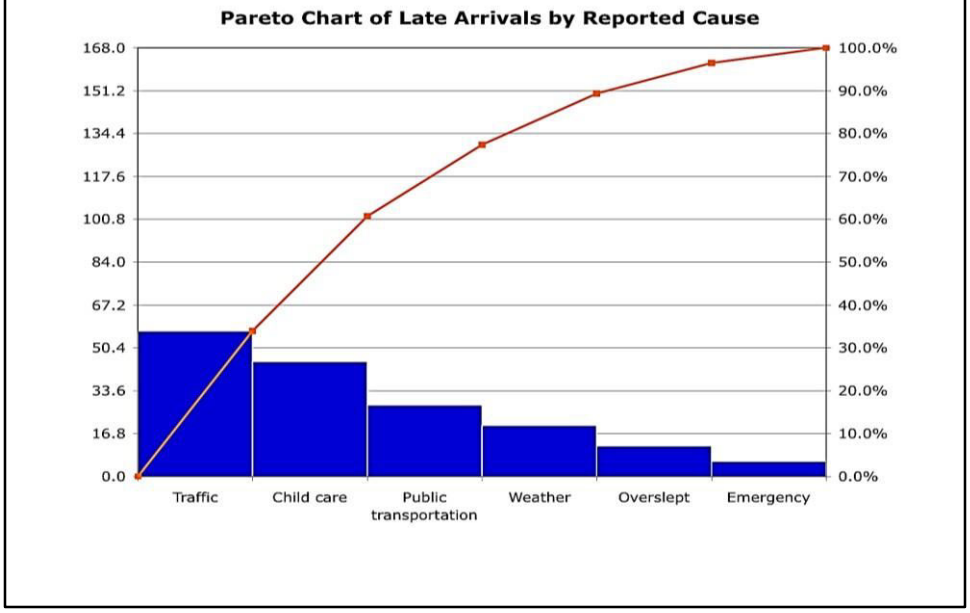
Pareto diagram is…

Quality Control Tools

- **Pareto diagram**
  - **It is used where data associated with a problem e.g. a defect, or mechanical failure, or a complaint from a customer are divided into smaller groups by cause or by phenomenon and sorted by the number of occurrences or the amount of money involved. (The name "Pareto" came from an Italian mathematician who created the diagram.)**
  - **Bar chart arranged in descending order of height from left to right, bars on left relatively more important than those on right**
  - **Separates the "vital few" from the "trivial many" (Pareto Principle)**

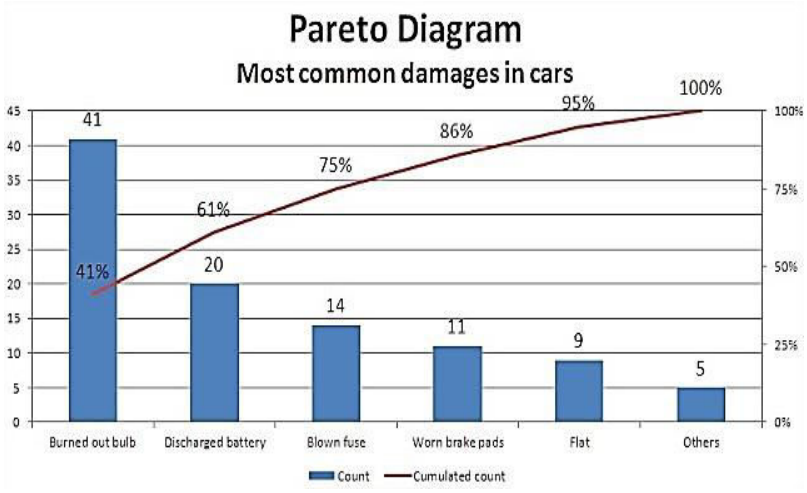
Pareto diagram is used ….

Quality Control Tools

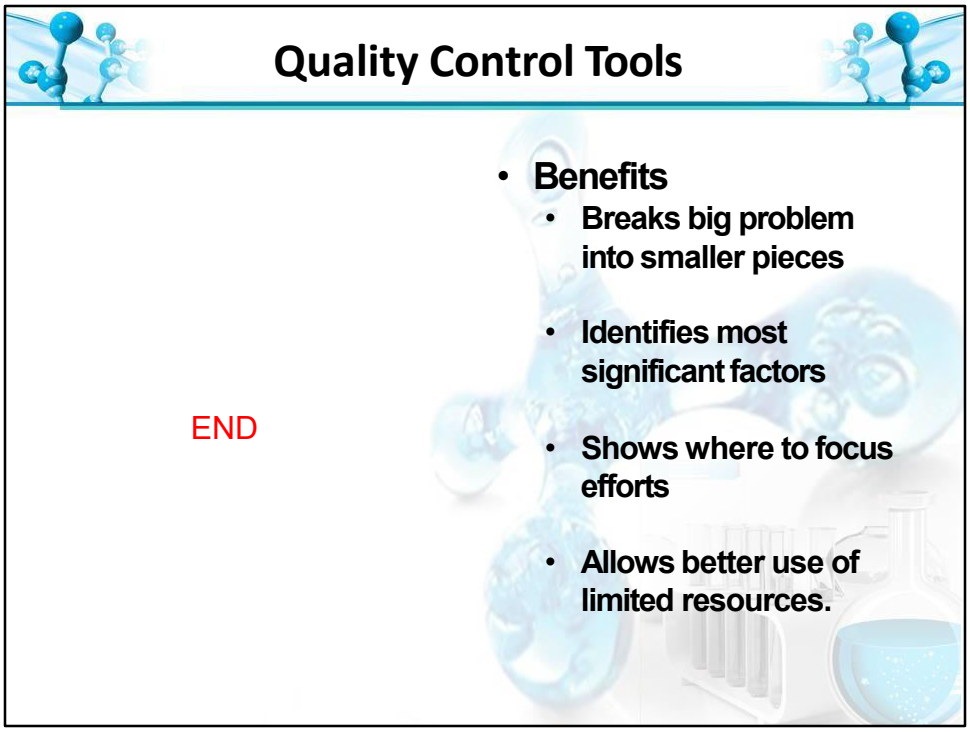


Examples…..

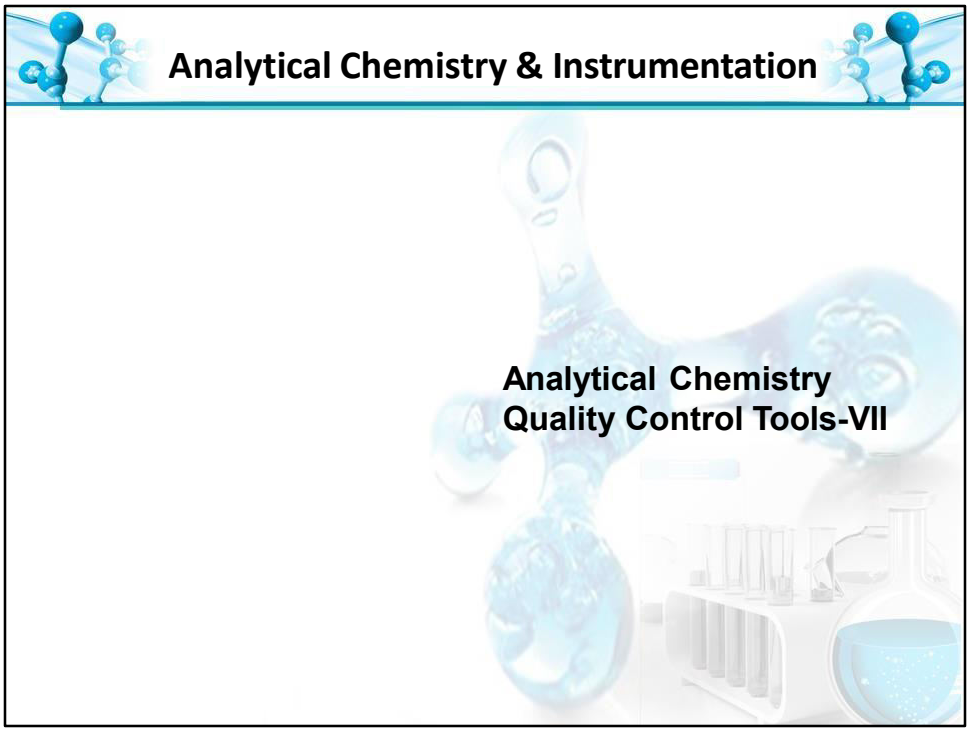
Quality Control Tools



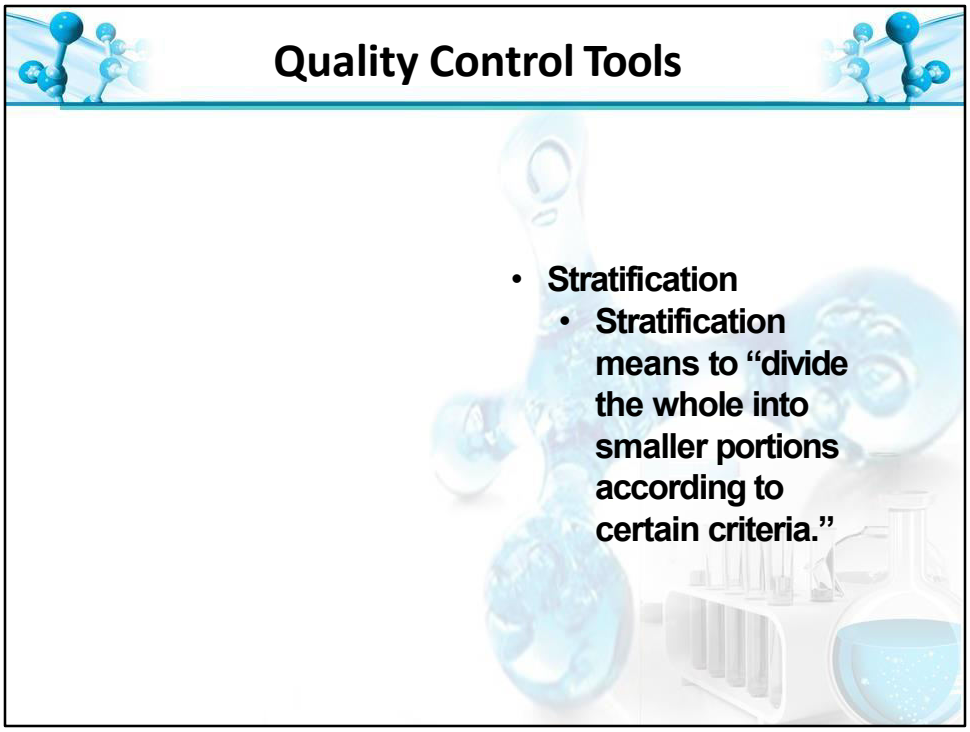
Examples…..


Quality Control Tools
END
• Benefits
• Breaks big problem into smaller pieces
• Identifies most significant factors
• Shows where to focus efforts
• Allows better use of limited resources.

Benefits….



The seventh QC tool is …..



Stratification means……

Quality Control Tools

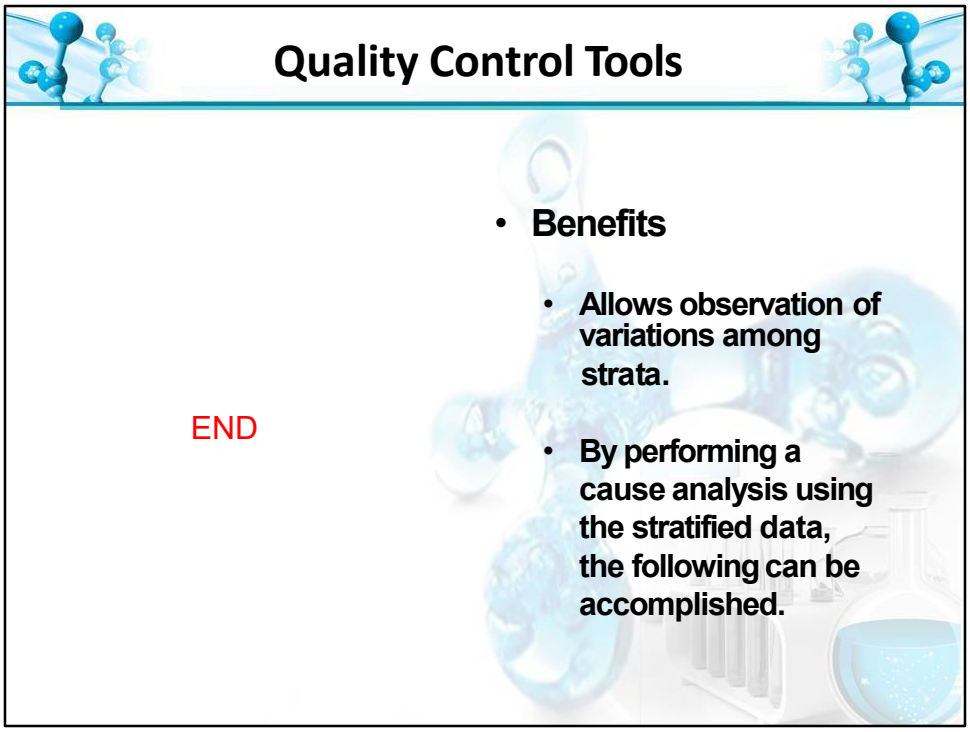
- **Stratification**
  - **In case of quality control, stratification generally means to divide data into several groups according to common factors or tendencies (e.g., type of defect and cause of defect).**
  - **Dividing into groups "fosters understanding of a situation." This represents the basic principle of quality control.**
  - **Stratification is a system or formation of layers, classes, or categories.**
  - **Stratification is used to describe a particular way of arranging seeds while planting, as well as the geological layers of rocks**

Use of stratification….

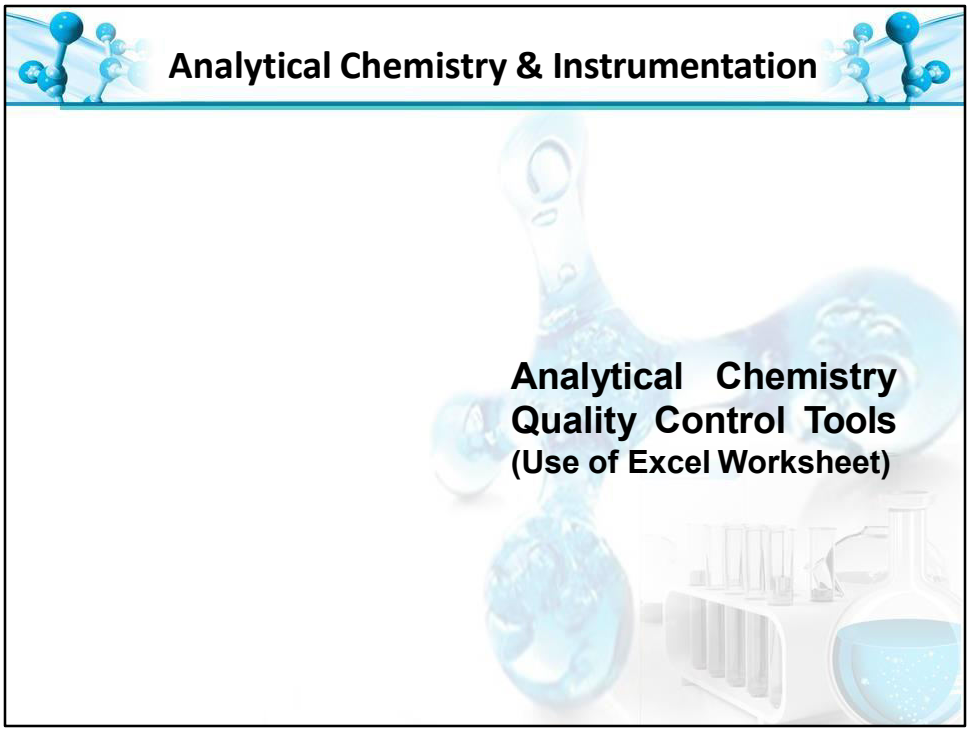
Quality Control Tools

Item	Method of Stratification
Elapse of time	Hour, a.m., p.m., immediately after start of work, shift, daytime, night time, day, week, month
Variations among workers	Worker, age, male, female, years of experience, shift, team, newly employed, experience
Variations among work methods	Processing method, work method, working conditions (temperature, pressure, and speed), temperature
Variations among measurement/ inspection methods	Measurement tool, person performing measurement, method of measurement, inspector, sampling, place of inspection

Examples…..


Quality Control Tools
END
• Benefits
• Allows observation of variations among strata.
• By performing a cause analysis using the stratified data, the following can be accomplished.

Benefits….



Here is summary of our discussion about QC tools.

Quality Control Tools

- **Summary**
  - **Pareto Diagram:** To identify the current status and issues
  - **Stratification:** Basic processing during data collection
  - **Scatter Diagram:** To identify the relationship between two variables
  - **Cause and Effect Diagram:** To identify the cause and effect relationship
  - **Histogram:** To see the distribution of data
  - **Check Sheet:** To record data collection
  - **Control Chart/Graph:** To find out abnormalities and identify the current status

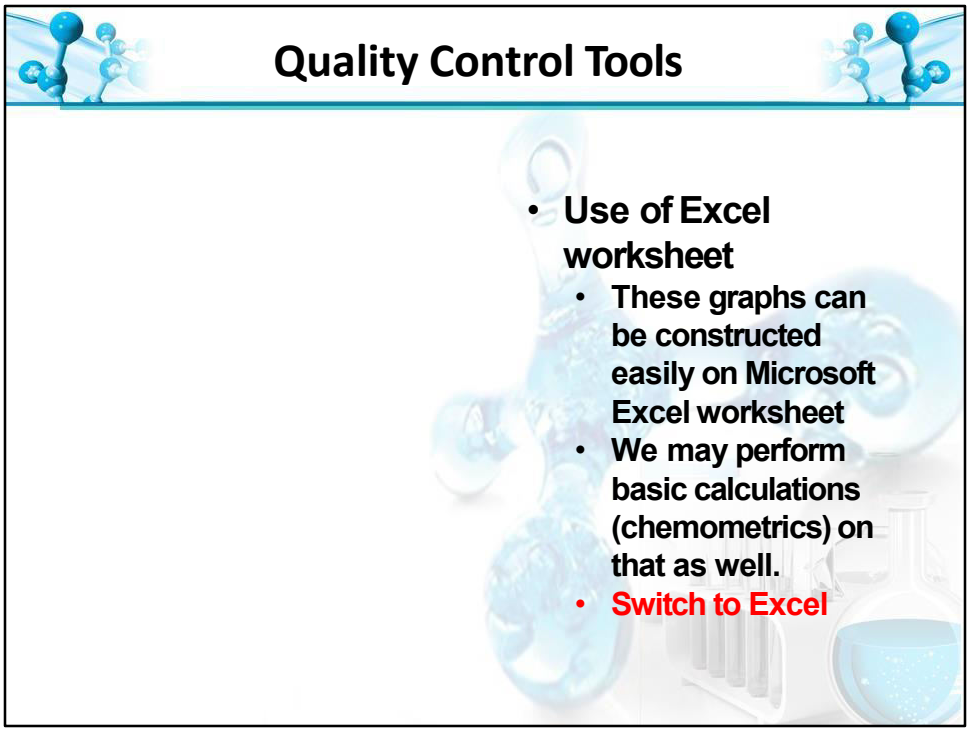
The purpose of different tools may be understood as…..

Quality Control Tools

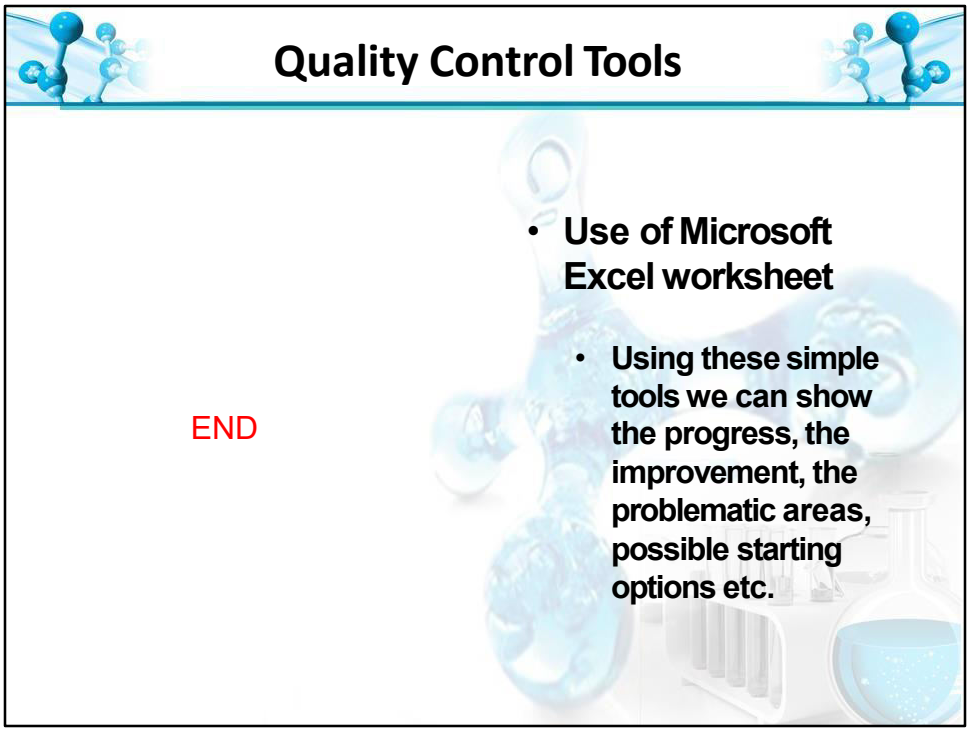
Review QC tools in Problem Solving

	Graphs	Check sheet	Stratification	Pareto Diagram	Cause & Effect Diagram	Histogram	Scatter Diagram	Control Chart
Identification of problem	◎			◎				
Defining the problem	○	◎	◎	◎		○		
Record of facts		◎	◎			◎		◎
Detecting causes of problem			○		◎	◎	◎	
Develop Improvement method (Solution)								
Implementation								
Evaluation of result	○		○		◎	◎	◎	
Process control (Standardization)								◎

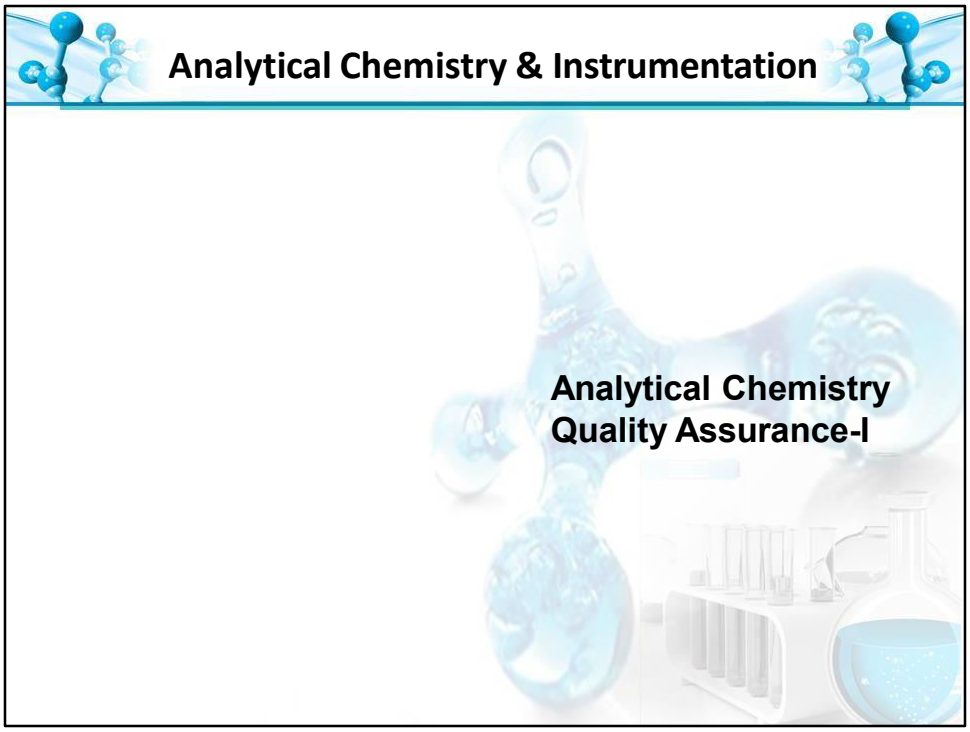
Review of QC tools in problem solving…..



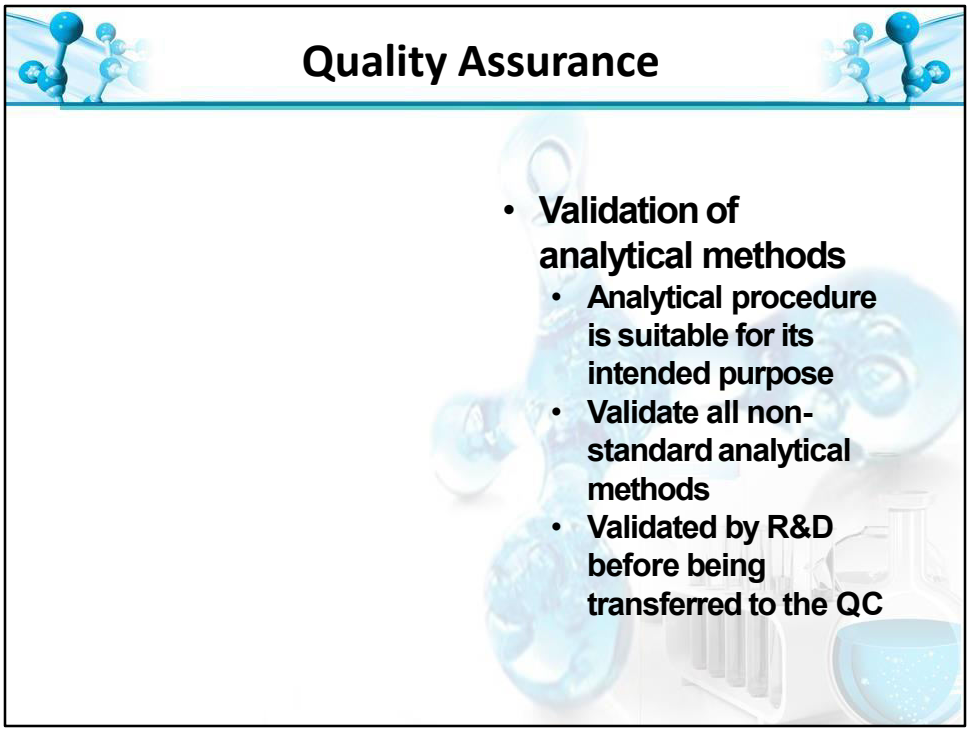
Use of Microsoft excel…..


Quality Control Tools
END
• Use of Microsoft Excel worksheet
• Using these simple tools we can show the progress, the improvement, the problematic areas, possible starting options etc.

Benefits…….



In this module we will discuss QA……



And we start this understanding with Validation of analytical methods…….

Quality Assurance

Considerations Prior to Method Validation

- **Suitability of Instrument**
  - Status of Qualification and Calibration
- **Suitability of Materials**
  - Status of Reference Standards, Reagents, etc.
- **Suitability of Analyst**
  - Status of Training and Qualification Records
- **Suitability of Documentation**
  - Written analytical procedure and proper approved protocol with pre-established acceptance criteria.

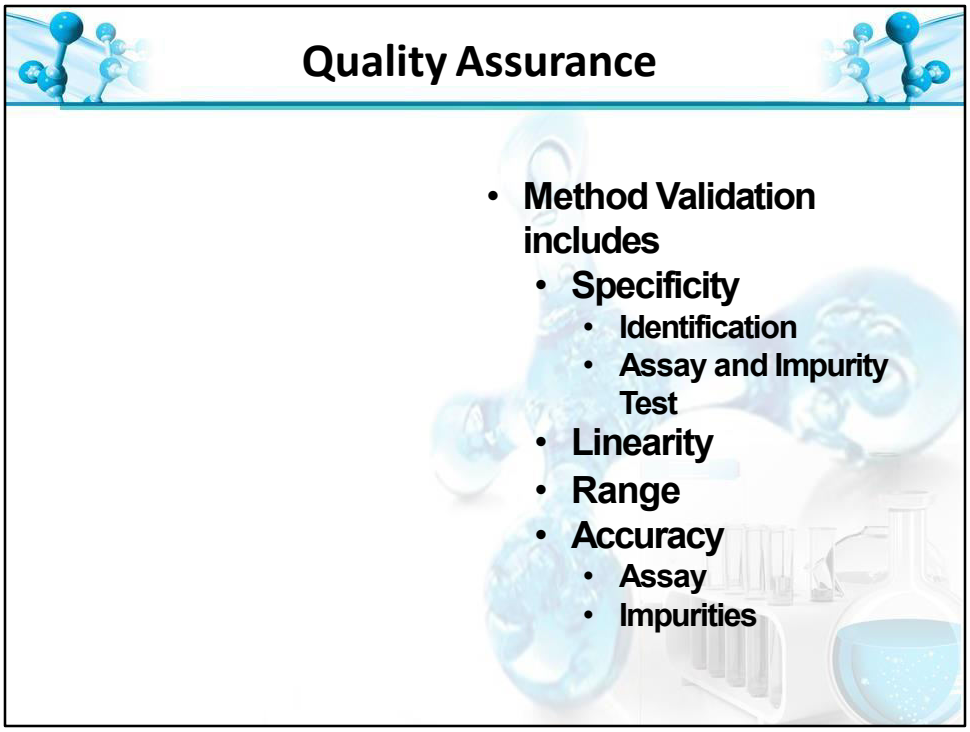
Consideration prior to method validation….

Quality Assurance

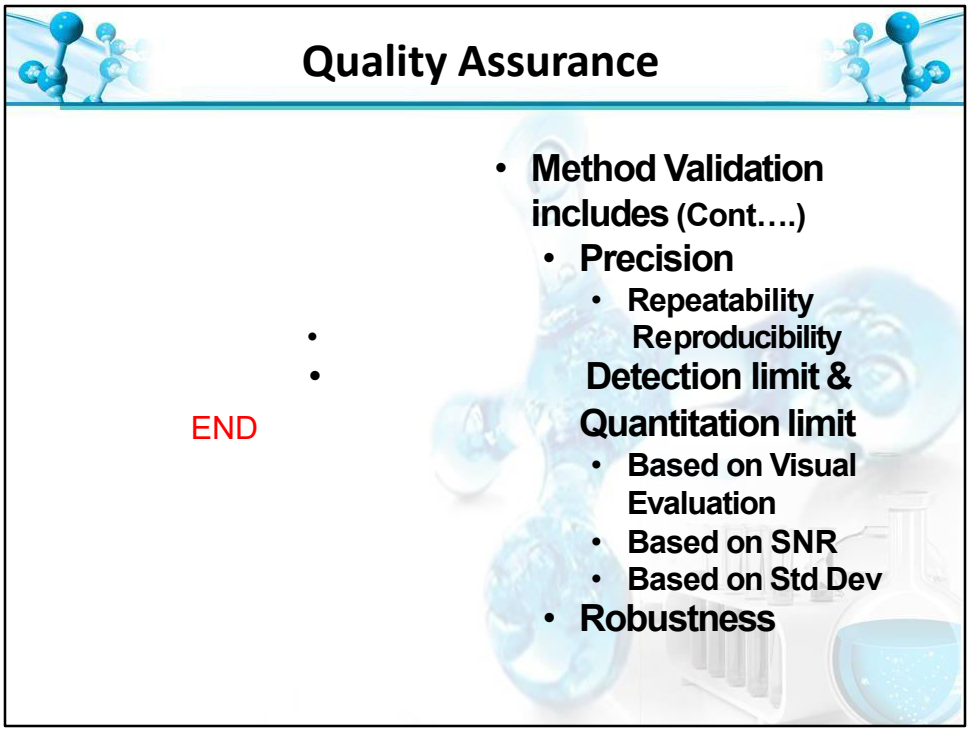
Method Validation Steps

- **Protocol:** includes procedures and acceptance criteria
- **Report:** documented results
  - Justification needed when non-pharmacopoeial methods are used (if pharmacopoeial methods are available). Justification to include data, e.g. comparisons with the pharmacopoeial or other methods
- **Detailed standard test methods include:**
  - chromatographic conditions, reagents and others

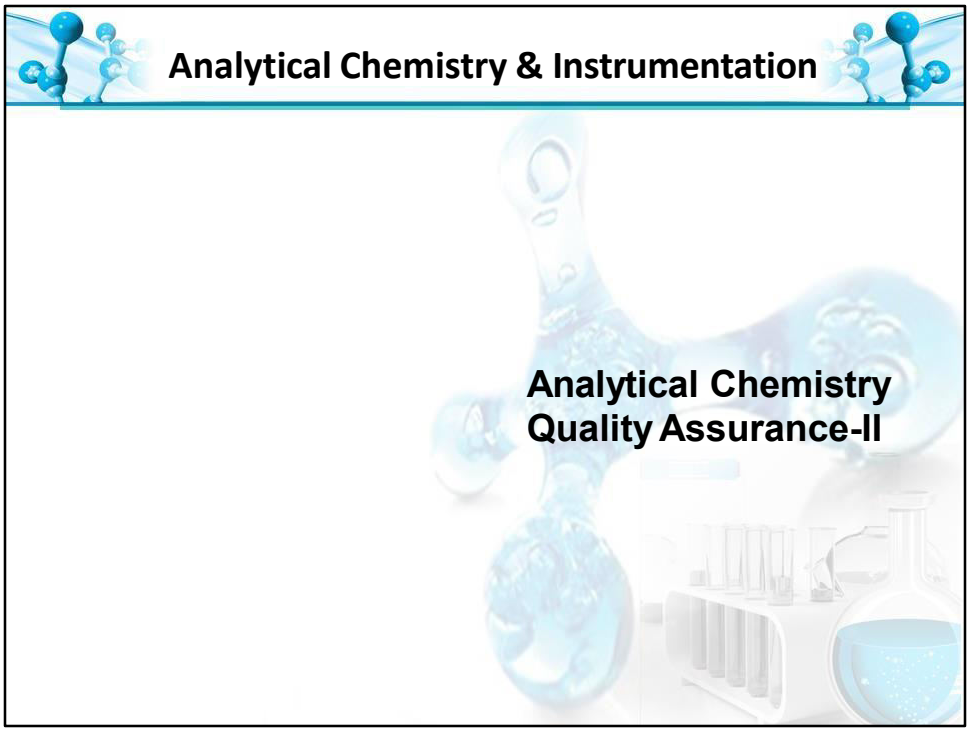
Protocol…..



The parameters that we study during method validation are …..



The parameters that we study during method validation are ….. Cont….



The first step of method validation is specificity or selectivity…..

Quality Assurance

- **Specificity / Selectivity**
  - **ability to measure the desired analyte**
  - **determination of impurities and assay.**
  - **If analytical procedure is not specific a combination of two or more analytical procedures is recommended**

A graph is…..

Quality Assurance

Identification

- Suitable tests that should be able to discriminate compounds (may be of closely related structures)
- The discrimination of a procedure may be confirmed by obtaining positive results by comparison with a known reference material (positive control and negative control).
- The identification test may be applied to materials structurally similar to or closely related to the analyte.
- The choice of such potentially interfering materials should be based on sensible scientific judgment with a consideration of the interferences that could occur.

Identification of method….

Quality Assurance

Assay and Impurities

- For chromatographic procedures, representative chromatograms should be used to demonstrate specificity, and individual components should be appropriately labeled.
- Critical separations in chromatography should be investigated at an appropriate level.
- In cases where a non-specific assay is used, other supporting analytical procedures should be used to demonstrate overall specificity. For example, where a titration is adopted to assay the drug substance for release, the combination of the assay and a suitable test for impurities can be used.

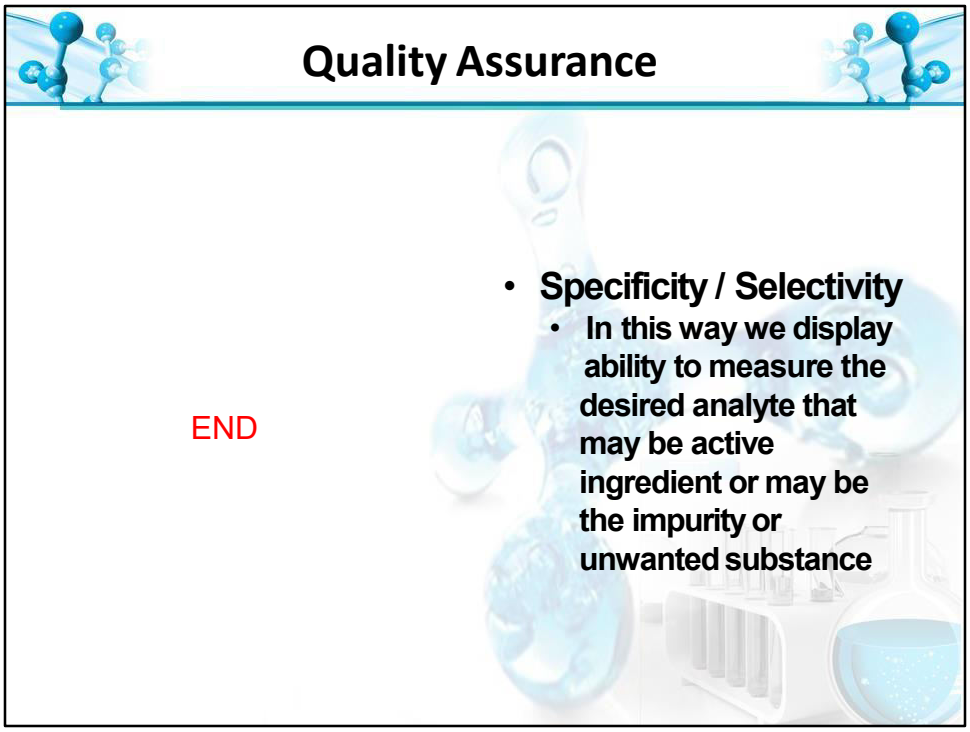
Assays or impurities determination….

Quality Assurance

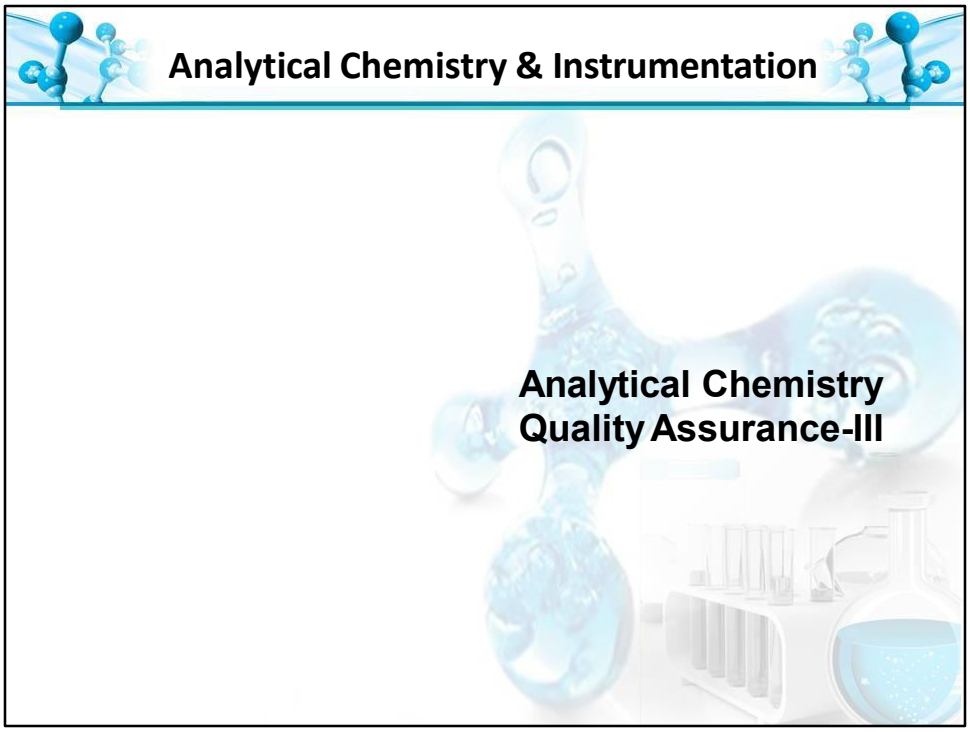
Examples

- **Total hardness test determined calcium and magnesium and reported as $CaCO_3$. For specifically Ca or Mg we may use atomic absorption spectroscopy.**
- **Free Fatty acids / Acid value determine through acid base titration.**
- **Crude protein estimated as Kjeldhal Nitrogen.**

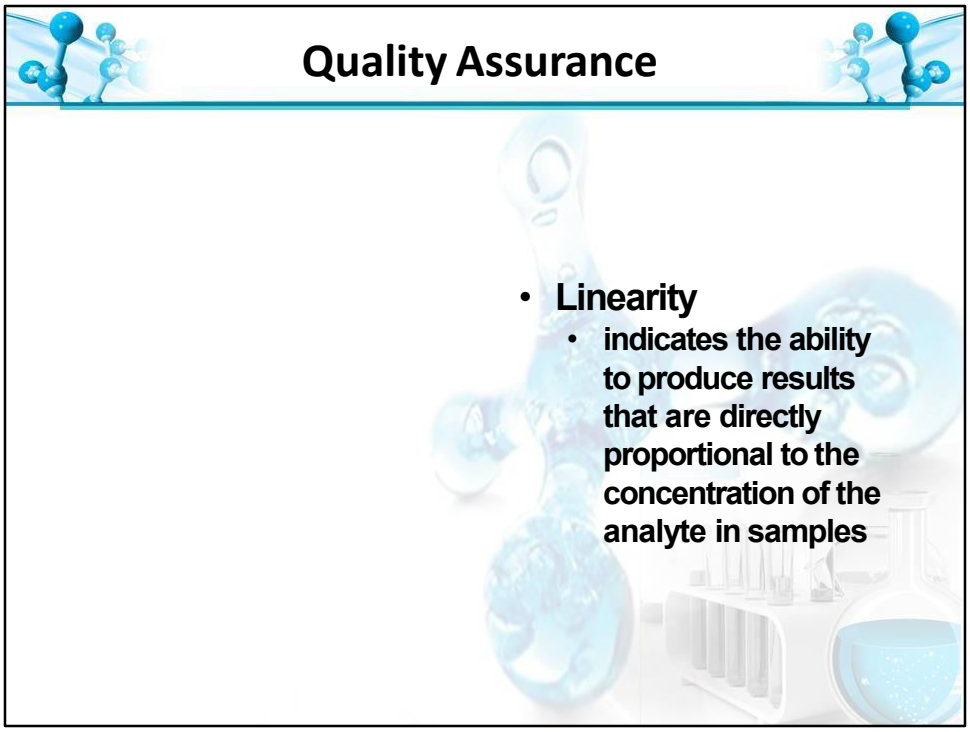
Examples……


Quality Assurance
END
• Specificity / Selectivity
• In this way we display ability to measure the desired analyte that may be active ingredient or may be the impurity or unwanted substance

Benefits…..



The second step in method validation is linearity….



Linearity indicates….

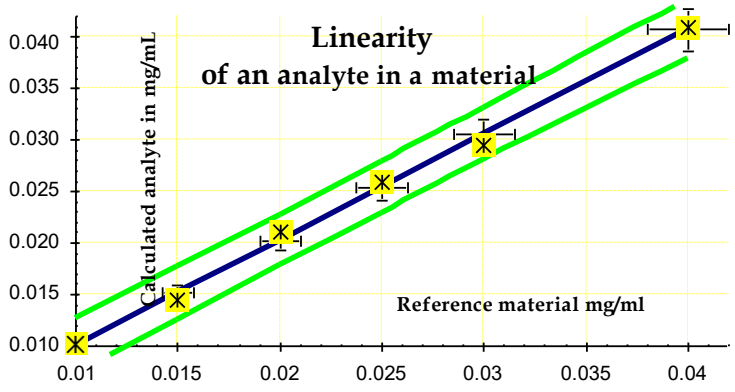
Quality Assurance

Linearity:

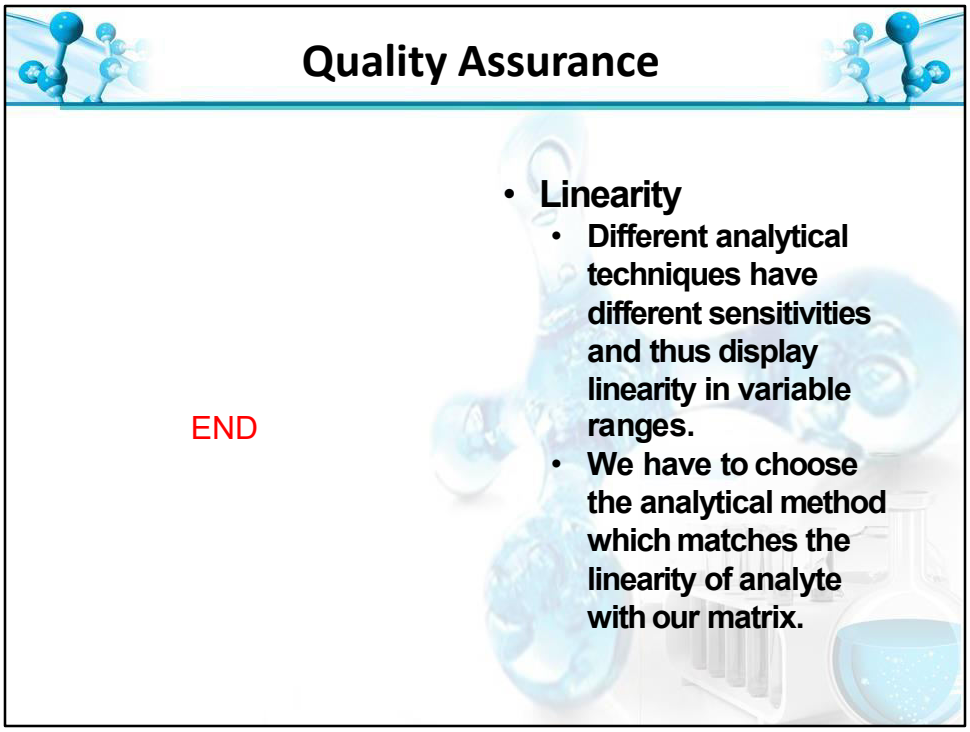
- A series of samples should be prepared in which the analyte concentrations span the claimed range of the procedure. If there is a linear relationship, test results should be evaluated by appropriate statistical methods.
- The correlation coefficient, y-intercept, slope of the regression line, and residual sum of squares should be submitted.
- For the establishment of linearity, a minimum of five concentrations is recommended.
- Some analytical procedures, such as immunoassays, do not demonstrate linearity.

How to proceed…..

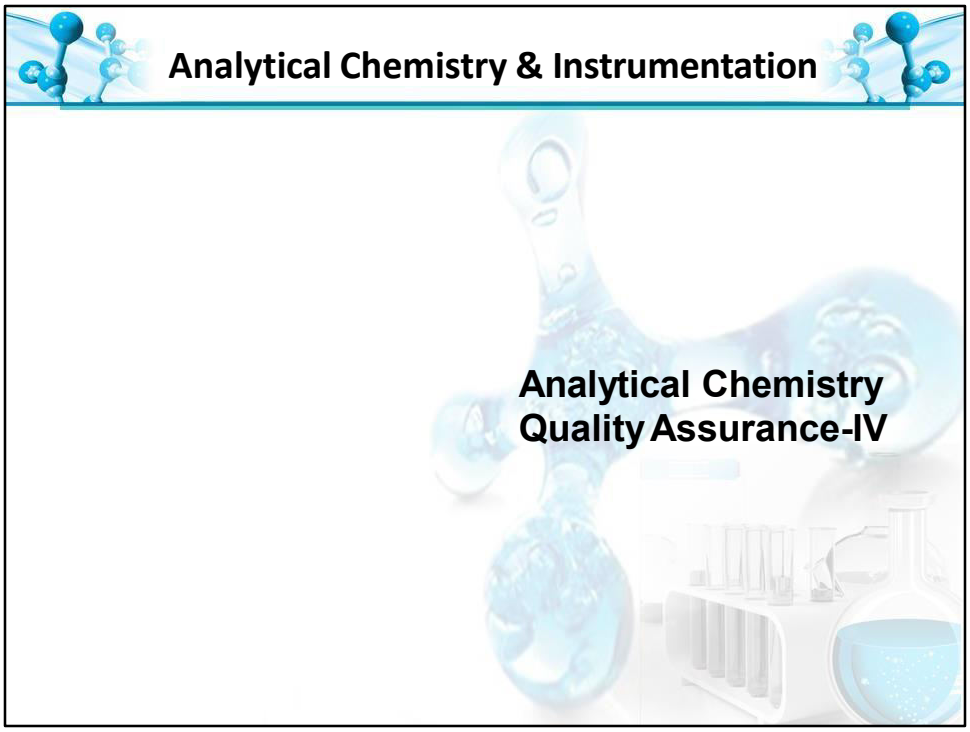
Quality Assurance



Linear graph, scatter diagram…..


Quality Assurance
END
• Linearity
• Different analytical techniques have different sensitivities and thus display linearity in variable ranges.
• We have to choose the analytical method which matches the linearity of analyte with our matrix.

Benefits….



The third step of validation is to determine range….

Quality Assurance

- **Range:**
  - **Range is an expression of the lowest and highest levels of analyte that have been demonstrated to be determinable for the product.**

A graph is…..

Quality Assurance

Range:

- **The specified range is normally derived from linearity studies and depends on the intended application of the procedure.**
- **It is established by confirming that the analytical procedure provides an acceptable degree of linearity, accuracy, and precision when applied to samples containing amounts of analyte within or at the extremes of the specified range of the analytical procedure.**

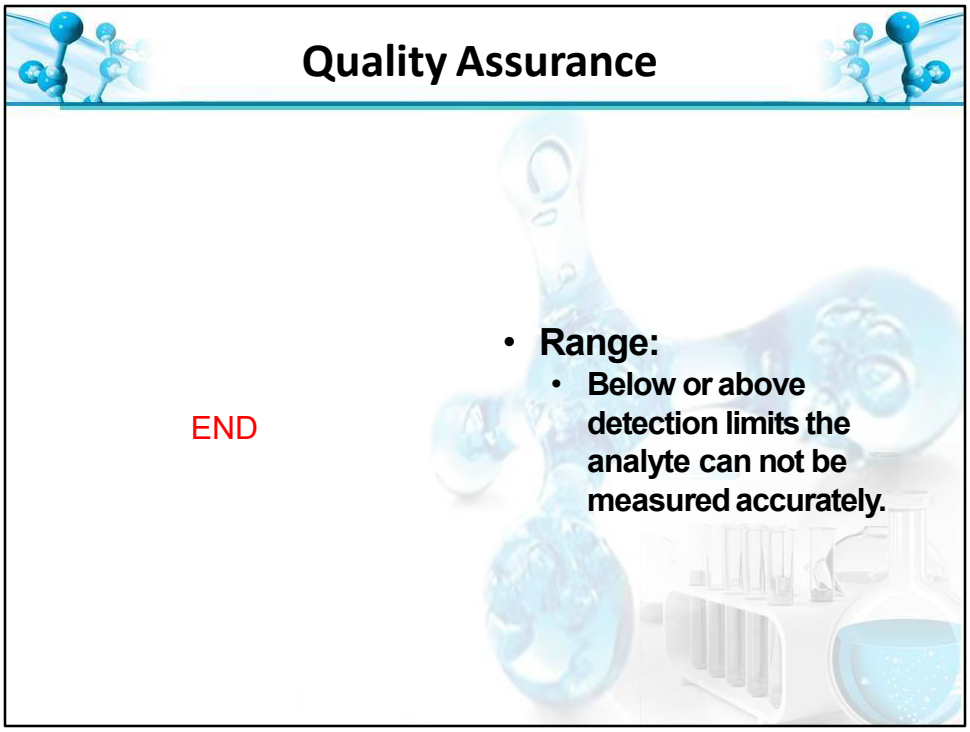
Range is …….

Quality Assurance

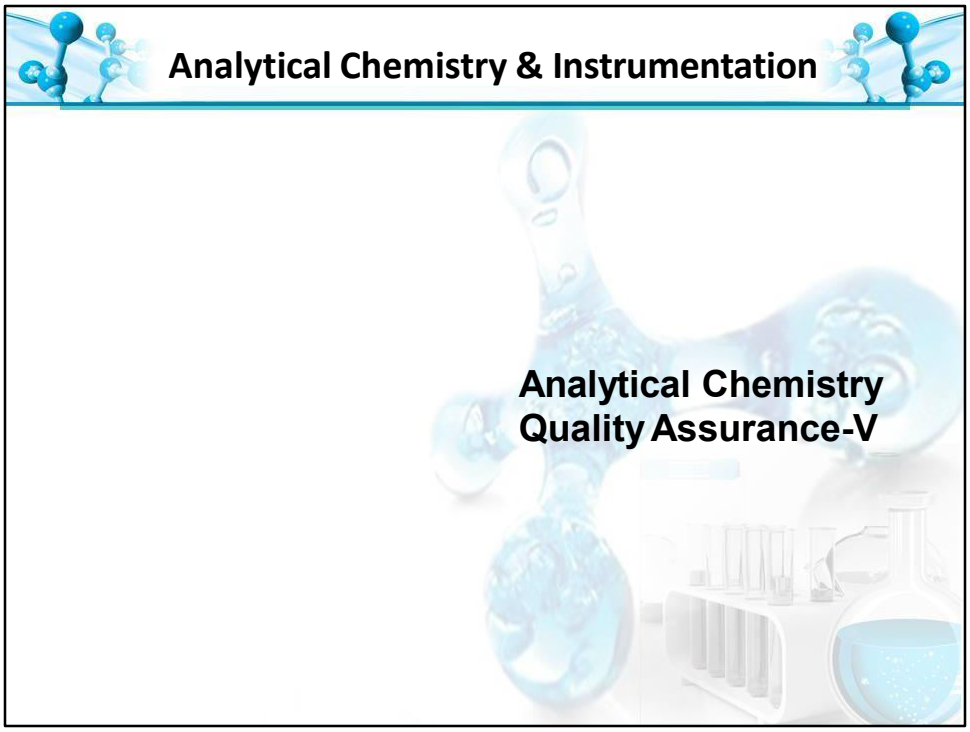
Minimum specified ranges that should be considered:

- **For the assay of a drug substance or a finished (drug) product:** Normally from 80 to 120 percent of the test concentration.
- **For content uniformity:** Covering a minimum of 70 to 130 percent of the test concentration.
- **For dissolution testing:** +/-20 percent over the specified range; e.g., if the specifications for a controlled released product cover a region from 20 percent, after 1 hour, up to 90 percent, after 24 hours, the validated range would be 0-110 percent of the label claim.
- **For the determination of an impurity:** From the reporting level of an impurity to 120 percent of the specification.

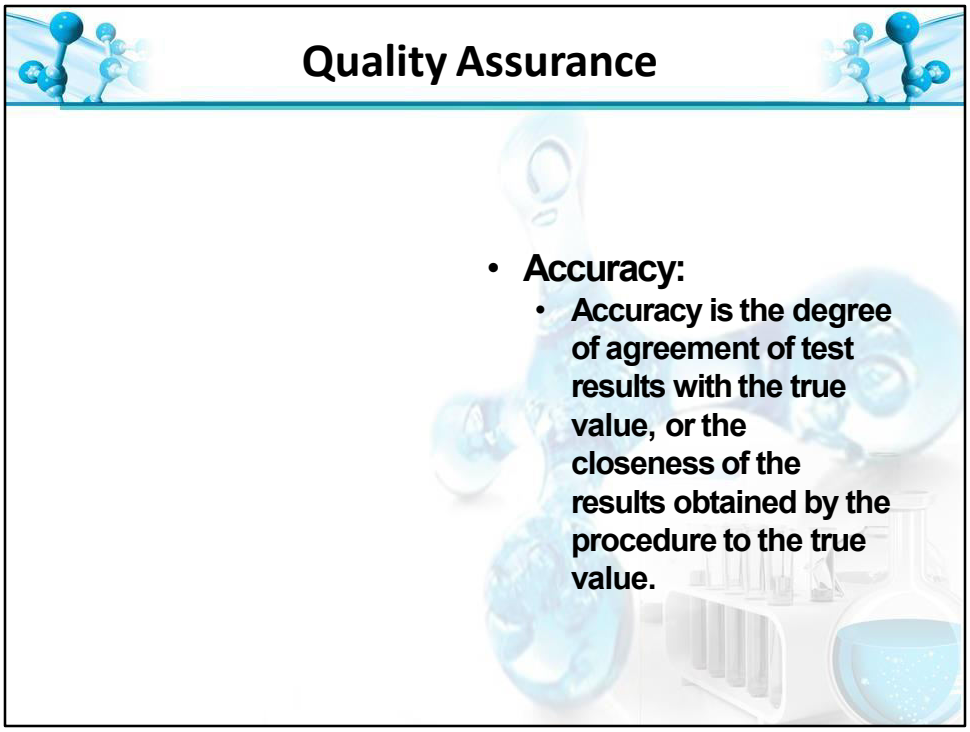
Minimum specified ranges are ……


Quality Assurance
END
• Range:
• Below or above detection limits the analyte can not be measured accurately.

Benefits…..



The fourth step of validation is to determine accuracy ….


Quality Assurance
• Accuracy:
• Accuracy is the degree of agreement of test results with the true value, or the closeness of the results obtained by the procedure to the true value.

Accuracy is…..

Quality Assurance

Accuracy:

- It is normally established on samples of the material to be examined that have been prepared to quantitative accuracy.
- Accuracy should be established across the specified range of the analytical procedure.
- It is acceptable to use a "spiked" placebo where a known quantity or concentration of a reference material is used.

Accuracy is …….

Quality Assurance

Assay of Drug substance

- **Application of an analytical procedure to an analyte of known purity (e.g., reference material);**
- **Comparison of the results of the proposed analytical procedure with those of a second well-characterized procedure.**

Assay of Drug product

- **Application of the analytical procedure to synthetic mixtures of the drug product components to which known quantities of the drug substance to be analyzed have been added.**

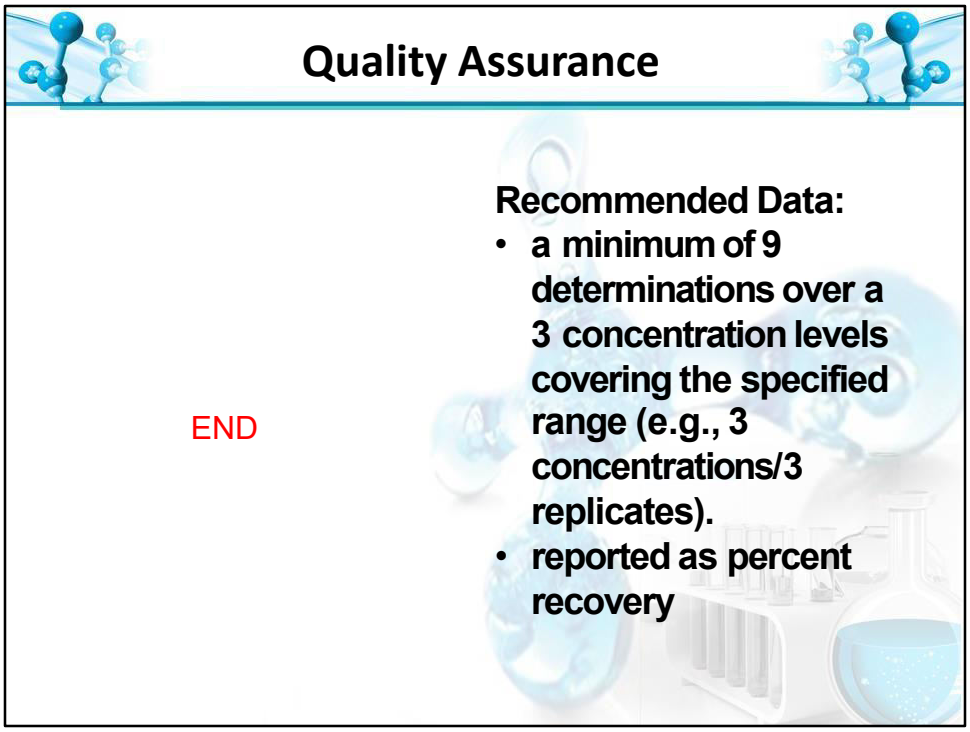
Accuracy may be considered of three type of products …….

Quality Assurance

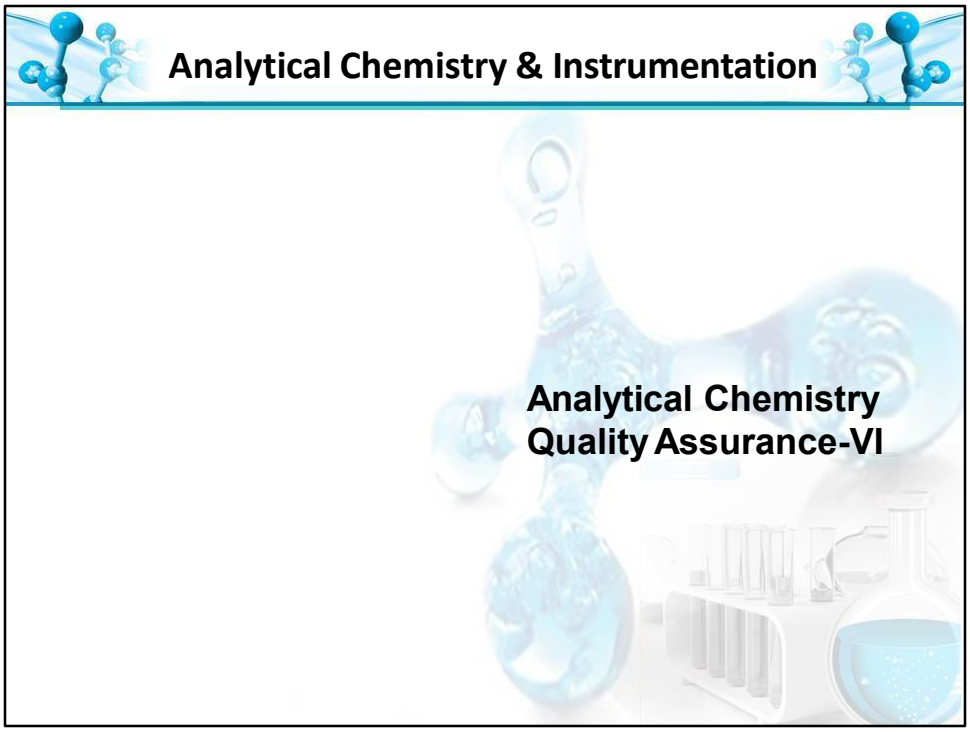
Accuracy (Impurities)

- **Accuracy should be assessed on samples (drug substance/drug product) spiked with known amounts of impurities.**
- **It should be clear how the individual or total impurities are to be determined, e.g., weight/weight or area percent, in all cases with respect to the major analyte.**
- **Accuracy may be inferred once precision, linearity, and specificity have been established.**

Accuracy is …….



Recommended data …..



The fifth step in validation of testing method will be to determine precision ….

Quality Assurance

- **Precision:**
  - **It is the degree of agreement among individual results.**
  - **The complete procedure should be applied repeatedly to separate and identical samples drawn from the same homogeneous batch of material. Reported as RSD.**

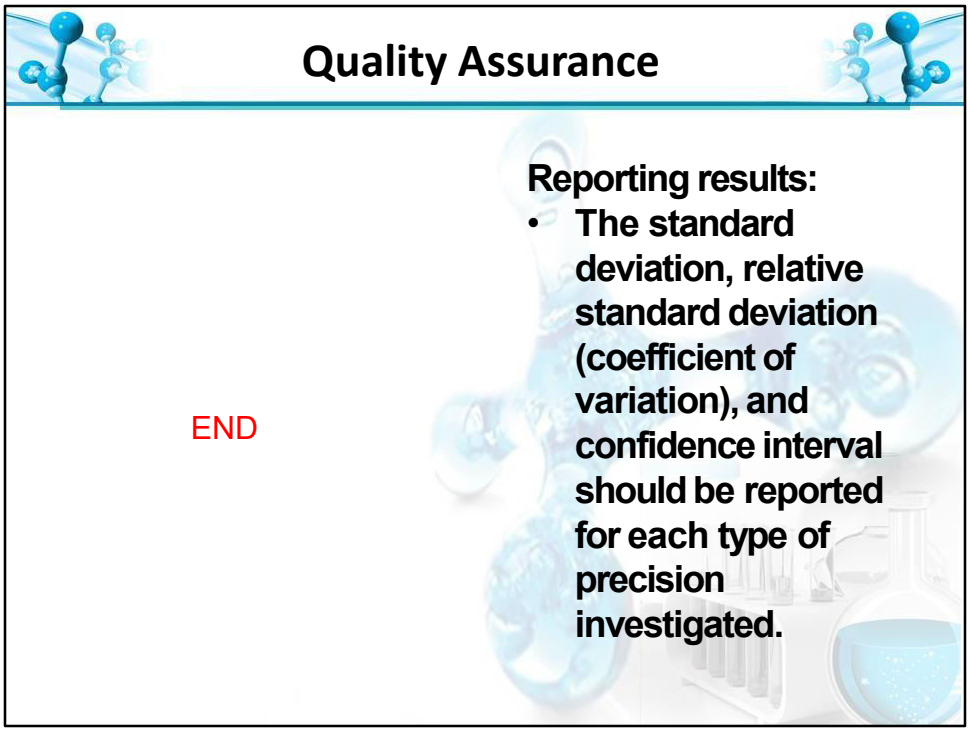
Precision is…..

Quality Assurance

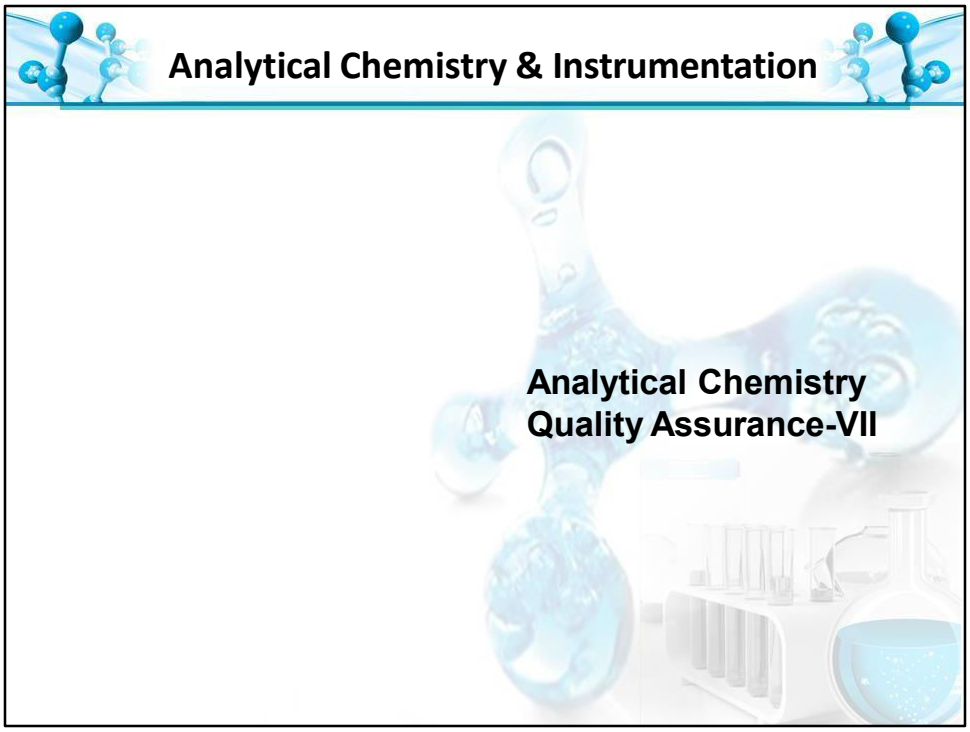
Precision:

- **Repeatability: A minimum of nine determinations covering the specified range for the procedure, e.g. three concentrations/three replicates each, or a minimum of six determinations at 100% of the test concentration**
- **Intermediate precision: Within-laboratory variations**
  - **usually on different days, analysts and equipment. (If reproducibility is assessed, a measure of intermediate precision is not required.)**
- **Reproducibility: Precision between laboratories**

To determine pricision …….



Reporting of results …..



The sixth step of validation is to determine LOD and LOQ ….

Quality Assurance

- **Limit of Detection (LOD):**
  - **It is the smallest quantity of an analyte that can be detected, and not necessarily determined in a quantitative fashion.**

A graph is…..

Quality Assurance

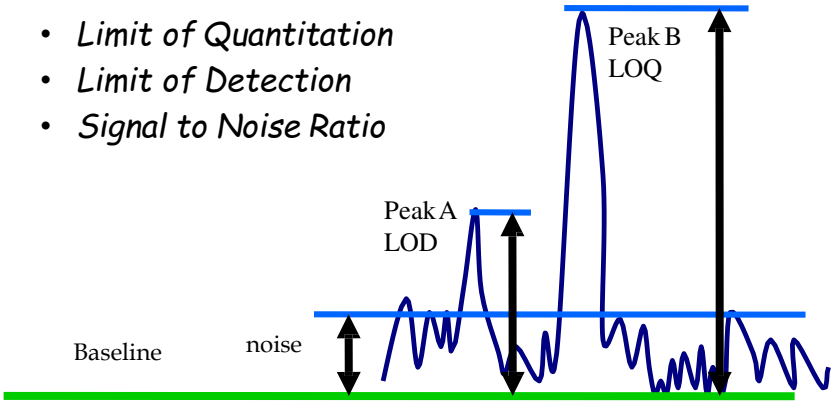
- **Limit of Quantitation (LOQ):**
  - **It is the lowest concentration of an analyte in a sample that may be determined with acceptable accuracy and precision.**

A graph is…..

Quality Assurance

LOD & LOQ:

- *Limit of Quantitation*
- *Limit of Detection*
- *Signal to Noise Ratio*



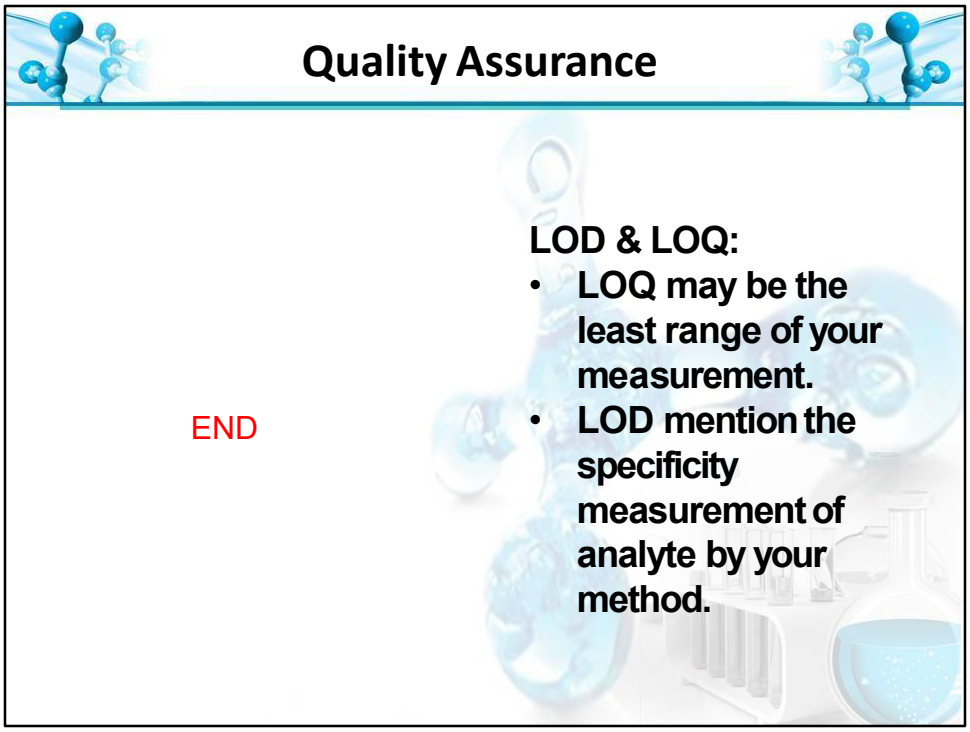
Range is …….

Quality Assurance

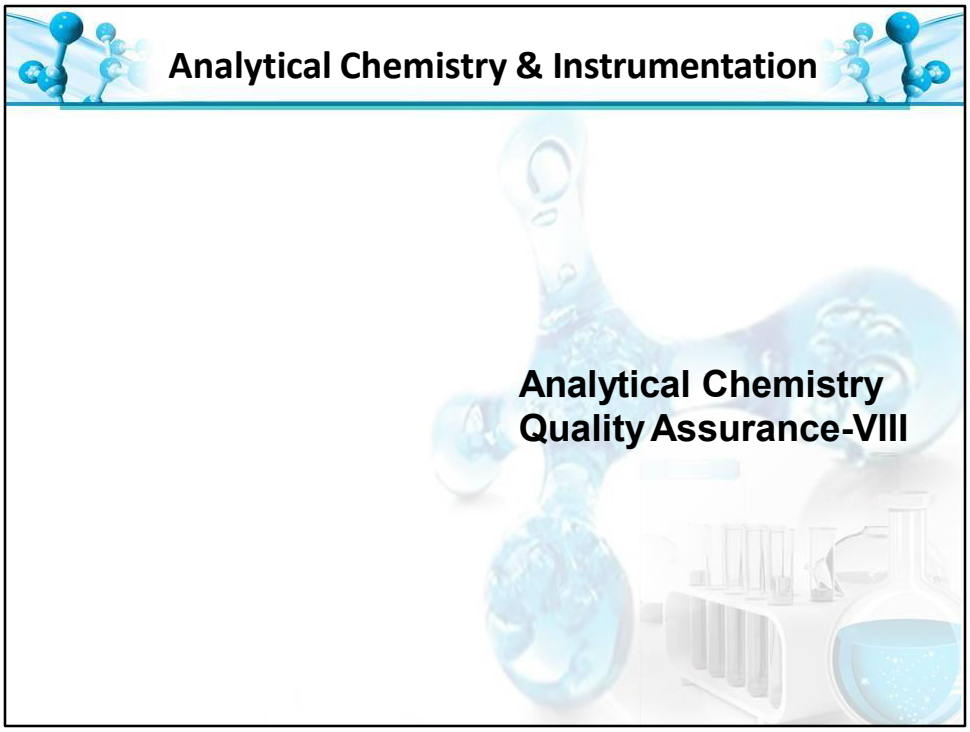
LOD & LOQ:

- **Approaches (instrumental or non-instrumental):**
  - **Visual Evaluation (mostly in non-instrumental methods)**
  - **Signal to Noise Ratio (3 or 2:1 is acceptable of LOD while 10:1 is generally acceptable for LOQ)**
  - **Standard Deviation of the Response and the Slope (3.3 σ/S for LOD & 10 σ/S for LOQ)**
  - **Standard Deviation of the Blank**
  - **Calibration Curve (The residual standard deviation of a regression line or the standard deviation of y-intercepts of regression lines may be used as the standard deviation).**

How to study LOQ and LOQ …….


Quality Assurance
END
LOD & LOQ:
• LOQ may be the least range of your measurement.
• LOD mention the specificity measurement of analyte by your method.

Benefits…..



The third step of validation is to determine range….

Quality Assurance

- **Robustness:**
  - **Robustness is the ability of the procedure to provide analytical results of acceptable accuracy and precision under a variety of conditions.**

A graph is…..

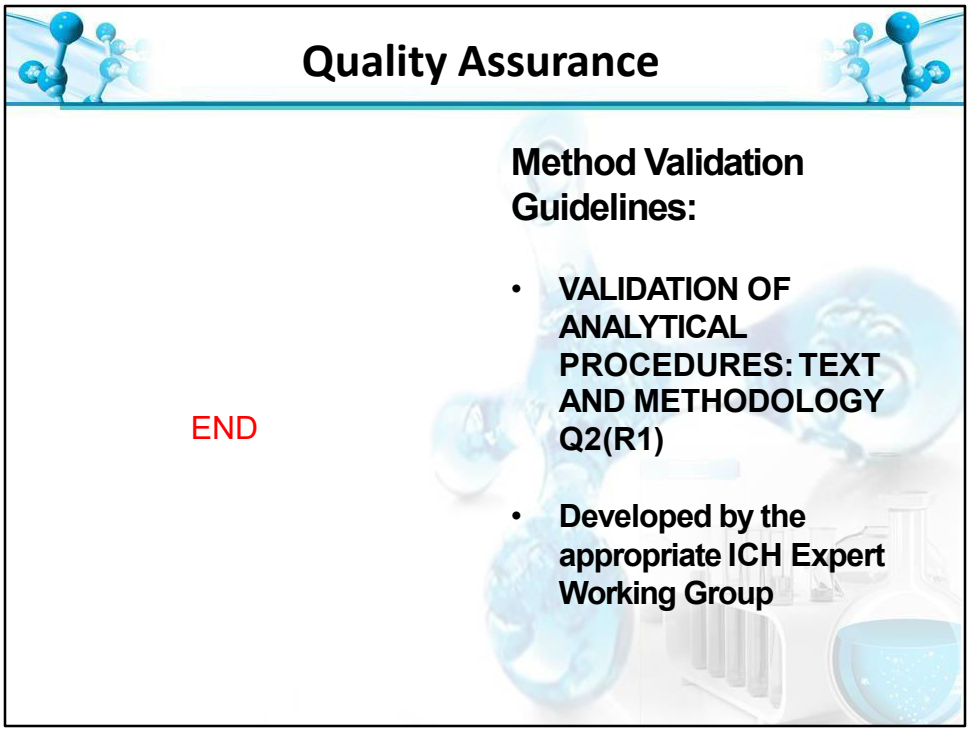
Quality Assurance

Examples of typical variations:

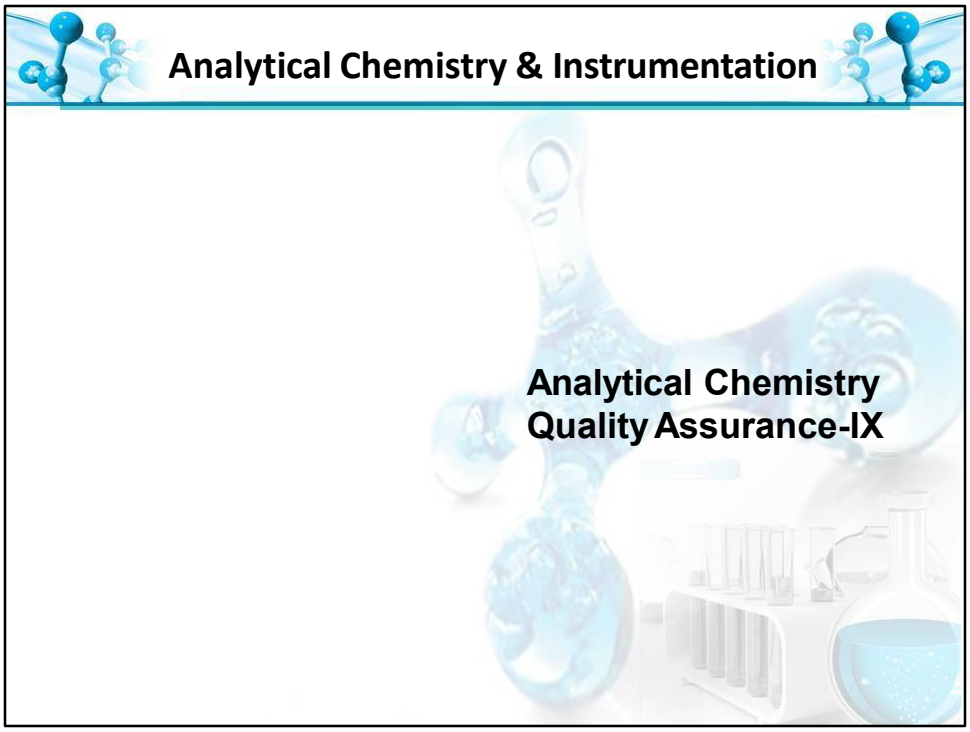
While performing chromatographic analysis:

- **stability of test and standard samples and solutions**
- **reagents (e.g. different suppliers)**
- **different columns (e.g. different lots and/or suppliers)**
- **extraction time**
- **variations of pH of a mobile phase**
- **variations in mobile phase composition**
- **temperature**
- **flow rate**

Range is …….


Quality Assurance
END
Method Validation Guidelines:
• VALIDATION OF ANALYTICAL PROCEDURES: TEXT AND METHODOLOGY Q2(R1)
• Developed by the appropriate ICH Expert Working Group

Benefits…..



The third step of validation is to determine range….

Quality Assurance

- **Measurement Uncertainty:**
  - **"Aparameter associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand".**

A graph is…..

Quality Assurance

Sources of uncertainty:

- **sampling,**
- **matrix effects and interferences,**
- **environmental conditions,**
- **uncertainties of masses**
- **volumetric equipment,**
- **reference values,**
- **approximations and assumptions incorporated in the measurement method and procedure,**
- **analyst**

Range is …….

Quality Assurance

Type A and Type B Uncertainty:

- **Uncertainty Type "A": Uncertainty component, evaluated experimentally from the dispersion of repeated measurements, using statistical methods.**
- **Uncertainty Type "B": Uncertainties that are always expressed and evaluated as probability distributions like normal, rectangular and triangular distributions and are called Type B, uncertainties.**
  - **Normal Distribution: 95% confidence level.**
  - **Rectangular Distribution: without specifying a level**
  - **Triangular Distribution**

Range is …….

Quality Assurance

Calculating Uncertainty:

- Formula for evaluating Uncertainty, (Type "A") $U_A = \sigma / N$ (Standard error of the mean)
- Component of type B
  - Type B = (Component 1)^2 + (Component 2)^2 +……..
  - All the components added must have same units and distribution.
- Formula for calculating the combined uncertainty (U_C)
  - $U_C = (U_A)^2 + (U_B)^2$ (Where U_A and U_B are the type "A" and type "B" Uncertainties)

Range is …….

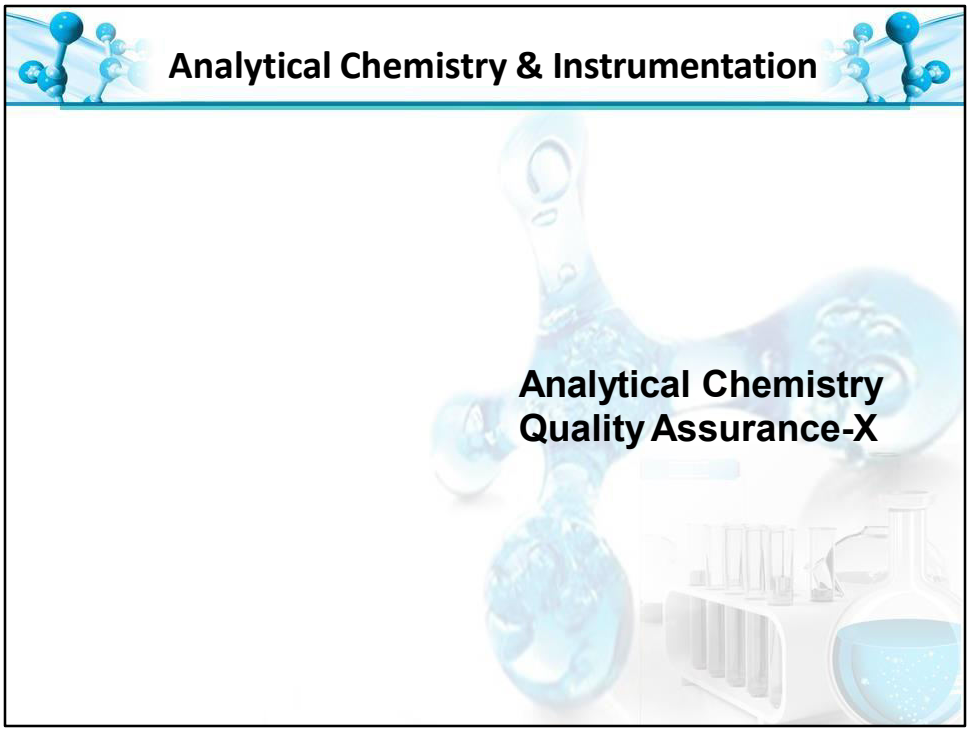
Quality Assurance

END

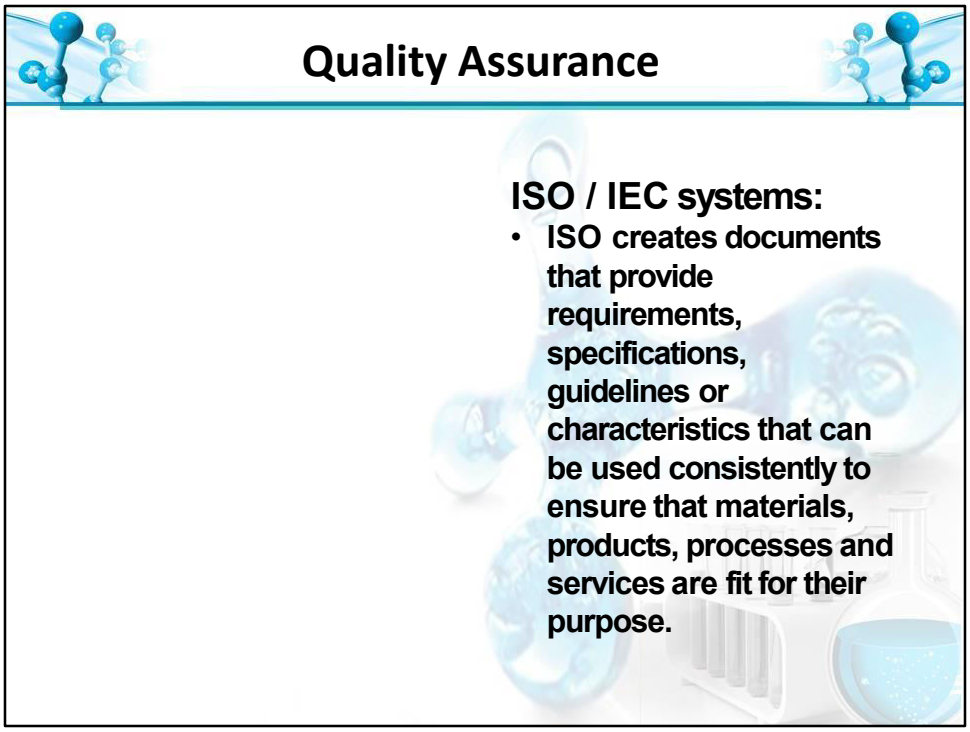
Reporting Uncertainty:

- **Expanded uncertainty $U_e = U_C$ X k (k = 2 Correspond to the confidence level of 95% in the normal distribution curve).**
- **Uncertainty is reported as $Y \pm U_e$ with confidence level of 95% (Where, Y = measurand)**

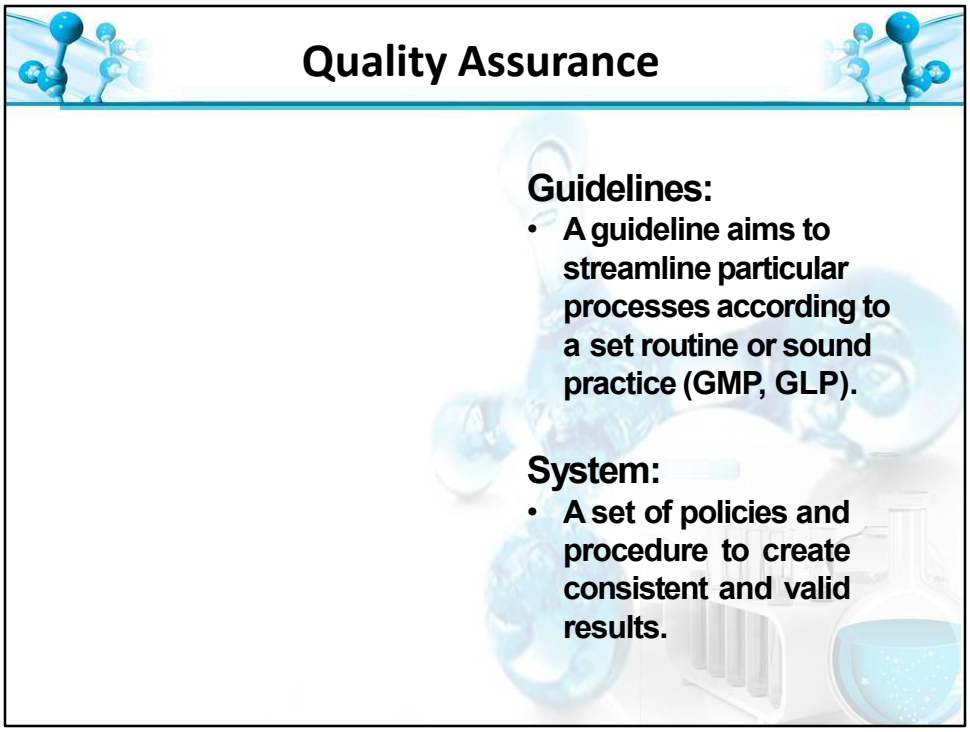
Benefits…..



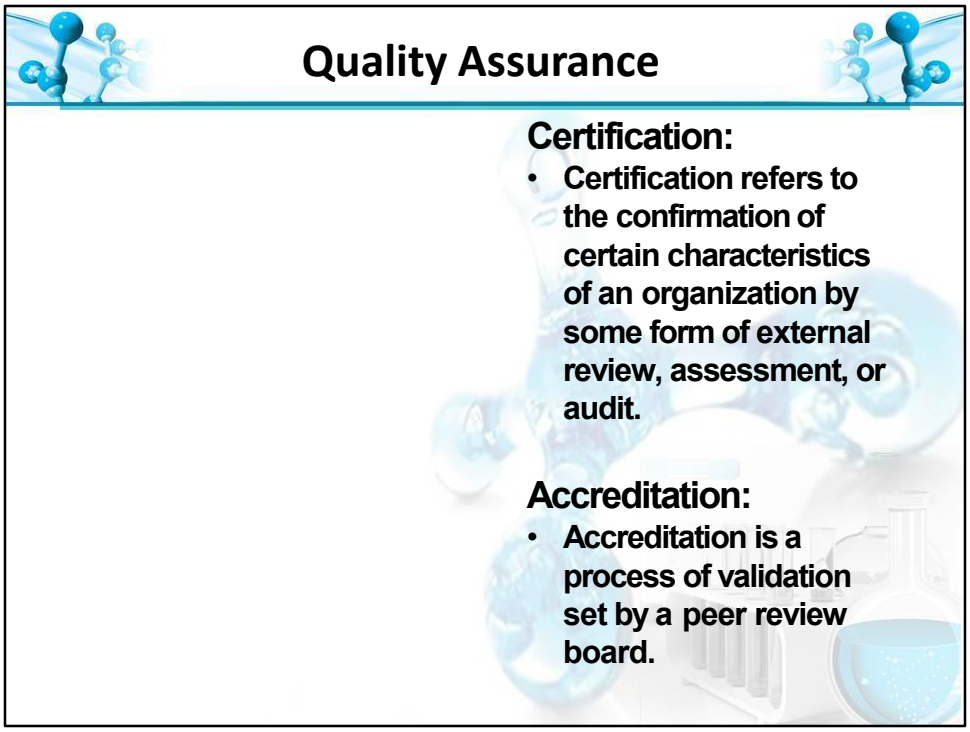
In quality assurance let's discuss another aspect, the ISO systems…..



International organization for standardization creates documents to ensure quality of process, environment, testing activities, human performance etc.



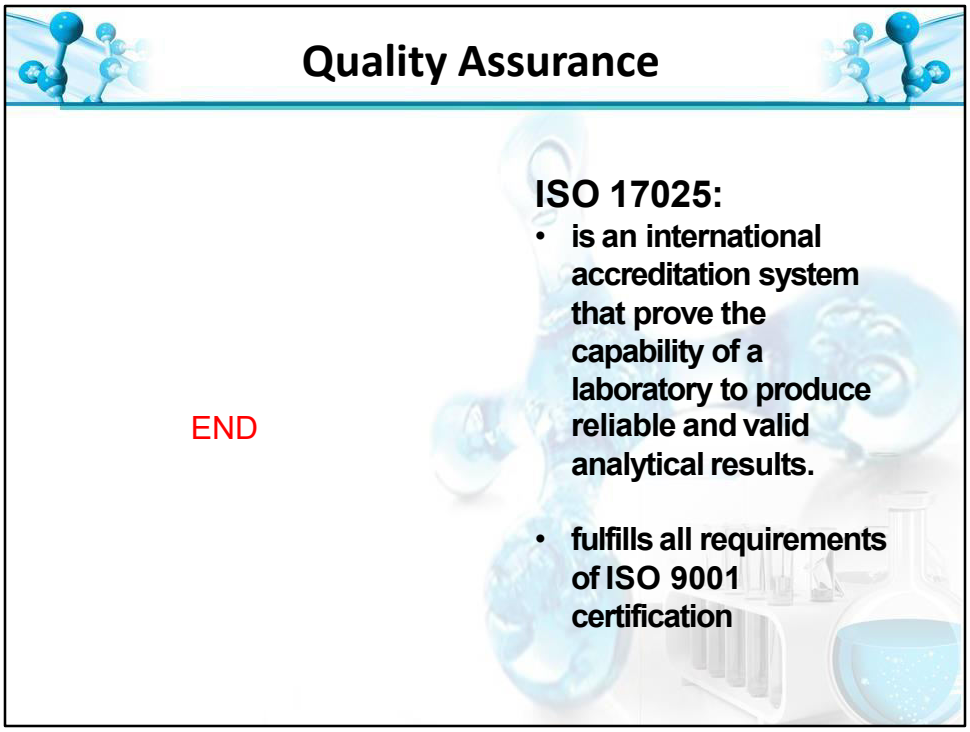
Guidelines and systems…….



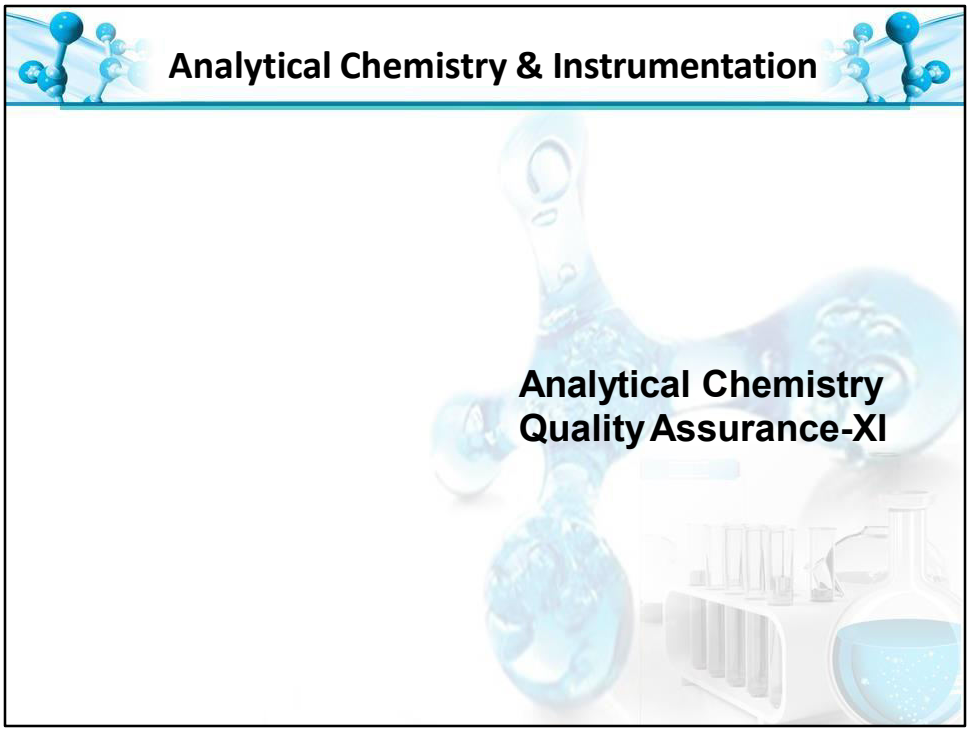
Certification and accreditation ……



Popular standards …….



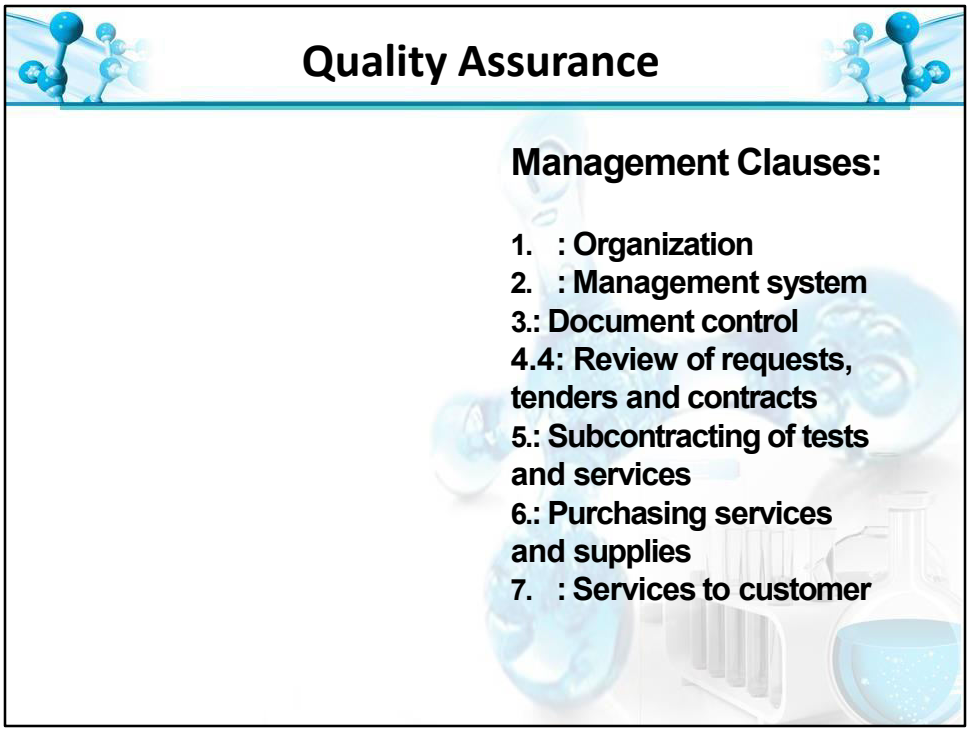
ISO 17025 significance …..



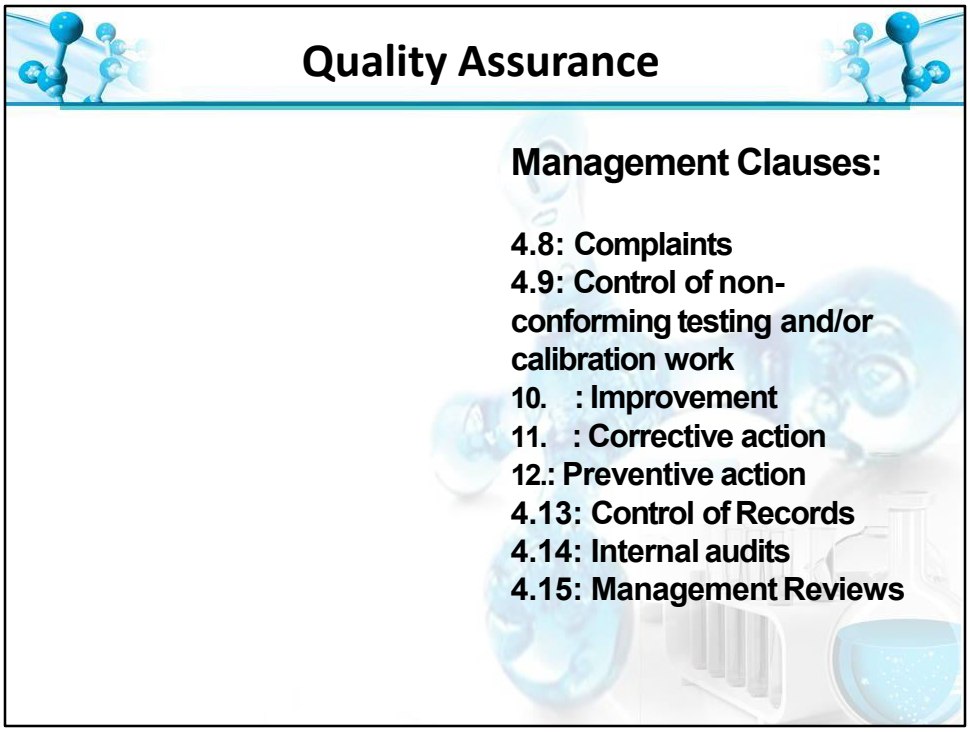
The third step of validation is to determine range….



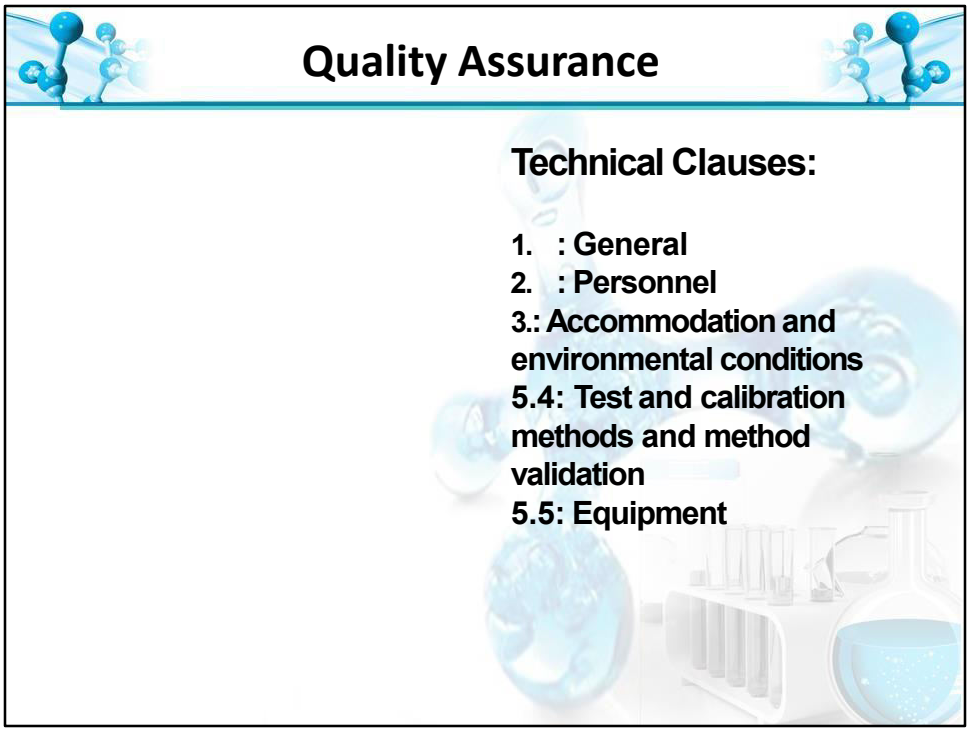
ISO 17025 comprises on 25 clauses ……Management clauses are to maintain management structure of an organization. You may understand it is the hardware. While the technical clauses are the software of the organization. The compliance of technical clauses mean that organization enhance technical competence of that lab or organization..



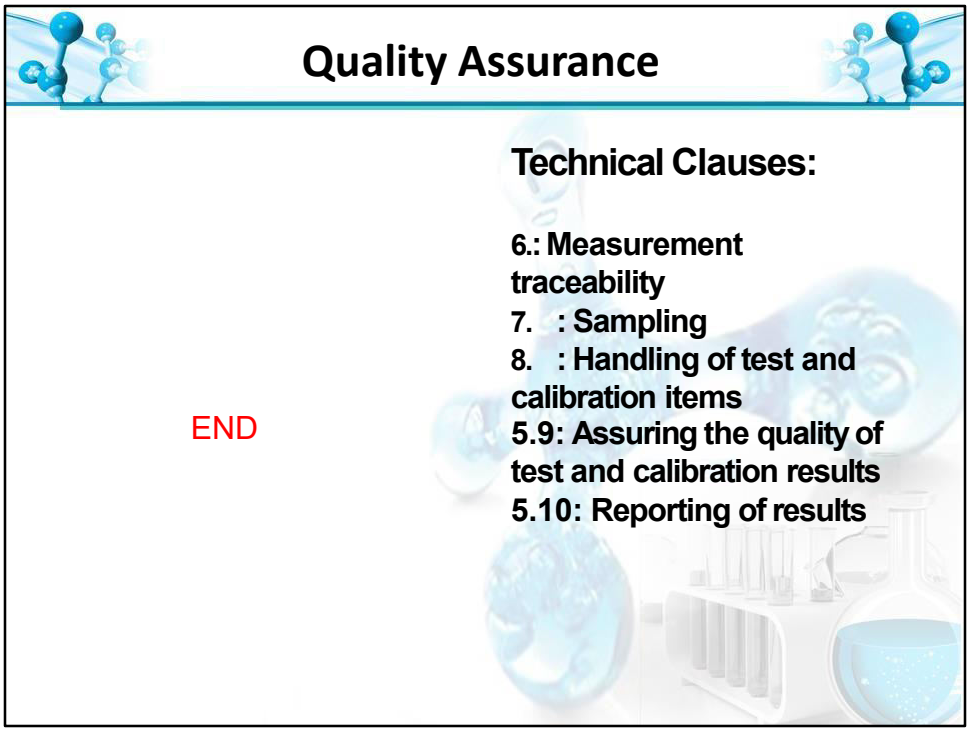
A review of clauses …..



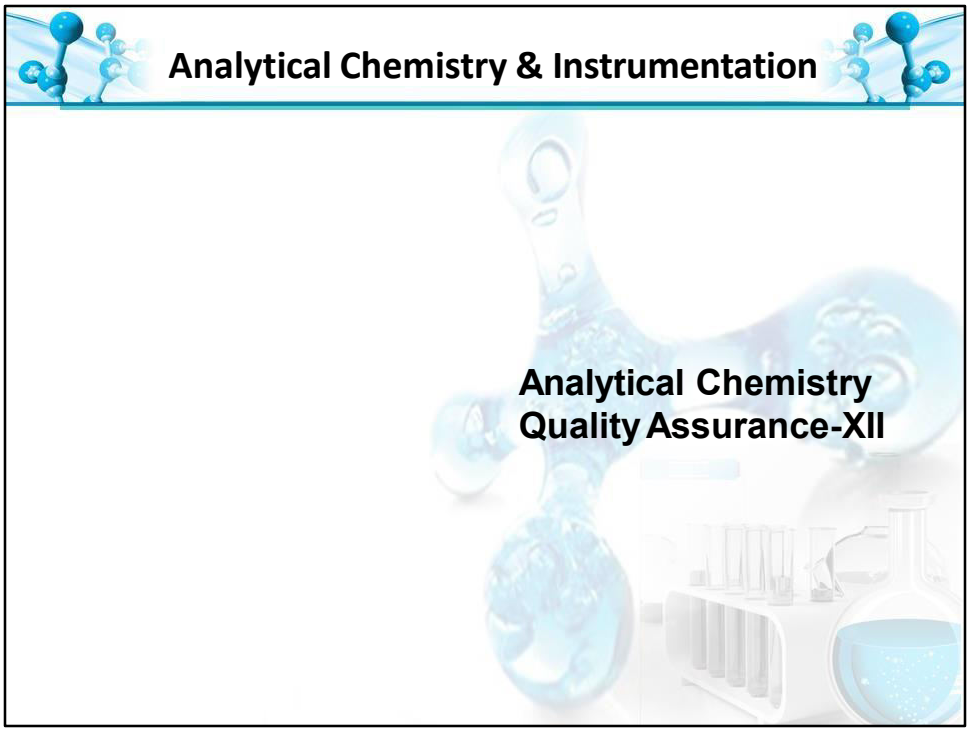
A review of clauses …..



A review of clauses …..



A review of clauses …..



Now we'll discuss the system clause wise…..



We will discuss clause 4.1 organization …….

Quality Assurance

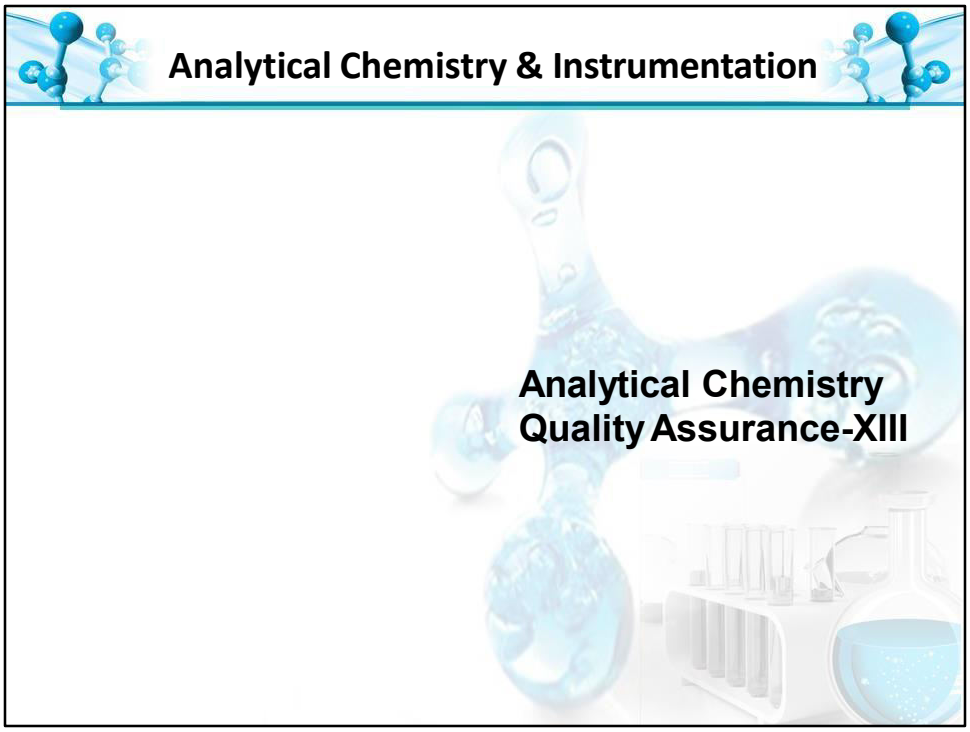
4.1 Organization

- **Legal entity**
- **Conflict of interest defined (Part of larger org.)**
- **Impartiality and freedom from commercial / financial pressures**
- **Have policies in place to manage impartiality, judgement or operational integrity**
- **Have Management and Technical staff.**
- **Technical Manager and Technical Management.**

Organization must be a legal entity.
Policies must be to reduce conflict of interest…



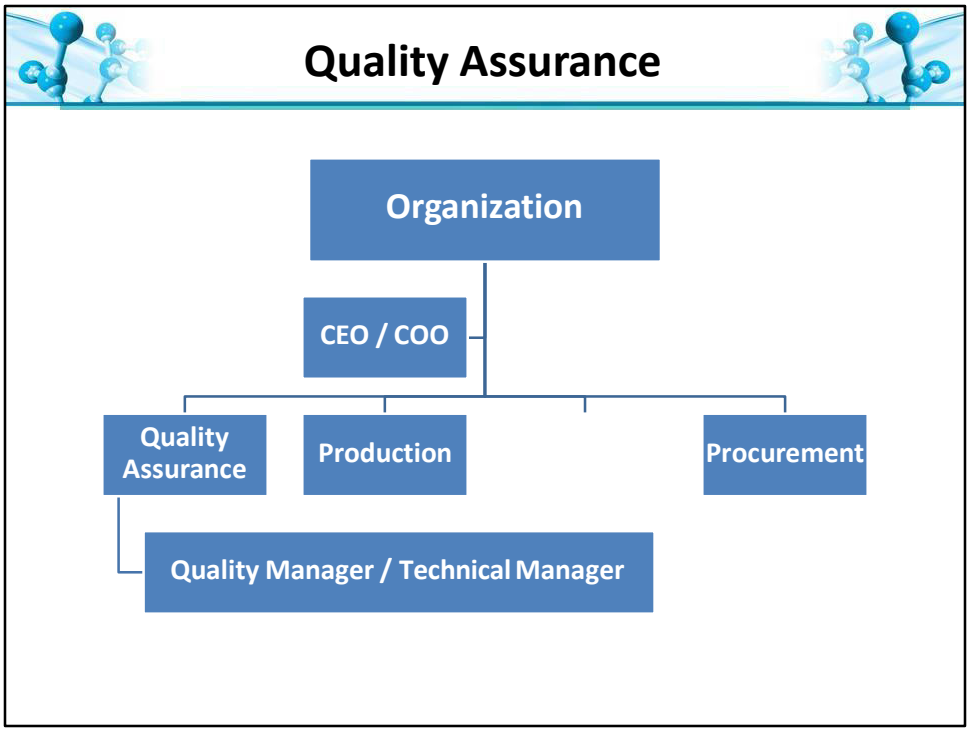
Organization must have technical manager and technical management



The third step of validation is to determine range….


Quality Assurance
4.2 Management system
• Organizational chart
• Policies
• Scope must be defined

Benefits…..



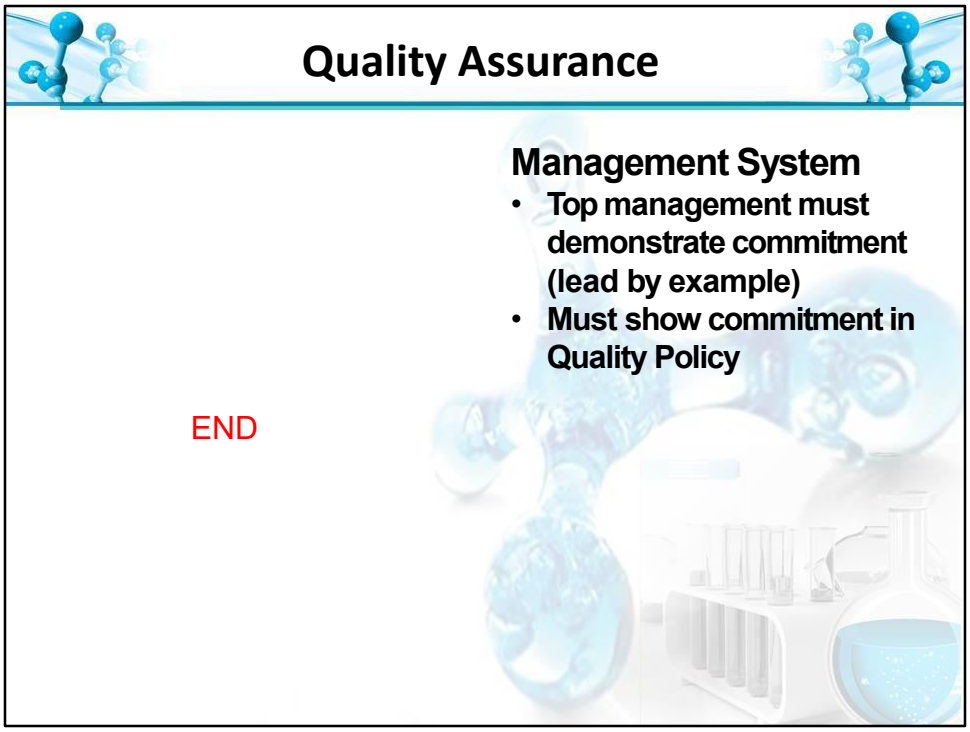
Organogram show hierarchy of an organization…. How big or small, what kind of variety it is producing, how many workers are there….

Quality Assurance

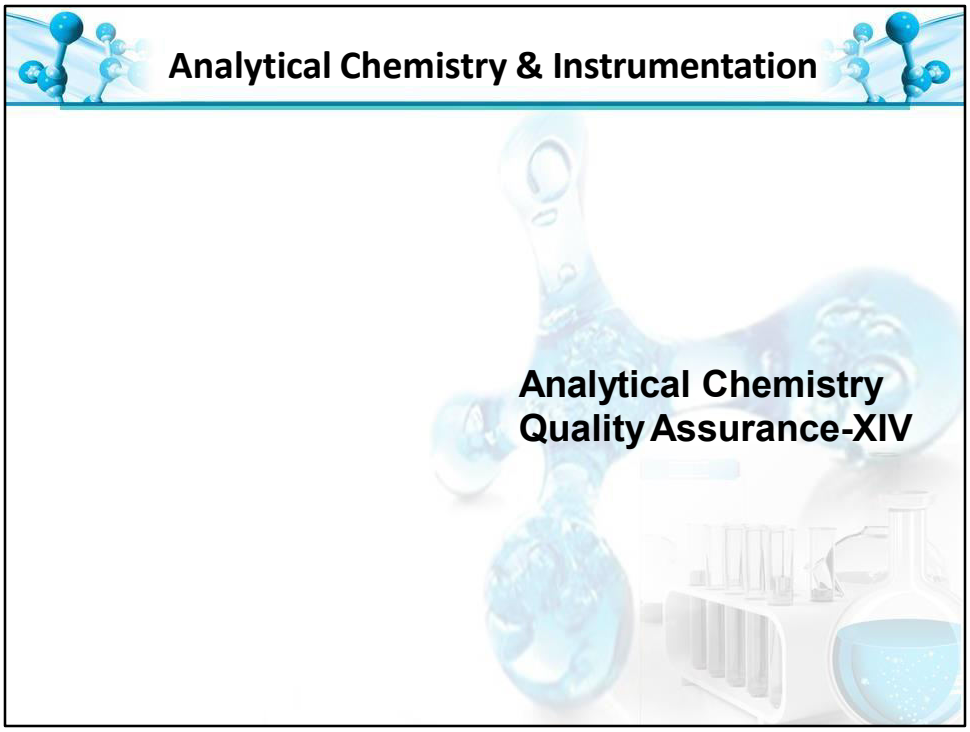
4.2 Management System

- **Must have implemented a MS appropriate for its scope and activities (documented to extent necessary to assure quality of result)**
- **Understood by all in organization and communicated to all**
- **System must be documented**
- **Top management must demonstrate commitment**

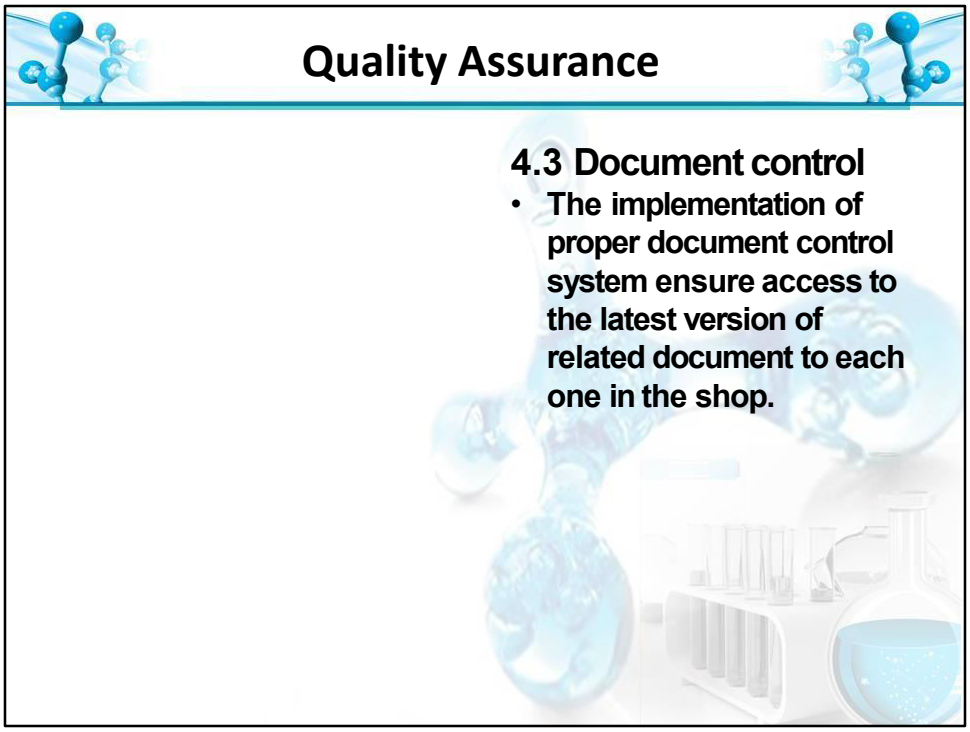
Management system must be implemented in proper sense…



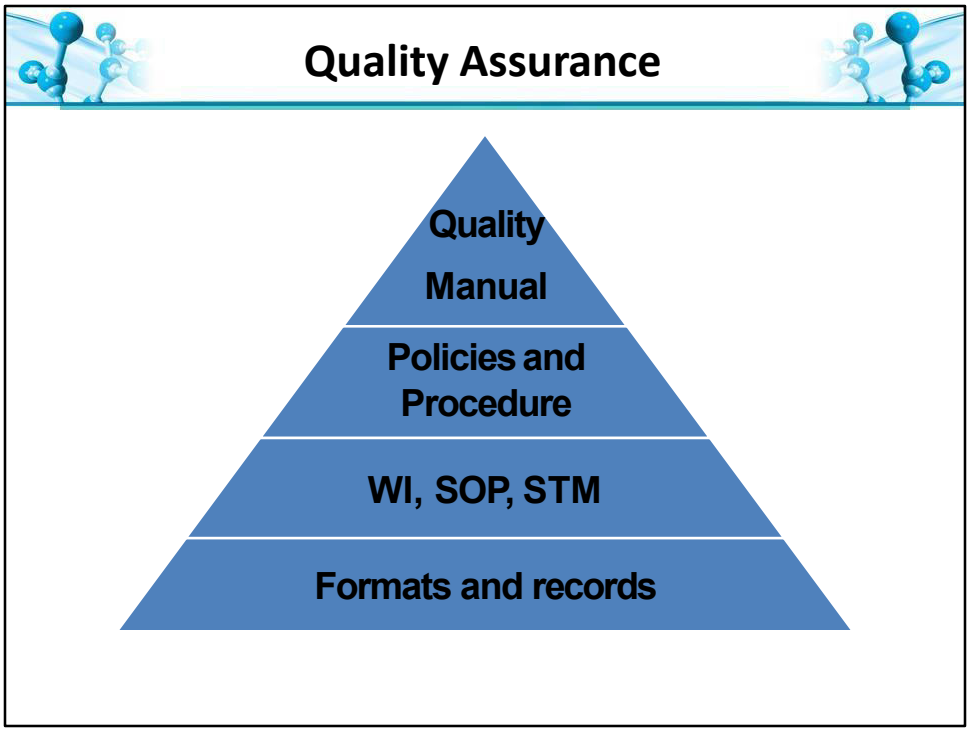
Top management must show commitment….



The next clause of this system is document control…



The implementation is must



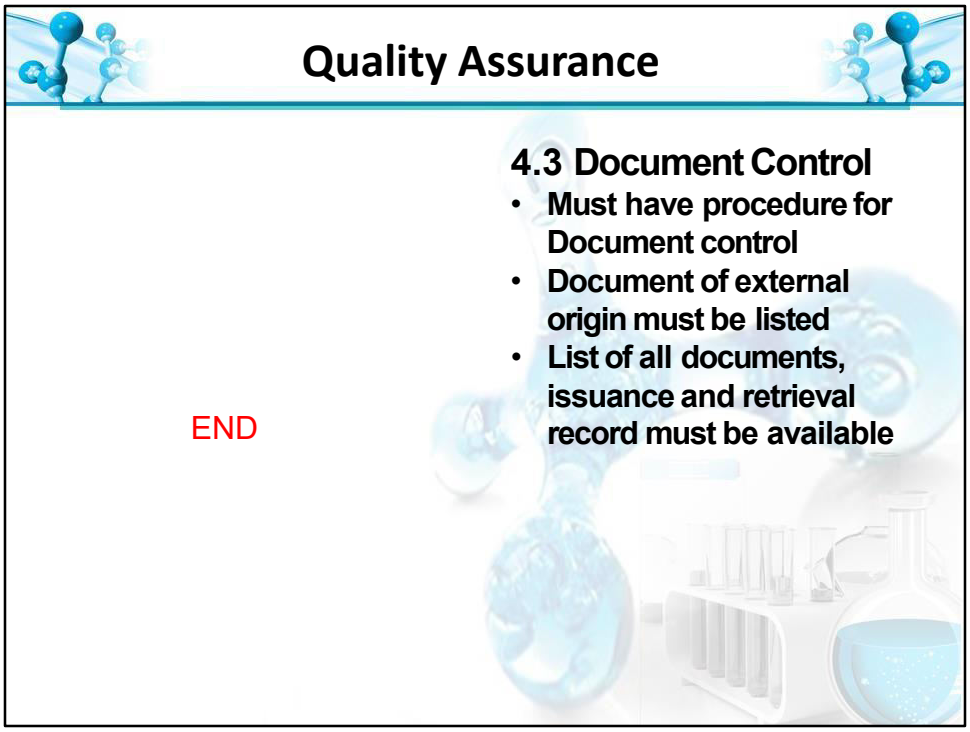
Levels of documents….

Quality Assurance

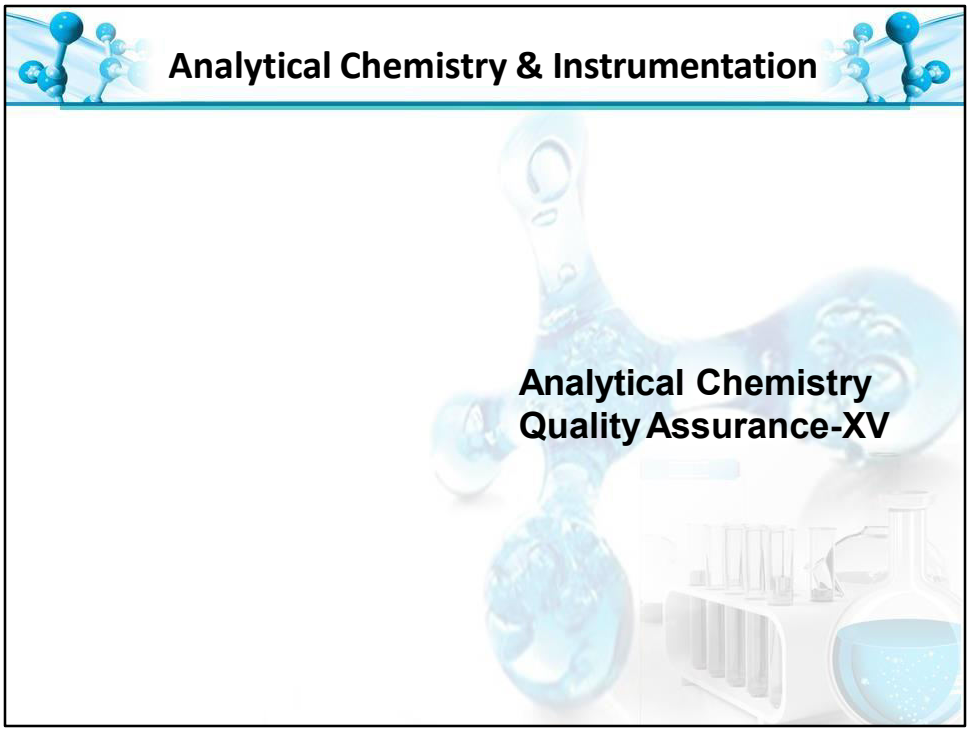
4.3 Document control

- **Documents must be uniquely identified**
- **Approval and issue – Reviewed and approved (uniquely identified)**
- **Document changes – by original function. Altered text highlighted**
- **Computerized data must be controlled**

System clauses……



Must have procedure to cover all internal and external documents….



The third step of validation is to determine range….



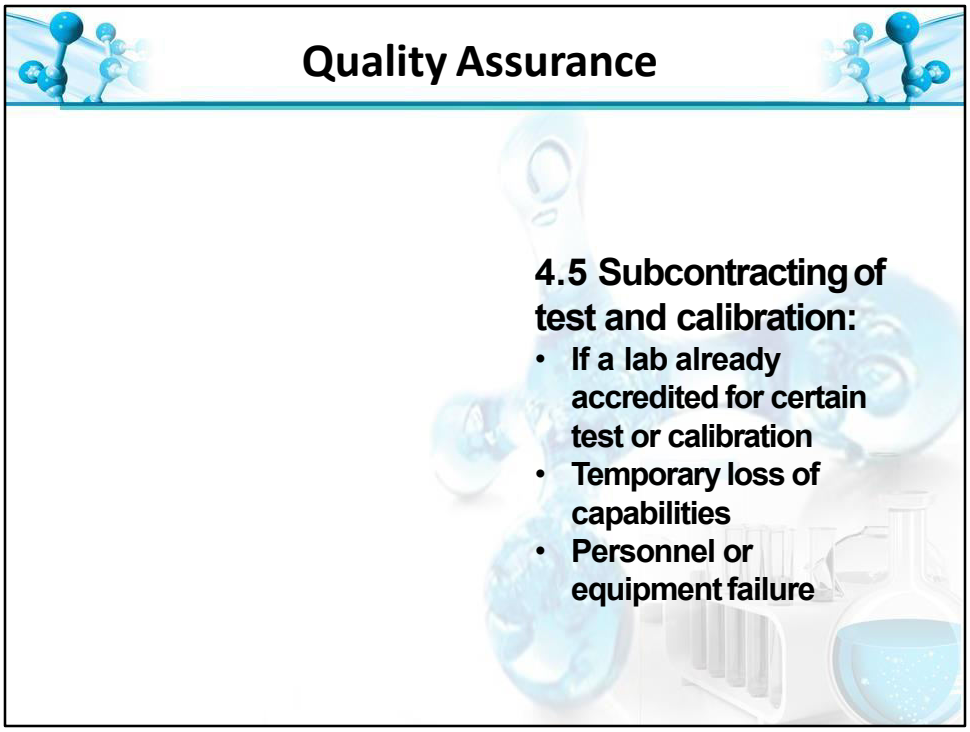
First define the words, request, tenders and contracts…

Quality Assurance

4.4 Review of requests, tenders and contracts

- **Have procedures in place for this (Records of review must be available)**
- **Have necessary resources**
- **Differences resolved before acceptance**
- **Can be oral or documented (contract)**
- **Review of simple or routine results**
- **Repetitive tests – Review at initial stage only as long as unchanged**
- **Customer shall be informed of any deviation**

Sub clauses of system…


Quality Assurance
4.5 Subcontracting of test and calibration:
• If a lab already accredited for certain test or calibration
• Temporary loss of capabilities
• Personnel or equipment failure

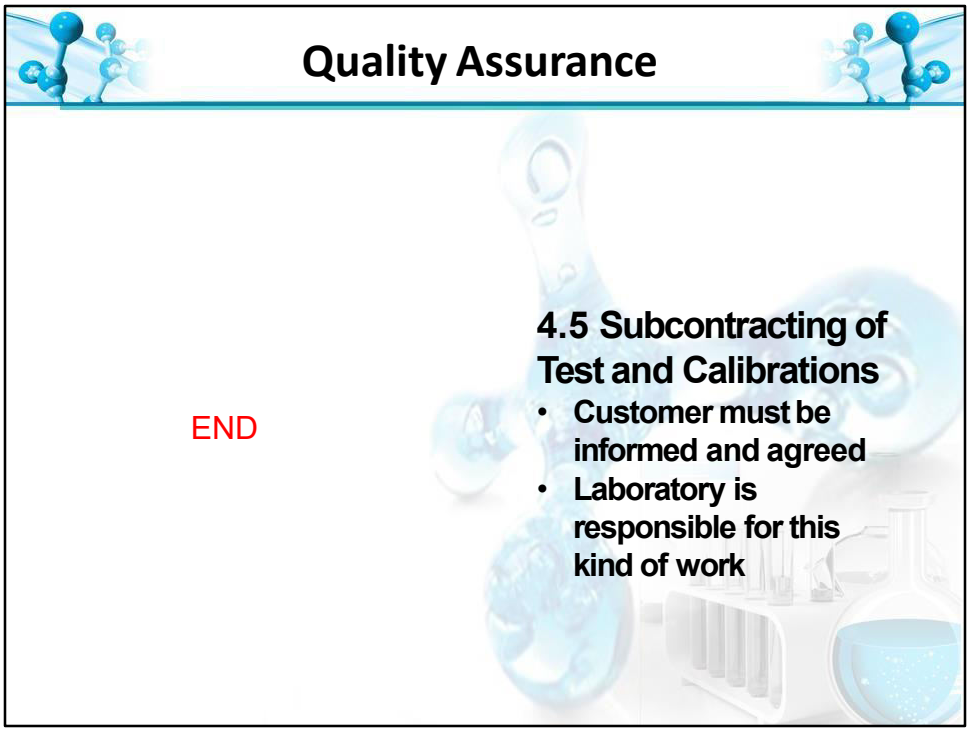
Benefits…..

Quality Assurance

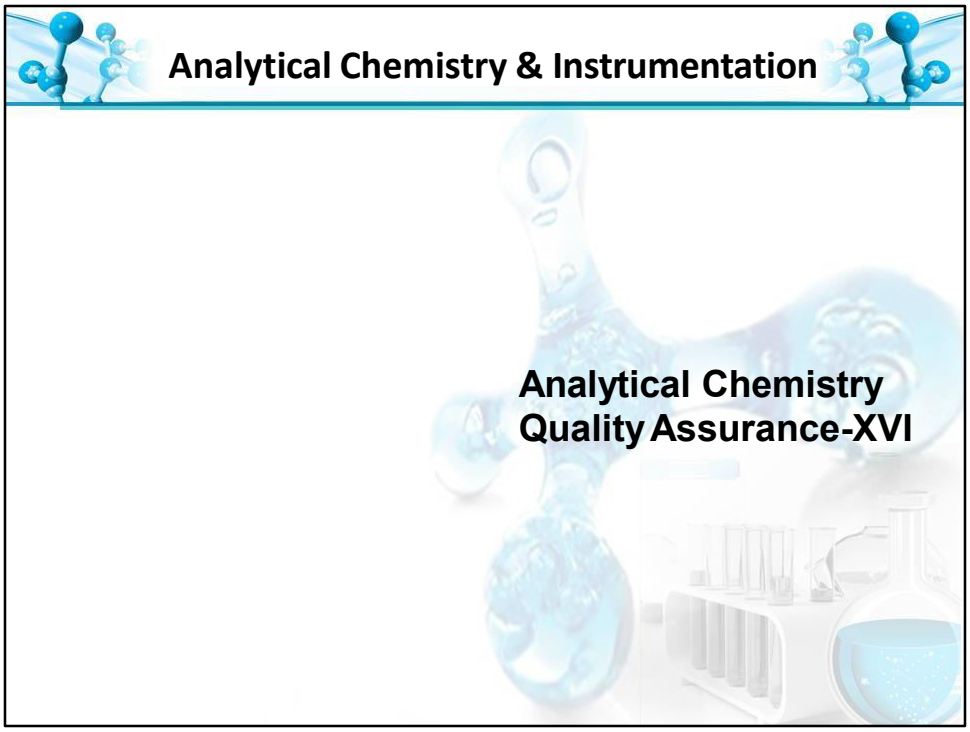
4.5 Subcontracting

- **Use of a competent subcontractor (could be one that complies with ISO/IEC 17025)**
- **Advise customer and obtain approval (in writing)**
- **Maintain a register of subcontractors**

Range is …….


Quality Assurance
END
4.5 Subcontracting of Test and Calibrations
• Customer must be informed and agreed
• Laboratory is responsible for this kind of work

Benefits…..



We will discuss next three clauses here, 4.6, 4.7 and 4.8



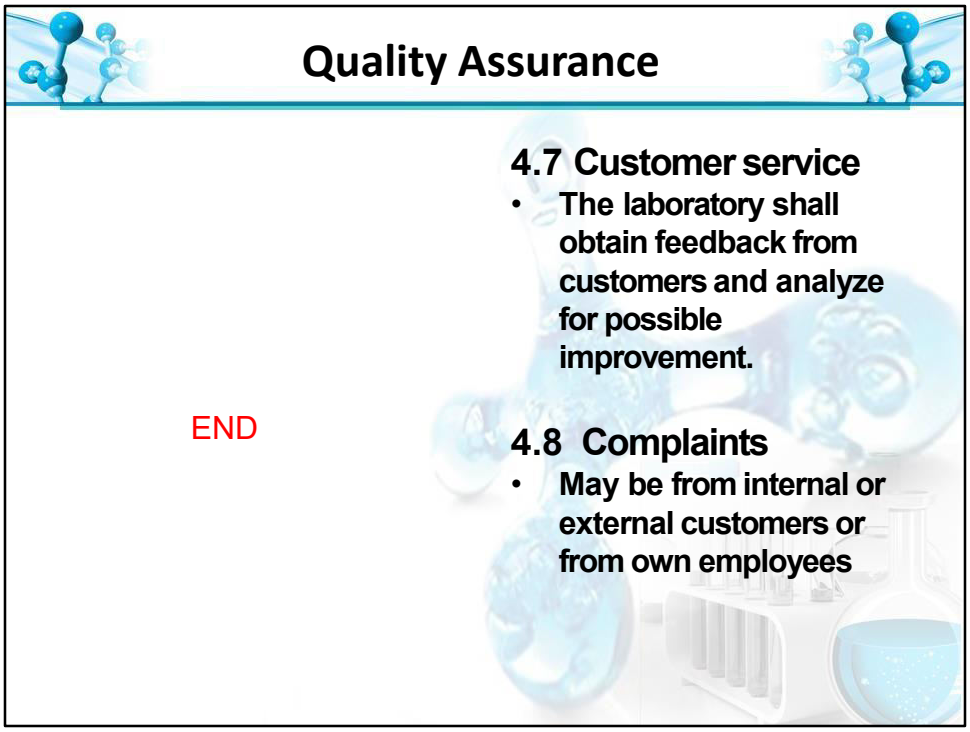
4.6 is purchasing of services and supplies

Quality Assurance

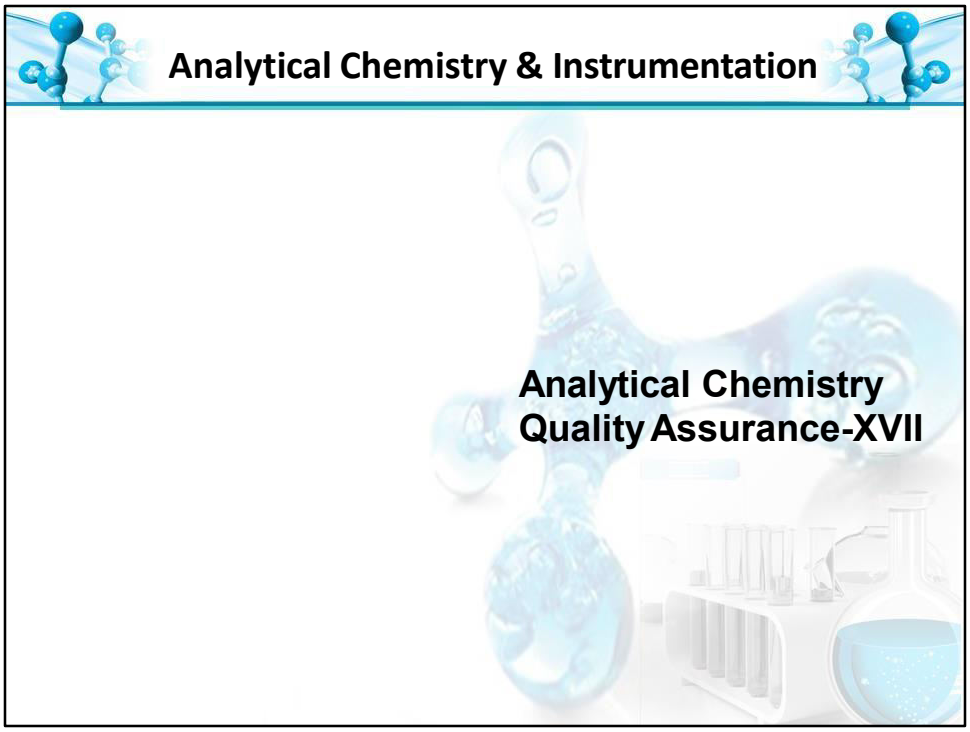
4.6 Purchasing of services and supplies

- **Must have procedure**
- **Supplier quality assurance (SQA)**
- **Maintain a register of suppliers**
- **Purchased items must be quarantine for inspection**
- **Properly segregated areas for accepted and rejected supplies**

Sub clauses…..



4.7 is customer service while 4.8 is complaints



Now we'll discuss clause 4.9 control of non-conforming testing work…..



What is non-conformance and what this procedure will cover..

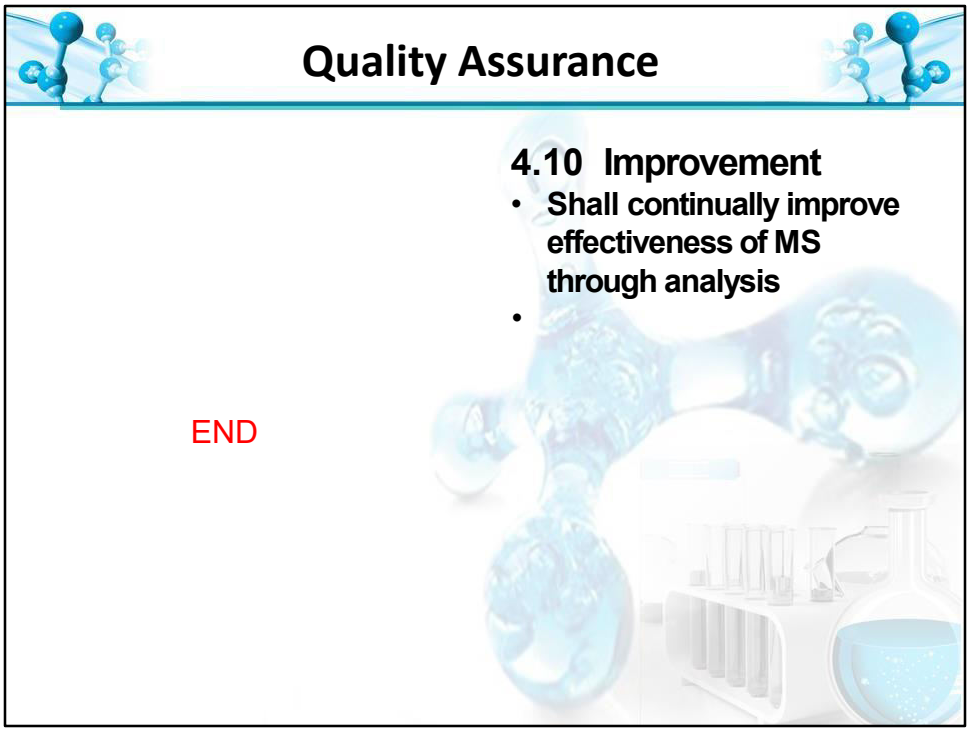
Quality Assurance

4.9 Control of nonconforming testing/calibration work

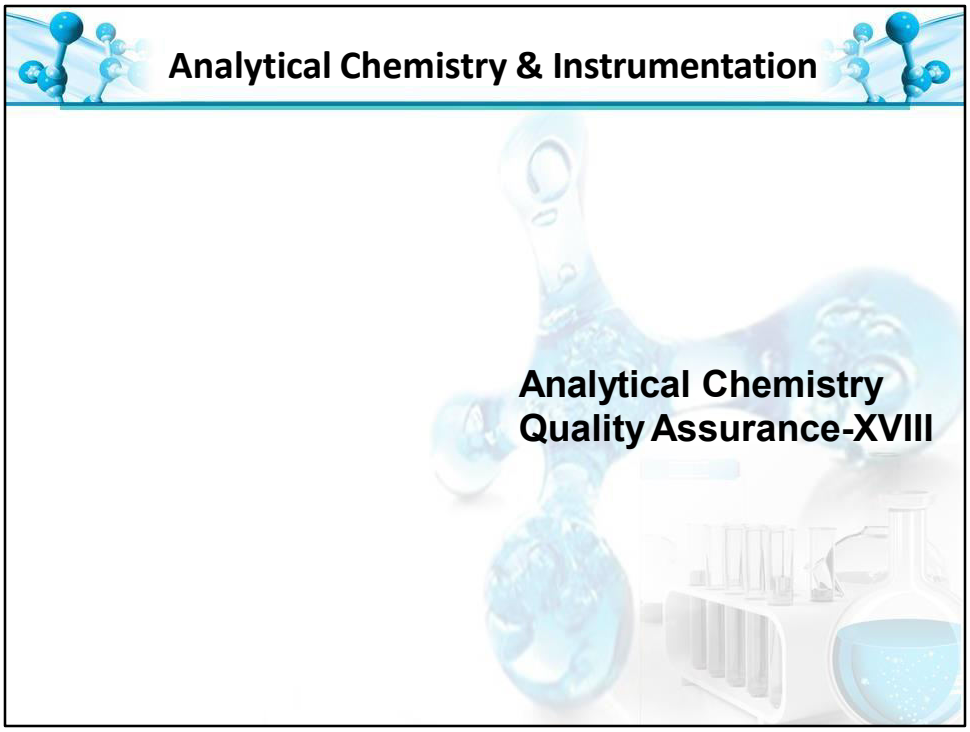
Policies – procedures implemented shall include

- **Responsibilities and authority**
- **Evaluation**
- **Correction taken "immediately". Where necessary the customer is notified and work is recalled**
- **Responsibility for authorizing resumption of work is defined**

Sub clauses…..



4.10 clause is improvement. ISO 9001 have concepts of quality circles, discussion forum, participation from workers to suggest improvement. Similarly incorporated in ISO 17025



Corrective and preventive action…..



One is reactive approach, while other is preventive approach

Quality Assurance

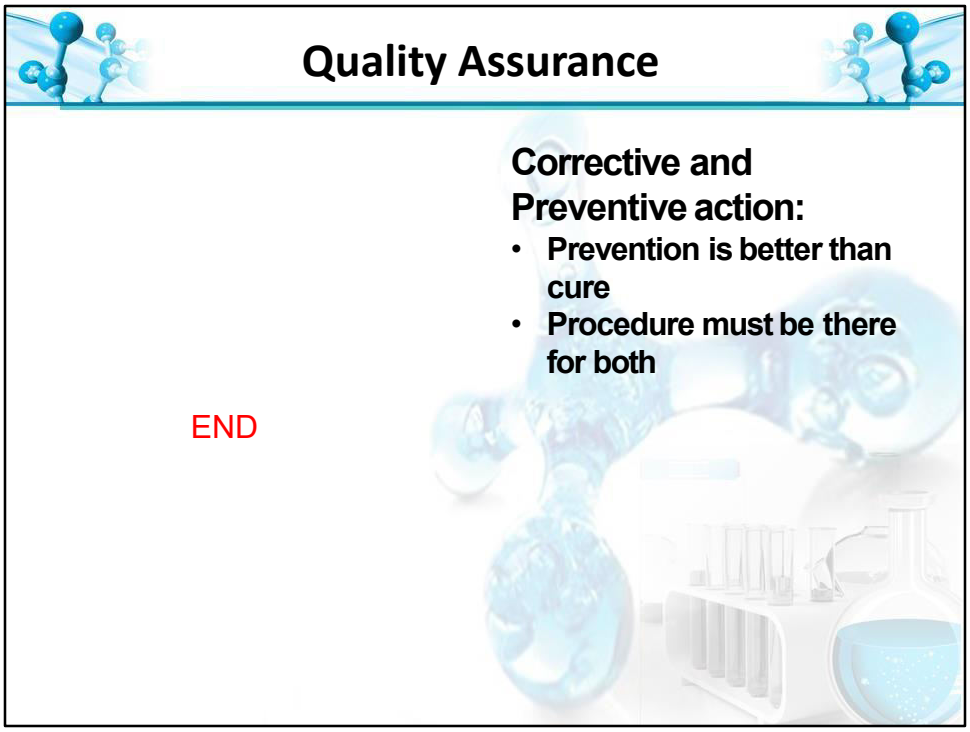
4.11 Corrective action

- **Shall designate and give authority for implementing Corrective Action**
- **Shall start with a cause analysis (analysis of all potential causes)**
- **Identify all potential corrective actions (select and implement)**
- **Shall monitor corrective actions effectiveness**

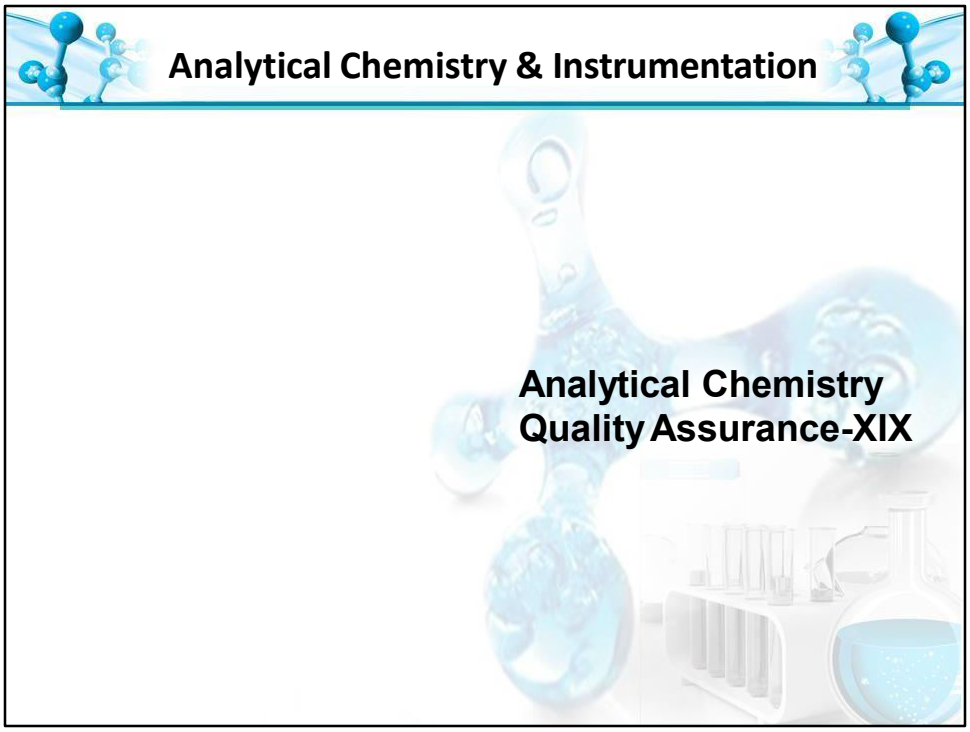
4.12 Preventative Actions

- **Needed improvements and potential sources of NC shall be identified**
- **Action plans developed when identified**

Sub clauses…


Quality Assurance
Corrective and Preventive action:
• Prevention is better than cure
• Procedure must be there for both
END

Benefits…..



4.13 clause is control of records..


Quality Assurance
4.13 Control of records:
• What are records?
• Readily retrievable
• Protection and back up needed

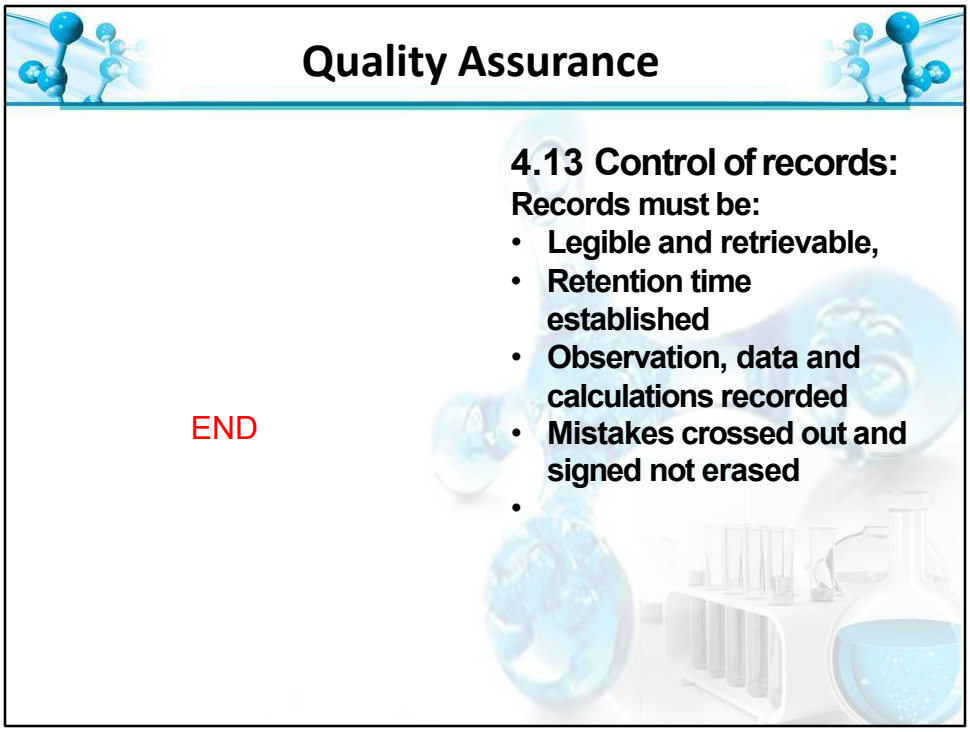
it means

Quality Assurance

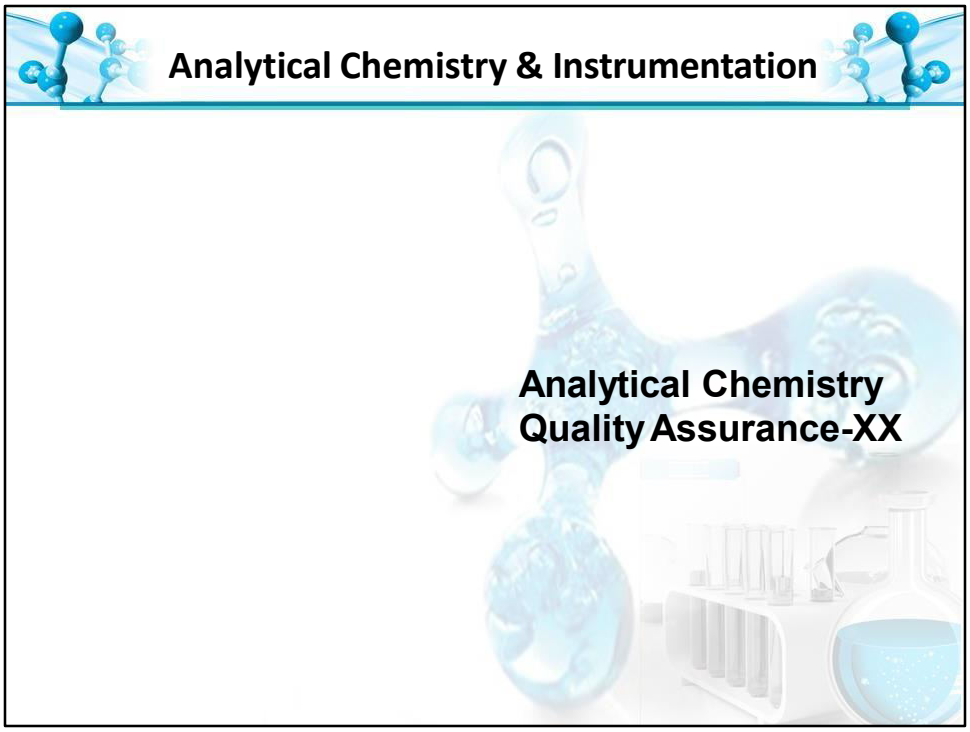
4.13 Control of records

- **Procedure for identification, storage, maintenance and disposal of records**
- **Quality records shall includes;**
  - **Test reports**
  - **Internal audit reports**
  - **Management reviews (meeting minutes)**
  - **Corrective and preventive action**
  - **Supplier Quality assurance**
  - **Sub contractors record**

Sub clauses…


Quality Assurance
4.13 Control of records:
Records must be:
• Legible and retrievable,
• Retention time established
• Observation, data and calculations recorded
• Mistakes crossed out and signed not erased
•
END

Must be …..



The last two management clauses are here..



Some basic definitions…

Quality Assurance

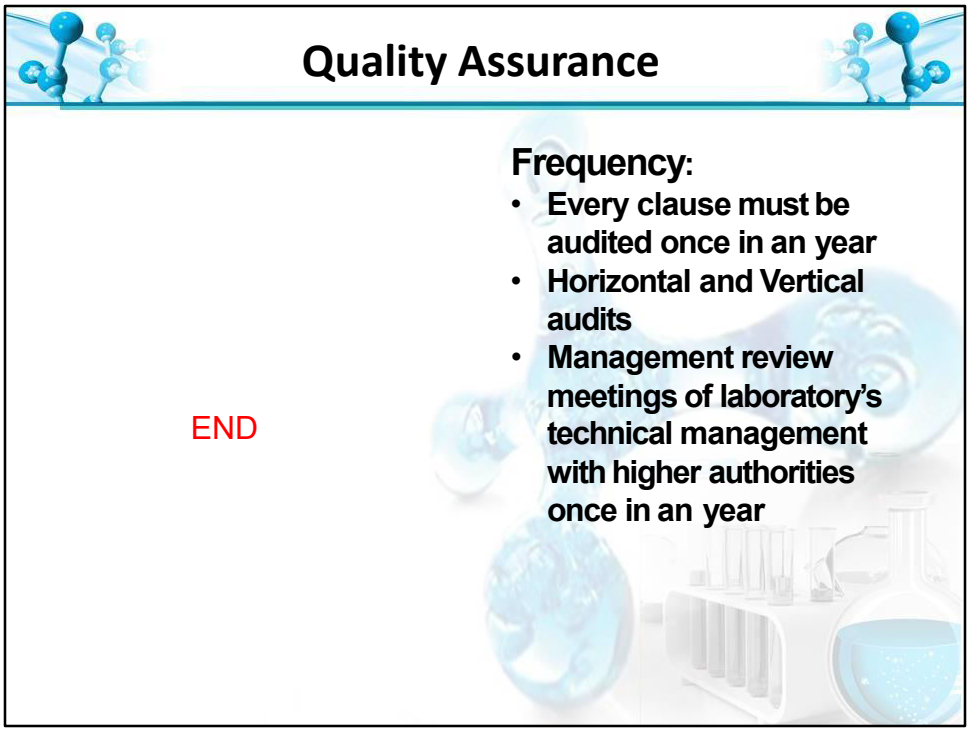
4.14 Internal audits

- **Done periodical – All elements of system annually**
- **Done by trained and qualified staff - by persons independent of activities to be audited**

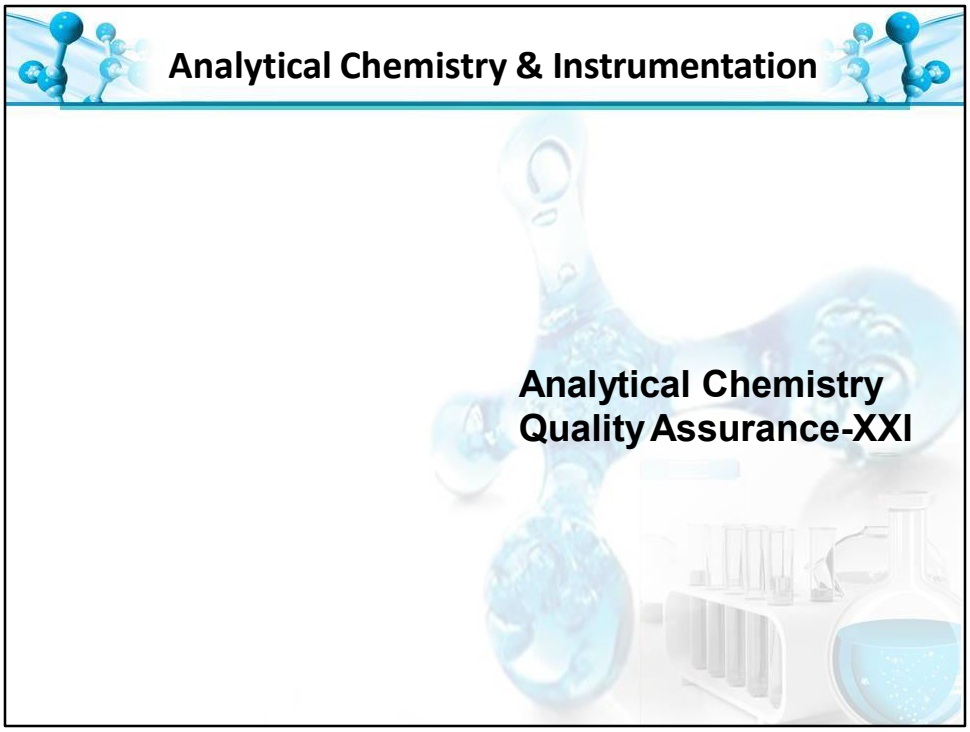
4.15 Management reviews

- **Suitability of policies and procedure**
- **Reports from laboratory personnel**
- **Outcome of Internal audit**
- **Assessment of external bodies**
- **Sub contractors and SQA record**
- **Results of ILC and PT**
- **Change in volume and type of work**
- **Customer complaints and feedback**

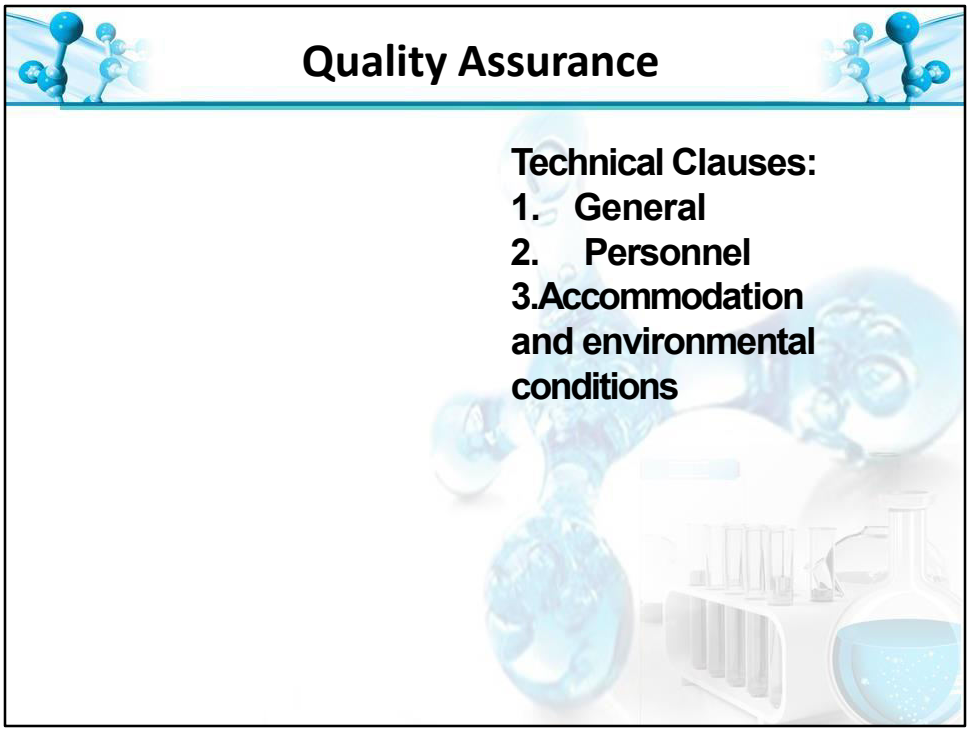
Sub clauses…



Frequency of audit and meeting



Technical clauses……



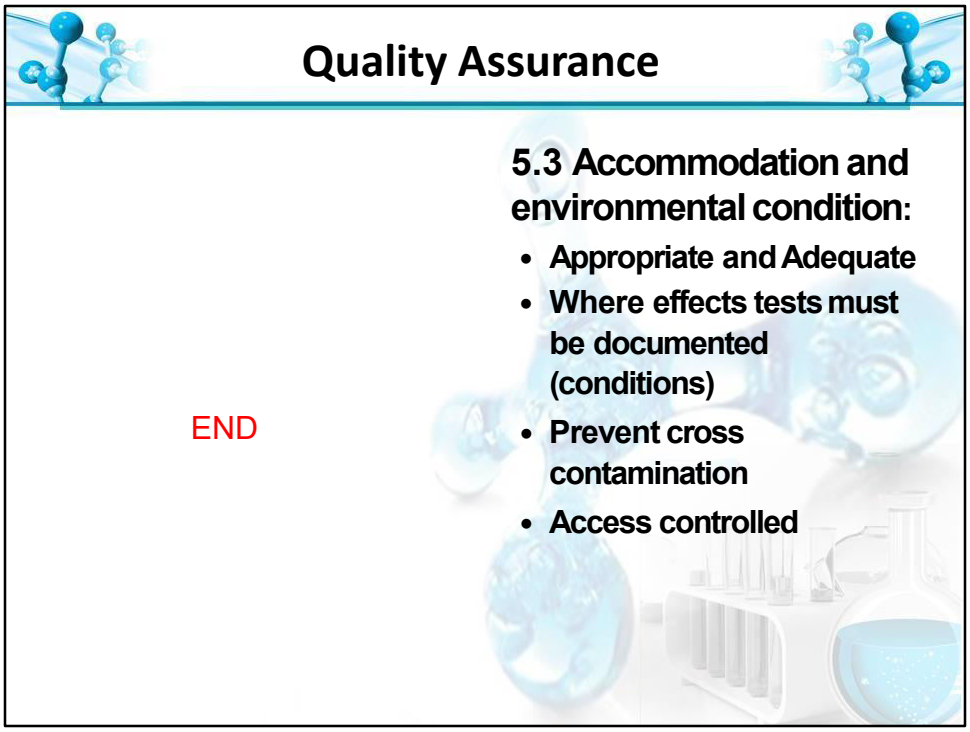
The first three clauses….

Quality Assurance

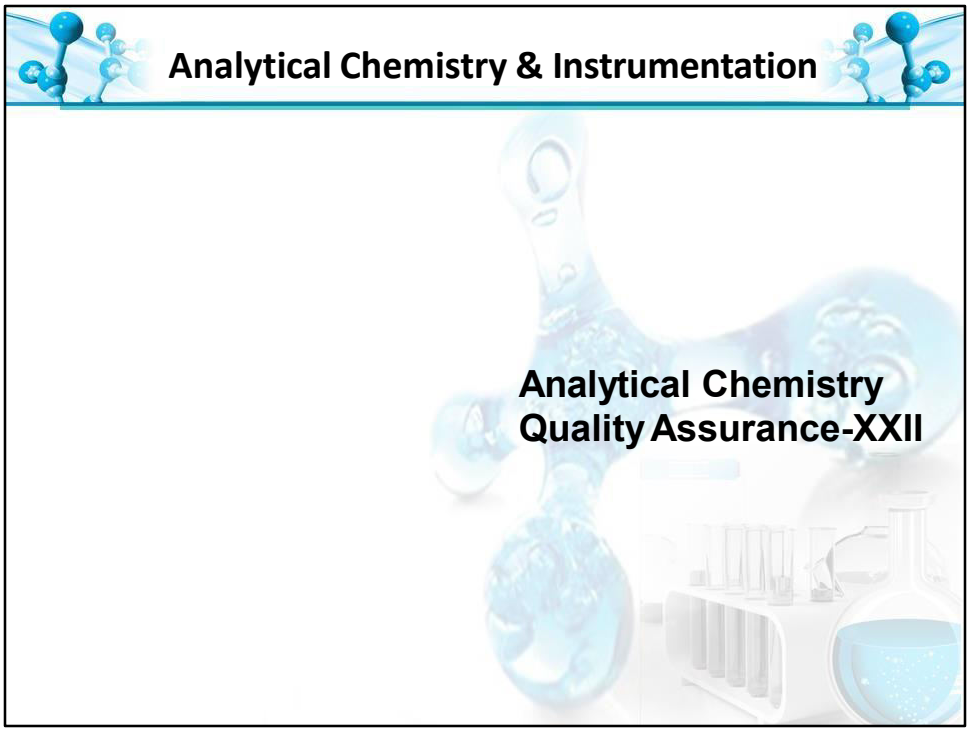
5.2 Personnel

- Competent qualified appropriately on basis of education, training, experience and demonstrated skills
- Management shall formulate the goals
- Persons who give opinions and interpretations of test results should have additional qualifications
- Policy to identify training needs
- Effectiveness of training shall be calculated
- Contractual worker properly supervised
- Maintain job description
- Authorize specific staff for specific work

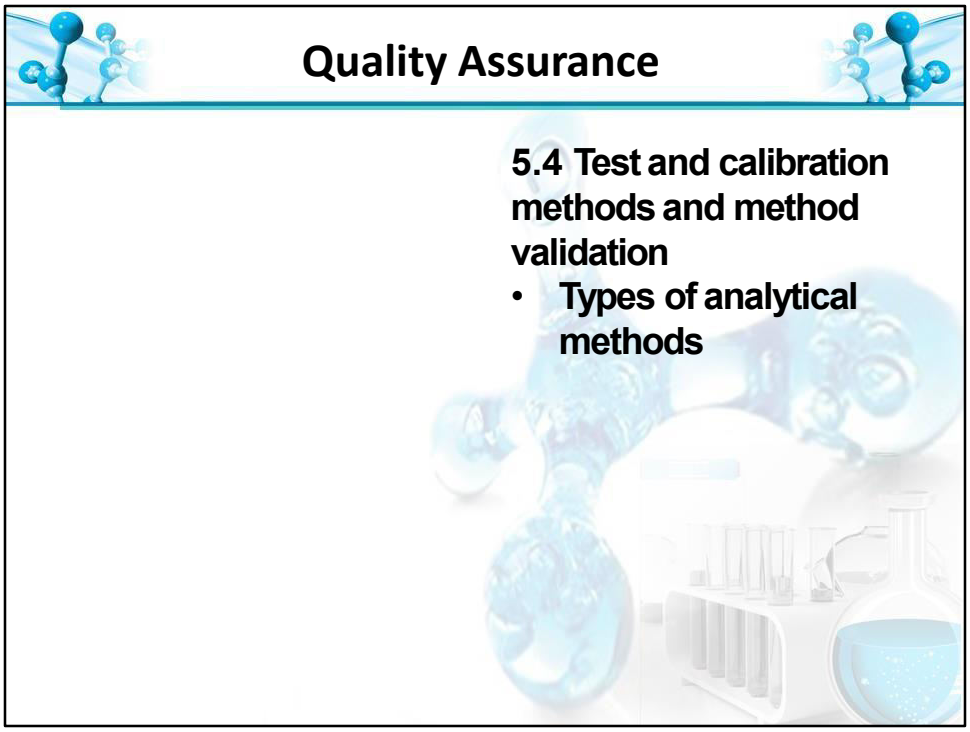
5.2 clause is….


Quality Assurance
5.3 Accommodation and environmental condition:
Appropriate and Adequate
Where effects tests must be documented (conditions)
Prevent cross contamination
Access controlled
END

5.3 clause is ….



5.4 clause is test and calibration methods….



Types of analytical methods

Quality Assurance

5.4 Test and calibration methods and …..

Validation of Methods:

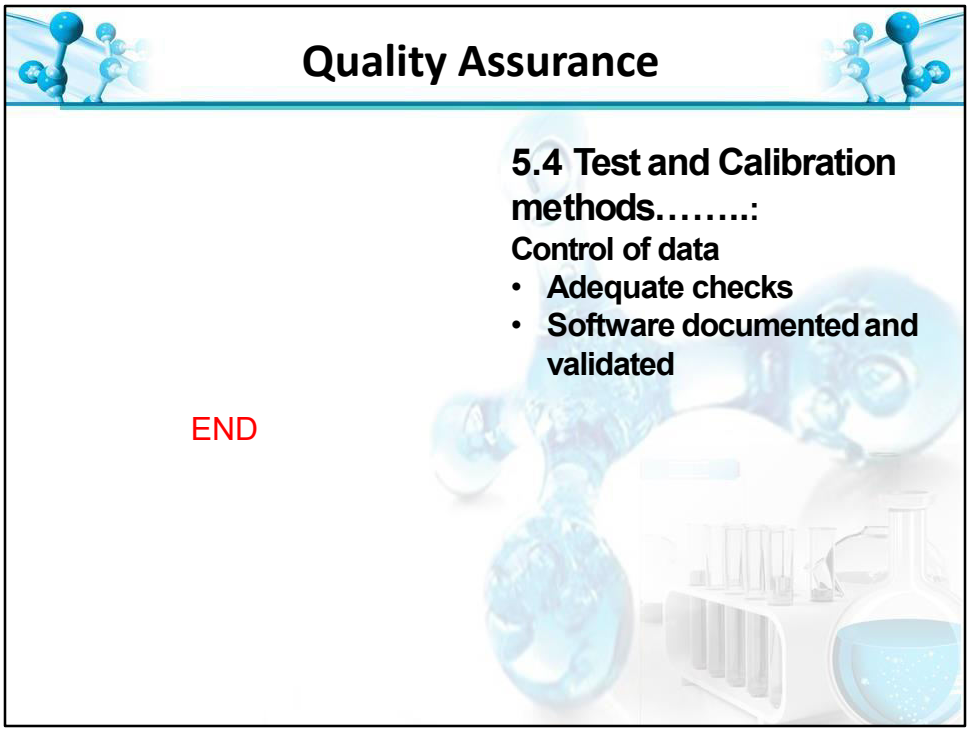
- **Validation requirements for a specific use**
- **Laboratories validate non standard methods, laboratories developed, standard methods used outside intended scope.**

•Record validation

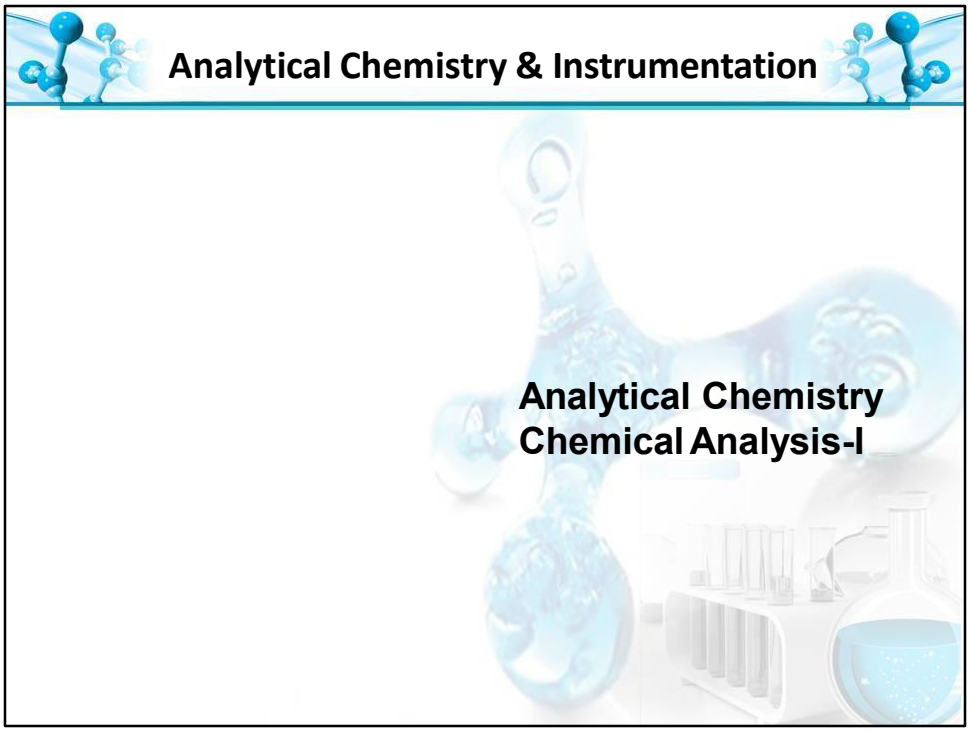
Estimation of Uncertainty

- **Procedure to estimate UOM**
- **Where methods includes and defines UOM then it is established**

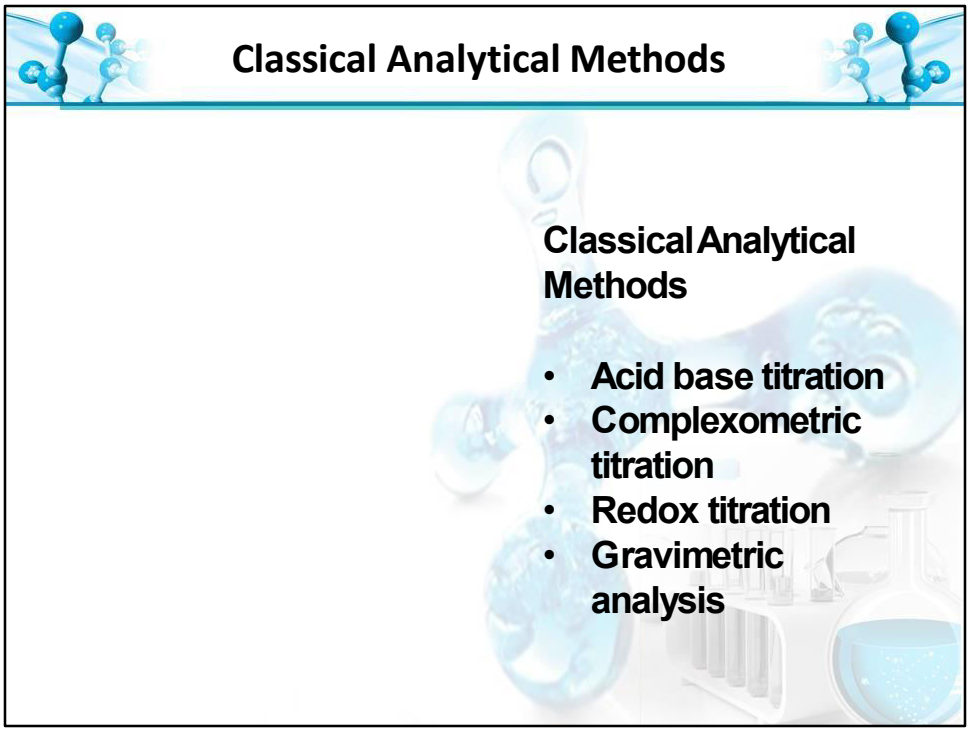
Sub clauses…..



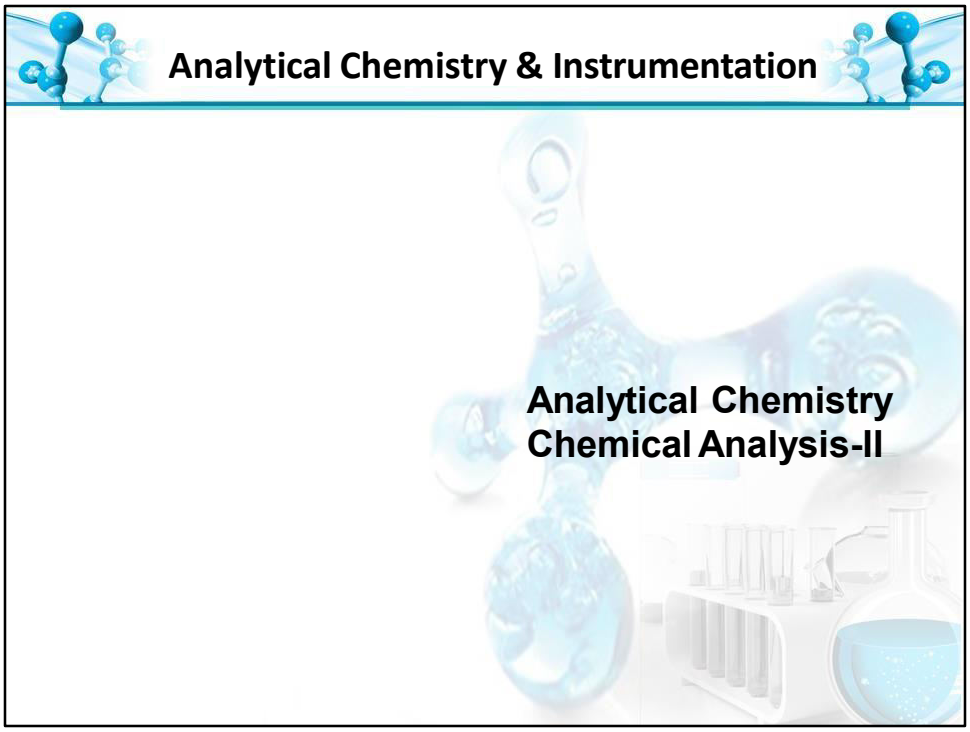
All data must be properly controlled..



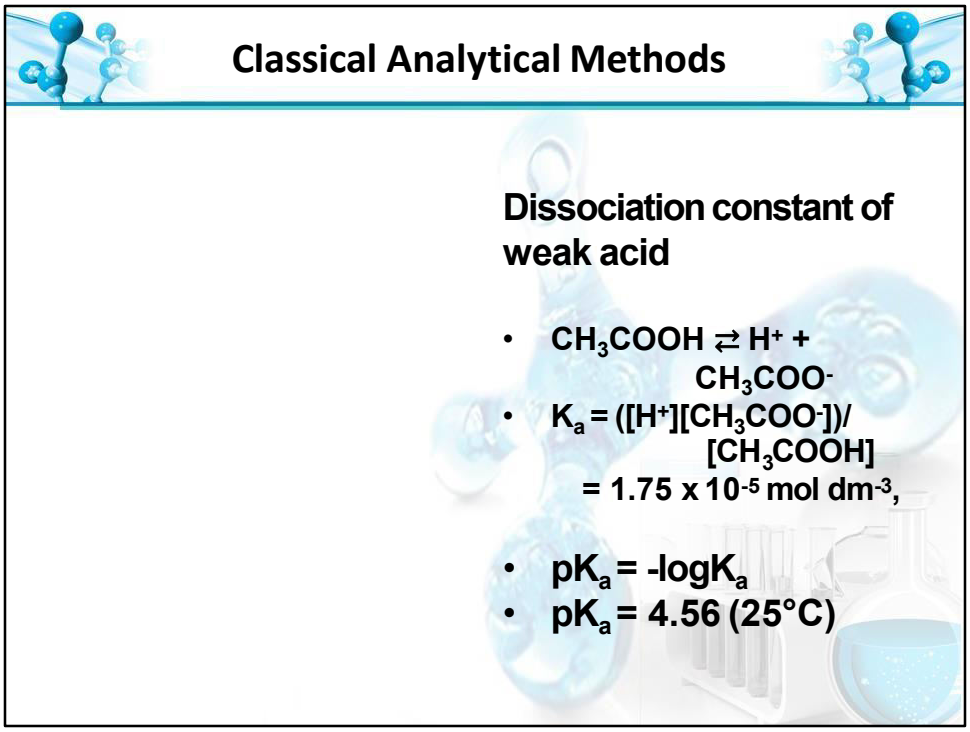
In this module we will discuss classical analytical techniques



Since ancient times these techniques are used, definitely with the time sophisticated and more accurate techniques replaced them but still they are most extensively sued techniques in analytical laboratories. They may also be considered as wet analysis techniques. They are …..



We will have some understanding about pKa value

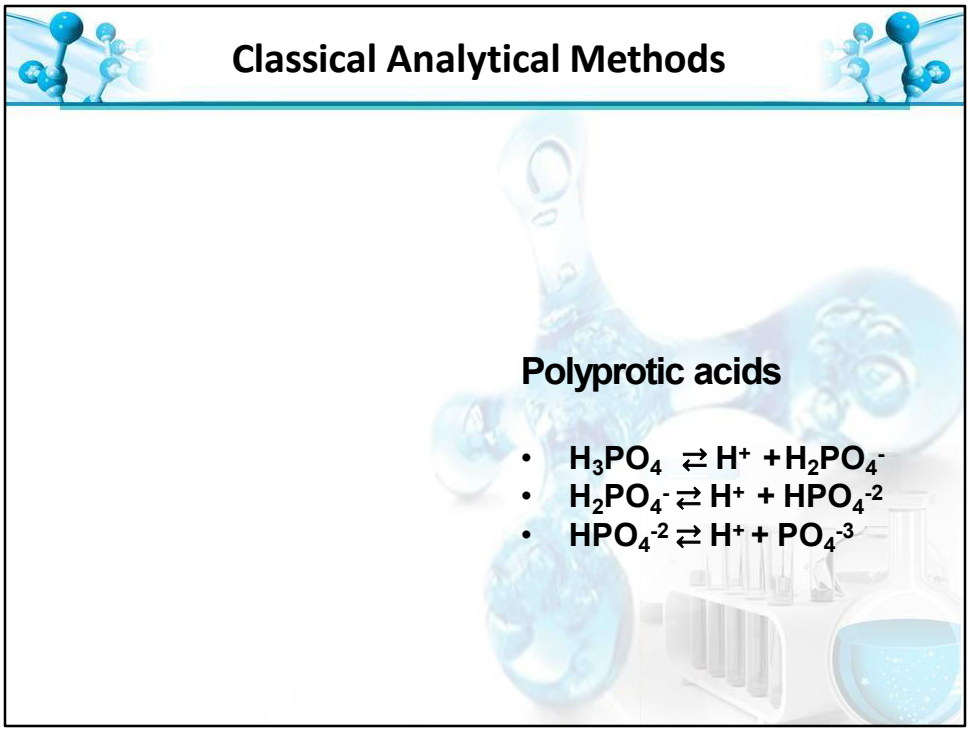


Dissociation of weak acid….

Classical Analytical Methods

Acids	Dissociation Constant	pK_a Dissociation Exponents
Formic acid (HCOOH)	1.77×10^{-4}	3.55
Acetic acid (CH_3COOH)	1.75×10^{-5}	4.56
Chloroacetic acid ($ClCH_2COOH$)	1.40×10^{-3}	2.68
Benzoic acid (C_6H_5COOH)	6.30×10^{-5}	4.20

Stronger and weak acids …….

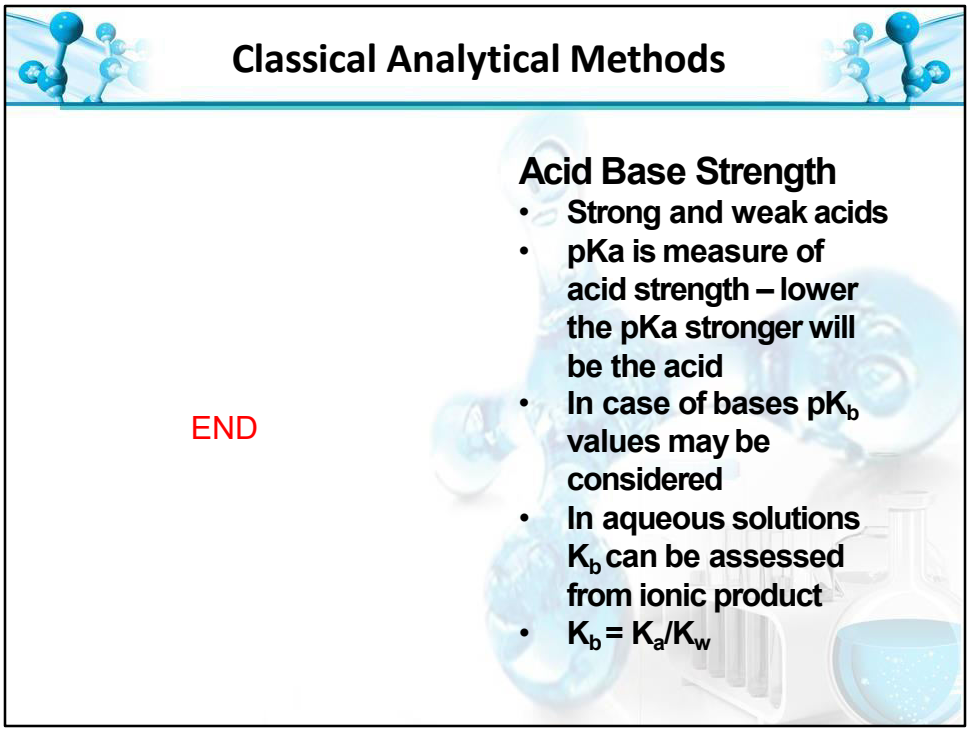


Polyprotic acid

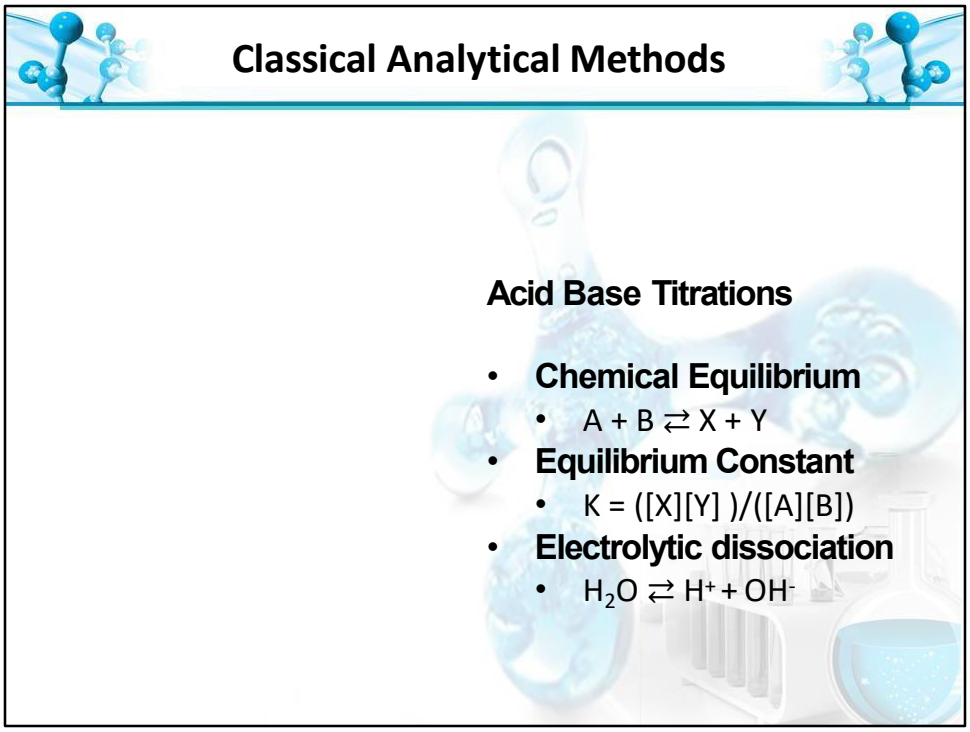
Classical Analytical Methods

Acids	Dissociation Constant	pK_a Dissociation Exponents
Carbonic acid (H_2CO_3)	$K_1 4.3 \times 10^{-7}$	6.35
	$K_2 5.6 \times 10^{-11}$	10.33
Hydrogen sulfide (H_2S)	$K_1 5.7 \times 10^{-8}$	7.02
	$K_2 1.2 \times 10^{-15}$	13.9
Phosphoric acid (H_3PO_4)	$K_1 7.5 \times 10^{-6}$	2.15
	$K_2 6.2 \times 10^{-8}$	7.20
	$K_3 4.8 \times 10^{-13}$	12.35

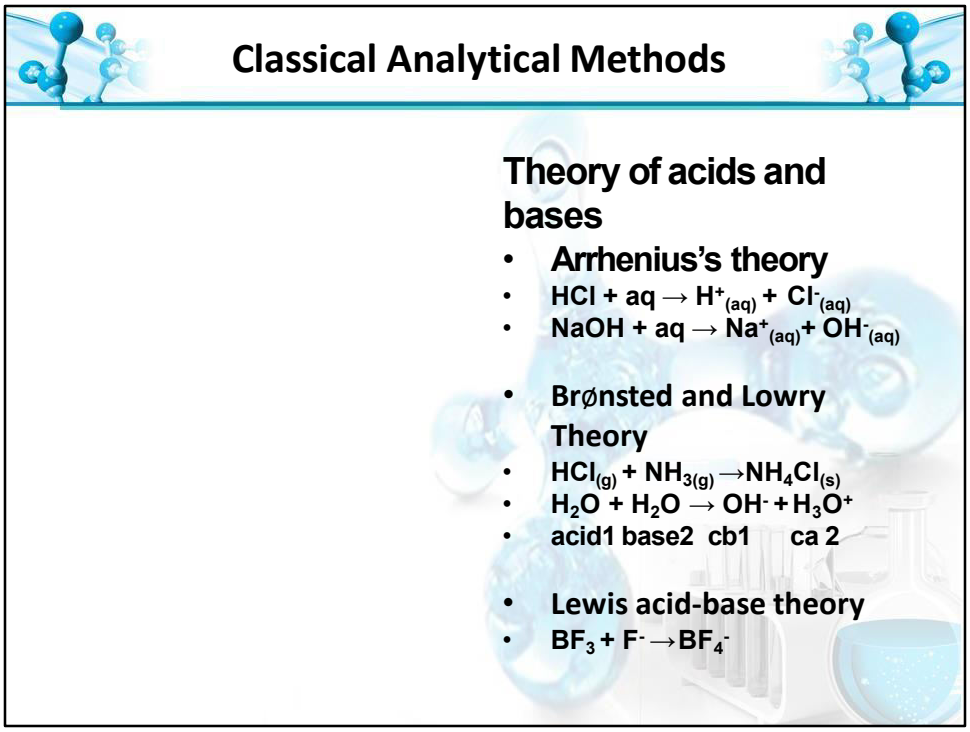
Dissociation constant of polyprotic acids



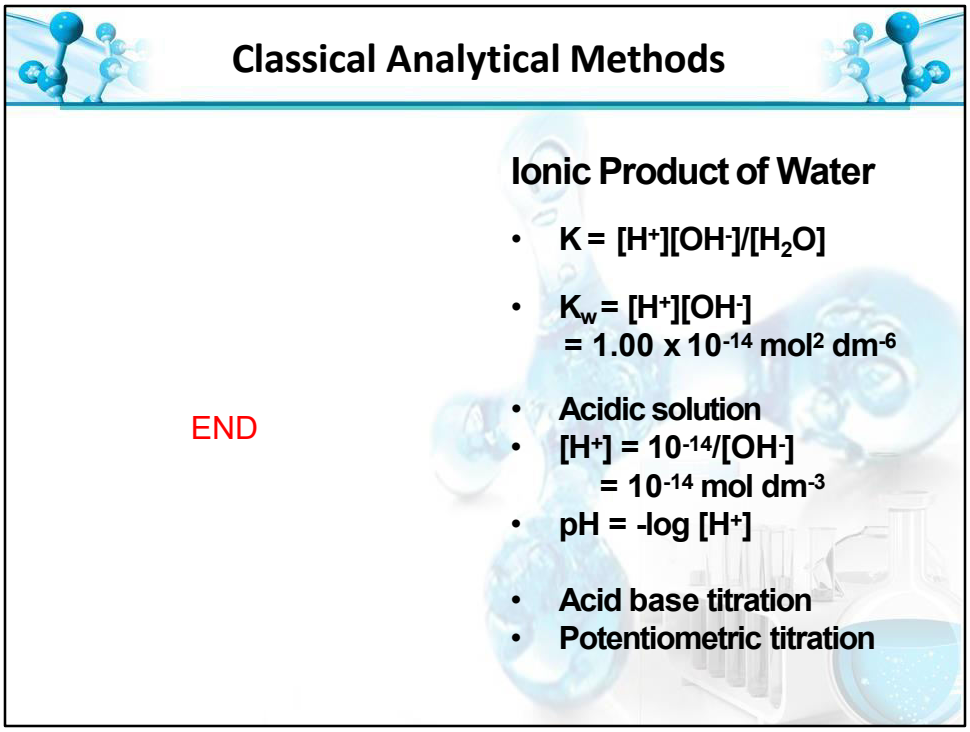
Strong and weak acids.



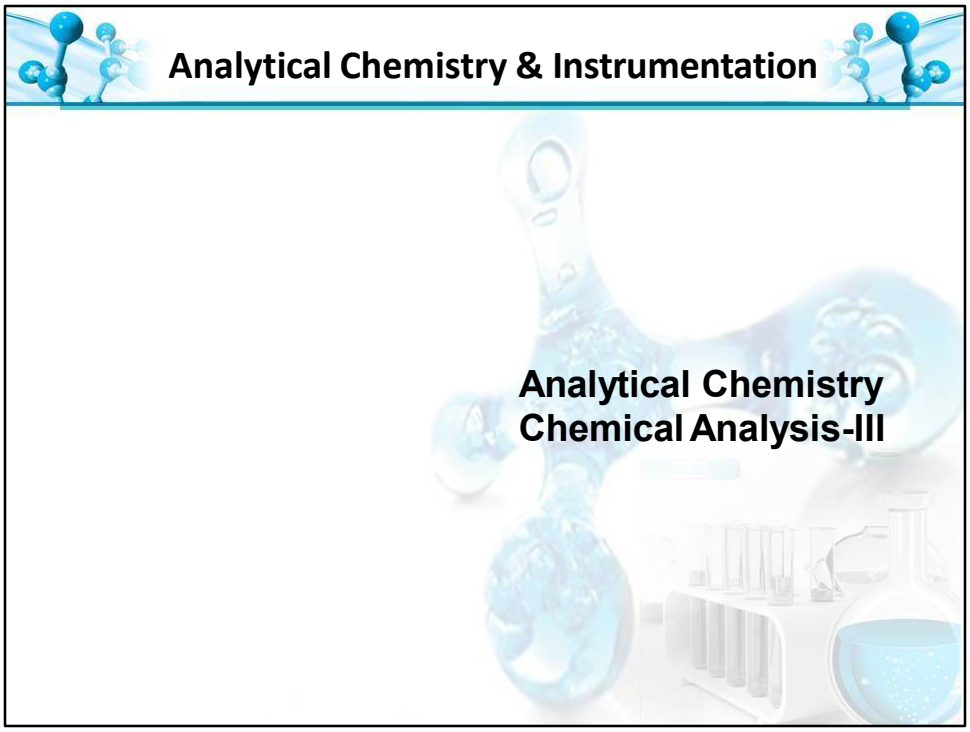
In a reversible reaction the reactants come into equilibrium with product with the time.
But in electric dissociations salts are ionized within no time, same is true for acid base reaction or salt formation.
However water dissociation is reversible reaction.



There are three theories of acids and bases….



Dissociation constant of water is written here.
Ionic product of water may be written as …..



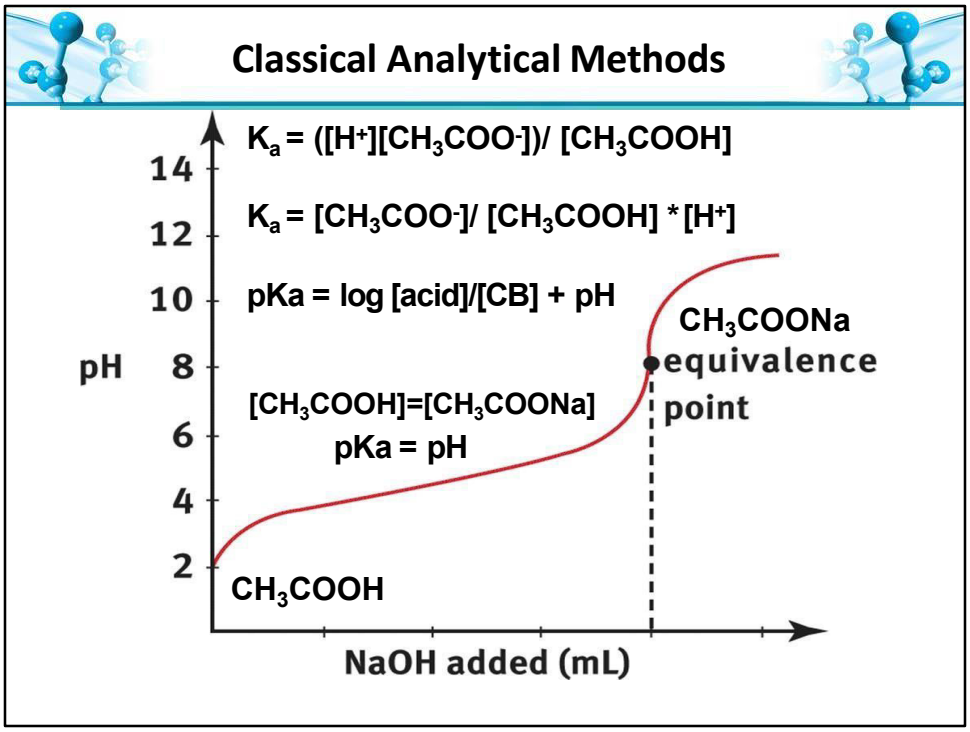
Titration curve are discussed here

Classical Analytical Methods

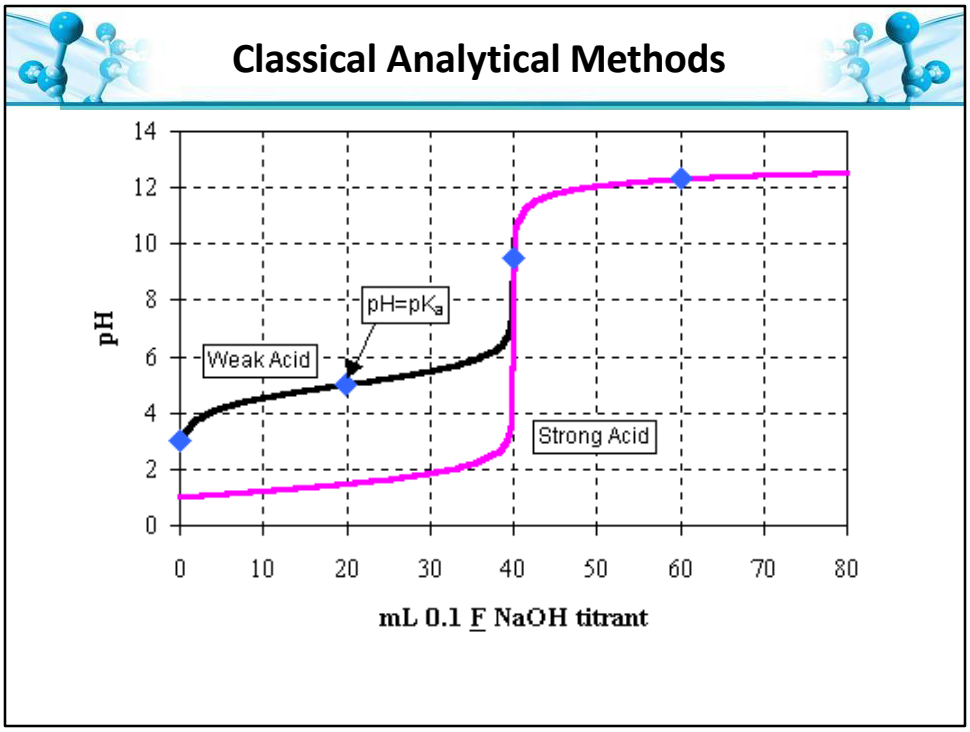
Acid base titration curve (Potentiometric titrations)

- **Consider titration of weak acid with a strong base**
- **$CH_3COOH \rightleftarrows H^+ + CH_3COO^-$**
- **$NaOH \rightarrow OH^- + Na^+$,**
- **Titration curve**

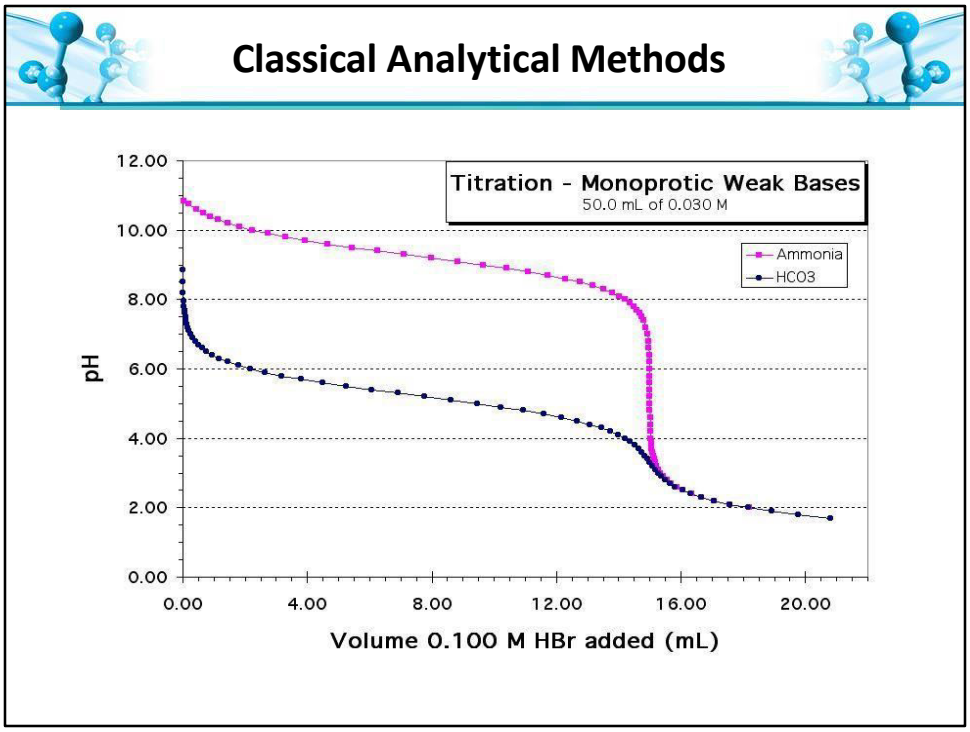
Potentiometric titration is a way to determine pKa values……



Titration curve of weak acid and strong base….



Titration curve of strong and weak acid



Titration of weak base with strong acid …..

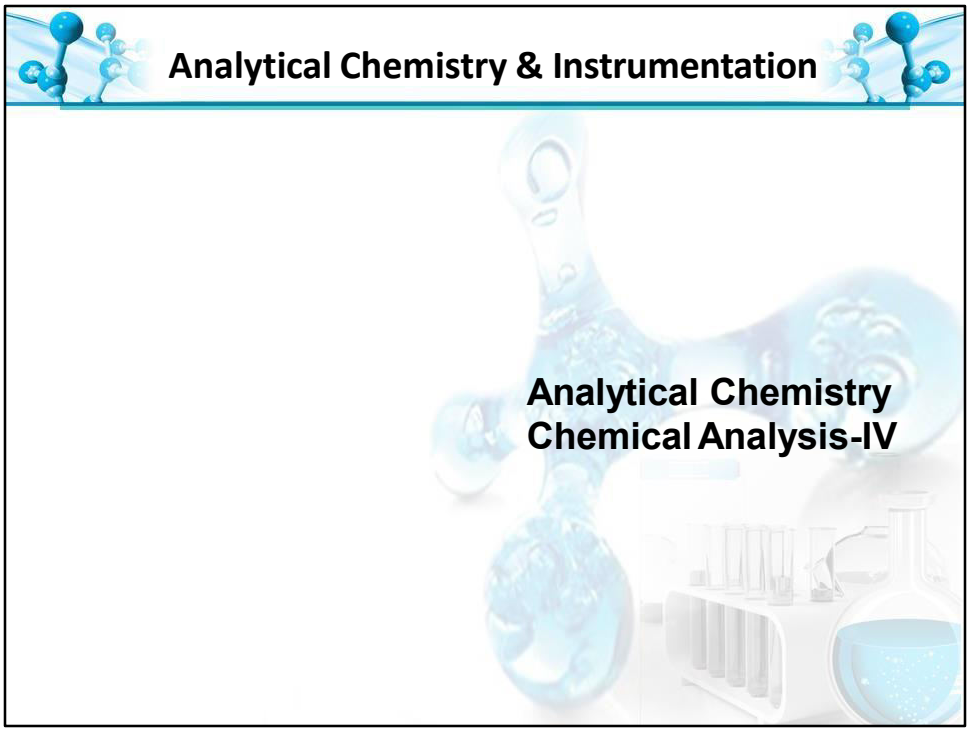
Classical Analytical Methods

END

Buffers

- $pH = pK_a + \log([salt]/[acid])$
- Phthalic acid + Potassium Phthalate (2.2 - 3.8)
- $CH_3COOH + CH_3COONa$ (3.7 - 5.6)
- $NaH_2PO_4 + Na_2HPO_4$ (5.8 - 8.0)
- $H_3BO_3 + Na_2B_4O_7$ (6.8 - 9.2)

Calculate pH of buffer

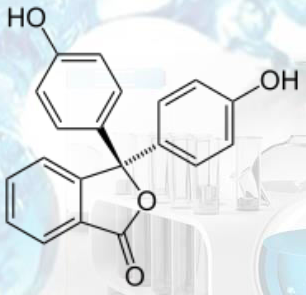


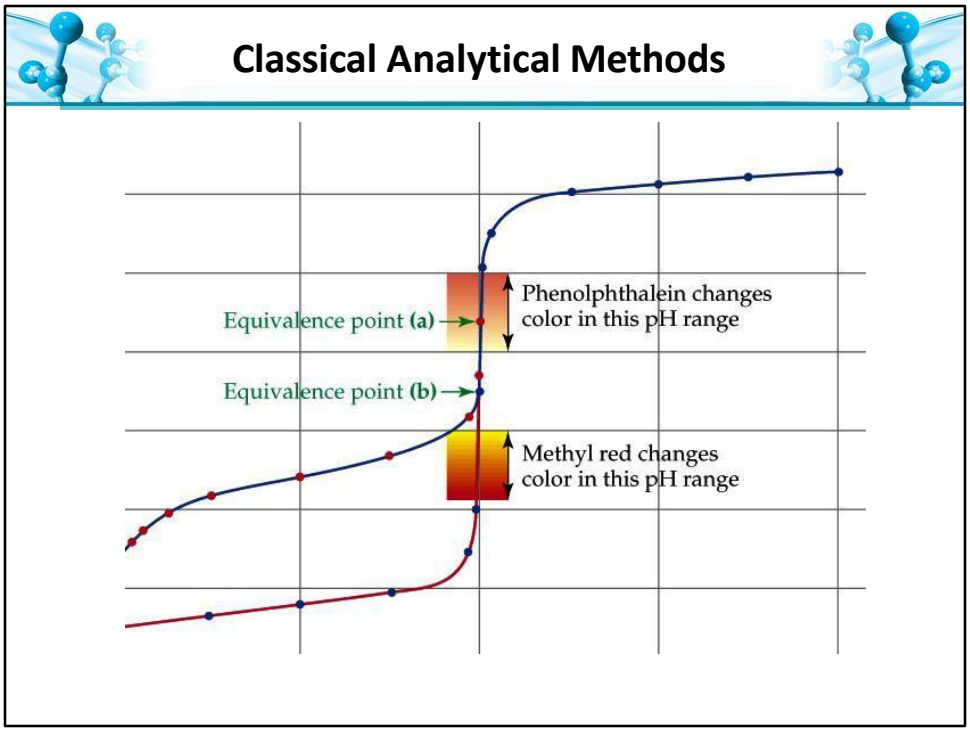
Chemistry of indicators

Classical Analytical Methods

Chemistry of indicators

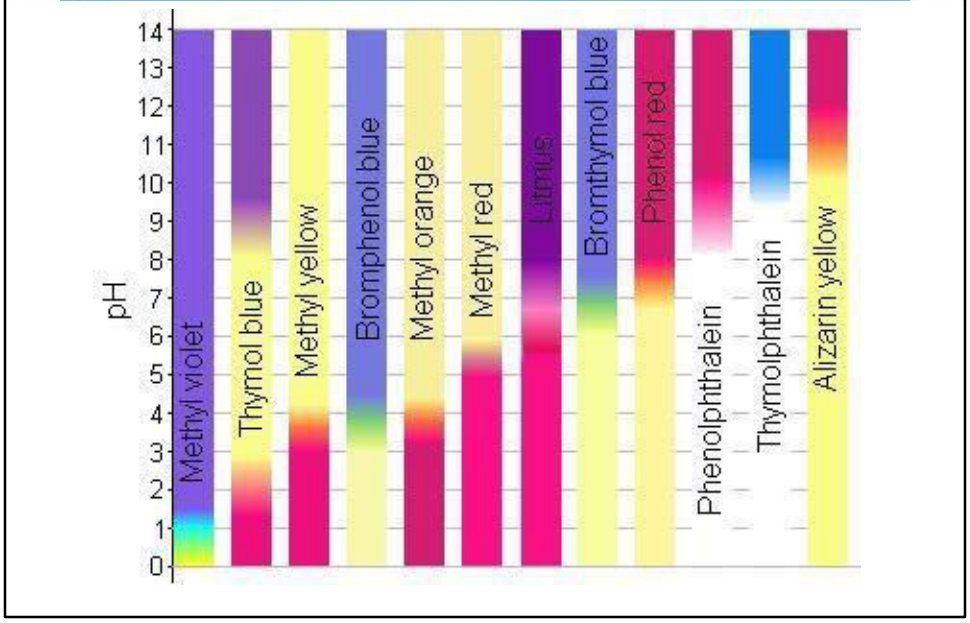
- $HIn \rightleftarrows H^+ + In^-$
- $K_I = [H^+][In^-]/[HIn]$
- Phenolphthalein





Colour change of buffer

Classical Analytical Methods

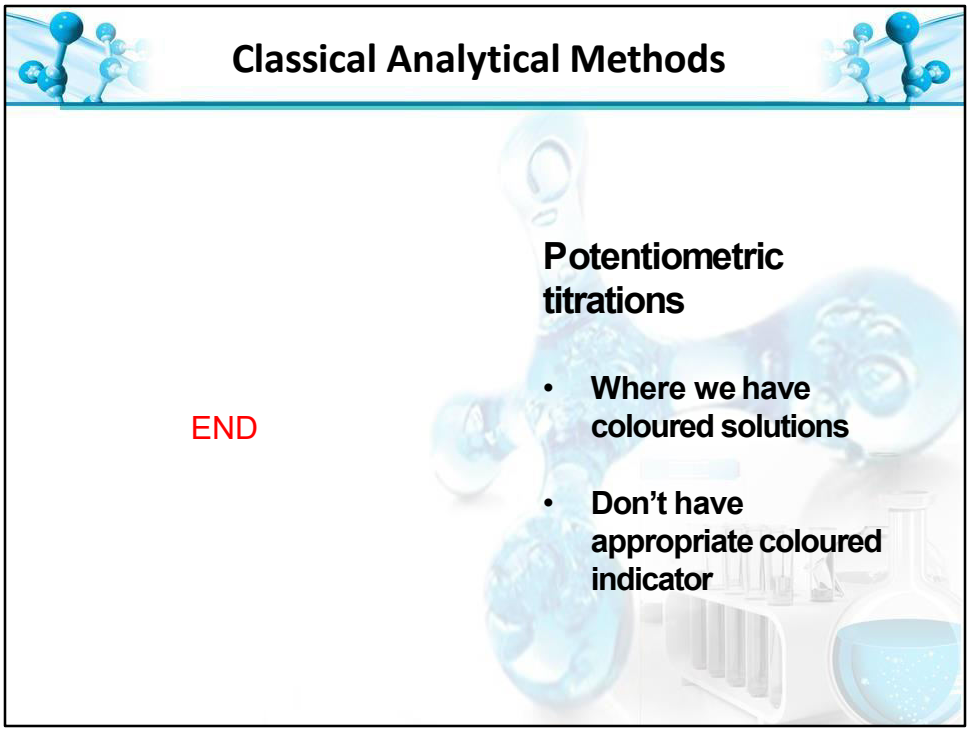


Colour change of indicators ….

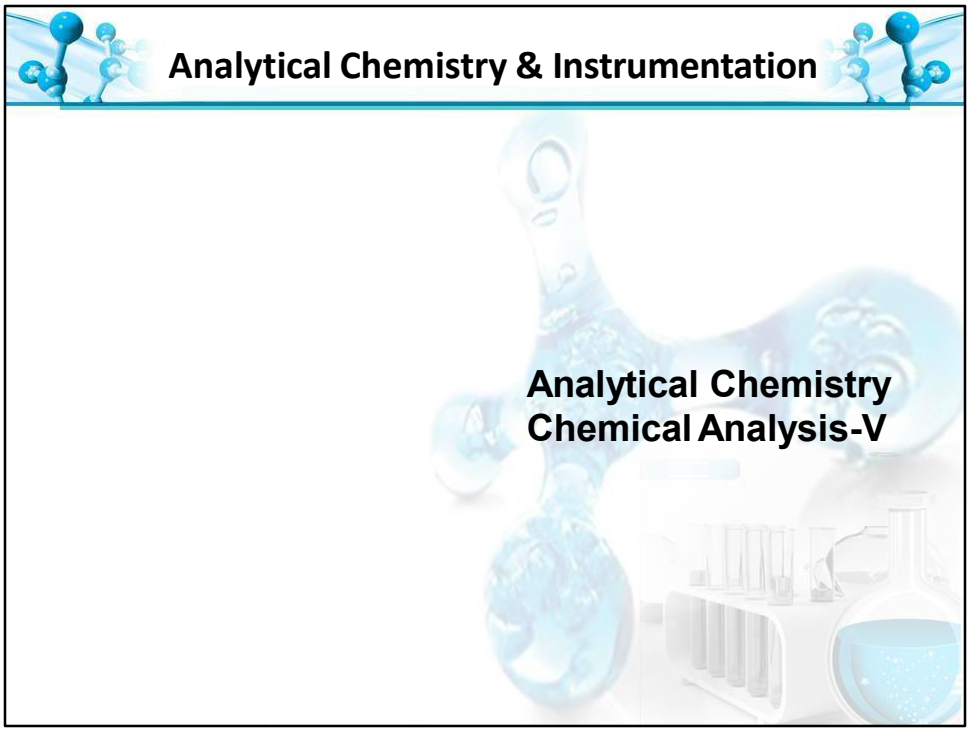
Classical Analytical Methods

Indicator	pH range	Colour change Acid → Base
thymol blue	1.2-2.8	red → yellow
methyl orange	3.1-4.4	red → yellow
methyl red	4.2-6.3	red → yellow
bromothymol blue	6.0-7.6	yellow → blue
cresol red	7.2-8.8	yellow → red
phenolphthalein	8.3-10.0	colorless → red
alizarin yellow	10.2-12.0	yellow → red

pH and colour change



Where to use potentiometric titrations…



Redox titrations….

Classical Analytical Methods

Redox reactions

- $2\ Zn(s) + O_2(g) \rightarrow 2\ ZnO(s)$

- $Zn \rightarrow Zn^{2+} + 2e^-$ or
- $2\ Zn \rightarrow 2\ Zn^{2+} + 4\ e^-$
- (Oxidation rex; Zn is reducing agent)

- $O + 2e^- \rightarrow O^{2-}$ or
- $O_2 + 4e^- \rightarrow 2\ O^{2-}$
- (Reduction rex)

What is reduction oxidation ….

Classical Analytical Methods

Assignment of Oxidation numbers:

- The oxidation number for each atom in a pure element is zero.
- The oxidation number of a monatomic ion is equal to its charge.
- When fluorine atoms are combined with atoms of other elements, their oxidation number is –1.
- When oxygen atoms are combined with atoms of other elements, their oxidation number is –2, with exception like in peroxides, H_2O_2, where their oxidation number is –1.
- The oxidation number for each hydrogen atom in a molecular compound or a polyatomic ion is +1.

Assignment of oxidation number …

Classical Analytical Methods

Which one is redox reaction?
(ammonium phosphate fertilizer manufacturing)

$$2\ Ca_3(PO_4)_2 + 6\ SiO_2 + 10\ C \rightarrow P_4 + 10\ CO + 6\ CaSiO_3$$

$$P_4 + 5\ O_2 + 6\ H_2O \rightarrow 4\ H_3PO_4$$

$$H_3PO_4 + NH_3 \rightarrow (NH_4)_3PO_4$$

Calculate oxidation number …

Classical Analytical Methods

Iodometric titrations (See reduction Oxidation)

$$2Cu^{+2} + 4I^{-} \rightarrow 2\,CuI + I_2$$

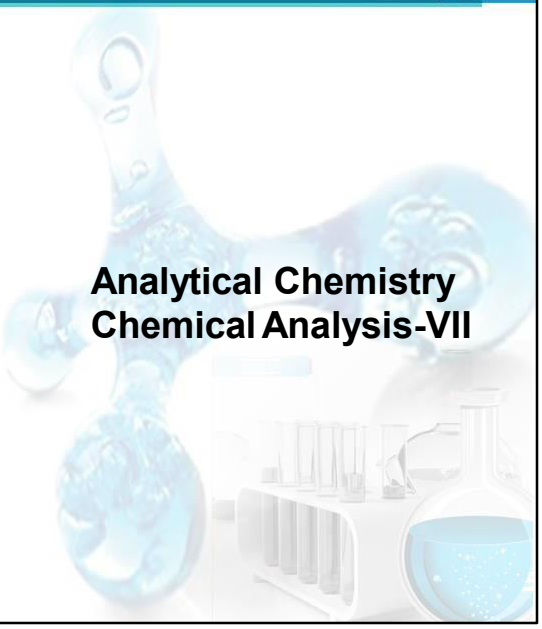
$$I_2 + 2S_2O_3^{-2} \rightarrow I^{-} + S_4O_6^{-2}$$

$$SO_3 + I_2 + H_2O \rightarrow SO_4^{-2} + 2I^{-} + 2H^{+}$$

END

Chemistry of Indicators

- Internal indicators
  - $K_2Cr_2O_7$
  - $KMnO_4$
  - I_2
- Change of colour due to oxidation or reduction



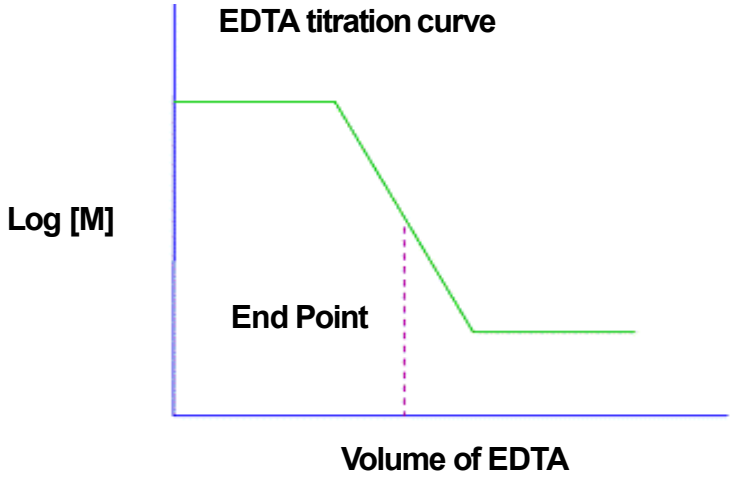
Analytical Chemistry
Chemical Analysis-VII

Classical Analytical Methods

Indicators used in Complexometric titrations

- **Indicators are ligands too**
- **Stability of Complexes**
- **pH of reaction**

Classical Analytical Methods



Classical Analytical Methods

Indicators used in Complexometric Titration

S.No.	Name of the Indicator	Colour change	pH range	Metals detected
1.	Mordant black II	Red to Blue	6-7	Ca, Ba, Mg, Zn, Cd, Mn, Pb, Hg
	Eriochrome blackT			
	Solochrome blackT			
2.	Murexide or Ammonium purpurate	Violet to Blue	12	Ca, Cu, Co
3.	Catechol-violet	Violet to Red	8-10	Mn, Mg, Fe, Co, Pb
4.	Methyl Blue	Blue to Yellow	4-5	Pb, Zn, Cd, Hg
	Thymol Blue	Blue to Grey	10-12	
5.	Alizarin	Red to Yellow	4.3	Pb, Zn, Co, Mg, Cu
6.	Sodium Alizarin sulphonate	Blue to Red	4	Al, Thorium
7.	Xylenol range	Lemon to Yellow	1-3	Bi, Thorium
			4-5	Pb, Zn
			5-6	Cd, Hg

Types of Complexometric titrations:

1.Direct Titration: standard chelon solution is added to the metal ion solution until the end point is detected e.g.-calcium gluconate injection, calcium lactate tablets. Interference observed due to presence of other ions.

2.Back Titration: Excess of EDTA standard is back titrated with a standard solution of a second metal ion. e.g. Mn cannot be assessed directly because of precipitation of $Mn(OH)_2$.

3.Replacement Titration: Where the metal may be determined by the displacement of an equivalent amount of Mg or Zn from a less stable complex.

$$Mn^{+2} + Mg\text{-}EDTA^{-2} \longrightarrow Mg^{+2} + Mn\text{-}EDTA^{-2}$$

Mn displaces Mg from Mn-EDTA solution. The freed Mg metal is then directly titrated with a standard EDTA solution.

Types of Complexometric titrations:

4. Indirect Titration: This is also known as Alkalimetric titration. It is used for the determination of ions such as anions, which do not react with EDTA chelate. Protons from disodium EDTA are displaced by a heavy metal and titrated with sodium alkali.

$$M^{+n} + H_2X^{-2} \rightarrow MX^{(n-4)} + 2H^{+}$$

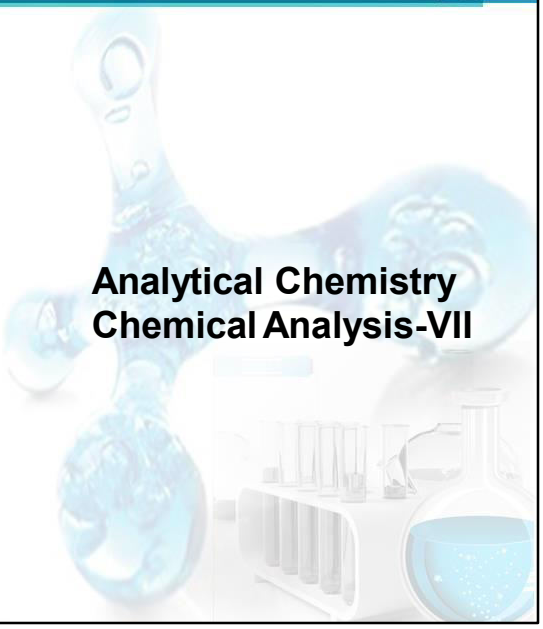
e.g. - Barbiturates do not react with EDTA but are quantitatively precipitated from alkaline solution by mercuric ions as 1:1 complex.

Classical Analytical Methods

END

Indirect titration

- Barbiturate to be analyzed is taken in a flask and heated with excess of mercury in alkaline solution.
- Hg-barbiturate complex is ppt, filtered and dissolved in excess of standard EDTA solution.
- Excess EDTA is back titrated with a standard Zn solution.



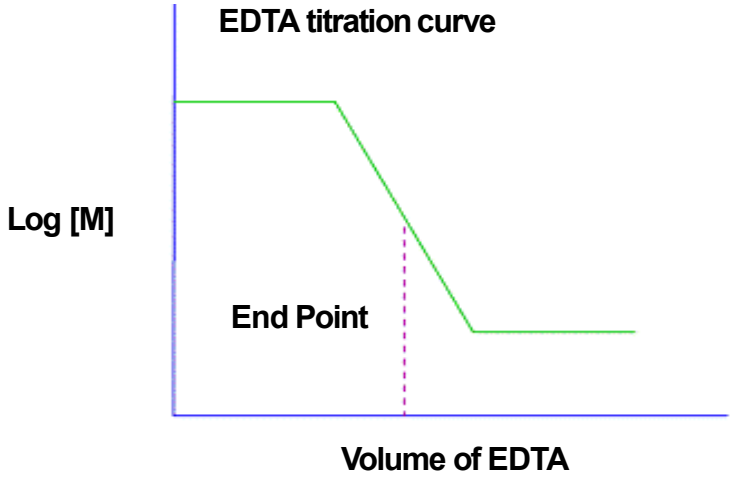
Analytical Chemistry
Chemical Analysis-VII

Classical Analytical Methods

Indicators used in Complexometric titrations

- **Indicators are ligands too**
- **Stability of Complexes**
- **pH of reaction**

Classical Analytical Methods



Classical Analytical Methods

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2.	Murexide or Ammonium purpurate	Violet to Blue	12	Ca, Cu, Co
3.	Catechol-violet	Violet to Red	8-10	Mn, Mg, Fe, Co, Pb
4.	Methyl Blue	Blue to Yellow	4-5	Pb, Zn, Cd, Hg
	Thymol Blue	Blue to Grey	10-12	
5.	Alizarin	Red to Yellow	4.3	Pb, Zn, Co, Mg, Cu
6.	Sodium Alizarin sulphonate	Blue to Red	4	Al, Thorium
7.	Xylenol range	Lemon to Yellow	1-3	Bi, Thorium
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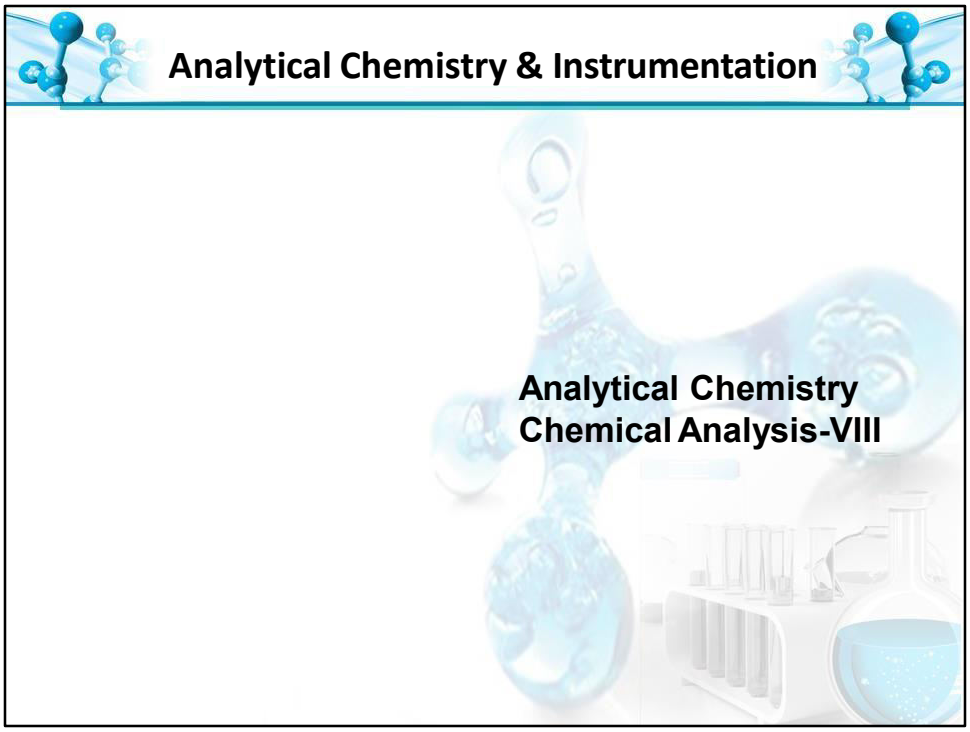
$$M^{+n} + H_2X^{-2} \rightarrow MX^{(n-4)} + 2H^{+}$$

e.g. - Barbiturates do not react with EDTA but are quantitatively precipitated from alkaline solution by mercuric ions as 1:1 complex.

END

Indirect titration

- Barbiturate to be analyzed is taken in a flask and heated with excess of mercury in alkaline solution.
- Hg-barbiturate complex is ppt, filtered and dissolved in excess of standard EDTA solution.
- Excess EDTA is back titrated with a standard Zn solution.



Here we are discussing last four technical clauses …..

Classical Analytical Methods

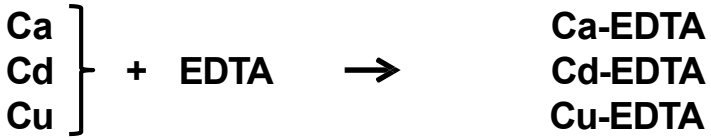
Masking and de-masking agents

- **Masking by precipitation**
- **Masking by complex formation**

Classical Analytical Methods

By using masking and demasking agents in complexometry, analysis of 3 metals, Cu, Cd and Ca can be done by following method

Step 1: All three metals are titrated.



Classical Analytical Methods

Step 2: Only Ca is titrated.

Cu + Cyanide ion → Cu-Cyanide complex
Cd + Cyanide ion → Cd-Cyanide complex
Ca + Cyanide ion → No reaction
Ca + EDTA → Ca-EDTA

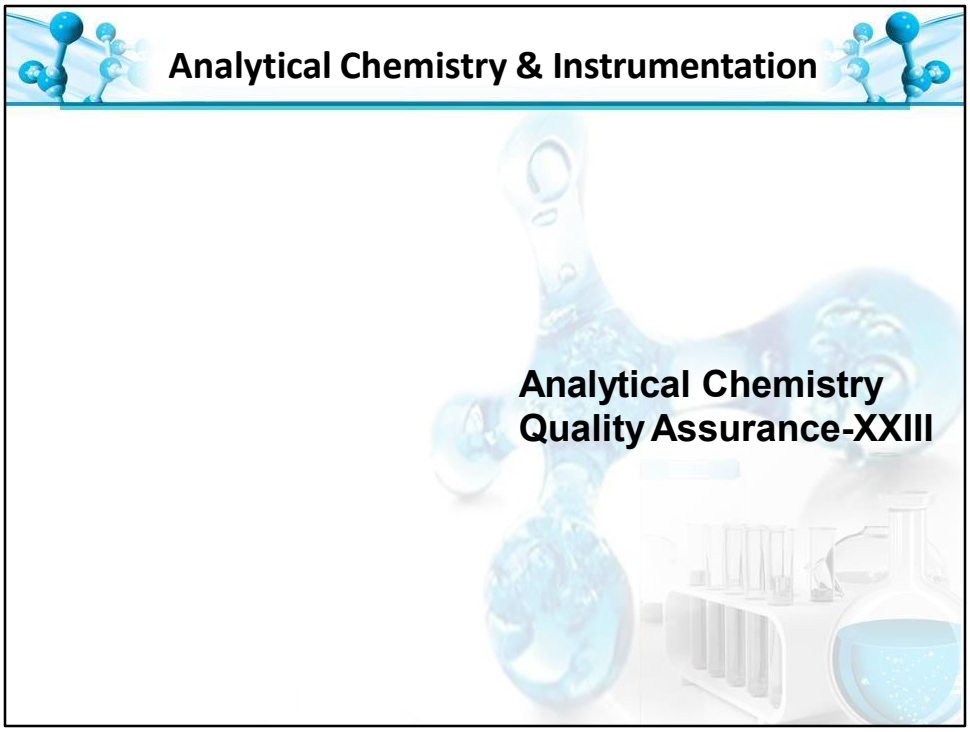
Step 3: Cd and Ca are titrated.

Cd-Cyanide complex + HCHO → Cd
Demasking agent Free

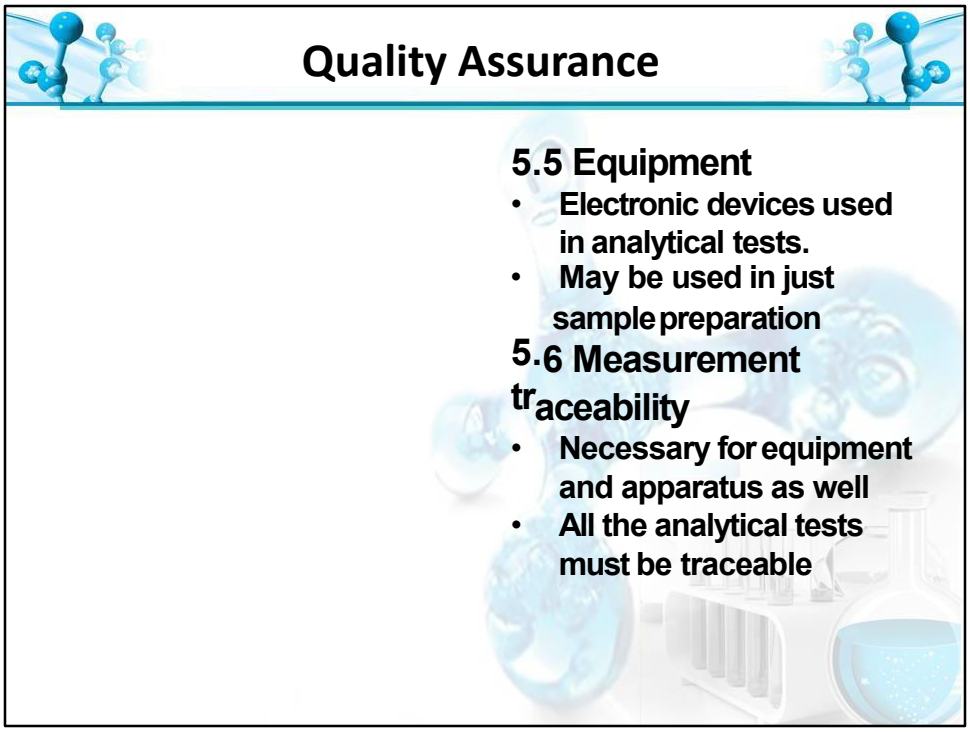
END

Titration Selectivity, Masking and Demasking Agents

- All three estimated in step 1 (A)
- Ca estimated in step 2 (B)
- Cd and Ca estimated in step 3 (C)
- Cd estimated = C- B
- Cu estimated = A - C



The next two clauses are equipment and measurement traceability….



What they cover….

Quality Assurance

5.5 Equipment

- Must cover all equipment that are used for tests. Allows use of equipment outside permanent control of lab
- Calibration must be scheduled
- Operated by authorized persons
- Maintenance and storage conditions
- Equipment labelled
- Safeguarded from adjustments which could invalidate it

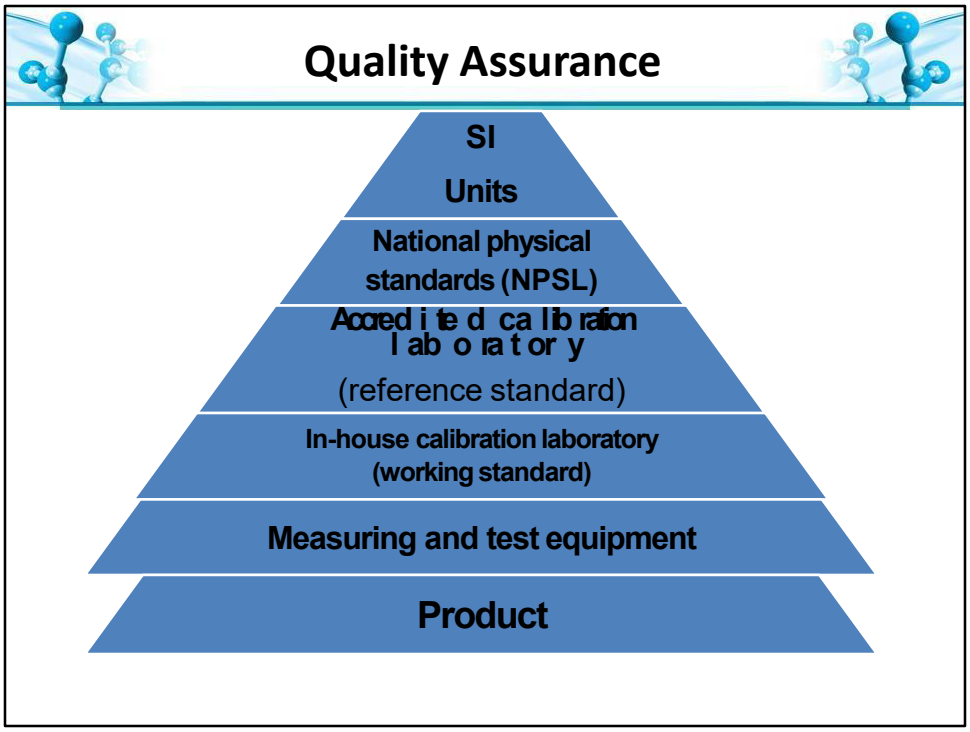
Sub clauses of equipment….

Quality Assurance

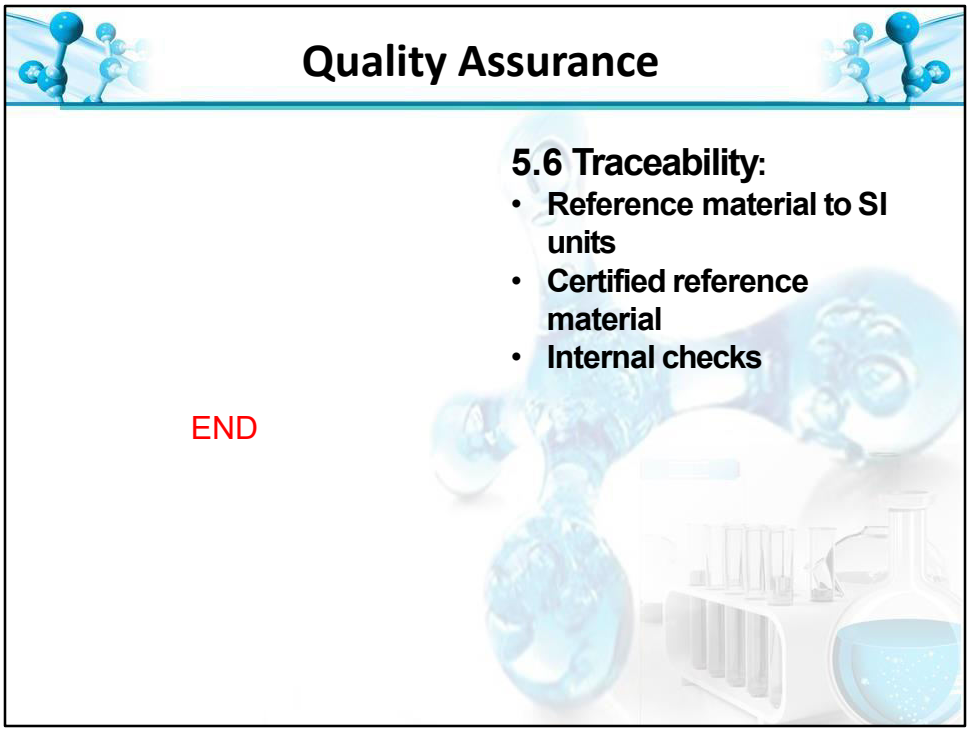
5.6 Measurement traceability

- **Calibrated before use**
- **Calibration traceable to SI System**
- **Calibration laboratories unbroken link to Primary Standards/Secondary Standards**
- **Testing laboratories: Reference Standards and Reference Materials, Calibration of Ref. Standards**
-

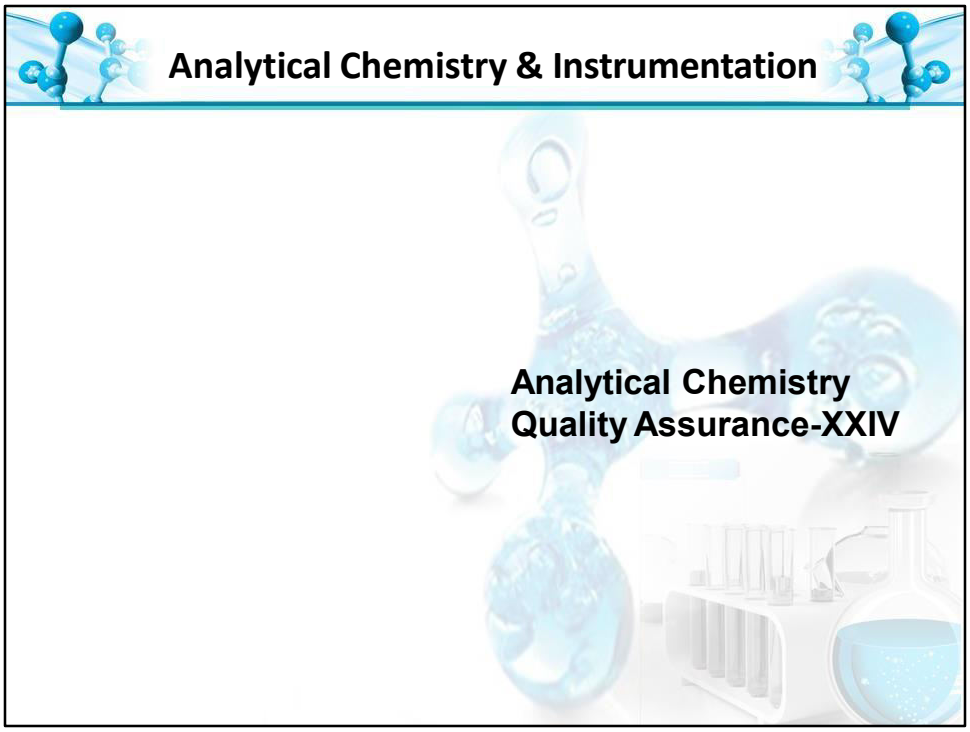
Traceability ……



Meaning of trceability…..



If not trceable to SI units we may establish data or objective evidence of analysis of ….



Here we are discussing last four technical clauses …..

Quality Assurance

7. Sampling
8. Handling of test and calibration items
9. Assuring the test and calibration results
10. Reporting of results

Quality Assurance

5.7 Sampling

- Sampling plan and procedures, when it carries out
- Sampling at location, where done.
- Based on statistical methods (where available)

5.8 Handling of test and calibration items

- Procedures for transport, receipt, handling, protection, storage, disposal
- Identifying test items

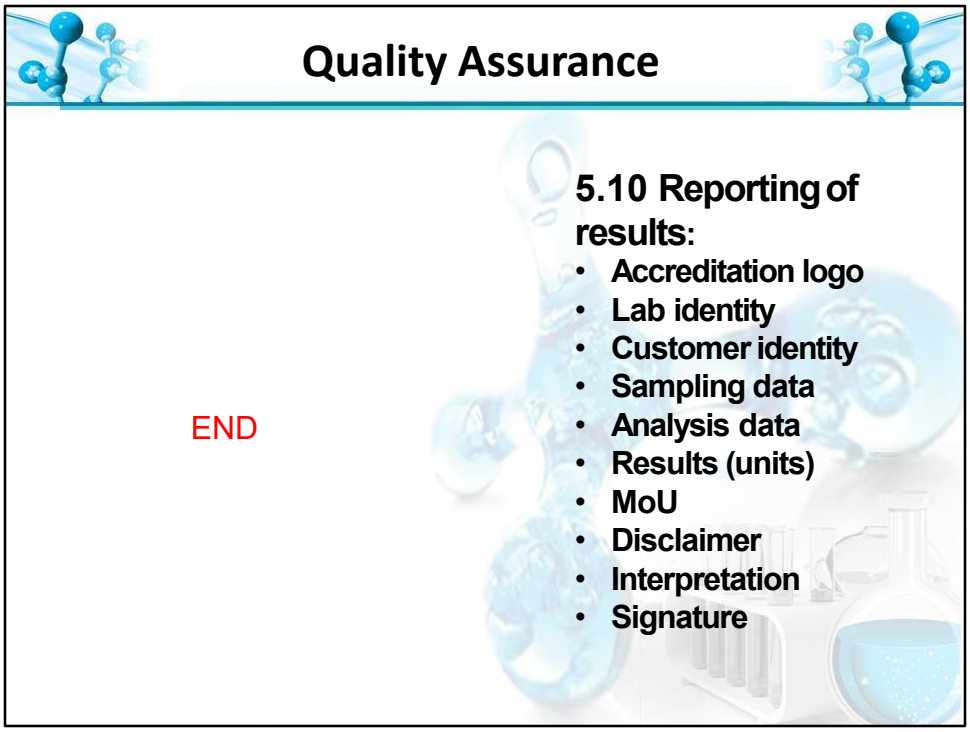
Sub clauses ….

Quality Assurance

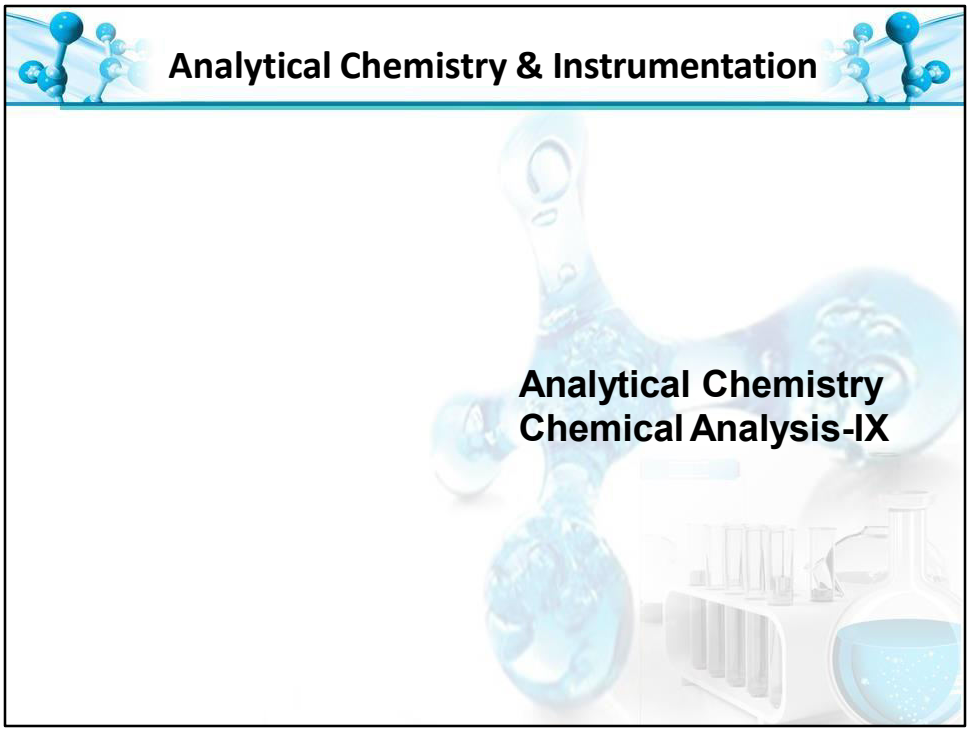
5.9 Assuring the testing and calibration results

- regular use of CRM and/or internal quality control (secondary reference materials)
- participation in inter laboratory comparison or proficiency-testing programs
- replicate tests or calibrations using the same or different methods
- retesting or recalibration of retained items
- correlation of results for different characteristics of an item.

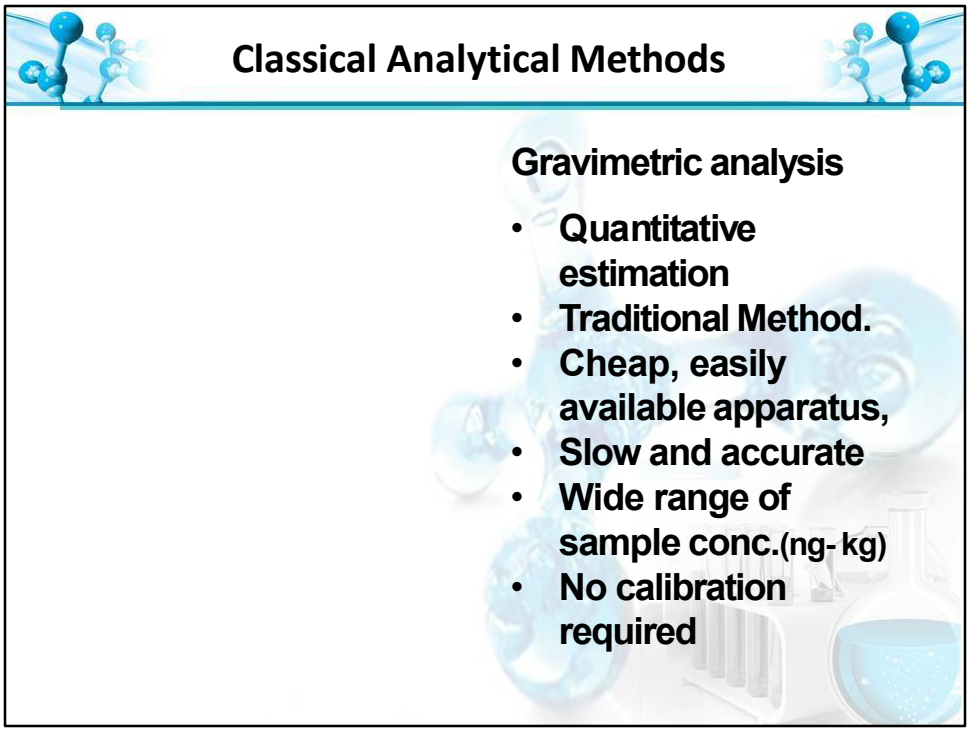
An important technical clause of ISO 17025 is …



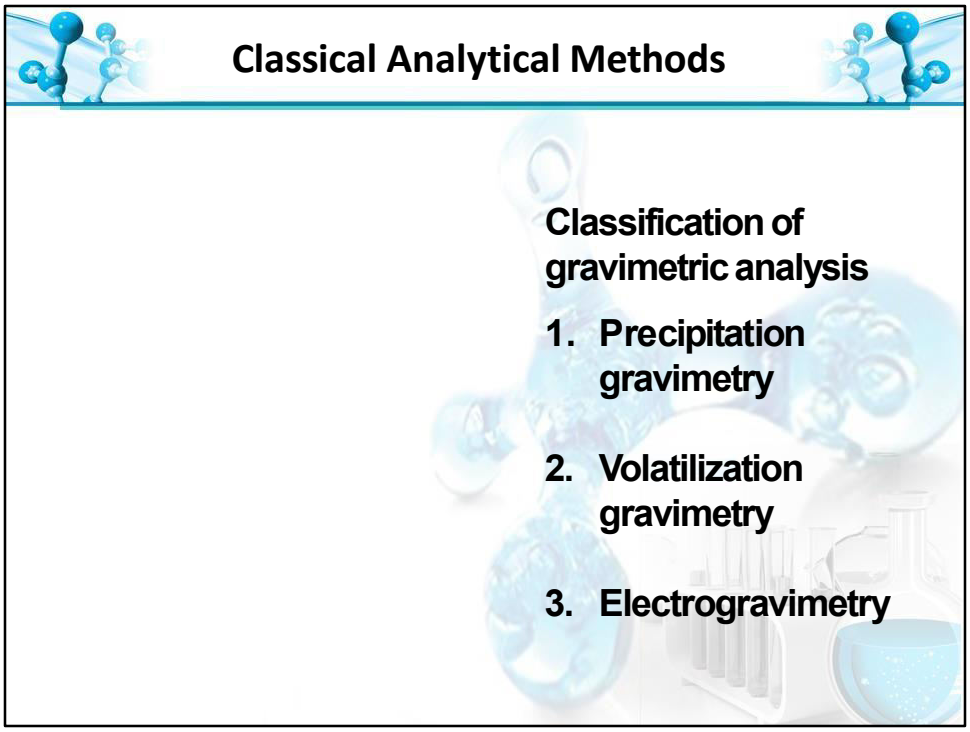
Reporting of analytical results…



Another chemical analytical technique is gravimetric analysis



Gravimetric methods are quantitative methods that are based on determining the mass of a pure compound to which the analyte is chemically related.



Precipitation gravimetry, the analyte is separated from a solution of the sample as a precipitate and is converted to a compound of known composition that can be weighed.

Volatilization gravimetry, the analyte is separated from other constituents of a sample by conversion to a gas of known chemical composition. The weight of this gas then serves as a measure of the analyte concentration.

Electrogravimetry, the analyte is separated by deposition on an electrode by an electrical current. The mass of this product then provides a measure of the analyte concentration.

Precipitation Gravimetry

- The analyte is converted to a sparingly soluble precipitate which are then filtered, washed (free of impurities), converted to a product of known composition by suitable heat treatment, and weighed.

- For example, a precipitation method for determining calcium in natural waters. The reactions are:

$$2NH_3 + H_2C_2O_4 \rightarrow 2NH^{4+} + C_2O_4^{-2}$$

$$Ca^{+2}_{(aq)} + C_2O_4^{-2}{}_{(aq)} \rightarrow CaC_2O_{4\,(s)}$$

Precipitation Gravimetry (Continued)

- The precipitates are filtered using a weighed filtering crucible, then dried and ignited to converts the precipitate to calcium oxide. The reaction is:

$$CaC_2O_{4\,(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_{(g)} + CO_2$$

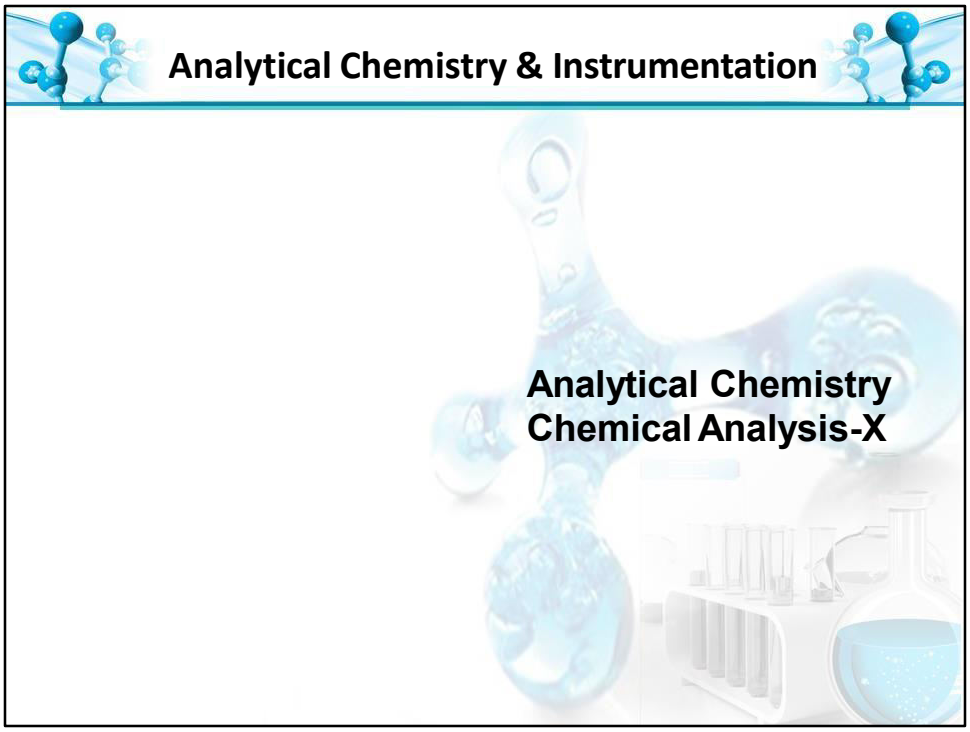
- After cooling, the crucible and precipitate are weighed, and the mass of calcium oxide is determined by subtracting the known mass of the crucible. The calcium content of the sample is then computed.

Classical Analytical Methods

END

Steps required in gravimetric analysis:

- **Preparation of the solution**
- **Precipitation**
- **Digestion**
- **Filtration**
- **Washing**
- **Drying or igniting**
- **Weighing**
- **Calculation**



Here we are discussing last four technical clauses …..

Classical Analytical Methods

Properties of good precipitates

- Should be filtered and washed easily
- Very low solubility (no significant loss during filtration / washing)
- Unreactive with constituents of the atmosphere
- Chemical composition must be known, if necessarily ignited.

Types of precipitates (on basis of particle size)

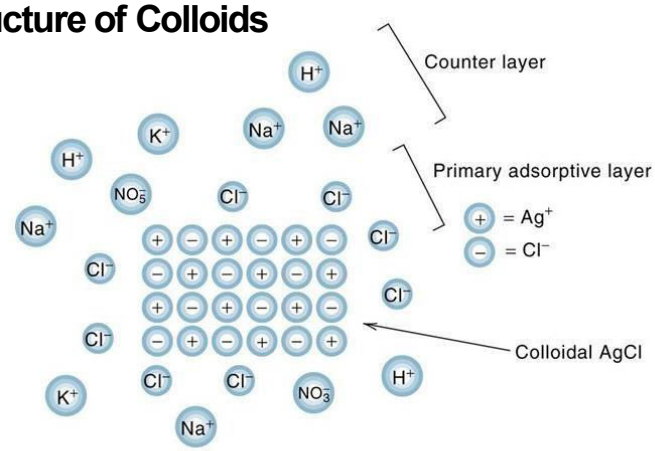
1-Colloidal suspensions

- Particles are invisible to the naked eye (10^{-7} - 10^{-4} cm in diameter).
- Colloidal particles show no tendency to settle from solution
- not easily filtered.

2-Crystalline suspension

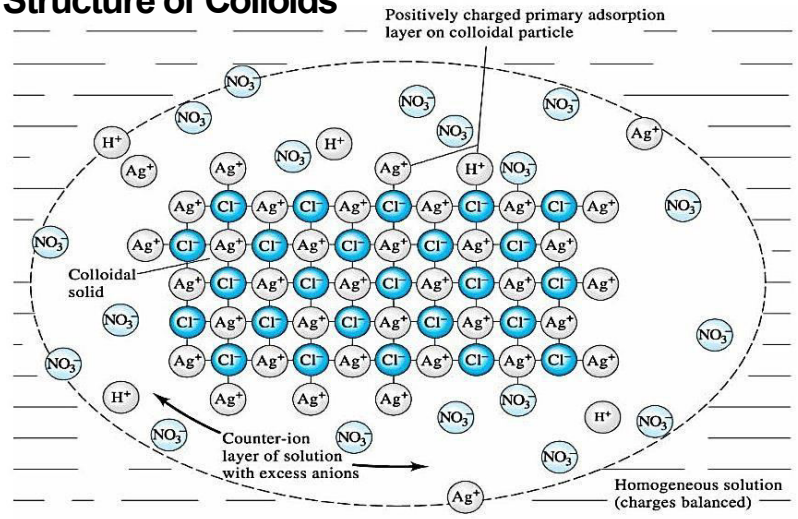
- particles with dimensions on the order of tenths of a millimeter or greater.
- The temporary dispersion of such particles of a tend to settle spontaneously,
- easily filtered.

Structure of Colloids



Representation of silver chloride colloidal particle and adsorptive layers when Cl^- is in excess.

Structure of Colloids

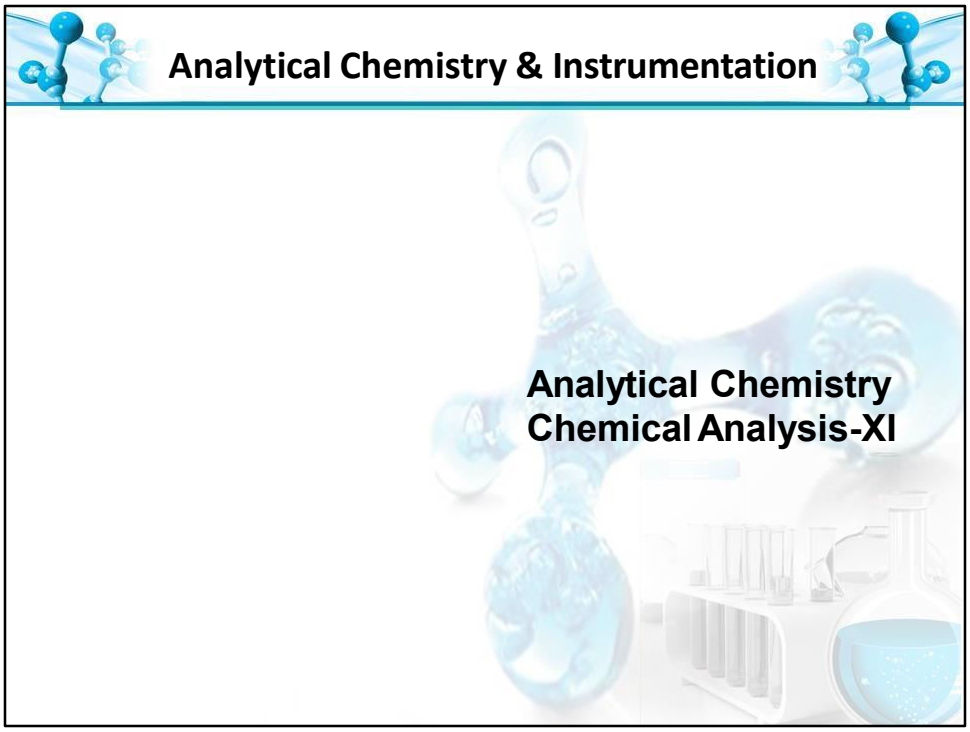


Classical Analytical Methods

END

Coagulation of Colloids:

- by short period of heating to decreases the number of adsorbed ions and thus the thickness of the double layer.
- Increase the electrolyte concentration of the solution
- In acidic or basic solution



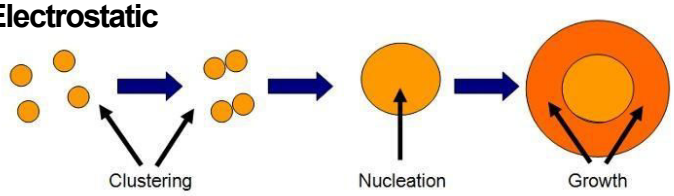
Here we are discussing last four technical clauses …..

Peptization of Colloids

- Peptization is the process by which a coagulated colloid reverts to its original dispersed state.
- Peptization can be avoided by washing with an electrolyte that can be volatized by heating.

Precipitation mechanism

1. Induction period.
2. Nucleation.
3. Particle growth to form larger crystal
4. Adsorption
5. Electrostatic



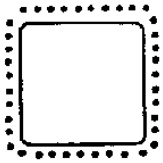
Note

- In nucleation, large number of small particles produced,
- If growth predominated, a smaller number of larger particles is produced

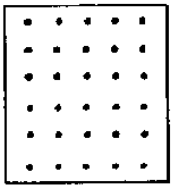
Classical Analytical Methods

Types of co-precipitation:

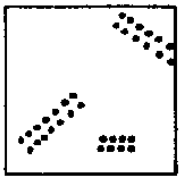
A. surface adsorption
B. mixed-crystal formation
C. occlusion
D. mechanical entrapment



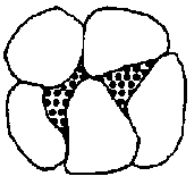
A



B



C



D

Classical Analytical Methods

END

Crystal Growth:

- Used for single crystal X-ray studies
- Separation of synthesized products
- Separation from impurities



Analytical Chemistry & Instrumentation



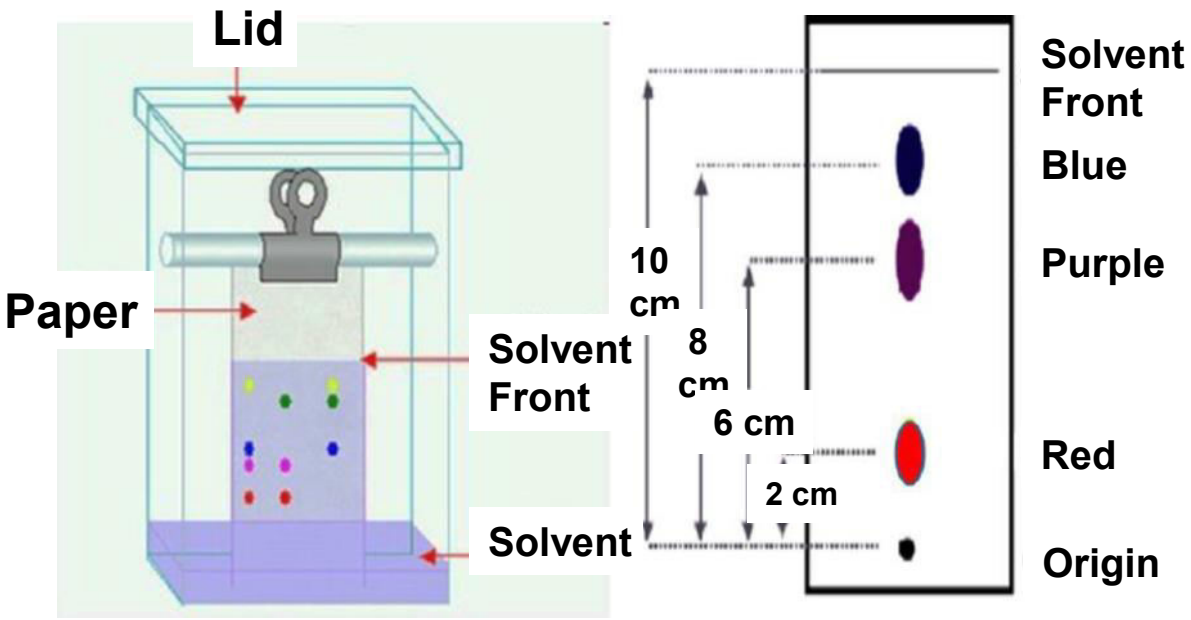
Instrumentation
Chromatography-III

Chromatography

Paper Chromatography

- **Stationary Phase used: Filter Paper, Cellulose**
- **Mobile Phase used: Water, water: alcohol or organic solvents**
- **Use chromatographic tank, beaker covered with petri dish, any jar with lid**

Chromatography



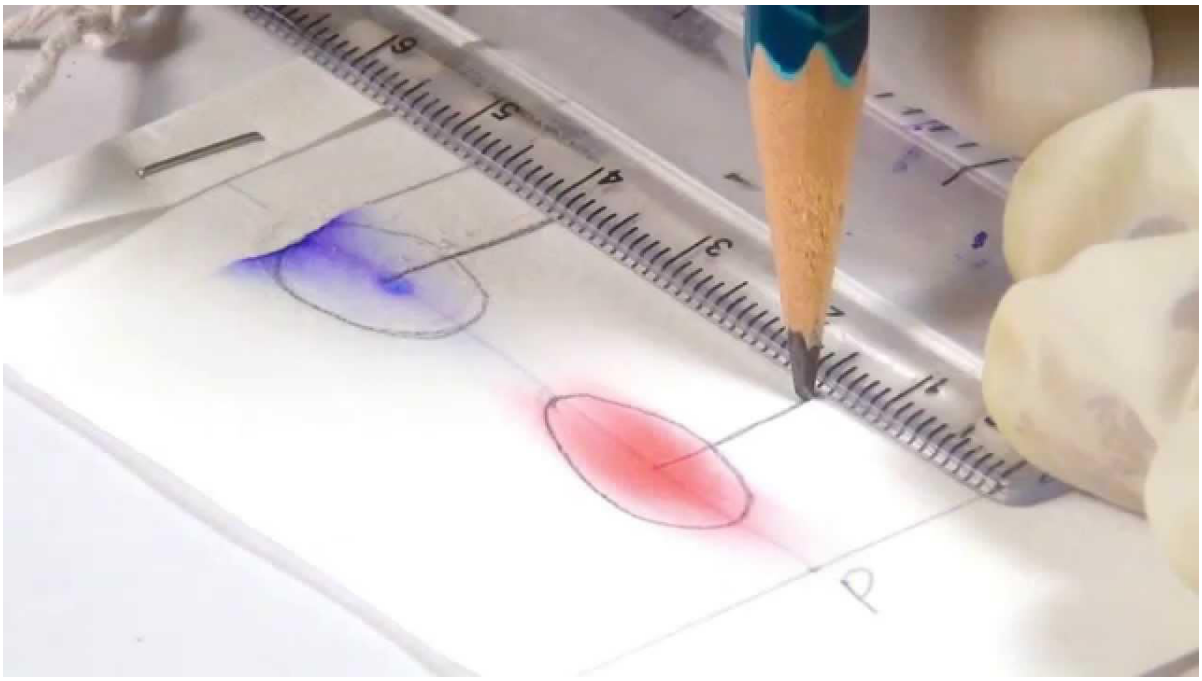
Chromatography

- **The moving components are depend on :**

  a. Solubility of solute in solvent
  b. Intermolecular forces
  c. Pore size of filter paper
  d. Size of solute molecule

- **With capillary action the solvent will move up to filter paper.**
- **At the end of process, components that are separated will emerge to different distance on filter paper.**
- **R_f values are used to identification of each the component.**

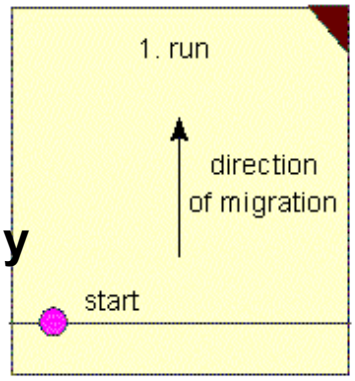
Chromatography

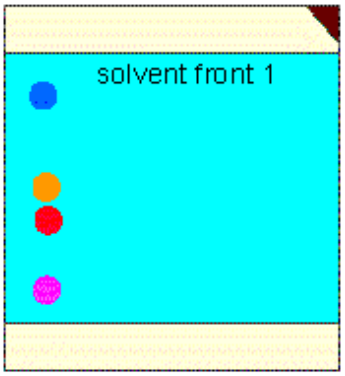


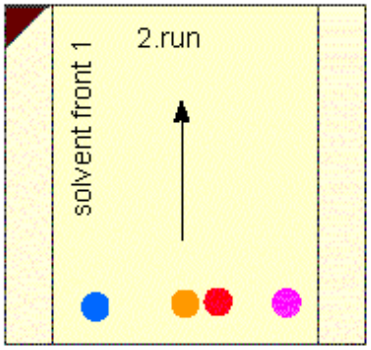
Chromatography

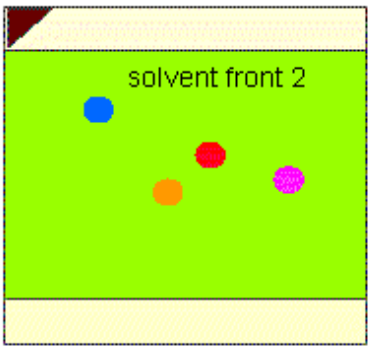
2 dimensional Chromatography

- Mark at one corner
- Start from one side
- Rotate at 45 degree

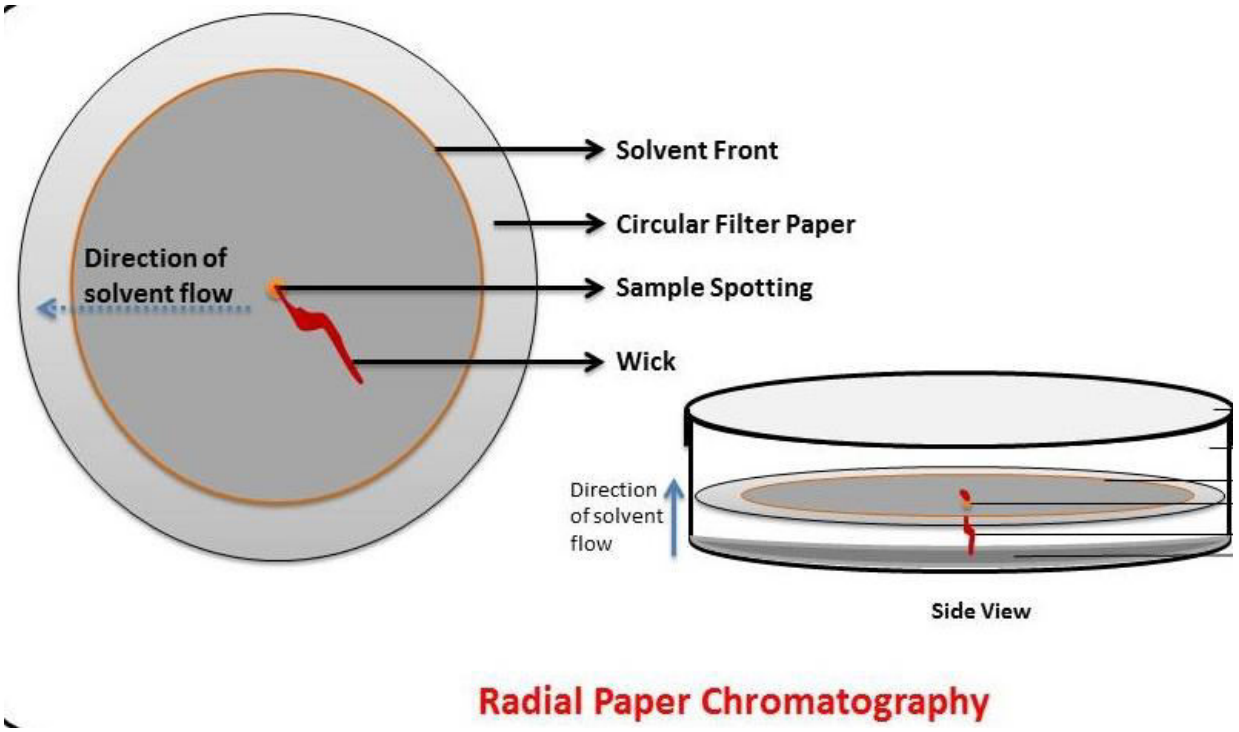








Chromatography



Radial Paper Chromatography

Chromatography

Paper Chromatography

- To analyze: Ink, steroids, food colours, pigments and drugs
- A qualitative technique to calculate and compare R_f values
- R_f is ratio between distance traveled by individual component divided by the distance travel by solvent

END

Final Term Syllabus
Lecture 01 to 64

Efforts By

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Contact us to join our esteemed group

“All About Biotech”

Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

“All About Biotech”

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